

Course Lecture Notes

James Guo

Fall 2025 & Spring 2026

Johns Hopkins University

This course develops the analytic foundations and core theory of partial differential equations, beginning with function spaces and functional analysis tools (weak derivatives, Sobolev and Hölder spaces, approximation/extension/trace theory, Sobolev inequalities and embeddings, compactness, and difference quotients connecting weak and classical derivatives). It then treats second-order linear elliptic PDEs via weak formulations, ellipticity, existence theorems, Fredholm theory, and eigenvalue problems, and builds a systematic regularity theory (interior, higher-order, and boundary regularity). From there it extends the linear framework to nonlinear settings, including second-order nonlinear/variational problems tied to Hilbert's XIX (with the minimal surface equation, existence/uniqueness of minimizers, and regularity results), and studies non-divergence form elliptic equations through harmonic function theory, maximum principles, the Hopf boundary point lemma, and a priori estimates. The course culminates with Schauder estimates (interior and global), continuity and Perron methods, Fredholm alternatives in non-divergence form, and quasilinear elliptic PDEs with applications again to minimal surface theory.

Table of Contents

1	Foundations for PDEs	1
I	Introduction	1
I.1	Partial Differential Equations	1
I.2	Function Spaces	2
I.3	Functional Analysis Preliminaries	4
2	Abstract Functional Spaces	8
II	Sobolev Space	8
II.1	Weak Derivative	8
II.2	Sobolev Space	11
II.3	Approximation of Sobolev Functions	14
III	Boundaries of Sobolev Space	18
III.1	Global Approximation of Sobolev Space	18
III.2	Extensions of Sobolev Functions	20
III.3	Trace of Sobolev Functions	22
III.4	Sobolev Inequalities	25
IV	Hölder Spaces	29
IV.1	Hölder Space	29
IV.2	Sobolev Embeddings	31

V	Compactness in Sobolev Space	37
V.1	Reflexivity	37
V.2	Weak Compactness	37
VI	Difference Quotients	41
VI.1	Difference Quotients	41
VI.2	Weak and Classical Derivatives	42
3	Second Order Linear PDEs	44
VII	Ellipticity and Existence	44
VII.1	Preliminaries and Weak Solutions	44
VII.2	Ellipticity	47
VII.3	The Existence Theorems	49
VII.4	Fredholm Theory	53
VII.5	Eigenvalues of Elliptic Operators	57
VIII	Regularity Theory	61
VIII.1	Intuitions on Regularity	61
VIII.2	Interior Regularity	61
VIII.3	Higher Interior Regularity	64
VIII.4	Boundary Regularity	66
IX	Extending from Second Order Linear PDEs to Nonlinear Cases	69
IX.1	Existence for a Nonlinear PDE	69
IX.2	Regularity for a Nonlinear PDE	71
4	Second Order Nonlinear PDEs	74
X	Hilbert's XIX-th Problem	74
X.1	Minimal Surface Equation	74
X.2	Existence and Uniqueness of Local Minimizer	75
X.3	Regularity for Local Minimizer	79
X.4	Further Results on Regularity	90
XI	Non-Divergence Forms	91
XI.1	Harmonic Functions	91
XI.2	Inequalities about Smooth Harmonic Functions	94
XI.3	The Maximum Principles	96
XI.4	Hopft Boundary Point	100
XI.5	A-Priori Estimates	103
XII	The Schauder Estimates	105
XII.1	Interior Schauder Estimates	105
XII.2	Boundary and Global Schauder Estimates	116

XII.3	The Continuity Method	119
XII.4	Perron's Method	123
XII.5	Fredholm Alternative for Non-divergence Form Elliptic Operators	129
XIII	Quasilinear Elliptic PDEs	132
XIII.1	From Elliptic to Quasilinear	132
XIII.2	Application to the Minimal Surface Equation	134
XIII.3	title	136

Acknowledgements:

- This document is part of the materials for AS.110.631–632 Partial Differential Equations sequence instructed by *Dr. Kobe Marshall-Stevens* at *Johns Hopkins University* in the 2025-26 school year.
- The document contains lecture contents, notes, and adapts contents from the following text:
 - *Partial Differential Equations* by Lawrence C. Evans.
 - *Regularity Theory for Elliptic PDE* by Xavier Fernández-Real and Xavier Ros-Oton.
 - *Elliptic Partial Differential Equations of Second Order* by David Gilbarg and Neil S. Trudinger.
 - *Schauder Estimates by Scaling* by Leon Simon.
- The document might contain minor typos or errors. Please point out any notable error(s).

Part 1

Foundations for PDEs

I Introduction

I.1 Partial Differential Equations

Definition I.1.1. Partial Differential Equation.

A **partial differential equation** (PDE) of order $k \in \mathbb{N} := \{0, 1, 2, \dots\}$ is an expression of the form:

$$F(x, u(x), Du(x), \dots, D^k u(x)) = 0, \quad (1)$$

where $x \in \Omega$, $\Omega \subset \mathbb{R}^n$ is open, $u : \Omega \rightarrow \mathbb{R}$, and $F : \Omega \times \mathbb{R} \times \mathbb{R}^n \times \dots \times \mathbb{R}^{n^k} \rightarrow \mathbb{R}$, since $D^m u(x)$ is the m -th derivative of $u(x)$ and is of dimension n^m .

Here, u is unknown. We solve the PDE (1) if we find all such $u : \Omega \rightarrow \mathbb{R}$. ┘

Remark I.1.2. Systems of PDE.

We can define **systems of PDE** by considering $u : \Omega \rightarrow \mathbb{R}^M$ for some $M \in \mathbb{N}^+$. ┘

Then, we give some examples of PDEs:

Example I.1.3.

- The **Possion equation** is:

$$-\Delta u = -\sum_{i=1}^n \frac{\partial^2 u}{\partial x_i^2} = f,$$

where the **Laplace equation** is when $f \equiv 0$.

- The **Heat equation** is:

$$\frac{\partial u}{\partial t} - \Delta u = 0,$$

which could be called a parabolic equation.

- The **Allen-Cahn equation** is:

$$\Delta u = u(u^2 - 1),$$

and we can note that this equation is nonlinear.

- The **Minimal Surface equation** is:

$$\operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) = 0,$$

which involves second order derivatives and nonlinear case. ◇

For this course, we use **multi-index** notion, *i.e.*, if $\alpha \in \mathbb{N}^n$ and $\alpha = (\alpha_1, \dots, \alpha_n)$, we write:

$$D^\alpha u = \frac{D^{|\alpha|} u}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}, \quad \text{where } |\alpha| = \sum_{i=1}^n \alpha_i.$$

Note that D^α here is an operator that acts on certain classes of functions.

Then, we classify PDEs in several classes:

Definition I.1.4. We say that a PDE is:

- **Linear** if it is of the form:

$$\sum_{|\alpha| \leq k} \underbrace{a_\alpha(x)}_{\text{coefficient}} D^\alpha u(x) = f(x) \quad \text{for given } a_\alpha, f : \Omega \rightarrow \mathbb{R}.$$

If $f \equiv 0$, we call the PDE **homogeneous**.

- **Semi-linear** if it is of the form:

$$\sum_{|\alpha|=k} a_\alpha(x) D^\alpha u(x) + a_0(x, u(x), \dots, D^{k-1} u(x)) = 0 \quad \text{for } a_\alpha, a_0 : \Omega \rightarrow \mathbb{R}.$$

- **Quasi-linear** if it is of the form:

$$\sum_{|\alpha|=k} a_\alpha(x, u(x), \dots, D^{k-1} u(x)) D^\alpha u(x) + a_0(x, u(x), \dots, D^{k-1} u(x)) = 0 \quad \text{for } a_\alpha, a_0 : \Omega \rightarrow \mathbb{R}.$$

- **Fully non-linear** if it depends non-linearly on its u -th derivatives, *e.g.*, $\left(\frac{\partial^2 u}{\partial x \partial y}\right)^2 = 0$. J

I.2 Function Spaces

Definition I.2.1. Let $\Omega \subset \mathbb{R}^n$ to be open, $k \in \mathbb{N}$, $p \in [1, \infty]$, we denote:

- $C^k(\Omega) = \{u : \Omega \rightarrow \mathbb{R} \mid \text{all partial derivatives of order } \leq k \text{ exist and are continuous}\}.$
- $C^k(\overline{\Omega}) = \{u \in C^k(\Omega) \mid \text{all partial derivatives of order } \leq k \text{ extend continuously to } \overline{\Omega}\}.$
- $C^\infty(\Omega) = \bigcap_{\ell \in \mathbb{N}} C^\ell(\Omega)$ and $C^\infty(\overline{\Omega}) = \bigcap_{\ell \in \mathbb{N}} C^\ell(\overline{\Omega})$.
- $C_c^k(\Omega) = \{u \in C^k(\Omega) \mid \overline{\{u \neq 0\}} \text{ is compact in } \Omega\}.$
- $C_c^\infty(\Omega) = \bigcap_{\ell \in \mathbb{N}} C_c^\ell(\Omega)$ is the class of test functions.

In fact, making the functions to be differential is very strong, and we need to have weaker condition.

- $L^p(\Omega) = \{u : \Omega \rightarrow \mathbb{R} \mid u \text{ is Lebesgue measurable and } \|u\|_{L^p(\Omega)} < \infty\} / \sim$, where:

$$\|u\|_{L^p(\Omega)} = \begin{cases} \left(\int_{\Omega} |u|^p \right)^{1/p} & \text{for } p \in [1, \infty), \\ \text{ess sup}_{\Omega} |u| = \inf\{s \in \mathbb{R} \mid |\{ |u| > s \}| = 0\} & \text{for } p = \infty. \end{cases}$$

and $u \sim v$ if and only if $u = v$ almost everywhere.

- $L^p_{\text{loc}}(\Omega) = \bigcap_{\tilde{\Omega} \Subset \Omega} L^p(\tilde{\Omega})$, where $\tilde{\Omega} \Subset \Omega$ if and only if there is a compact set $K \subset \Omega$ such that $\tilde{\Omega} \subset K \subset \Omega$. ┘

Example I.2.2. $\sqrt{x} \in C^\infty((0, \infty))$, $\sqrt{x} \in C^0([0, \infty))$, but $\sqrt{x} \notin C^1([0, \infty))$.

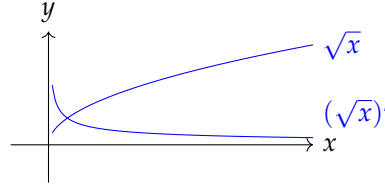


Figure I.1. Graph of $\sqrt{x}$ and its derivative.

Note that the equation $(\sqrt{x})'$ extend smoothly to 0^+ , but the first derivative of $\sqrt{x}$ ($1/2\sqrt{x}$) is not extending to 0^+ smoothly. ◇

By setting $\|u\|_{C^0(\bar{\Omega})} = \sup_{\Omega} |u|$ (supremum norm) and $\|u\|_{C^k(\bar{\Omega})} = \sum_{|\alpha| \leq k} \|D^\alpha u\|_{C^0(\bar{\Omega})}$ (C^k -norm), using the notation $\|D^\alpha u\|_* = \|D^\alpha u\|_{C^0(\bar{\Omega})}$.

Then, $(C^k(\bar{\Omega}), \|\cdot\|_{C^k(\bar{\Omega})})$ is Banach if Ω is bounded or we restrict to functions with bounded C^k -norm.

For $p \in [1, \infty]$, $(L^p(\Omega), \|\cdot\|_{L^p(\Omega)})$ is Banach and for $p = 2$, by setting the L^2 -inner product to be:

$$(u, v)_{L^2(\Omega)} = \int_{\Omega} u \cdot v \quad \text{for } u, v \in L^2(\Omega).$$

Specifically, $\|u\|_{L^2(\Omega)}^2 = (u, u)_{L^2(\Omega)}$, and $(L^2(\Omega), (\cdot, \cdot)_{L^2(\Omega)})$ is Hilbert.

Lemma I.2.3. Hölder Inequality.

For $p, q \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$, then if $\Omega \subset \mathbb{R}^n$ is open, $u \in L^p(\Omega)$, $v \in L^q(\Omega)$, we have:

$$\|u \cdot v\|_{L^1(\Omega)} \leq \|u\|_{L^p(\Omega)} \cdot \|v\|_{L^q(\Omega)}.$$

When $p = q = 2$, this is the **Cauchy-Schwartz inequality**.

The *proof* of this could utilize the **Young's inequality** ($ab \leq \frac{a^p}{p} + \frac{b^q}{q}$).

Lemma I.2.4. Fundamental Lemma of Calculus of Variation (FLCV).

Suppose $\Omega \subset \mathbb{R}^n$ is open and $v \in L^1_{\text{loc}}(\Omega)$, then:

$$v = 0 \text{ almost everywhere in } \Omega \iff \int_{\Omega} v \cdot \varphi = 0 \text{ for every } \varphi \in C_c^\infty(\Omega).$$

The *proof* would be requiring the *mollifiers*.

Example I.2.5. Consider the following functions:

- $u(x) = e^x$ on $\mathbb{R}$, we have $u \in C^\infty(\mathbb{R})$, $\|u\|_{C^0(\mathbb{R})} = \infty$, $\|u\|_{L^p(\mathbb{R})} = \infty$, but $u \in L^1_{\text{loc}}(\mathbb{R})$.
- $v(x) = \begin{cases} 1+x, & x < 0, \\ 1-x, & x \geq 0 \end{cases}$, which can be visualized as follows:

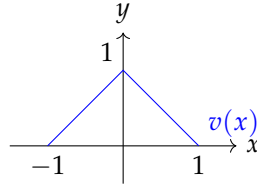


Figure I.2. Graph of $v(x)$.

we have $v \in C^0((-1, 1) \setminus (\bigcup_{k \geq 1} ((-1, 1))))$, $\|v\|_{C^0((-1, 1))} = 1$, and $\|v\|_{L^p((-1, 1))} < \infty$.

- Now, consider $\epsilon > 0$ to be small and fixed, with the following functions:

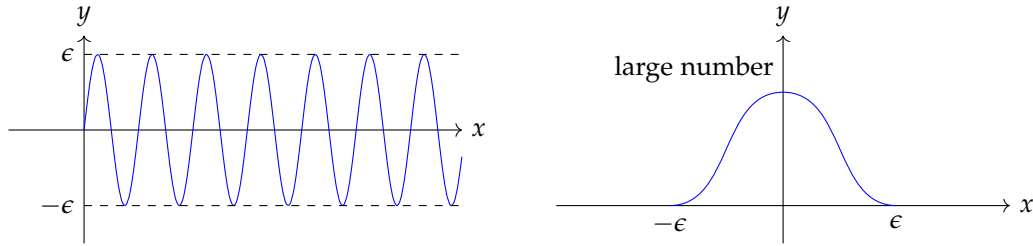


Figure I.3. Graph of two example functions.

The left function has small $\|\cdot\|_{C^0}$ and large $\|\cdot\|_{C^1}$, whereas the right function has small $\|\cdot\|_{L^p}$ and large $\|\cdot\|_{C^0}$ or $\|\cdot\|_{C^1}$ $\diamond$

I.3 Functional Analysis Preliminaries

Recall that a **Banach space** is a complete “normed vector space.”

- Some examples include $(\mathbb{R}, |\cdot|)$ or $(L^p, \|\cdot\|_{L^p})$.

A **Hilbert space** is a complete “inner product space.” A Hilbert space is a Banach space.

- Some examples include $(\mathbb{R}^n, \cdot)$, in which $\cdot(a, b) = \sum_{i=1}^n a_i b_i$, or $(L^2, (\cdot, \cdot)_{L^2})$, in which $(u, v)_{L^2} = \int uv$.

Definition I.3.1. $L : (X, \|\cdot\|_X) \rightarrow (Y, \|\cdot\|_Y)$ is a bounded linear map between Banach spaces X and Y if:

$$\|L\| = \sup_{\|x\|_X \leq 1} \{\|L(x)\|_Y\} < \infty.$$

┘

Proposition I.3.2. A linear map is bounded if and only if it is continuous.

Definition I.3.3. Dual Space.

The **dual space**, denoted $(X^*, \|\cdot\|_{X^*})$, endowed with the operator norm, of $(X, \|\cdot\|_X)$ is the collection of bounded linear maps $L : (X, \|\cdot\|_X) \rightarrow (\mathbb{R}, |\cdot|)$. $\lrcorner$

Remark I.3.4.

- If $\dim(X) < \infty$, this (continuous) dual space is the algebraic dual of X . But every infinite dimensional Banach space has unbounded linear maps.
- Radon measures are dual space to $(C_c^0(\Omega), \|\cdot\|_{C^0(\Omega)})$. $\lrcorner$

Proposition I.3.5. If $\Omega \subset \mathbb{R}^n$ is open and $p \in [1, \infty)$, then:

$$(L^p(\Omega))^* = L^q(\Omega), \quad \text{where } \frac{1}{p} + \frac{1}{q} = 1.$$

This “=” is an isomorphism, towing $v \in L^q(\Omega)$ to the linear map $L_v : L^p(\Omega) \rightarrow \mathbb{R}$ defined by:

$$L_v(u) = \int_{\Omega} u \cdot v \quad \text{for } u \in L^p(\Omega).$$

Note that L_v is bounded since by **Lemma I.2.3** Hölder inequality that:

$$|L_v(u)| = \left| \int_{\Omega} u \cdot v \right| \leq \int_{\Omega} |uv| \leq \|v\|_{L^p(\Omega)} \cdot \|v\|_{L^q(\Omega)},$$

and hence we have:

$$\|L_v\| \leq \|v\|_{L^1(\Omega)}.$$

Remark I.3.6. Note that $L^1(\Omega) \subsetneq (L^\infty(\Omega))^*$. $\lrcorner$

Now, consider the restriction to Hilbert space.

Theorem I.3.7. Riesz Representation Theorem.

Let $(\mathcal{H}, (\cdot, \cdot))$ be a Hilbert space, then for each $L \in \mathcal{H}^*$ there is a unique $u \in \mathcal{H}$ such that:

$$L(v) = (u, v) \quad \text{for any } v \in \mathcal{H}.$$

Moreover, $\ker L := \{v \in \mathcal{H} \mid L(v) = 0\} = u^\perp = \{v \in \mathcal{H} \mid (u, v) = 0\}$.

Example I.3.8. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is linear, $f \in (\mathbb{R}^n)^*$. Consider $f(x) = a \cdot x$ for each $x \in \mathbb{R}^n$ and $a \in \mathbb{R}^n$. $\diamond$

Now, we consider a blueprint for solving PDEs.

Example I.3.9. Poisson Equation.

Let $B \subset \mathbb{R}^n$ be the unit open ball and $f \in C^\infty(\overline{B})$ be given. We want to solve the Poisson's equation on B with zero (or Dirichlet condition) on boundary.

Hence, we seek $u \in C^\infty(\overline{B})$ which solves the PDE:

$$\begin{cases} -\Delta u = f, & \text{on } B, \\ u = 0, & \text{on } \partial B. \end{cases}$$

- (i) Reformulation of the equation. If we solved the equation then for each $\varphi \in C_c^\infty(B)$, we would have:

$$\int_B f \cdot \varphi = - \int_B \Delta u \cdot \varphi = \int_B \nabla u \cdot \nabla \varphi,$$

since we have:

$$\operatorname{div}(\varphi \nabla u) = \nabla u \cdot \nabla \varphi + \varphi \cdot \Delta u,$$

and $\varphi|_{\partial B} = 0$.

Let's instead restrict to finding $u \in C^\infty(\overline{B})$ solving:

$$\begin{cases} \int_B f \varphi = \int_B \nabla u \cdot \nabla \varphi & \text{for all } \varphi \in C_c^\infty(B), \\ u = 0 & \text{on } \partial B. \end{cases} \quad (2)$$

But then we turned the original equation into (2), since if it holds, then for any $\varphi \in C_c^\infty(B)$, we have:

$$\int_B f \varphi - \int_B \nabla v \cdot \nabla \varphi = \int_B (f + \Delta v) \varphi = 0,$$

thus by [Lemma I.2.4](#), we have:

$$f + \Delta v = 0 \text{ on } B \text{ almost everywhere.}$$

Note that we only need 1 derivative of u for (2) to make sense, even if f and ∇u are only L^p functions, *i.e.*, defined almost everywhere.

- (ii) Hilbert space formulation of (2). Then we try to find some function u solving the integral equality in (2). We define on $C_c^\infty(B)$ the H_0^1 inner product as:

$$\langle u, v \rangle_{H_0^1(B)} = \int_B \nabla u \cdot \nabla v \quad \text{for } u, v \in C_c^\infty(B).$$

This inner product induces the H_0^1 norm as:

$$\|v\|_{H_0^1(B)}^2 = \langle v, v \rangle_{H_0^1(B)} = \|\nabla v\|_{L^2(B)}^2.$$

If we define, for $f \in L^2(B)$, the linear map $L_f : (C_c^\infty(B), \|\cdot\|_{H_0^1(B)}) \rightarrow \mathbb{R}$ by setting:

$$L_f(\varphi) = \int_B f \cdot \varphi.$$

By Hölder's inequality (Lemma I.2.3), we have:

$$|L_f(\varphi)| = \left| \int_B f \varphi \right| \leq \int_V |f \varphi| \leq \|f\|_{L^2(B)} \cdot \|\varphi\|_{L^2(B)}.$$

Now, if we had an estimate of the form:

$$\|\varphi\|_{L^2(B)} \leq C_p \|\varphi\|_{H_0^1(B)},$$

which is known as the Poincaré inequality, then:

$$|L_f(\varphi)| \leq \|f\|_{L^2(B)} \cdot \|\varphi\|_{L^2(B)} \leq \|f\|_{L^2(B)} \cdot c C_p \|\varphi\|_{H_0^1(B)} \leq C_p \|f\|_{L^2(B)}.$$

Hence L_f is bounded.

However, $(C_c^\infty(B), \langle \cdot, \cdot \rangle_{H_0^1(B)})$ is not a Hilbert space, so we cannot apply Riesz representation.

If it were a Hilbert space, Riesz would give a unique $u \in ?$ such that:

$$\int_B f \varphi = L_f(\varphi) = \langle u, \varphi \rangle_{H_0^1(B)} = \int_B \nabla u \cdot \nabla \varphi.$$

To get around, this, we take the Cauchy completion with respect to $\|\cdot\|_{H_0^1(B)}$ to get a Hilbert space as:

$$(H_0^1(B), \langle \cdot, \cdot \rangle_{H_0^1(B)}),$$

which turns out as a **Sobolev Space**. Hence, we would have Riesz giving a *weak solution* $u \in H_0^1(B)$:

$$\int_B f \varphi = L_f(\varphi) = \langle u, \varphi \rangle_{H_0^1(B)} = \int_B \nabla u \cdot \nabla \varphi.$$

(iii) Regularity. Given a weak solution $u \in H_0^1(B)$ to the Poisson equation, it is equivalent to that:

$$\int_B f \varphi = \int_B \nabla v \cdot \nabla \varphi \quad \text{for every } \varphi \in C_c^\infty(B),$$

which made sense for $f \in L^2(B)$, and we will show that $f \in C^k(\overline{B})$ implies that $u \in C^{k+1}(\overline{B})$ by estimating the derivative of u by the derivatives of f . (This is stronger than $f \in C^\infty$ implies $u \in C^\infty$.)

◇

Part 2

Abstract Functional Spaces

II Sobolev Space

II.1 Weak Derivative

Throughout this course, we use the principle that “*strong theorems* become weak definition.”

Recall when we tried to solve the Poisson equation in [Example I.3.9](#), we used integration by parts to obtain:

$$\begin{cases} -\Delta u = f \text{ on } B \\ u = 0 \text{ on } \partial B \end{cases} \iff \begin{cases} \int \nabla u \cdot \nabla \varphi = \int f \varphi \text{ for all } \varphi \in C_c^\infty(B) \text{ on } B \\ u = 0 \text{ on } \partial B \end{cases}$$

Then, we will expand the concept of integration by parts as weak derivatives.

Definition II.1.1. Weak Derivatives.

If $\Omega \subset \mathbb{R}^n$ is open, $u \in L^1_{\text{loc}}(\Omega)$, and $\alpha \in \mathbb{N}^k$, then $v \in L^1_{\text{loc}}(\Omega)$ is a **weak derivative** for u corresponding to α if:

$$\int_{\Omega} u \cdot D^\alpha \varphi = (-1)^{|\alpha|} \int_{\Omega} v \varphi \quad \text{for all } \varphi \in C_c^\infty(\Omega).$$

We then write $D^\alpha u = v$ almost everywhere on Ω . ┘

In other words, we can say that integration by parts holds for u and v .

Remark II.1.2. If $u \in C^k(\Omega)$, then $D^\alpha u$, in the usual sense, is a weak derivative of u corresponding to α for $|\alpha| \leq k$. ┘

Lemma II.1.3. Properties of Weak Derivatives.

If $\Omega \subset \mathbb{R}^n$ is open, $\alpha \in \mathbb{N}^k$, and $u, \tilde{u} \in L^1_{\text{loc}}(\Omega)$, then:

- (i) Weak derivatives are unique almost everywhere, so $D^\alpha u$ is unambiguous as an $L^1_{\text{loc}}(\Omega)$ function.
- (ii) Taking weak derivatives is a linear operator, i.e.:

$$D^\alpha(u + \lambda \tilde{u}) = D^\alpha u + \lambda D^\alpha \tilde{u}.$$

Proof. (i) Suppose $v, \tilde{v}$ are weak derivative of u , by definition:

$$(-1)^{|\alpha|} \int_{\Omega} v \cdot \varphi = \int_{\Omega} u \cdot D^\alpha \varphi = (-1)^{|\alpha|} \int_{\Omega} \tilde{v} \cdot \varphi$$

for any test function $\varphi \in C_c^\infty(\Omega)$, so we have:

$$\int_{\Omega} v \cdot \varphi = \int_{\Omega} \tilde{v} \cdot \varphi \quad \text{for any } \varphi \in C_c^\infty(\Omega).$$

Hence, we equivalently have:

$$\int_{\Omega} (v - \tilde{v}) \cdot \varphi = 0 \quad \text{for any } \varphi \in C_c^\infty(\Omega),$$

and so by the fundamental lemma of calculus on variation ([Lemma I.2.4](#)), $v - \tilde{v} = 0$ almost everywhere.

(ii) Linearity follows from the linearity of *integration*. □

Now, we will see some examples of weak derivatives, in which they are badly behaving, but not too badly.

Example II.1.4. Consider $|x| \in C^0(-1, 1) \setminus C^1(-1, 1)$.

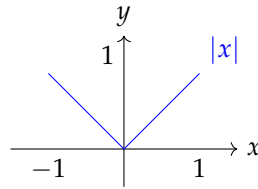


Figure II.1. Graph of $|x|$.

Away from 0, we have $|x|$ differentiable with the derivative that:

$$v(x) = \begin{cases} -1 & \text{if } -1 < x < 0, \\ 1 & \text{if } 0 < x < 1. \end{cases}$$

Here, we can illustrate it as:

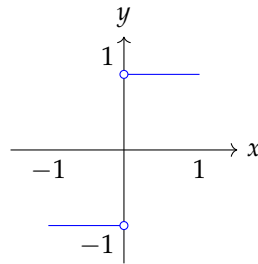


Figure II.2. Graph of $v(x)$, candidate of the weak derivative of $|x|$.

Let's now verify that v is the first weak derivative of u . For any $\varphi \in C_c^\infty(-1, 1)$, we have:

$$\begin{aligned} \int_{-1}^1 |x| \cdot \varphi'(x) &= - \int_{-1}^0 x \varphi'(x) + \int_0^1 x \varphi'(x) \\ &= \left(\left[-x \varphi(x) \right]_{-1}^0 + \int_{-1}^0 \varphi(x) \right) + \left(\left[x \varphi(x) \right]_0^1 - \int_0^1 \varphi(x) \right) \\ &= \int_{-1}^0 \varphi(x) - \int_0^1 \varphi(x) = - \left(- \int_{-1}^0 \varphi(x) + \int_0^1 \varphi(x) \right) = - \int_{-1}^1 v(x) \cdot \varphi(x), \end{aligned}$$

and hence v is the weak derivative of $|x|$.

Then, we could see if $D^2|x| = Dv$ exists weakly, as we saw $D|x| = v$. But if w was a weak derivative of v , since if $\phi \in C_c^\infty(0,1)$ then:

$$\int_0^1 w(x)\phi(x) = \int_{-1}^1 w(x)\phi(x) = - \int_{-1}^1 v(x)\phi'(x) = - \int_0^1 v(x)\phi(x) = - \int_0^1 \phi'(x) = \phi(0) - \phi(1) = 0,$$

by the fundamental theorem of calculus, and since ϕ vanishes on the boundary. Again, by the fundamental lemma on calculus of variation, $w = 0$ almost everywhere on $(0,1)$.

Similarly, we can choose $\phi \in C_c^\infty(-1,0)$, then $w = 0$ almost everywhere on $(-1,0)$. Thus, w is almost everywhere on $(-1,1)$.

Now, if we choose some $\varphi \in C_c^\infty(-1,1)$ with $\varphi(0) \neq 0$, then we have:

$$0 = \int_{-1}^1 w(x)\varphi(x) = - \int_{-1}^1 v(x)\varphi'(x) = -2 \int_0^1 \varphi'(x) = -2 \int_0^1 \varphi'(x) = 2\varphi(0) \neq 0,$$

and this leads to a contradiction. Hence w is not weak derivative of v . Thus v is not weakly differentiable.

Aside, v do have a weaker derivative, called a distributional derivative, which turns out to be the dirac delta function with a infinite point mass at 0.

Hence, we would have Lipchitz function to be weakly differentiable, but not step functions.

There are continuous functions on $(0,1)$ which are classically/strongly differentiable almost everywhere, but not weakly differentiable.

Example II.1.5. Cantor/Devil's Staircase.

Consider the ternary cantor set $\mathcal{C}$, on each level construction, we set the middle 1/3 part to middle of the left and right endpoints with middle connected with straight lines.

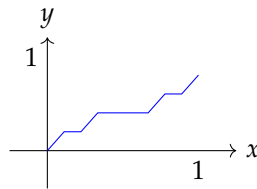


Figure II.3. Graph of Cantor Staircase equation at level 2.

Thus, the function is continuous and monotone, while is it differentiable almost everywhere as 0, but it is not weakly differentiable. ◇

Another example is corresponding to the spike function with respect to the Sobolev space.

Example II.1.6. If we let $\gamma \in (0,\infty)$, let $u : B \rightarrow \mathbb{R}$ by setting $u(x) = |x|^{-\gamma}$ on $B \setminus \{0\}$ and $u(0) = 0$, and this γ tells me how spike this function is.

In particular, we can show that $u \in L^1(B)$ if and only if $\gamma < n$ for $B \subset \mathbb{R}^n$.

When away from 0, we have:

$$\frac{\partial}{\partial x_i} |x|^{-\gamma} = -\gamma \cdot \frac{x_i}{|x|^{\gamma+1}} \quad \text{for } i = 1, \dots, n.$$

Noting that $\left| \frac{\partial}{\partial x_i} u \right| \leq \gamma \cdot \frac{1}{|x|^{\gamma+1}}$, which leads to $\frac{\partial}{\partial x_i} u \in L^1(B)$ which is if and only if $\gamma + 1 < n$. Hence, if $\varphi \in C_c^\infty(B)$, by the divergence theorem, we have:

$$\int u \frac{\partial \varphi}{\partial x_i} = \lim_{\epsilon \rightarrow 0} \left(\int_{B \setminus B_\epsilon} u \frac{\partial \varphi}{\partial x_i} \right) = \lim_{\epsilon \rightarrow 0} \left(\int_{\partial B_\epsilon} u \varphi \cdot v_i - \int_{B \setminus B_\epsilon} \frac{\partial u}{\partial x_i} \varphi \right).$$

Now, we have:

$$\left| \int_{\partial B_\epsilon} u \varphi v_i \right| \leq \int_{\partial B_\epsilon} |u \varphi| \leq \|\varphi\|_{L^\infty(B)} \int_{\partial B_\epsilon} |x|^{-\gamma} = \|\varphi\|_{L^\infty(B)} \epsilon^{-\gamma} \cdot \omega_{n-1} \epsilon^{n-1}.$$

Hence, if $n - 1 - \gamma > 0$, or equivalently $n > \gamma + 1$, then:

$$\int_{\partial B_\epsilon} u \varphi v_i \rightarrow 0 \quad \text{as } \epsilon \rightarrow 0.$$

So provided $\gamma + 1 < n$, then we have:

$$\int_B u \frac{\partial \varphi}{\partial x_i} = - \int_B \frac{\partial u}{\partial x_i} \varphi \quad \text{for any } \varphi \in C_c^\infty(B).$$

Therefore, $-\gamma \frac{x_i}{|x|^{\gamma+2}}$ is the weak derivative of $|x|^{-\gamma}$ if $\gamma + 1 < n$. ◇

Remark II.1.7. This is precise by the dominated convergence theorem. ┘

II.2 Sobolev Space

Definition II.2.1. If $\Omega \subset \mathbb{R}^n$ is open, $k \in \mathbb{N}$, $p \in [1, \infty]$, we then define the **Sobolev space** as:

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega) \text{ such that all weak derivatives of } u \text{ up to order } k \text{ exists, and are in } L^p(\Omega)\}.$$

With the norm defined as:

$$\|u\|_{W^{k,p}(\Omega)} = \begin{cases} \left(\sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^p(\Omega)}^p \right)^{1/p}, & \text{for } p \in [1, \infty), \\ \sum_{|\alpha| \leq k} \|D^\alpha u\|_{L^\infty(\Omega)}, & \text{for } p = \infty. \end{cases}$$
┘

Note that we can have $\|f\|_* := \| |f| \|_*$ for $f \in \Omega \rightarrow \mathbb{R}^M$.

Remark II.2.2. $W^{0,p}(\Omega) = L^p(\Omega)$ for $p \in [1, \infty]$ with $\|\cdot\|_{W^{0,p}(\Omega)} = \|\cdot\|_{L^p(\Omega)}$. $\lrcorner$

Theorem II.2.3. If $\Omega \subset \mathbb{R}^n$ is open, $k \in \mathbb{N}$, and $p \in [1, \infty]$, then $W^{k,p}(\Omega)$ is Banach.

Proof. We genuinely consider the case when $k = 1$, and $k > 1$ follows by the same reasoning.

We define an inclusion map:

$$I : W^{1,p}(\Omega) \rightarrow \prod_{i=1}^{n+1} L^p(\Omega), \quad u \mapsto (u, D^1 u, \dots, D^n u)$$

for $u \in W^{1,p}(\Omega)$ where $D_k u = D^\alpha u$ for $\alpha = e_k = (0, \dots, 0, 1, 0, \dots, 0)$ in which 1 is at the k -th space.

It should be easy to observe that this map I is linear. We will see that $I(W^{1,p}(\Omega))$ is closed in $\prod_{i=1}^{n+1} L^p(\Omega)$, which itself is a Banach space, hence $W^{1,p}(\Omega)$ will be seen to be closed, i.e., I is an embedding.

If $(u_i, D_1 u_i, \dots, D_n u_i) \rightarrow (v_0, v_1, \dots, v_n) \in \prod_{i=1}^{n+1} L^p(\Omega)$, then we have that $u_i \rightarrow v_0$ and $D_u v_i \rightarrow v_u$ in $L^p(\Omega)$ for $k = 1, \dots, n$. We now show that $v_u = D_u v_0$ so that $(v_0, v_1, \dots, v_n) = (v_0, D_1 v_0, \dots, D_n v_0) \in I(W^{1,p}(\Omega))$.

By Hölder's inequality, if $\eta_i \rightarrow \eta$ in $L^p(\Omega)$ and $\varphi \in C_c^\infty(\Omega)$, then:

$$\int_{\Omega} \eta_i \cdot \varphi \rightarrow \int_{\Omega} \eta \cdot \varphi \quad \text{as } i \rightarrow \infty.$$

Hence, by the above, for $\varphi \in C_c^\infty(\Omega)$, we have:

$$\int_{\Omega} v_0 D_u \varphi = \lim_{i \rightarrow \infty} \int_{\Omega} u_i D_u \varphi = - \lim_{i \rightarrow \infty} \int_{\Omega} D_u v_i \varphi = - \int_{\Omega} v_k \varphi.$$

So by definition, $D_k v_0 = v_k$. Hence $I(W^{1,p}(\Omega)) \subset X$ is closed, thus as I is an embedding, we have that $W^{1,p}(\Omega)$ is Banach. $\square$

Definition II.2.4. If $\Omega \subset \mathbb{R}^n$ is open, we set:

$$\mathcal{H}^k(\Omega) = W^{k,2}(\Omega),$$

which are Hilbert spaces with respect to the inner produce:

$$\langle u, v \rangle_{\mathcal{H}^k(\Omega)} = \sum_{|\alpha| \leq k} \langle D^\alpha u, D^\alpha v \rangle_{L^2(\Omega)} = \sum_{|\alpha| \leq k} \int_{\Omega} D^\alpha u D^\alpha v.$$

Recall in [Example I.3.9](#), we defined a space without weak derivatives, but we can then define another class of functions.

Definition II.2.5. If $\Omega \subset \mathbb{R}^n$ is open, $k \in \mathbb{N}$, and $p \in [1, \infty]$, then:

$$W_0^{k,p}(\Omega) = \overline{C_c^\infty(\Omega)} \text{ in } (W^{k,p}(\Omega), \|\cdot\|_{W^{k,p}(\Omega)}),$$

and set $H_0^k = W_0^{k,2}(\Omega)$. ┘

Remark II.2.6. $W_0^{k,p}(\Omega) \subset W^{k,p}(\Omega)$, but later we will see that $u \in W_0^{k,p}(\Omega)$ if and only if $u \in W^{k,p}(\Omega)$ and $u = 0$ almost everywhere on $\partial\Omega$ (a.e. for $\partial\Omega$ is smooth enough) in some sense (trace). ┘

Again, if we consider our old example in [Example I.2.5](#), we will be using mollifiers to construct a sequence of $W^{1,p}(\Omega)$ that converges.

Example II.2.7. Recall in [Example II.1.4](#), we consider that $|x|$ on $(-1, 1) \subset \mathbb{R}$ has one weak derivative, but not two, which lay in $L^p((-1, 1))$ for any $p \in [1, \infty]$. Hence $|x| \in W^{1,p}((-1, 1)) \setminus W^{2,p}((-1, 1))$.

Similarly, in $|x|^{-\gamma}$ on $B \subset \mathbb{R}^n$ from [Example II.1.6](#), it has $L^1(B)$ weak derivative if $(\gamma + 1) < n$, so $|x|^{-\gamma} \in W^{1,p}(B)$ if $(\gamma + 1)p < n$.

By modifying this example, we can define:

$$u(x) = \sum_{i=1}^{\infty} \frac{1}{2^i} |x - x_i|^{-\gamma} \in W^{1,p}(B),$$

where $\{x_i\}$ is dense in B (for example, $\mathbb{Q}^n \cap B$) if $(\gamma + 1)p < n$.

In fact, $|x|^{-\gamma} \in L^{\frac{np}{n-p}}(B)$ since $(\gamma + 1)p < n$ but $\frac{np}{n-p} > p$, so being in $W^{1,p}(\Omega)$ gives more integrability.

This is a special instance of Sobolev embedding. ◇

If $n = 1$, $|x|^{-\gamma}$ fails to be in $W^{1,p}(B)$ for any $\gamma \in (0, \infty)$. In fact, we can characterize $W^{1,p}$ if $n = 1$ and $p \in [1, \infty]$ in a clean way.

Definition II.2.8. Absolute Continuity.

A function $u : [a, b] \rightarrow \mathbb{R}$ is called absolutely continuous if it is differentiable almost everywhere on (a, b) with $u' \in L^1((a, b))$ and for each $x \in [a, b]$, we have:

$$u(x) = u(a) + \int_a^x u'(s) ds. \quad \text{┘}$$

Remark II.2.9. u' is then the weak derivative of u by the fundamental lemma of calculus on variation, since $u\varphi$ is absolutely continuous for $\varphi \in C_c^\infty((a, b))$.

Hence, u being absolutely continuous and $u' \in L^p((a, b))$ implies that $u \in W^{1,p}((a, b))$. ┘

Theorem II.2.10. If $(a, b) \subset \mathbb{R}$ and $p \in [1, \infty]$, then $u \in W^{1,p}((a, b))$ if and only if $u = \bar{u}$ almost everywhere on (a, b) for $\bar{u}$ being absolutely continuous and $\bar{u}' \in L^p((a, b))$.

Proof. ($\Leftarrow$;) **Remark II.2.9.**

($\Rightarrow$;) We define:

$$\tilde{u}(x) = \int_a^x u'(s) ds,$$

hence $\tilde{u}$ is absolutely continuous. Then $\tilde{u}$ and u have the same weak derivative, so $(\tilde{u} - u)' = 0$ almost everywhere in (a, b) , and hence $u = \tilde{u} + C$ almost everywhere on (a, b) for some $C \in \mathbb{R}$. Hence we set $\bar{u} = \tilde{u} + C$ to conclude. $\square$

II.3 Approximation of Sobolev Functions

We would need a class of new tools to have Sobolev function approximated.

Remark II.3.1. It is true that if $u \in L^p(\Omega)$ where $p \in [1, \infty)$ and $\Omega \subset \mathbb{R}^n$ is open, that u can be approximated by some sequence of $\{u_i\}_{i \in \mathbb{N}^+} \subset C_c^0(\Omega) \cap L^p(\Omega)$, i.e., $\|u - u_i\|_{L^p(\Omega)} \rightarrow 0$ as $i \rightarrow \infty$.

- With $p = \infty$, the issues is with approximating an indicator function over an open interval. $\lrcorner$

To improve the above to a sequence $\{u_i\}_{i \in \mathbb{N}^+} \subset C_c^\infty(\Omega) \cap L^p(\Omega)$, we need a new tool.

Definition II.3.2. Mollifier.

A **mollifier** is a nonnegative test function $\eta \in C_c^\infty(\mathbb{R}^n)$ such that:

$$\int_{\mathbb{R}^n} \eta = 1.$$

A **standard mollifier**, $\eta \in C_c^\infty(B)$ is defined by:

$$\eta(x) = \begin{cases} c_n \exp\left(\frac{1}{4|x|^2-1}\right), & \text{if } |x| < \frac{1}{2}, \\ 0, & \text{if } |x| \geq \frac{1}{2}, \end{cases}$$

where c_n is chosen so that $\int_B \eta = 1$. $\lrcorner$

Remark II.3.3. Let η be defined as the standard mollifier:

- η is radially symmetric (i.e., $\eta(x) = \eta(y)$ if and only if $|x| = |y|$).
- η is not real analytic. $\lrcorner$

Often, we “rescale” η and for $\epsilon > 0$ we set:

$$\eta_\epsilon(x) = \frac{1}{\epsilon^n} \eta\left(\frac{x}{\epsilon}\right) \in C_c^\infty(B_\epsilon), \quad (3)$$

where B_ϵ is the ball of radius ϵ centered at $0 \in \mathbb{R}^n$.

Proposition II.3.4. η_ϵ is also a mollifier, i.e., $\int_{\mathbb{R}^n} \eta_\epsilon = 1$.

Definition II.3.5. If $\Omega \subset \mathbb{R}^n$ is open and $\epsilon > 0$, we let:

$$\Omega_\epsilon := \{x \in \Omega : \text{dist}(x, \partial\Omega) > \epsilon\}.$$

For $v \in L^1_{\text{loc}}(\Omega)$, we define the **mollification of u** at scale ϵ to be:

$$u_\epsilon(x) = (u * \eta_\epsilon)(x) = \int_{B_\epsilon(x)} \eta_\epsilon(x-y)u(y)dy = \int_{B_\epsilon(0)} \eta_\epsilon(y)u(x-y)dy,$$

so that $u_\epsilon = u * \eta_\epsilon \in C^\infty(\Omega_\epsilon)$. ┘

Remark II.3.6.

- One does not need the standard mollifier to define (3).
- To see why $u_\epsilon \in C^\infty(\Omega_\epsilon)$, we note that:

$$D^\alpha(\eta_\epsilon * u)(x) = \int_{\Omega} D^\alpha \eta_\epsilon(x-y)u(y)dy, \quad \text{for } x \in \Omega_\epsilon.$$

- $f * g$ is the convolution of f and g . ┘

Lemma II.3.7. If $\Omega \subset \mathbb{R}^n$ is open, and $u \in L^1_{\text{loc}}(\Omega)$, then:

- (i) $u_\epsilon \rightarrow u$ almost everywhere in Ω as $\epsilon \searrow 0$.
- (ii) If $u \in C^0(\Omega)$, $u_\epsilon \rightarrow u$ uniformly on compact subsets of Ω .
- (iii) If $u \in L^p(\Omega)$ for $p \in [1, \infty)$, $u_\epsilon \rightarrow u$ in L^p on compact subsets of Ω , i.e., in $L^p_{\text{loc}}(\Omega)$.

Proof. The theorem relies on the Lebesgue differentiation theorem, which says that if $u \in L^1_{\text{loc}}(\Omega)$, then:

$$\frac{1}{|B_r(x)|} \int_{B_r(x)} |u(y) - u(x)|dy \rightarrow 0 \quad \text{as } r \rightarrow 0 \text{ for almost every } x \text{ in } \Omega. \quad \square$$

Lemma II.3.8. Local Approximation.

If Ω is open, $n \in \mathbb{N}$, and $p \in [1, \infty)$, and $u \in W^{k,p}(\Omega)$, then:

- (i) $\|u_\epsilon\|_{W^{k,p}(\Omega)} \leq \|u\|_{W^{k,p}(\Omega)},$
- (ii) $u_\epsilon \rightarrow u$ in $W^{k,p}$ on compactly contained subsets of Ω , i.e., $W^{k,p}_{\text{loc}}(\Omega)$.

Proof. (i) By the definition of mollifiers, Fubini and Hölder after establishing the following. We show that for $\alpha \in \mathbb{N}^\alpha$ with $|\alpha| \leq k$, we have:

$$D^\alpha(u_\epsilon) = \eta_\epsilon * (D^\alpha u) = (D^\alpha u)_\epsilon, \quad (4)$$

then the $L^1_{\text{loc}}(\Omega)$ convergence follows by the previous lemma part 3.

To check (4), for each $x \in \Omega_\epsilon$, we have:

$$\begin{aligned} D^\alpha(u_\epsilon)(x) &= \int_{\Omega} D^\alpha_x \eta_\epsilon(x - y) u(y) dy \\ &= (-1)^{|\alpha|} \int_{\Omega} D^\alpha_y \eta_\epsilon(x - y) u(y) dy = (-1)^{|\alpha|} (-1)^{|\alpha|} \int_{\Omega} \eta_\epsilon(x - y) D^\alpha_y u(y) dy. \end{aligned}$$

□

Theorem II.3.9. Interior Approximation.

If $\Omega \subset \mathbb{R}^n$ is open, $k \in \mathbb{N}$, $p \in [1, \infty)$, and $u \in W^{k,p}(\Omega)$, then there is a sequence $\{u_i\} \subset C^\infty(\Omega) \cap W^{k,p}(\Omega)$ such that $u_i \rightarrow u$ in $W^{k,p}(\Omega)$.

Proof. We first find a sequence of bounded open sets, $\{\Omega_i\} \Subset \Omega$ such that $\Omega = \bigcup_{i \geq 1} \Omega_i$ and $\Omega_i \Subset \Omega_{i+1}$ for $i \geq 1$, with $\Omega_0 = \emptyset$, which is a compact exhaustion of Ω .

For bounded cases, Evans uses $\Omega_i = \{x \in \Omega \mid \text{dist}(x, \partial\Omega) > 1/i\}$ as a concrete construction.

Setting $\tilde{\Omega}_i = \Omega_{i+1} \setminus \overline{\Omega_i}$ for $i \geq 1$, we see that $\{\tilde{\Omega}_i\}$ forms a locally finite open cover of Ω , i.e., $\Omega = \bigcup_{i \geq 1} \tilde{\Omega}_i$. Therefore, we can find a **partition of unity** subordinate to the $\{\tilde{\Omega}_i\}$, i.e., a sequence $\{\mathcal{P}_i\} \subset C_c^\infty(\Omega)$ with $\text{supp}(\mathcal{P}_i) \Subset \tilde{\Omega}_i$ for $0 \leq \mathcal{P}_i \leq 1$ and $\sum_{i=1}^\infty \mathcal{P}_i(x) = 1$ for each $x \in \Omega$.

Noting that $\mathcal{P}_i \in W^{k,p}(\Omega)$ which equals 0 on $\Omega \setminus \tilde{\Omega}_i$ with $\text{supp}(\mathcal{P}_i) \Subset \tilde{\Omega}_i$ for each $i \geq 1$.

For each $\epsilon > 0$, we can find a sequence $\{\epsilon_i\} \rightarrow 0$ being small enough, such that for $i \geq 2$:

$$v_i = \eta_{\epsilon_i} * (\mathcal{P}_i u) \in C_c^\infty(\Omega_{i+1} \setminus \overline{\Omega_{i-2}})$$

is such that $\|v_i - (\mathcal{P}_i u)\|_{W^{k,p}(\Omega)} \leq \frac{\epsilon}{2^i}$.

For $x \in \Omega$ we then let:

$$v(x) = \sum_{i=2}^\infty v_i(x) \in C^\infty(\Omega) \cap W^{k,p}(\Omega).$$

Then, $\|v - r\|_{W^{k,p}(\Omega)} \leq \|\sum_{i=2}^\infty (u \mathcal{P}_i - v_i)\|_{W^{k,p}(\Omega)} \leq \sum_{i=1}^\infty \frac{\epsilon}{2^i} = \epsilon$, we then set $u_i = v^{\epsilon_i}$ for each ϵ_i . □

Remark II.3.10. The $\{u_i\}_{i=1}^\infty \subset W^{k,p}(\Omega)$ in particular ensures the control of weak derivatives near boundary of Ω in the L^p sense. ┘

Lemma II.3.11. Chain Rule.

If $f \in C^1(\mathbb{R})$ with $f' \in L^\infty(\mathbb{R})$, $\Omega \subset \mathbb{R}^n$ is open, and $u \in L^1_{\text{loc}}(\Omega)$ which is weakly differentiable in Ω ,

then $f \circ u \in L^1_{\text{loc}}(\Omega)$ and has weak derivatives:

$$D_i(f \circ v) = (f' \circ u)D_i u \quad \text{for } i = 1, \dots, n.$$

Moreover, if $u \in W^{1,p}_{\text{loc}}(\Omega)$, then $(f \circ u) \in W^{1,p}_{\text{loc}}(\Omega)$ for $p \in [1, \infty]$.

Proof. Show $(f \circ u) \in L^1_{\text{loc}}(\Omega)$ directly, then show approximations converge locally. □

III Boundaries of Sobolev Space

III.1 Global Approximation of Sobolev Space

Definition III.1.1. Boundary Regularity.

If $\Omega \subset \mathbb{R}^n$ is open and bounded, $k \in \mathbb{N}$, then $\partial\Omega$ is said to be C^k **regular** if for each $x \in \partial\Omega$, there is some $r > 0$ and a function $\gamma \in C^k(\mathbb{R}^{n-1})$ such that (up to relabeling the coordinate axes) we have:

$$\Omega \cap B_r(x) = \{y \in B_r(x) : y > \gamma(y_1, \dots, y_{n-1})\}.$$

In particular, $\partial\Omega \cap B_r(x) = \{y \in B_r(x) : y = \gamma(y_1, \dots, y_{n-1})\}.$ ┘

We can illustrate the graph as follows:

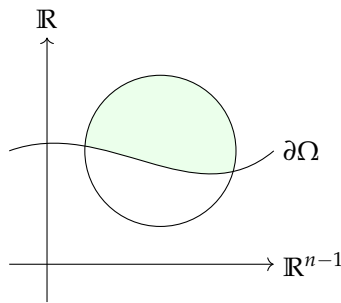


Figure III.1. The region above the curve (shaded).

Remark III.1.2. Consider the above definition:

- This works for **Lipschitz/Hölder** γ .
- This, equivalently, says that $\partial\Omega$ is an embedded $(n-1)$ -manifold of class C^k .

With this definition, it rules out cusps, immersions/self-intersection, slitted domains, or polygonal domains from candidate of γ .

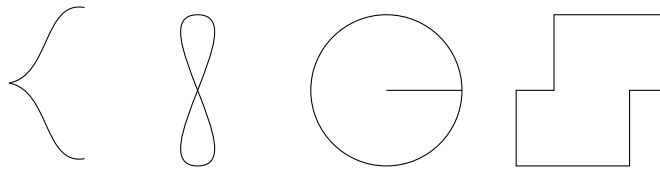


Figure III.2. Examples of cusps, immersions/self-intersection, slitted domains, or polygonal domains (from left to right). ┘

Theorem III.1.3. Global Approximation.

If $\Omega \subset \mathbb{R}^n$ is open and bounded, with $\partial\Omega$ is C^1 regular (or Lipschitz), $k \in \mathbb{N}$, $p \in [1, \infty)$, and $u \in W^{k,p}(\Omega)$, then there is a sequence $\{u_i\}_{i=1}^\infty \in C^\infty(\overline{\Omega})$ such that $u_i \rightarrow u$ in $W^{k,p}(\Omega)$.

Example III.1.4. Consider on $S = B_1(0) \setminus \{x_1 > 0, x_2 = 0\} \subset \mathbb{R}^2$, then we have $f(r, \theta) = \theta \in W^{1,2}(S)$, and we cannot achieve the regularity as requested. ◇

Proof of Theorem III.1.3. First, if $x \in \partial\Omega$, as $\partial\Omega$ is C^1 , there is some radius $r_x > 0$ such that:

$$\Omega \cap B_{r_x}(x) = \{y \in B_{r_x}(x) \mid y_n > \gamma(y_1, \dots, y_{n-1})\}$$

of some $\gamma_x \in C^1(\mathbb{R}^{n-1})$.

We set $V_x = \Omega \cap B_{r_x/2}(x)$, and for each $y \in V_x$ and $\lambda, \epsilon > 0$, we set:

$$y_\epsilon = y + \lambda \cdot \epsilon \cdot e_n,$$

and for fixed $\lambda > 0$ large enough, we have that:

$$B_\epsilon(y_\epsilon) \subset B_{r_x}(x)$$

for all $y \in V_x$ and ϵ sufficiently small. We then set:

$$u^\epsilon(y) = u(y_\epsilon) \quad \text{on } V_x,$$

i.e., translate u by $\lambda \epsilon e_n$, and let:

$$v_x^{\tilde{\epsilon}} - \eta_{\tilde{\epsilon}} * u^\epsilon \in C^\infty(V_x) \quad \text{for all } \tilde{\epsilon} \in (0, \epsilon).$$

Now, we show that, as $\epsilon, \tilde{\epsilon} \rightarrow 0$, that:

$$v_x^{\tilde{\epsilon}} \rightarrow u \quad \text{in } W^{k,p}(V_x).$$

To see this, we note that by the triangle inequality:

$$\|v_x^{\tilde{\epsilon}} - u\|_{W^{k,p}(V_x)} \leq \|v_x^{\tilde{\epsilon}} - u^\epsilon\|_{W^{k,p}(V_x)} + \|u^\epsilon - u\|_{W^{k,p}(V_x)}.$$

By the continuity of translation in L^p for $p \neq \infty$, we have that $u^\epsilon \rightarrow u$ in $W^{k,p}(V_x)$ as $\epsilon \searrow 0$.

Now, for a given tolerance $\delta > 0$, we choose $\epsilon > 0$ small enough to ensure $\|u^\epsilon - u\|_{W^{k,p}(V_x)} > \frac{\delta}{2}$. Then, for this fixed $\epsilon > 0$, we choose $\tilde{\epsilon} > 0$ sufficiently small so that as $v_x^{\tilde{\epsilon}} = \eta_{\tilde{\epsilon}} * u^\epsilon \rightarrow u^\epsilon$ in $W^{k,p}(V_x)$ as $\tilde{\epsilon} \rightarrow 0$, which is extending to be 0 outside Ω , then resulting in $\|v_x^{\tilde{\epsilon}} - u^\epsilon\|_{W^{k,p}(V_x)} < \frac{\delta}{2}$ also.

Therefore, since $\delta > 0$ is arbitrary, $v_x^{\tilde{\epsilon}} \rightarrow u$ in $W^{k,p}(V_x)$.

As $\partial\Omega$ is closed and bounded (as Ω is bounded), it is compact, and so for some $\{x_1, \dots, x_N\} \subset \partial\Omega$ we have that:

$$\partial\Omega \subset \bigcup_{i=1}^N B_{\frac{r_{x_i}}{2}}(x_i) \quad \text{for } r_{x_i} \text{ as above.}$$

By the previous reasoning, for any $\delta > 0$, we can find $v_i \in C^\infty(\overline{V_{x_i}})$ for $i = 1, \dots, N$ such that $\|v_i - u\|_{W^{k,p}(V_{x_i})} < \delta$. We apply local/interior approximation to some $V_0 \subset \Omega$ that is open with $\Omega = V_0 \cup V_{x_1} \cup \dots \cup V_{x_N}$ to get $v_0 \in C^\infty(\overline{V_0})$ with $\|v_0 - u\|_{W^{k,p}(V_0)} < \delta$.

We can then choose a partition of unity subordinate to $\{V_0, V_{x_1}, \dots, V_{x_N}\}$ on $\overline{\Omega}$, call it $\{\mathcal{P}_i\}_{i=0}^N$, i.e., $\text{supp}(\mathcal{P}_i) \Subset V_{x_i}$ for $i = 1, \dots, N$, $\text{supp}(\mathcal{P}_0) \Subset V_0$, and $\mathcal{P}_i \in C^\infty(\overline{B + r_{x_i}(x_i)})$ for $i = 1, \dots, N$, $\mathcal{P}_0 \in C^\infty(\overline{V_0})$, and for each $x \in \overline{\Omega}$, $\sum_{i=1}^N \mathcal{P}_i(x) = 1$.

We then define for this $\delta > 0$ that:

$$v = \sum_{i=0}^N \mathcal{P}_i \cdot v_i \in C^\infty(\overline{\Omega}).$$

Since we can write:

$$u = \sum_{i=0}^N \mathcal{P}_i \cdot u \quad \text{on } \Omega.$$

Hence, we can use the product rule for weak derivatives to see that for $\alpha \in \mathbb{N}^n$ with $|\alpha| \leq u$, we have:

$$\|D^\alpha v - D^\alpha u\|_{L^p(\Omega)} \leq \sum_{i=0}^N \|D^\alpha(\mathcal{P}_i v_i) - D^\alpha(\mathcal{P}_i u)\|_{L^p(V_i)} \leq C_k \sum_{i=1}^N \|v_i - u\|_{W^{k,p}(V_i)} \leq C_k(N+1)\delta,$$

where C_k is a constant depending on the supremum of the derivative of the $\{\mathcal{P}_i\}_{i=0}^N$ of order at most k . Hence, for each $\delta > 0$, we can find $v \in C^\infty(\bar{\Omega})$ with $\|v - u\|_{W^{k,p}(\Omega)} < \tilde{\delta}$. Then, we let $u_i = v$ for $\tilde{\delta} = \frac{1}{i}$, then $\{u_i\}_{i=1}^\infty \subset C^\infty(\bar{\Omega})$ with $u_i \rightarrow u$ in $W^{k,p}(\Omega)$. $\square$

III.2 Extensions of Sobolev Functions

Then, we think about how to extend the Sobolev functions to the whole domain.

Example III.2.1. Consider $u = 1 \in W^{k,p}((-1,1))$ for all $u \in \mathbb{N}$ and $p \in [1, \infty]$.

- Of course, we can extend u to $\mathbb{R}$ by setting $u = 0$ on $\mathbb{R} \setminus (-1,1)$. However, this extension does not lie in $W^{k,p}(\mathbb{R})$ for any $k \geq 1$.

Hence, we would need to

$\diamond$

Theorem III.2.2. Extension on $W^{1,p}$.

If $\Omega, \tilde{\Omega} \subset \mathbb{R}^n$ are open, bounded, with $\partial\Omega \in C^1$, $\Omega \Subset \tilde{\Omega}$, and $p \in [1, \infty)$, then there is a bounded linear map (called the **extension operator**):

$$E : W^{1,p}(\Omega) \rightarrow W^{1,p}(\mathbb{R}^n),$$

such that for each $u \in W^{1,p}(\Omega)$ the following hold:

- (i) $E(u)|_\Omega = u$ almost everywhere in Ω ,
- (ii) $\text{supp}(E(u)) = \overline{\{u \neq 0\}} \subset \tilde{\Omega}$, i.e., $u = 0$ almost everywhere on $\mathbb{R}^n \setminus \tilde{\Omega}$,
- (iii) $\|E(u)\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|u\|_{W^{1,p}(\Omega)}$ for a constant $C > 0$ depending only on $\Omega, \tilde{\Omega}$, n , and p . (Note that C does not depend on u .)

Example III.2.3. Continued from Example III.2.1...

Consider the original equation, we can consider $\tilde{\Omega} = (-2,2)$ so we have $E(u) \in W^{1,p}(\mathbb{R})$.

$\diamond$

Remark III.2.4. This works for $W^{k,p}$ and $p = \infty$ but the construction is more delicate, provided we assume that $\partial\Omega$ is in C^k .

In fact, this really shows the existence of an extension operator:

$$\hat{E} : W^{k,p}(\Omega) \rightarrow W_0^{k,p}(\tilde{\Omega})$$

if $\partial\Omega$ is C^k . J

Proof of Theorem III.2.2. Let's first prove it in the case that $\partial\Omega = \{x_n = 0\}$ in a ball, and $u \in C^\infty(\overline{\Omega})$.

(i) **Step 1:** Flat $\partial\Omega$.

Fix $x \in \partial\Omega$ and suppose that in $B = B_r(x)$ for some $r > 0$, we have:

$$\begin{aligned} B^+ &:= B \cap \{y \in \mathbb{R}^n \mid y_n \geq 0\} \subset \Omega, \\ B^+ &:= B \cap \{y \in \mathbb{R}^n \mid y_n \leq 0\} \subset \mathbb{R}^n \setminus \Omega. \end{aligned}$$

Since $u \in C^\infty(\overline{\Omega})$ is defined on B^+ , we reflect u in $\{x_n = 0\}$ and for $y \in \partial B^+ = \partial B^-$, let:

$$\bar{u}(y) = \begin{cases} u(y), & \text{if } y \in B^+, \\ -3u(y_1, \dots, y_{n-1}, -y_n) + 4u(y_1, \dots, y_{n-1}, -\frac{y_n}{2}) & \text{if } y \in B \setminus B^+, \end{cases}$$

then $\bar{u} \in C^1(B)$ (check by direct computation) and $\|\bar{u}\|_{W^{1,p}(B)} \leq C\|u\|_{W^{1,p}(B^+)}$, as C is not defined on u .

(ii) **Step 2:** Straighten $\partial\Omega$

If the $\partial\Omega$ is not $\{x_n = 0\}$, since the $\partial\Omega \in C^1$, we can find some diffeomorphism (C^1 map with inverse), such that if $y \in B_r(x)$ for some $r > 0$:

$$\Phi_x(y) = (y_1, \dots, y_{n-1}, y_n - \gamma(y_1, \dots, y_{n-1})).$$

If $\gamma_x \in C^1(\mathbb{R}^{n-1})$ with $\partial\Omega \cap B_r(x) = \{y \in B_r(x) \mid y_n > \gamma(y_1, \dots, y_{n-1})\}$. So $\partial\Omega \cap B_r(x)$ maps to $\{x_n = 0\}$ with Φ_x .

And since Φ_x is invertible, there is an open set $x \in W \subset B_r(x)$ with $\Phi_x(W) = B_s(z)$ and $\Phi_x(\Omega \cap W)$ where $x = \Phi_x(x) \in \mathbb{R}^{n-1} \times \{0\}$ and $s > 0$.

We then set $u' = u \circ \Phi_x^{-1}$ on $B_s(z)$.

Hence, $u' \in C^1(\overline{B_s(z)})$ so we can apply step 1 to extend u' to a function $\bar{u}' \in C^1(\overline{B_s(z)})$ with:

$$\|\bar{u}'\|_{W^{1,p}(B_s(z))} \leq C\|u'\|_{W^{1,p}(B_s(z))}.$$

Then, $\bar{u}' \circ \Phi_x \in C^1(W)$ and as Φ_x is C^1 , the chain rule gives:

$$\|\bar{u}\|_{W^{1,p}(W)} \leq \tilde{C}\|u\|_{W^{1,p}(W \cap \Omega)} \leq \tilde{C}\|u\|_{W^{1,p}(\Omega)}$$

(iii) **Step 3:** Patching up.

As $\partial\Omega$ is compact, there are finitely many $\{x_1, \dots, x_n\} \subset \partial\Omega$ for which we have open sets $\{W_{x_1}, \dots, W_{x_n}\} \subset \mathbb{R}^n$ with $x_i \in W_i$ for each $i = 1, \dots, N$ and such that $\partial\Omega \subset \bigcup_{i=1}^N W_{x_i}$, i.e., by **step 2** for each $x \in \partial\Omega$.

To each x_i , we have an associated local boundary extension, $\bar{u}_i \in C^1(W_i)$, $\bar{u}_i = u$ on $W_i \cap \Omega$ almost everywhere.

Now, we choose some $\Omega_0 \Subset \Omega$ as open such that:

$$\overline{\Omega} \subset \bigcup_{i=1}^N W_i, \quad \text{with } W_i = W_{x_i}.$$

Then, we fix a partition of unity subordinate to this cover, $\{\mathcal{P}_i\}_{i=0}^N$, so $\text{supp}(\mathcal{P}_i) \Subset W_i$, and $\sum_{i=0}^N \mathcal{P}_i(x) = 1$.

We then set:

$$\bar{u} = \sum_{i=0}^N \mathcal{P}_i \bar{u}_i \in C^1(\mathbb{R}^n),$$

and we have:

$$\|\bar{u}\|_{W^{1,p}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\Omega)}.$$

Thus, $\bar{u}$ satisfies (i) and (iii) if $u \in C^1(\overline{\Omega})$, by considering $\chi \cdot \bar{u}$ for some $\chi \in C_c^\infty(\tilde{\Omega})$ with $\chi = 1$ in Ω (where one way of construction is $\chi = \eta_\epsilon * \mathbb{1}_{\Omega_\epsilon}$), then $\chi \cdot \bar{u}$ satisfies (i), (ii), and (iii).

We then define $E(u) = \chi \cdot \bar{u}$ if $u \in C^1(\overline{\Omega})$, which we note is linear and bounded. ($\overline{u+v} = \bar{u} + \bar{v}$).

By global approximation, if $u \in W^{1,p}(\Omega)$, there is a sequence:

$$\{u_i\} \subset C^\infty(\overline{\Omega}) \subset C^1(\overline{\Omega}),$$

with $u_i \rightarrow u$ in $W^{1,p}(\Omega)$.

As $\{E(u_i)\} \subset W^{1,p}(\mathbb{R}^n)$ we have:

$$\|E(u_i) - E(u_j)\|_{W^{1,p}(\mathbb{R}^n)} = \|E(u_i, u_j)\|_{W^{1,p}(\mathbb{R}^n)} \leq C \|u_i - u_j\|_{W^{1,p}(\Omega)}.$$

as $u_i \rightarrow u$ in $W^{1,p}(\Omega)$ that $\{E(u_i)\}$ is Cauchy, hence convergent as $W^{1,p}(\tilde{\Omega})$ is Banach, and we denote $E(u) := \lim_{i \rightarrow \infty} E(u_i)$ in $W^{1,p}(\tilde{\Omega})$.

One can check that this limit is independent of the approximating sequence. Hence, $E(u)$ is well-defined for any $u \in W^{1,p}(\Omega)$ and satisfies (i), (ii), and (iii). $\square$

Remark III.2.5. If $\partial\Omega \in C^k$, we can use:

$$\sum_{i=1}^k c_i \cdot u \left(y_1, \dots, y_{n-1}, -\frac{y_n}{i} \right)$$

in the flat case (step 1) where $\sum_{i=1}^k c_i \left(-\frac{1}{i} \right)^m = 1$ for $m = 1, 2, \dots, k-1$. $\lrcorner$

III.3 Trace of Sobolev Functions

Let $\Omega \subset \mathbb{R}^n$ be open. If $u \in C^0(\overline{\Omega})$, then $u|_{\partial\Omega} : \partial\Omega \rightarrow \mathbb{R}$ is well defined. But if $u \in L^p$, since $L^n(\partial\Omega) = 0$ (provided $\partial\Omega$ is C^1), $u|_{\partial\Omega}$ doesn't have a unique or canonical choice.

We want to define $u|_{\partial\Omega}$ for Sobolev functions:

Theorem III.3.1. Trace on $W^{1,p}$.

If $\Omega \subset \mathbb{R}^n$ is open, bounded, $\partial\Omega$ is C^1 , and $p \in [1, \infty)$, then there is a bounded linear map:

$$T : W^{1,p}(\Omega) \rightarrow L^p(\partial\Omega),$$

called the **trace operator**, such that for each $u \in W^{1,p}(\Omega)$:

- $T(u) = u|_{\partial\Omega}$ if $u \in W^{1,p}(\Omega) \cap C^0(\overline{\Omega})$ almost everywhere, and
- $\|T(u)\|_{L^p(\partial\Omega)} \leq C\|u\|_{W^{1,p}(\Omega)}$, where C is independent of u .

Remark III.3.2.

- We call $T(u) \in L^p(\partial\Omega)$ the trace of u .
- If $u \in W^{k,p}(\Omega)$, $T(D^\alpha u)$ is well defined for all $|\alpha| \leq k - 1$.
- There is no trace operator $T : L^p(\Omega) \rightarrow L^p(\partial\Omega)$. For example, we let $\Omega = B_1(0) \subset \mathbb{R}^n$ and $u = (1 - |x|^\alpha)^{-1}$ for $\alpha p < n$.
- This also works for $p = \infty$.
- Since the trace operator is bounded linear operator, we can utilize this to show that $C_c^\infty(\Omega)$ cannot be dense in $W^{1,p}(\Omega)$. ┘

Proof of Theorem III.3.1. We prove it locally, and then “patch up” ad for the extension operator.

Let $u \in C^1(\overline{\Omega})$, $x \in \partial\Omega$, and set $B = B_r(x)$, $\hat{B} = B_{\frac{r}{2}}(x)$ under the assumption that $\partial\Omega \cap B = \{y_n = 0\} \cap B$. We fix $\mathcal{P} \in C_c^\infty(B)$ with $\mathcal{P} \geq 1$ on $\hat{B}$. We denote $\Gamma := \partial\Omega \cap \hat{B}$. Then, we apply the divergence theorem:

$$\begin{aligned} \int_{\Gamma} |u|^p dy' &\leq \int_{\{y_n=0\}} \mathcal{P} |u|^p dy' = - \int_{B^+ := B \cap \{y_n > 0\}} \frac{\partial}{\partial y_n} [\mathcal{P} |u|^p] dy \\ &= \int_{B^+} \left(\frac{\partial \mathcal{P}}{\partial y_n} \cdot |u|^p + \mathcal{P} \cdot pu^{p-1} \cdot \text{sgn}(u) \cdot \frac{\partial u}{\partial y_n} \right) dy. \end{aligned}$$

Then, we use the Young's inequality that $|ab| \leq \frac{|a|^p}{p} + \frac{|b|^q}{q}$ with $p := \frac{p}{p-1}$ and $q := p$ on $|u|^{p-1} \frac{\partial u}{\partial y_n}$:

$$\int_{B^+} \left(\frac{\partial \mathcal{P}}{\partial y_n} \cdot |u|^p + \mathcal{P} \cdot pu^{p-1} \cdot \text{sgn}(u) \cdot \frac{\partial u}{\partial y_n} \right) dy \leq C \int_{\Omega} (|u|^p + |Du|^p) dy.$$

Hence, by straightening $\partial\Omega$, we obtain that:

$$\int_{\Gamma} |u|^p \leq C \int_{\Omega} (|u|^p + |Du|^p) dy \quad \text{for any } \Gamma \subset \partial\Omega \text{ being open.}$$

By compactness of $\partial\Omega$, we have:

$$\|u|_{\partial\Omega}\|_{L^p(\partial\Omega)} \leq C\|u\|_{W^{1,p}(\Omega)}.$$

So we set $T(u) = u|_{\partial\Omega}$ if $u \in C^1(\overline{\Omega})$, then this satisfies the two conditions.

By using the global approximation for $u \in W^{1,p}(\Omega)$, as in the proof of extension operator, we can then

define that:

$$T(u) = \lim_{i \rightarrow \infty} T(u_i) \quad \text{if } u_i \rightarrow u \text{ in } W^{1,p}(\Omega).$$

Since approximations coverage uniformly if $u \in W^{1,p}(\Omega) \cap C^0(\overline{\Omega})$ which implies that $T(u) = u|_{\partial\Omega}$ is this case, hence satisfying the conditions. $\square$

Note that when we have the **Dirichlet's condition**, we automatically obtain that the trace operator is zero.

Theorem III.3.3. Trace Zero in $W^{1,p}$.

If $\Omega \subset \mathbb{R}^n$ is open, bounded, $\partial\Omega \in C^1$ and $p \in [1, \infty)$, then:

$$W_0^{1,p}(\Omega) = \{u \in W^{1,p}(\Omega) \mid T(u) = 0\}.$$

Proof. ($\subset$): if $u \in W_0^{1,p}(\Omega)$, then there are $\{u_i\} \subset C_c^\infty(\Omega)$ with $u_i \rightarrow u$ in $W^{1,p}(\Omega)$, then as $T(u_i) = 0$ and T is bounded, then $T(u) = 0$ as well.

($\supset$): If $u \in W^{1,p}(\Omega)$ and $T(u) = 0$. By construction, there are $\{u_i\} \subset C^\infty(\overline{\Omega}) \cap W^{1,p}(\Omega)$ with $T(u_i) \rightarrow 0$ in $L^p(\partial\Omega)$. Again, by straightening $\partial\Omega$ and patching up, we can prove the case $u \in W^{1,p}(\mathbb{R}_+^n)$ with $\{u \neq 0\}$ compact, where $\mathbb{R}_+^n = \{y \in \mathbb{R}^n \mid y_n > 0\}$ and $T(u) = 0$. Therefore, for $x' \in \mathbb{R}^{n-1}$ and $x_n \geq 0$, by the fundamental theorem of calculus, we have:

$$|u_i(x', x_n)| \leq |u_i(x', 0)| + \left| \int_0^{x_n} \frac{\partial}{\partial x_n} u_i(x, s) ds \right|.$$

Therefore, by integration and using $(a+b)^p \leq \frac{a^p}{2} + \frac{b^p}{2}$, we have:

$$\begin{aligned} 2 \int_{\mathbb{R}^{n-1}} |u_i(x', x_n)|^p dx' &\leq \int_{\mathbb{R}^{n-1}} |u_i(x', 0)|^p dx' + \int_{\mathbb{R}^{n-1}} \left[\int_0^{x_n} \left| \frac{\partial}{\partial x_n} u_i(x, s) \right| ds \right]^p dx' \\ &\leq \int_{\mathbb{R}^{n-1}} |u_i(x', 0)|^p dx' + \int_{\mathbb{R}^{n-1}} \left[\left(\int_0^{x_n} 1^{\frac{p}{p-1}} ds \right) \left(\int_0^{x_n} \left| \frac{\partial}{\partial x_n} u_i(x', s) \right|^p ds \right) \right] dx' \\ &\leq \underbrace{\int_{\mathbb{R}^{n-1}} |u_i(x', 0)|^p dx'}_{\|u_i\|_{L^p(\partial\Omega)}^p \rightarrow 0} + \int_{\mathbb{R}^{n-1}} x_n^{p-1} \int_0^{x_n} |Du_i(x', s)|^p ds dx'. \end{aligned}$$

Hence, by sending $i \rightarrow \infty$, we deduced that:

$$\int_{\mathbb{R}^{n-1}} |u(x', x_0)|^p dx' \leq \frac{x_n^{p-1}}{2} \int_{\mathbb{R}^{n-1}} \int_0^{x_0} |Du(x', s)|^p ds dx'.$$

Now, let $\mathcal{P} \in C^\infty([0, \infty))$ with $\mathcal{P} = 1$ on $[0, 1]$, $\mathcal{P} = 0$ on $[2, \infty)$ and $0 \leq \mathcal{P} \leq 1$. We set $\mathcal{P}_m(x) = \mathcal{P}(mx_n)$ for $m \geq 1$, $x \in \mathbb{R}_+^n$ and define $w_m(x) = u(x) \cdot (1 - \mathcal{P}_m(x))$. Then, we have that:

$$\begin{cases} \frac{\partial w_m}{\partial x_n} = \frac{\partial u}{\partial x_n} \cdot (1 - \mathcal{P}_m) - m\mathcal{P}'(m_-)u, \\ D_{x'} w_m = D_{x'} u (1 - \mathcal{P}_m). \end{cases}$$

Hence, we estimate:

$$\begin{aligned} \int_{\mathbb{R}_+^n} |Dw_m - Du|^p &= \int_{\mathbb{R}_+^n} \left| D_{x'} u(-\mathcal{P}_m) + \frac{\partial}{\partial x_n}(-\mathcal{P}_m) - mu\mathcal{P}'(m_-) \right|^p \\ &\leq \underbrace{\frac{1}{2} \int_{\mathbb{R}_+^n} |Du|^p |\mathcal{P}_m|^p}_{\rightarrow 0 \text{ when } m \rightarrow \infty} + \frac{1}{2} m^p \|\mathcal{P}'\|_{C^0(\mathbb{R})}^p \int_0^{\frac{2}{m}} \int_{\mathbb{R}^{n-1}} |u|^p dx' ds. \end{aligned}$$

Now, we just consider the remaining part. Note that we can set $C \geq \frac{1}{2} \|\mathcal{P}'\|^p$, so we have:

$$\begin{aligned} \frac{1}{2} m^p \|\mathcal{P}'\|_{C^0(\mathbb{R})}^p \int_0^{\frac{2}{m}} \int_{\mathbb{R}^{n-1}} |u|^p dx' ds &\leq m^p C \int_0^{\frac{2}{m}} \int_{\mathbb{R}^{n-1}} |u(x', s)|^p dx' ds \\ &\leq C m^p \int_0^{\frac{2}{m}} \left[\frac{s^{p-1}}{2} \int_{\mathbb{R}^{n-1}} \int_0^s |Du(x', t)|^p dx dt \right] ds \leq C m^p \left(\int_0^{\frac{2}{m}} \frac{s^{p-1}}{2} ds \right) \left(\int_{\mathbb{R}^n} \int_0^{\frac{2}{m}} |Du(x', t)|^p ds dt \right) \\ &= C m^p \left(\left(\frac{2}{m} \right)^p \cdot \frac{1}{p} \right) \left(\int_0^{\frac{2}{m}} \int_{\mathbb{R}^{n-1}} |Du(x', t)|^p dx' dt \right) \leq \tilde{C} \left(\int_{\mathbb{R}_+^n} |Du|^p - \int_{\frac{2}{m}}^\infty \int_{\mathbb{R}^{n-1}} |Du|^p \right) \rightarrow 0 \end{aligned}$$

as $m \rightarrow \infty$.

Hence, $Dw_m \rightarrow Dv$ in $L^p(\mathbb{R}_+^n)$ and as $w_m \rightarrow u$ in $L^p(\mathbb{R}_+^n)$, we have that $w_m \rightarrow u$ in $W^{1,p}(\mathbb{R}_+^n)$.

Noting that $w_m = 0$ for $x_n \in [0, \frac{1}{m}]$, by having $v_m = \eta_{\frac{1}{2m}} * w_m \in C_c^\infty(\mathbb{R}_+^n)$ with $v_m \rightarrow v$ in $W^{1,p}(\mathbb{R}_+^n)$, and hence $v \in W_0^{1,p}(\mathbb{R}_+^n)$, as desired. $\square$

III.4 Sobolev Inequalities

We saw that $|x|^{-\gamma} \in W^{1,p}(B)$ implies that $|x|^{-\gamma} \in L^{\frac{np}{n-p}}(B)$, i.e., it gained integrability as $p < \frac{np}{n-p}$ for $p \neq \infty$. This is a general principle, namely belonging to a Sobolev space means the function belongs to other (ideally more regular spaces).

However, we need to take care of the cases:

$$1 \leq p < n, \quad p = n, \quad \text{or} \quad n < p \leq \infty.$$

Remark III.4.1. Gagliardo-Nirenberg-Sobolev Inequality.

For $1 \leq p < n$, we try to see that if:

$$\|u\|_{L^q(\mathbb{R}^n)} \leq C \|Du\|_{L^p(\mathbb{R}^n)}$$

for some $C > 0$ independent of $u \in C_c^\infty(\mathbb{R}^n)$. We see that q is pre-determined by p .

To see this, if $u \neq 0$, we set:

$$u_\lambda(x) = u(\lambda x) \text{ for } \lambda > 0 \text{ and } x \in \mathbb{R}^n.$$

Therefore, for any u_λ , we have:

$$\|u_\lambda\|_{L^q(\mathbb{R}^n)} \leq C \|Du_\lambda\|_{L^p(\mathbb{R}^n)},$$

but we have:

$$\begin{cases} \|u_\lambda\|_{L^q(\mathbb{R}^n)}^q = \int_{\mathbb{R}^n} |u_\lambda|^q = \int_{\mathbb{R}^n} |u(\lambda x)|^q dx = \int_{\mathbb{R}^n} \lambda^{-n} |u|^q dy = \lambda^{-n} \|u\|_{L^q(\mathbb{R}^n)}^q, \\ \|Du_\lambda\|_{L^p(\mathbb{R}^n)}^p = \int_{\mathbb{R}^n} |Du_\lambda|^p = \int_{\mathbb{R}^n} |D(u(\lambda x))|^p dx = \int_{\mathbb{R}^n} \lambda^p |Du(\lambda x)|^p dx = \int_{\mathbb{R}^n} |Du(y)|^p dy = \lambda^{p-n} \|Du\|_{L^p(\mathbb{R}^n)}^p. \end{cases}$$

Hence, we have that $\lambda^{-\frac{n}{q}} \|u\|_{L^q(\mathbb{R}^n)} \leq c \lambda^{1-\frac{n}{p}} \|Du\|_{L^p(\mathbb{R}^n)}$, which is equivalently that:

$$\|u\|_{L^2(\mathbb{R}^n)} \leq \lambda^{1-\frac{n}{p}+\frac{n}{q}} \|Du\|_{L^p(\mathbb{R}^n)}.$$

If $1 - \frac{n}{p} + \frac{n}{q} \neq 0$, by sending $\lambda \nearrow \infty$ or $\searrow 0$, we get a contradiction to $u \neq 0$. Thus if the assumption holds, we have $1 - \frac{n}{p} + \frac{n}{q} = 0$, this is equivalently $q = \frac{np}{n-p}$. $\lrcorner$

Definition III.4.2. Sobolev Conjugate.

For $p \in [1, n)$, the **Sobolev conjugate** of p is:

$$p^* = \frac{np}{n-p},$$

so that we have $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}$ and $p^* > p$. $\lrcorner$

Theorem III.4.3. Gagliardo-Nirenberg-Sobolev Inequality (GNS).

If $p \in [1, n)$, then there exists a constant $C > 0$ such that:

$$\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|Du\|_{L^p(\mathbb{R}^n)} \quad \text{for all } u \in C_c^1(\mathbb{R}^n).$$

Proof. The idea is to use the fundamental theorem of calculus and the general Hölder inequality.

Let's assume $p = 1$ as $u \in C_c^1(\mathbb{R}^n)$ for each $i = 1, \dots, n$ and $x \in \mathbb{R}^n$ we have by the fundamental theorem of calculus that:

$$|u(x)| \leq \int_{-\infty}^{x_i} \left| \frac{\partial u}{\partial x_i}(x_1, \dots, x_{i-1}, y_i, x_{i+1}, \dots, x_n) \right| dy_i.$$

Noting that $1^* = \frac{n \cdot 1}{n-1} = \frac{n}{n-1}$ so we have:

$$|u(x)|^{\frac{n}{n-1}} = (|u(x)|^n)^{\frac{1}{n-1}} \leq \prod_{i=1}^n \left[\int_{-\infty}^{\infty} |Du(x_1, \dots, y_i, \dots, x_n)| dy_i \right]^{\frac{1}{n-1}}.$$

Then, when we integrate with respect to x_1 , we obtain:

$$\int_{-\infty}^{\infty} |u(x)|^{\frac{n}{n-1}} \leq \left(\int_{-\infty}^{\infty} |Du| dy_1 \right)^{\frac{1}{n-1}} \int_{-\infty}^{\infty} \prod_{i=2}^n \left[\int |Du| dy_i \right]^{\frac{1}{n-1}} dx_1.$$

Consider generalized Hölder for $p_i = n-1$, then we have:

$$\sum_{i=1}^{n-1} \frac{1}{p_i} = \frac{n-1}{n-1} = 1,$$

so we have have that:

$$\int_{-\infty}^{\infty} \prod_{i=2}^n \left[\int |f_i| dy_i \right]^{\frac{1}{n-1}} dx_1 \leq \prod_{i=2}^n \left(\int_{|f|_i}^{n-1} dx_1 \right)^{\frac{1}{n-1}}.$$

We use $f_i = (\int |Du| dx_i)^{\frac{1}{n-1}}$, and with Fubinni, we have:

$$\begin{aligned} \int |u(x)|^{\frac{n}{n-1}} dx_1 &\leq \left(\int |Du| dy_1 \right)^{\frac{1}{n-1}} \int \prod_{i=2}^n \left[\int |Du| dy_i \right]^{\frac{1}{n-1}} dx_1 \\ &\leq \left(\int |Du| dy_1 \right)^{\frac{1}{n-1}} \cdot \left[\prod_{i=2}^n \int |Du| dx_1 dy_i \right]^{\frac{1}{n-1}}. \end{aligned}$$

Then, we integrate with respect to x_2 to obtain that:

$$\iint |u(x)|^{\frac{n}{n-1}} dx_1 dx_2 \leq \left(\iint |Du| dx_1 dy_2 \right)^{\frac{1}{n-1}} \int \left(\int |Du| dy_1 \right)^{\frac{1}{n-1}} \cdot \prod_{i=3}^n \left(\iint |Du| dx_1 dy_i \right)^{\frac{1}{n-1}} dx_2.$$

For simplicity, we use $I_i := f_i$ so again, by generalized Hölder and Fubinni, we have:

$$\iint |u|^{\frac{n}{n-1}} dx_1 dx_2 \leq \left(\iint |Du| dy_1 dx_2 \right)^{\frac{1}{n-1}} \left(\iint |Du| dx_1 dy_2 \right)^{\frac{1}{n-1}} \left(\prod_{i=3}^n \iint |Du| dx_1 dx_2 dy_i \right)^{\frac{1}{n-1}}.$$

Repeat inductively, we have:

$$\int_{\mathbb{R}^n} |u|^{\frac{n}{n-1}} \leq \prod_{i=1}^n \left(\int |Du| dx_1 \cdots dx_{i-1} dy_i dx_{i+1} \cdots dx_n \right)^{\frac{1}{n-1}} = \|Du\|_{L^1(\mathbb{R}^n)}^{\frac{n}{n-1}}.$$

Hence, this holds for $p = 1$ case.

Now, for any $p \in (1, n)$, of we consider $|u|^\gamma$, we go from the $p = 1$ case that:

$$\begin{aligned} \left(\int_{\mathbb{R}^n} |u|^{\frac{\gamma n}{n-1}} \right)^{\frac{n-1}{n}} &= \| |u|^\gamma \|_{L^{\frac{n}{n-1}}(\mathbb{R}^n)}^{\frac{n-1}{n}} \leq \|D|u|^\gamma\|_{L^1(\mathbb{R}^n)} = \int_{\mathbb{R}^n} D(|u|^\gamma) = \int_{\mathbb{R}^n} \gamma \cdot |u|^{\gamma-1} |Du| \\ &\leq \gamma \|Du\|_{L^p(\mathbb{R}^n)} \cdot \| |u|^{\gamma-1} \|_{L^{\frac{p}{p-1}}(\mathbb{R}^n)}. \end{aligned}$$

If we choose γ such that:

$$\frac{\gamma n}{n-1} = (\gamma-1) \frac{p}{p-1} \iff \gamma = \frac{p(n-1)}{n-p} > 1.$$

Hence, we have $\frac{\gamma n}{n-1} = \cdots = \frac{np}{n-p}$, then if $\| |u|^{\frac{np}{n-p}} \|_{L^{\frac{p}{p-1}}(\mathbb{R}^n)} \neq 0$, i.e., when $u \neq 0$, we can divide by $\| |u|^{\frac{np}{n-p}} \|_{L^{\frac{p}{p-1}}(\mathbb{R}^n)}$ on both sides, to obtain that:

$$\|u\|_{L^{p^*}(\mathbb{R}^n)} = \left(\int |u|^{\frac{np}{n-p}} \right)^{\frac{n-1}{n} - \frac{p-1}{p}} \leq \frac{p(n-1)}{n-p} \cdot \|Du\|_{L^p(\mathbb{R}^n)}.$$

□

Remark III.4.4.

- We need $u \in C_c^1(\mathbb{R}^n)$, as for example, we have $u = 1$ fails for the inequality.

- The constant $C = \frac{p(n-1)}{n-p}$ is not optimal in the proof. In fact, if $p = 1$, GNS is equivalent to the isoperimetric inequality.

If we choose $u \in C_c^1(\mathbb{R}^n)$ with $\begin{cases} u = 1 \text{ on embedding of } \Omega, \Omega_\epsilon \\ |\nabla u| \sim \frac{1}{\epsilon} \text{ on } \Omega_\epsilon \setminus \Omega, \end{cases}$ and then we have:

$$\|\nabla u\|_{L^1(\mathbb{R}^n)} \sim \text{Area of } \partial\Omega,$$

or we have that:

$$\|u\|_{L^{\frac{n}{n-1}}(\mathbb{R}^n)} \sim (\text{Volume of } \Omega)^{\frac{n-1}{n}}.$$

」

Theorem III.4.5. GNS for $W^{1,p}$.

If $\Omega \subset \mathbb{R}^n$ is open, bounded, $\partial\Omega$ is C^1 , $p \in [1, n)$, then:

$$W^{1,p}(\Omega) \subset L^{p^*}(\Omega)$$

and there exists a $C > 0$, depending on Ω, p, n such that:

$$\|u\|_{L^{p^*}(\Omega)} \leq C \|u\|_{W^{1,p}(\Omega)} \quad \text{for all } u \in W^{1,p}(\Omega).$$

Proof of sketch. We extend u to $W^{1,p}(\mathbb{R}^n)$ function, mollify it, and apply GNS for $\mathbb{R}^n$. □

Theorem III.4.6. Poincaré Inequality.

If $\Omega \subset \mathbb{R}^n$ is open, bounded, $p \in [1, n)$, and $q \in [1, p^*]$, there is a constant $c > 0$, depending on Ω, p, q, n such that:

$$\|u\|_{L^q(\Omega)} \leq C \|Du\|_{L^p(\Omega)} \quad \text{for } u \in W_0^{1,p}(\Omega)$$

In particular, as $p \leq p^*$, which implies that $\|u\|_{L^p(\Omega)} \leq C \|Du\|_{L^p(\Omega)}$.

Proof of sketch. As $u \in W_0^{1,p}(\Omega)$, this implies that $\{u_i\} \subset C_c^\infty(\Omega) \subset C_c^\infty(\mathbb{R}^n)$, and with $u_i \rightarrow u$ in $W^{1,p}$, we can then apply GNS to u_i . □

IV Hölder Spaces

IV.1 Hölder Space

We treat the case when $p > n$.

Definition IV.1.1. Hölder Continuous.

If $\Omega \subset \mathbb{R}^n$ is open, we say that $u : \Omega \rightarrow \mathbb{R}$ is **Hölder continuous** with exponents $\gamma \in (0, 1]$ if there is a constant $C > 0$ such that:

$$|u(x) - u(y)| \leq C \cdot |x - y|^\gamma \quad \text{for all } x, y \in \Omega.$$

When $\gamma = 1$, we have the Lipschitz condition. ┘

Definition IV.1.2. The γ -Hölder semi-norm is defined as:

$$[u]_{C^{0,\gamma}(\overline{\Omega})} = \sup_{\substack{x, y \in \Omega \\ x \neq y}} \left(\frac{|u(x) - u(y)|}{|x - y|^\gamma} \right).$$
┘

The Hölder semi-norm is not a norm since for the constant function 1, we have:

$$[1]_{C^{0,\gamma}(\overline{\Omega})} = 0,$$

but 1 is not the zero function.

Definition IV.1.3. The γ -Hölder norm is defined using the semi-norm:

$$\|u\|_{C^{0,\gamma}(\overline{\Omega})} = \|u\|_{C^0(\overline{\Omega})} + [u]_{C^{0,\gamma}(\overline{\Omega})}.$$
┘

The Hölder space, for $k \in \mathbb{N}$, are then:

$$C^{k,\gamma}(\overline{\Omega}) = \{u \in C^k(\overline{\Omega}) \mid \|D^\alpha u\|_{C^{0,\gamma}(\overline{\Omega})} < \infty \text{ for } |\alpha| \leq k\},$$

in which if we equip with the norm, we have:

$$\|u\|_{C^{k,\gamma}(\overline{\Omega})} = \|u\|_{C^k(\overline{\Omega})} + \sum_{|\alpha| \leq k} [D^\alpha u]_{C^{0,\gamma}(\overline{\Omega})}.$$

Remark IV.1.4.

- $(C^{k,\gamma}(\overline{\Omega}), \|\cdot\|_{C^{k,\gamma}(\overline{\Omega})})$ is Banach (by Arzelá-Ascoli).
- If $\gamma > 1$, then the γ -Hölder functions are constant.
- If Ω is bounded and convex, $u \in C^{k+1}(\overline{\Omega})$ implies that $u \in C^{k,\gamma}(\overline{\Omega})$ for all $\gamma \in (0, 1]$. ┘

Example IV.1.5. If $\gamma \in (0, 1]$, we let $u(x) = |x|^\gamma$ on $(-1, 1)$ is $C^{0,\gamma}$ but not $C^{0,\tilde{\gamma}}$ for $\tilde{\gamma} \in (\gamma, 1]$. $\diamond$

Theorem IV.1.6. Morrey Inequality.

If $p \in (n, \infty]$ then there is a constant $C > 0$ depending on p and n such that for $u \in C^1(\mathbb{R}^n)$ we have:

$$\|u\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)},$$

where $\gamma = 1 - \frac{n}{p}$ (where $\gamma = 1$ if $p = \infty$).

Proof. The idea is to bound $\|u\|_{C^0(\bar{\Omega})}$ and $[u]_{C^{0,\gamma}(\bar{\Omega})}$ by the fundamental theorem of calculus and Hölder inequality.

Fix $r > 0$, $x \in \mathbb{R}^n$, and for $\omega \in \partial B_1(0) = \mathbb{S}^{n-1}$, we have by the fundamental theorem of calculus that for $0 < s < r$, we have:

$$u(x + s\omega) - u(x) \leq \int_0^s \left| \frac{d}{dt}(u(x + t\omega)) \right| dt = \int_0^s |Du(x + t\omega) \cdot \omega| dt \leq \int_0^s |Du(x + t\omega)| dt.$$

Then, by using Fubini, we have:

$$\int_{\mathbb{S}^{n-1}} |u(x + s\omega) - u(x)| dS \leq \int_{\mathbb{S}^{n-1}} \int_0^s |Du(x + t\omega)| dt dS = \int_0^s \int_{\mathbb{S}^{n-1}} |Du(x + t\omega)| \frac{t^{n-1}}{t^n} dt dS.$$

Now, we set $y = x + t\omega$ and $t = |x - y|$ (since $|\omega| = 1$) with $dy = t^{n-1} dS dt$, we have:

$$\int_0^s \int_{\mathbb{S}^{n-1}} |Du(x + t\omega)| \frac{t^{n-1}}{t^n} dt dS \leq \int_{B_r(x)} |u(y) - u(x)| dy.$$

Then, by integrating from 0 to r after multiplying by $\mathbb{S}^{n-1}$, we have:

$$\begin{aligned} \int_0^r \int_{\mathbb{S}^{n-1}} |u(x + s\omega) - u(x)| dS s^{n-1} ds &= \int_{B_r(x)} |u(y) - u(x)| dy \leq \int_0^r s^{n-1} \int_{B_r(x)} \frac{|Du(y)|}{|x - y|^{n-1}} dy ds \\ &= \frac{r^n}{n} \int_{B_r(x)} \frac{|Du(y)|}{|x - y|^{n-1}} dy. \end{aligned} \quad (5)$$

By denoting $f_A f = \frac{1}{m(A)} \int_A f$, we have:

$$\begin{aligned} |u(x)| &= \oint_{B_1(x)} |u(x)| \leq \oint_{B_1(x)} |u(x) - u(y)| dy + \oint_{B_1(x)} |u(y)| dy \\ &\leq \frac{m(B_1(0))}{n} \int_{B_1(x)} \frac{|Du(y)|}{|x - y|^{n-1}} dy + \oint_{B_1(x)} |u(y)| \\ &\leq C \|Du\|_{L^p(B_1(x))} \cdot \left(\int_{B_1(x)} |x - y|^{(1-n) \cdot \frac{p}{p-1}} \right)^{\frac{p-1}{p}} + C \|u\|_{L^p(B_1(x))} \leq C \|u\|_{W^{1,p}(B_1(x))}, \end{aligned}$$

as we have $\frac{(n-1)p}{p-1} < n$ and we can consider this as a Sobolev spike, i.e.:

$$\|u\|_{C^0(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)}.$$

Now, if $x, y \in \mathbb{R}^n$, $r = |x - y|$, and we set $W := N_r(x) \cap B_r(y)$, so we have:

$$|u(x) - u(y)| = \int_W |u(x) - u(z)| dz + \int_W |u(z) - u(y)| dz.$$

Hence, by (5) and Hölder inequality, we have:

$$\begin{aligned} \int_{B_r(x)} |u(x) - u(z)| dz &= \frac{C}{r^n} \int_{B_r(x)} |u(x) - u(z)| dz \leq C \int \frac{|Du(z)|}{|x - z|^{n-1}} dz \\ &\leq C \|Du\|_{L^p(\mathbb{R}^n)} \left(\int_{B_r(x)} |x - z|^{(1-n) \cdot \frac{p}{p-1}} dz \right)^{\frac{p-1}{p}} \\ &\leq C \|Du\|_{L^p(\mathbb{R}^n)} \left(r^{n-(n-1) \frac{p}{p-1}} \right)^{\frac{p-1}{p}} = C \|Du\|_{L^p(\mathbb{R}^n)} \cdot r^{1-\frac{n}{p}}. \end{aligned}$$

Hence, we have:

$$|u(x) - u(y)| \leq C \cdot r^{1-\frac{n}{p}} \|Du\|_{L^p(\mathbb{R}^n)} = C |x - y|^{1-\frac{n}{p}} \|Du\|_{L^p(\mathbb{R}^n)}.$$

So we have that:

$$[u]_{C^{0,1-\frac{n}{p}}} \leq C \cdot \|Du\|_{L^p(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)},$$

and hence we have $\|u\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\mathbb{R}^n)}$. $\square$

Theorem IV.1.7. Morrey for $W^{1,p}$.

If $\Omega \subset \mathbb{R}^n$ is open, bounded, $\partial\Omega$ is C^1 , and $p \in (n, \infty]$, then there is a constant $c > 0$ depending on p, n , and Ω such that if $u \in W^{1,p}(\Omega)$, then $u = \tilde{u}$ almost everywhere for some $\tilde{u} \in C^{0,\gamma}(\overline{\Omega})$ where $\gamma = 1 - \frac{n}{p}$ (as well $\gamma = 1$ if $p = \infty$), and $\|\tilde{u}\|_{C^{0,\gamma}(\overline{\Omega})} \leq C \|u\|_{W^{1,p}(\Omega)}$.

Proof of sketch. Same as for GNS in $W^{1,p}$, we extend and mollify/approximate u is $p < \infty$. $p = \infty$ case we will treat later. $\square$

IV.2 Sobolev Embeddings

We have not addressed $W^{1,n}$, but note that as $p \nearrow n$, we have:

$$p^* = \frac{np}{n-p} \rightarrow \infty,$$

so we may hope that $W^{1,n} \subset L^\infty$.

Example IV.2.1. If $u(x) = \log\left(\log\left(1 + \frac{1}{|x|}\right)\right)$, then $u \in W^{1,n}(B) \setminus L^\infty(B)$. $\diamond$

Remark IV.2.2. One can show that $W^{1,n} \subset \text{BMO}$ (bounded mean oscillation) by establishing the Poincaré type inequality. Hence, we have BMO replaces L^∞ if $p = n$. $\lrcorner$

Theorem IV.2.3. Sobolev Embedding.

If $\Omega \subset \mathbb{R}^n$ is open, bounded, $\partial\Omega$ is C^1 , $p \in [1, \infty)$, $k \in \mathbb{N}$, and $u \in W^{k,p}(\Omega)$, then:

(i) If $k < \frac{n}{p}$, then $u \in L^q(\Omega)$ where $\frac{1}{q} = \frac{1}{p} - \frac{k}{n}$ and:

$$\|u\|_{L^q(\Omega)} \leq C \|u\|_{W^{k,p}(\Omega)}$$

for $C > 0$ dependent on k, p, n, Ω .

(ii) If $k > \frac{n}{p}$, then $u \in C^{k-1-\lfloor \frac{n}{p} \rfloor, \gamma}(\overline{\Omega})$ where:

$$\gamma = \begin{cases} \text{any value in } (0, 1), & \text{if } \frac{n}{p} \in \mathbb{N}, \\ \lfloor \frac{n}{p} \rfloor - \frac{n}{p} + 1 & \text{if } \frac{n}{p} \notin \mathbb{N}, \end{cases}$$

$$\text{and } \|u\|_{C^{k-1-\lfloor \frac{n}{p} \rfloor, \gamma}(\overline{\Omega})} \leq C \|u\|_{W^{k,p}(\Omega)}$$

for $C > 0$ dependent on k, n, p, γ, Ω .

Remark IV.2.4. For the above theorem, when $k = 1$, then (1) is GNS and (2) is Poincaré. ┘

Proof. If $k < \frac{n}{p}$, as $D^\alpha u \in L^p(\Omega)$ for all $|\alpha| \leq k$, so we can apply GNS inequality (Theorem III.4.3) for $W^{1,p}$ to $D^\beta u \in W^{1,p}(\Omega)$ for $|\beta| \leq k-1$ implies that:

$$\|D^\beta u\|_{L^{p^*}(\Omega)} \leq C \|u\|_{W^{k,p}(\Omega)},$$

so we have $u \in W^{k-1, p^*}(\Omega)$. Note that $p^* > p$.

We apply the GNS inequality again to obtain that $u \in W^{k-2, p^{**}}(\Omega)$, where we have:

$$\frac{1}{p^{**}} = \frac{1}{p^*} - \frac{1}{n} = \frac{1}{p} - \frac{1}{n} - \frac{1}{n} = \frac{1}{p} - \frac{2}{n}.$$

Doing this k times, we see that:

$$u \in W^{0, p^{***}}(\Omega) = L^{p^{***}}(\Omega),$$

where we have $\frac{1}{p^{***}} = \frac{1}{p} - \frac{k}{n}$, so we have that $u \in L^{\frac{1}{\frac{1}{p} - \frac{k}{n}}}(\Omega) = L^q(\Omega)$, so (1) case holds.

For the other case, consider $k > \frac{n}{p}$. If $\frac{n}{p} \in \mathbb{N}^+$, let's set $\ell = \frac{n}{p} - 1 \in \mathbb{N}$, applying GNS inequality gives us that $u \in W^{k-\ell, r}(\Omega)$ where:

$$\frac{1}{r} = \frac{1}{p} - \frac{\ell}{n} \iff r = \frac{np}{n - p\ell} = n,$$

i.e., $u \in W^{k-\ell, n}(\Omega) \subset W^{k-\ell, s}(\Omega)$ for all $s \in [1, n)$ (Bounds for Hölder inequality) since Ω is bounded.

Now, we apply GNS inequality for $W^{1,p}$ to get $D^\alpha u \in L^{s^*}(\Omega)$ for $|\alpha| \leq k - \ell - 1 = k - \frac{n}{p}$ and noting that $s \in [1, n)$ implies that $s^* \in (n, \infty)$, we have, by the Morrey inequality, for $W^{1,p}$ implies that $D^\beta u \in C^{0, 1-\frac{n}{q}}(\overline{\Omega})$ for any $q \in (n, \infty)$, $|\beta| \leq k - 1 - \frac{n}{p}$.

Eventually, if $k > \frac{n}{p}$ and $\frac{n}{p} \notin \mathbb{N}$, we repeat as before with $\ell = \lfloor \frac{n}{p} \rfloor$ implying that $u \in W^{k-\ell, r}(\Omega)$ with $r = \frac{np}{n - p\ell} > n$ for $\ell p < n$. Therefore, by Morray again, we have $D^\beta u \in C^{0, 1-\frac{n}{r}}(\overline{\Omega})$ for $|\beta| \leq k - 1 - \lfloor \frac{n}{p} \rfloor$

and as $1 - \frac{n}{r} = 1 - \frac{n}{p} + \ell = \lfloor \frac{n}{p} \rfloor - \frac{n}{p} + 1$, i.e.:

$$u \in C^{k-1-\lfloor \frac{n}{p} \rfloor, \lfloor \frac{n}{p} \rfloor - \frac{n}{p} + 1}(\overline{\Omega}),$$

as desired. $\square$

Remark IV.2.5. Consider $u \in W^{k,p}(\Omega)$ for all $k \geq 1$ implies that $u \in C^\infty(\overline{\Omega})$. $\lrcorner$

Then, we will be discuss about the compactness in Sobolev space.

Theorem IV.2.6. Arzelá-Ascoli.

If $K \subset \mathbb{R}^n$ is compact and $\{f_n\}_{n=1}^\infty$ is a sequence of continuous function on K such that it is:

- (i) Pointwise bounded, i.e., $|f_n| \leq \phi$ for some $\phi : K \rightarrow \mathbb{R}$ for all $n \geq 1$,
- (ii) Equicontinuous, for $\epsilon > 0$, there is a $\delta > 0$ such that for all $x, y \in K$ in which $|x - y| < \delta$, then:

$$|f_n(x) - f_n(y)| < \epsilon \quad \text{for all } n \geq 1.$$

Then $\{f_n\}$ is uniformly bounded and has a uniformly convergent subsequence.

Remark IV.2.7. If $\{f_n\} \subset C^1(\mathbb{R}^n)$ almost everywhere bounded, then $f_n \rightarrow f$ uniformly on compact sets up to a subsequence. $\lrcorner$

Definition IV.2.8. If X and Y are Banach, with $X \subset Y$, we have a **continuous embedding**, denoted:

$$X \hookrightarrow Y$$

if $\|x\|_Y \leq C\|x\|_X$ for all $x \in X$ and $C > 0$ independent of x .

We denote $X \xhookrightarrow{c} Y$ if $X \hookrightarrow Y$ and every bounded sequence in X has a convergent subsequence in Y , in which we say X is compactly embedded in Y . $\lrcorner$

Example IV.2.9.

- By Arzelá-Ascoli ([Theorem IV.2.6](#)), $C^1(\overline{\Omega}) \xhookrightarrow{c} C^0(\overline{\Omega})$ for $\Omega \in \mathbb{R}^n$ being bounded.
- By GNS inequality ([Theorem III.4.3](#)), we have $W^{1,p}(\Omega) \hookrightarrow L^{p^*}(\Omega)$ for $p < n$.
- By Morrey ([Theorem IV.1.6](#)), we have $W^{1,p}(\Omega) \hookrightarrow C^{0,1-\frac{n}{p}}(\overline{\Omega})$ for $p > n$. $\diamond$

Theorem IV.2.10. Rellich-Kondrachov Compactness.

If $\Omega \subset \mathbb{R}^n$ is open, bounded, $\partial\Omega$ is C^1 , and $p \in [1, n)$, then:

$$W^{1,p}(\Omega) \xhookrightarrow{c} L^q(\Omega) \quad \text{for } q \in [1, p^*).$$

If instead $p \in [1, \infty]$, we have:

$$W^{1,p}(\Omega) \hookrightarrow L^p(\Omega).$$

And, if $\partial\Omega$ is not necessarily C^1 , we have:

$$W_0^{1,p}(\Omega) \hookrightarrow L^p(\Omega).$$

Example IV.2.11. $W^{1,p}(\Omega) \hookrightarrow L^{p^*}(\Omega) = L^{\frac{np}{n-p}}(\Omega)$ is never compact.

To see this, let $\Omega = B := B_1(0) \subset \mathbb{R}^n$, and $u \in W_0^{1,p}(\Omega) \setminus \{0\}$ and $\|u\|_{W^{1,p}(\Omega)} > 0$. In particular, $u \in W^{1,p}(\mathbb{R}^n)$ by extension, with $u = 0$ on $\mathbb{R}^n \setminus B$. By scaling, for $\lambda \in (0, 1]$, we have that:

$$u_\lambda(x) = u\left(\frac{x}{\lambda}\right) \quad \text{so we have} \quad \begin{cases} \|u_\lambda\|_{L^{p^*}} = \lambda^{\frac{n}{p^*}} \|u\|_{L^{p^*}(B)} > 0, \\ \|Du_\lambda\|_{L^p(B)} = \lambda^{\frac{n}{p}-1} \|Du\|_{L^p(B)}. \end{cases}$$

Then, we have $\lambda^{-\frac{n}{p^*}} u_\lambda \in W_0^{1,p}(B)$ with $\|\lambda^{-\frac{n}{p^*}} u_\lambda\|_{L^{p^*}(B)} = \|u\|_{L^{p^*}(B)} > 0$, then $(\lambda^{-\frac{n}{p^*}} u_\lambda)$ is bounded in $W^{1,p}(B)$.

Note that $\lambda^{-\frac{n}{p^*}} u_\lambda \rightarrow 0$ almost everywhere in B as $\lambda \rightarrow 0$ as $\frac{x}{\lambda} \notin B$ for some small λ .

Hence, if $\lambda^{-\frac{n}{p^*}} u_\lambda \rightarrow v$ in $L^{p^*}(B)$, then $v \equiv 0$. But $\|\lambda^{-\frac{n}{p^*}} u_\lambda\|_{L^{p^*}(B)} > 0$ so $\|v\|_{L^{p^*}(B)} > 0$. Therefore, it has no convergent subsequence, so $W^{1,p}(B) \hookrightarrow L^{p^*}(B)$ is not compact. $\diamond$

Proof of Theorem IV.2.10. Fix $q \in [1, p^*)$, and since Ω is bounded:

$$W^{1,p}(\Omega) \hookrightarrow L^1(\Omega).$$

(If $\{u_i\} \subset W^{1,p}(\Omega)$ is bounded, by the extension theorem (since $\partial\Omega$ is C^1), we can assume $u_i \in W^{1,p}(\tilde{\Omega})$ for some $\Omega \Subset \tilde{\Omega}$ with:

$$\sup_i \|u_i\|_{W^{1,p}(\tilde{\Omega})} < +\infty.$$

Hence, we consider $u_i^\epsilon = \eta_\epsilon * u_i$ for $\epsilon > 0$ sufficiently small (determined by $\tilde{\Omega}$), so that $\text{supp}(u_i^\epsilon) \Subset \tilde{\Omega}$.

If $u_i \in C_c^1(\tilde{\Omega})$ for some $i \geq 1$, then for $x \in \tilde{\Omega}$, we have:

$$\begin{aligned} u_i^\epsilon(x) - u_i(x) &= \int_{B_\epsilon(x)} \eta_\epsilon(x-y)(u_i(y) - u_i(x)) dy \\ &= \int_{B_1(0)} \eta(z)(u_i(x-\epsilon z) - u_i(x)) dz \\ &= -\epsilon \int_{B_1(0)} \eta(z) \left[\int_0^1 \nabla u_i(x - \epsilon tz) z dt \right] dz \quad (\text{FTC}). \end{aligned}$$

By Fubinni, we have:

$$\int_{\tilde{\Omega}} |u_i^\epsilon(x) - u_i(x)| dx \leq \epsilon \int_{B_1(0)} \eta(z) \left[\int_0^1 \int_{\tilde{\Omega}} |Du_i(x - \epsilon tz) z| dx dt \right] dz \leq \epsilon \|Du_i\|_{L^1(\tilde{\Omega})}.$$

By global approximation, for $u_i \in W^{1,p}(\tilde{\Omega})$, we have:

$$\|u_i^\epsilon - u_i\|_{L^1(\tilde{\Omega})} \leq \epsilon \|Du_i\|_{L^1(\tilde{\Omega})} \leq \epsilon \cdot c \|Du_i\|_{L^1(\tilde{\Omega})} \leq \epsilon \cdot C \cdot \sup_i \|u_i\|_{W^{1,p}(\tilde{\Omega})}.$$

Thus, we have $u_i^\epsilon \rightarrow u_i$ in $L^1(\tilde{\Omega})$ as $\epsilon \rightarrow 0$ uniformly with i . By the interpolation of L^p spaces, if $\lambda \in (0,1)$, with $\frac{1}{q} = \lambda + \frac{1-\lambda}{p^*}$, then:

$$\begin{aligned} \|u_i^\epsilon - u_i\|_{L^q(\tilde{\Omega})} &\leq \|u_i^\epsilon - u_i\|_{L^1(\tilde{\Omega})}^\lambda \cdot \|u_i^\epsilon - u_i\|_{L^{p^*}(\tilde{\Omega})}^{1-\lambda} \leq C \|u_i^\epsilon - u_i\|_{L^1(\tilde{\Omega})}^\lambda \|Du_i^\epsilon - Du_i\|_{L^p(\tilde{\Omega})}^{1-\lambda} \\ &\leq C \|u_i^\epsilon - u_i\|_{L^1(\tilde{\Omega})}^\lambda [\|Du_i^\epsilon\|_{L^p(\tilde{\Omega})} + \|Du_i\|_{L^p(\tilde{\Omega})}]^{1-\lambda}, \end{aligned}$$

and as $\sup_i \|u_i\|_{W^{1,p}(\tilde{\Omega})} < \infty$, then we have $u_i^\epsilon \rightarrow u_i$ in $L^q(\tilde{\Omega})$ as $\epsilon \rightarrow 0$ uniformly in i .

Hence, if $x \in \tilde{\Omega}$, we have:

$$|u_i^\epsilon(x)| \leq \int_{B_\epsilon(x)} \eta_\epsilon(x-y) |u_i(y)| dy \leq \epsilon^{-n} \|u_i\|_{L^1(\tilde{\Omega})}.$$

Therefore, we have $\{u_i^\epsilon\}$ being uniformly bounded on $\tilde{\Omega}$, so:

$$|Du_i^\epsilon(x)| \leq \int_{B_\epsilon(x)} |D\eta_\epsilon(x-y)| |u_i(y)| dy \leq C\epsilon^{-(n+1)}.$$

Hence, $\{u_i^\epsilon\}$ has uniformly bounded C^1 norm, so by the mean value theorem, we have $\{u_i^\epsilon\}$ being equicontinuous. Thus, by Arzelà-Ascoli, there is up to a subsequence that we have uniform convergence. So, if $\delta > 0$, there is some $N \geq 1$ with:

$$\|u_{i_j}^\epsilon - u_{i_k}^\epsilon\|_{L^q(\tilde{\Omega})} < \frac{\delta}{3} \quad \text{for } j, k \geq N.$$

Also, as $u_i^\epsilon \rightarrow u_i$ in $L^q(\tilde{\Omega})$, we have both:

$$\begin{cases} \|u_{i_j}^\epsilon - u_{i_j}\|_{L^q(\tilde{\Omega})} < \frac{\delta}{3}, \\ \|u_{i_k}^\epsilon - u_{i_k}\|_{L^q(\tilde{\Omega})} < \frac{\delta}{3}, \end{cases} \quad \text{for } j, k \geq \tilde{N}.$$

Hence, by the triangle inequality, we have:

$$\|u_{i_j} - u_{i_k}\|_{L^q(\tilde{\Omega})} \leq \|u_{i_j} - u_{i_j}^\epsilon\|_{L^q(\tilde{\Omega})} + \|u_{i_j}^\epsilon - u_{i_k}^\epsilon\|_{L^q(\tilde{\Omega})} + \|u_{i_k}^\epsilon - u_{i_k}\|_{L^q(\tilde{\Omega})} < \delta.$$

By choosing $\delta = \frac{1}{\ell}$ for $\ell \geq 1$, we have $\{u_{i_j}\}$ being Cauchy in $L^2(\tilde{\Omega})$, and hence the space is complete. As $L^1(\tilde{\Omega})$ is complete, we obtain a convergent subsequence of $\{u_i\}$.

Hence, we have $W^{1,p}(\Omega) \xhookrightarrow{c} L^q(\Omega)$.

Noting that $p < p^*$ and $p^* \rightarrow \infty$ as $p \rightarrow n^-$, so we're done if $p \in [1, n)$, we just need to consider the case in which $p \geq n$.

- If $p^* > n$, then $W^{1,n}(\Omega) \hookrightarrow W^{1,p}(\Omega) \xhookrightarrow{c} L^n(\Omega)$.

For $p \in (n, \infty]$, we have the Morrey and Arzelà-Ascoli giving us that:

$$W^{1,p}(\Omega) \hookrightarrow C^{0,1-\frac{n}{p}}(\overline{\Omega}) \xhookrightarrow{c} L^p(\Omega).$$

If $\partial\Omega$ is not C^1 , as in the first part, choose $\Omega \Subset \tilde{\Omega}$ with $\tilde{\Omega}$ bounded, $\partial\tilde{\Omega}$ is C^1 to apply the above and get:

$$W_0^{1,p}(\Omega) \hookrightarrow W_0^{1,p}(\tilde{\Omega}) \xhookrightarrow{c} L^p(\Omega)$$

for $p \in [1, \infty]$. □

Remark IV.2.12. Also, this theorem fails if Ω is unbounded, as we can translate test functions to infinity.

┘

V Compactness in Sobolev Space

V.1 Reflexivity

Recall that if X is Banach, X^* is the space of all bounded linear functions on X , i.e., $L : X \rightarrow \mathbb{R}$ which is bounded (or *continuous*).

Note that X^* is also a Banach space with respect to the operator norm $\|\cdot\|_{\text{op}}$.

Then $(X^*)^*$ is the *double* dual consisted of bounded linear functionals on X^* . Hence, if $x \in X$ and $L \in X^*$, then $L(x) \in \mathbb{R}$. Then, we define $x^{**} : X^* \rightarrow \mathbb{R}$ by setting:

$$x^{**}(L) := \text{ev}_x(L) = L(x).$$

Therefore, also, we have $\|x^{**}\|_{**} = \|x\|_X$.

Thus, $\Lambda : X \rightarrow (X^*)^*$ defined by $\Lambda(x) = x^{**}$ is a bounded linear injection/embedding (which we can interpret it as $X \subset X^{**}$).

Definition V.1.1. If $\Lambda : X \rightarrow (X^*)^*$ is a linear isomorphism, we say that X is reflexive, i.e., it Λ is surjective. ┘

Remark V.1.2. Hilbert space is *reflexive* by the Riesz representation theorem. ┘

Example V.1.3. $(L^p)^* = L^q$ if $p \in (1, \infty)$ and $\frac{1}{p} + \frac{1}{q} = 1$. Hence, L^p is reflexive. However, since $L^1 \subsetneq (L^\infty)^*$ so L^1, L^∞ are not reflexive. ◇

Remark V.1.4. There is example of $X \simeq X^{**}$ but not being reflexive, e.g. see the James' space. ┘

Remark V.1.5. $W^{k,p}$ is also reflexive for $p \in (1, \infty)$. ┘

V.2 Weak Compactness

Recall that $\dim(X) < \infty$ if and only if the unit ball in X is compact.

To resolve this lack of compactness when $\dim(X) = \infty$, we introduce weak compactness.

Definition V.2.1. Weak Compactness.

If X is Banach and $\{x_n\} \subset X$, we say that $x_n \rightharpoonup x$ or $\{x_n\}$ **weakly converges** to $x \in X$ if:

$$L(x_n) \rightarrow L(x) \quad \text{for all } L \in X^*.$$

┘

Remark V.2.2. If $x_n \rightarrow x$ in X , then $x_n \rightharpoonup x$ since:

$$|L(x_n) - L(x)| \leq \|L\|_{\text{op}} \cdot \underbrace{\|x_n - x\|_X}_{\rightarrow 0} \rightarrow 0$$

for every $L \in X^*$.

If $x_n \rightharpoonup x$, then $\{x_n\}$ is bounded. The proof uses **principle of uniform boundedness** and implies, by the **Hahn-Banach**, that:

$$\|x\|_X \leq \liminf_{n \rightarrow \infty} \|x_n\|_X,$$

with lower semi-continuity with respect to weak convergence. ┘

Example V.2.3. In a Hilbert space $(\mathcal{H}, (\cdot, \cdot))$, **Riesz representation theorem** implies that $x_n \rightharpoonup x$ if and only if $(x_n, y) \rightarrow (x, y)$ for each $y \in \mathcal{H}$.

Thus, in $H_0^1(\Omega)$, if $f_n \rightharpoonup f$ in $\mathcal{H}_0^1$, then:

$$(f_n, g)_{H_0^1(\Omega)} \rightarrow (f, g)_{H_0^1(\Omega)} \quad \text{for } g \in H_0^1(\Omega),$$

which is equivalently:

$$\int_{\Omega} Df_n \cdot Dg \rightarrow \int_{\Omega} Df \cdot Dg \quad \text{for all } g \in H_0^1(\Omega). \quad \diamond$$

Example V.2.4. If $p \in [1, \infty)$ as $(L^p)^* = L^q$ for $\frac{1}{p} + \frac{1}{q} = 1$, so we have $f_n \rightharpoonup f$ in L^p if and only if $\int f_n \cdot g \rightarrow \int f \cdot g$ for all $g \in L^q$. $\diamond$

Theorem V.2.5. Weak Compactness.

If X is a reflexive Banach space, every bounded sequence in X has a weakly convergent subsequence.

Proof. Consequence of Banach-Alaoglu theorem. $\square$

Example V.2.6. This applies in $(H^k, \langle \cdot, \cdot \rangle_{H^k})$ which will be useful in PDEs.

- In $L^2((0,1))$ the bounded sequence $\{\sin(nx)\}_{n=1}^{\infty}$ has no (strongly) convergent subsequence, but $\sin(nx) \rightharpoonup 0$.

Then, we can use this to (weakly) solve the Poisson's equation (again) using weak compactness by the **direct method** (of the calculus of variations).

Recall we want to solve the PDE:

$$\begin{cases} -\Delta u = f, & \text{on } B, \\ u = 0, & \text{on } \partial B, \end{cases} \quad \text{for some } u \in C^\infty(\overline{B}).$$

We similarly (from [Example I.3.9](#)) want to solve that:

$$\int_B \nabla u \cdot \nabla \varphi = \langle u, \varphi \rangle_{H_0^1(B)} = \int_B f \varphi$$

for all $\varphi \in C_c^\infty(B)$ and for some $u \in H_0^1(B)$.

Now, suppose we solve for $f \in L^2(B)$, we can define a new concept.

Definition V.2.7. Dirichlet Energy.

We define the **Dirichlet energy** $E : H_0^1(B) \rightarrow \mathbb{R}$ by:

$$E(u) = \int_B \left(\frac{1}{2} |\nabla u|^2 - fu \right) \quad \text{for } u \in H_0^1(B).$$

We will minimize E over $H_0^1(B)$ to solve.

By Hölder, we have:

$$E(u) = \frac{1}{2} \|\nabla u\|_{L^2(B)}^2 - \int_B fu \geq \frac{1}{2} \|\nabla u\|_{L^2(B)}^2 - \|f\|_{L^2(B)} \|u\|_{L^2(B)}.$$

Then, by the Poincaré inequality, we have:

$$E(u) \geq \frac{1}{2} \|\nabla u\|_{L^2(B)}^2 - C_p \|f\|_{L^2(B)} \cdot \|\nabla u\|_{L^2(B)} \sim \frac{x^2}{2} - ax = x \left(\frac{x}{2} - a \right).$$

By assimilating this with quadratic, we have the minimum at a with value $-\frac{a^2}{2}$, and thus the equation is bounded below by:

$$E(u) \geq x \left(\frac{x}{2} - a \right) \geq -\frac{C_p^2 \|f\|_{L^2(B)}^2}{2} > -\infty.$$

Therefore, $E(u)$ is bounded below on $H_0^1(B)$.

So, consider $\{u_n\} \subset H_0^1(B)$ such that:

$$\lim_{n \rightarrow \infty} E(u_n) = \inf_{u \in H_0^1(B)} E(u) \geq -\infty.$$

Note that this infimum is $< \infty$ as $E(0) = 0$. Thus, eventually in n , $E(u_n) < 1$, therefore:

$$\frac{1}{2} \|\nabla u_n\|_{L^2(B)}^2 - c_p \|f\|_{L^2(B)} \cdot \|\nabla u_n\|_{L^2(B)} - 1 < 0,$$

so note that $\frac{1}{2}x^2 - ax - 1 < 0$ has roots given by:

$$y_{\pm} = c_p \|f\|_{L^2(B)} \pm \sqrt{2c_p \|f\|_{L^2(B)}^2 + 2} < \infty,$$

and hence, we have:

$$\|u_n\|_{H_0^1(B)} = \|\nabla u_n\|_{L^2(B)} < y_+ < \infty \quad \text{for large } n \geq 1.$$

Thence, we have $\{u_n\}$ bounded in $H_0^1(B)$ so weak compactness implies that $u \rightharpoonup u \in H_0^1(B)$ for some u (up to a subsequence, not relabeled). By lower semi-continuity of norm under weak convergence, we have:

$$\|u\|_{H_0^1(B)} \leq \liminf_{n \rightarrow \infty} \|u_n\|_{H_0^1(B)},$$

and so we have $\int_B |\nabla u|^2 \leq \liminf_{n \rightarrow \infty} \int_B |\nabla u_n|^2$.

By the Rellich-Kondrachov, $H_0^1(B) \xhookrightarrow{c} L^2(B)$, so we have $u_n \rightarrow u$ in $L^2(B)$.

Therefore, we have $\int f u_n \rightarrow \int f u$ as $n \rightarrow \infty$:

$$\int |f u_n - f u| = \int |f| |u_n - u| \leq \|f\|_{L^2} \|u_n - u\|_{L^2}.$$

Thus, we have that:

$$E(u) = \int_B \left(\frac{1}{2} |\nabla u|^2 - f u \right) \leq \liminf_{n \rightarrow \infty} \int_B \left(\frac{1}{2} |\nabla u_n|^2 - f u_n \right) = \liminf_{n \rightarrow \infty} E(u_n) = \inf_{\tilde{u} \in H_0^1(B)} E(\tilde{u}),$$

so we have u minimizes E on $H_0^1(B)$.

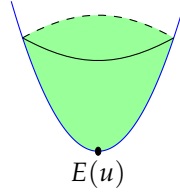


Figure V.1. The minimizer of the energy.

Now, if $\varphi \in H_0^1(B)$ and $t \in \mathbb{R}$ with $g(0) \leq g(t)$, so we have $E(u) \leq E(u + t\varphi)$ since $u + t\varphi \in H_0^1(B)$.

As a function $t \mapsto E(u + t\varphi) = g(t)$, and we have:

$$g(t) = \int_B \frac{1}{2} |\nabla(u + t\varphi)|^2 - f(u + t\varphi) = E(u) + t \left(\int_B (\nabla u \cdot \nabla \varphi - f \varphi) \right) + \frac{t^2}{2} \int_B |\nabla \varphi|^2.$$

Then g is smooth in t , so we have:

$$\begin{cases} g'(t) = 0 + \int_B (Du \cdot D\varphi - f\varphi) + t \int_B |\nabla \varphi|^2, \\ g''(t) = \int_B |\nabla \varphi|^2 \geq 0. \end{cases}$$

But we have $g'(0) = \int_B (\nabla u \cdot \nabla \varphi - f\varphi)$ and $g'(0) = 0$. Therefore, we have:

$$\int_B \nabla u \cdot \nabla \varphi = \int_B f\varphi \quad \text{for all } \varphi \in H_0^1(B).$$

◇

VI Difference Quotients

VI.1 Difference Quotients

Definition VI.1.1. Difference Quotients.

If $\Omega \subset \mathbb{R}^n$ is open, $\tilde{\Omega} \Subset \Omega$, and $u : \Omega \rightarrow \mathbb{R}$, then the i -th difference quotient of u (for $i = 1, \dots, n$) of size h , where $0 < |h| < \text{dist}(\tilde{\Omega}, \partial\Omega)$ is:

$$D_i^h u(x) = \frac{u(x + he_i) - u(x)}{h},$$

where e_i is the i -th basis vector.

Specifically, we write $D^h u(x) = (D_1^h u(x), \dots, D_n^h u(x))$. ┘

Remark VI.1.2. This is a *discrete* derivative and if $u \in C_c^1(\Omega)$, then $D_i^h u \rightarrow D_i u$ uniformly as $h \rightarrow 0$. ┘

Lemma VI.1.3. Properties of D_i^h .

If $\Omega \subset \mathbb{R}^n$ is open, $\tilde{\Omega} \Subset \Omega$, $u, v : \Omega \rightarrow \mathbb{R}$, and $0 < |h| < \text{dist}(\tilde{\Omega}, \partial\Omega)$, then for each $i = 1, \dots, n$, we have:

(i) For all $x \in \tilde{\Omega}$:

$$D_i^h(uv)(x) = u(x)D_i^h v(x) + D_i^h u(x)v(x + he_i).$$

(ii) If $\frac{1}{p} + \frac{1}{q} = 1$, $u \in L^p(\Omega)$, $v \in L^q(\Omega)$ and $\text{supp}(v) \subset \{x \in \Omega \mid \text{dist}(x, \partial\Omega) > 2h\}$, then:

$$\int_{\Omega} (D_i^h u)v = - \int_{\Omega} u \cdot D_i^{-h} v.$$

Lemma VI.1.4. D_i^h to D .

If $\Omega \subset \mathbb{R}^n$ is open, $\tilde{\Omega} \Subset \Omega$ and $0 < |h| < \frac{1}{2} \text{dist}(\tilde{\Omega}, \partial\Omega)$, then:

(i) If $p \in [1, \infty)$ and $u \in W^{1,p}(\Omega)$, we have:

$$\|D^h u\|_{L^p(\tilde{\Omega})} \leq C \|Du\|_{L^p(\Omega)}$$

for a constant $C > 0$ dependent on n and p .

(ii) If $p \in (1, \infty)$ and $u \in L^p(\Omega)$ with $\|D^h u\|_{L^p(\tilde{\Omega})} \leq M$ for some $M > 0$ and all such h as above. Therefore, $u \in W^{1,p}(\tilde{\Omega})$ and $\|Du\|_{L^p(\tilde{\Omega})} \leq M$.

Remark VI.1.5. Case (ii) fails for $p = 1$. We consider the Heaviside function:

$$H(x) = \begin{cases} 1, & \text{if } x \geq 0, \\ 0, & \text{if } x < 0. \end{cases}$$

Then, we have:

$$D^h(x) = \begin{cases} \frac{1}{h}, & \text{for } x \in [-h, 0), \\ 0, & \text{otherwise.} \end{cases}$$

If $h > 0$, we have that $\|D^h H\|_{L^1(\mathbb{R})} = 1$, but H has no weak derivative as we have shown that $H \notin W^{1,1}(\mathbb{R})$ from [Example II.1.4](#). $\square$

Proof. (i) If $u \in C^1(\Omega)$, then for $x \in \tilde{\Omega}$, the fundamental theorem of calculus gives us that:

$$u(x + he_i) - u(x) = \int_0^h \frac{d}{dt} (u(x + te_i)) dt.$$

Hence, we have:

$$|D_i^h u(x)| \leq \int_0^h \frac{1}{h} |D_i u(x + te_i)| dt \leq \frac{h^{\frac{1}{q}}}{h} \left(\int_0^h |D_i u(x + te_i)|^p dt \right)^{\frac{1}{p}}.$$

Therefore, we have:

$$\|D_i^h u\|_{L^p(\tilde{\Omega})}^p \leq \frac{1}{h} \int_{\tilde{\Omega}} \left(\int_0^h |D_i u(x + te_i)|^p dt \right) dx \leq \frac{1}{h} \int_0^h \|D_i u\|_{L^p(\Omega)}^p dt = \|D_i u\|_{L^p(\Omega)}^p.$$

Then, we have:

$$\|D^h u\|_{L^p(\tilde{\Omega})} = \int_{\tilde{\Omega}} \left(\sum_{i=1}^n |D_i^h u(x)|^2 \right)^{\frac{p}{2}} dx \leq \int_{\tilde{\Omega}} \left(\sum_{i=1}^n |D_i^h u(x)| \right)^p dx \leq \sum_{i=1}^n \|D_i^h u\|_{L^p(\tilde{\Omega})}^p \leq n \|Du\|_{L^p(\Omega)}^p.$$

Hence, $\|D^h u\|_{L^p(\tilde{\Omega})} \leq n \|Du\|_{L^p(\Omega)}$ for all $u \in C^1(\Omega) \cap L^p(\Omega)$.

By approximation, we have the statement hold for $u \in W^{1,p}(\Omega)$.

(ii) Since L^p is reflexive for $p \in (1, \infty)$, if $h_u \rightarrow 0$, as $\|D_i^{h_k} u\|_{L^p(\tilde{\Omega})} \leq \|D^{h_k} u\|_{L^p(\tilde{\Omega})} \leq M$. Thus, up to a subsequence (not relabeled), we have $D_i^{h_k} u \rightarrow v_i$ in $L^p(\tilde{\Omega})$.

If $\varphi \in C_c^\infty(\tilde{\Omega})$, we have that:

$$\int_{\tilde{\Omega}} u D_i \varphi = \int_{\Omega} u \cdot D_i \varphi = \lim_{h_k \rightarrow 0} \int_{\Omega} u \cdot D_i^{-h_k} \varphi = - \lim_{h_k \rightarrow 0} \int_{\Omega} D_i^{h_k} u \cdot \varphi = - \int_{\Omega} v_i \cdot \varphi.$$

Thus, $D_i u = v_i$ for $i = 1, \dots, n$, hence $u \in W^{1,p}(\tilde{\Omega})$ and $\|Du\|_{L^p(\tilde{\Omega})} \leq M$ by lower semi-continuity of norms under weak convergence. $\square$

VI.2 Weak and Classical Derivatives

Theorem VI.2.1. $W^{1,\infty} = C^{0,1}$.

If $\Omega \subset \mathbb{R}^n$ is open, bounded, and $\partial\Omega$ is C^1 , then $u \in W^{1,\infty}(\Omega)$ if and only if a version of u is Lipschitz, i.e., $u \in C^{0,1}(\bar{\Omega})$.

Proof. ($\implies$): Morrey inequality (Theorem IV.1.6) implies that $u \in C^{0,1}(\overline{\Omega})$ if $u \in W^{1,\infty}(\Omega)$.

($\impliedby$): As we have $|u(x + he_i) - u(x)| \leq \text{Lip}(u) \cdot |h|$, so we have $|D_i^h u(x)| \leq \text{Lip}(u)$. So $\|D_i^h u\|_{L^\infty(\Omega)} \leq \text{Lip}(u)$.

Hence, $(\nabla_i^{h_k} u)$ is bounded in $L^2(\Omega)$ as $h_k \rightarrow 0$. So, by weak compactness for L^2 , up to a subsequence (n, k related), we have $D_i^{h_k} u \rightharpoonup v_i$ for some $v_i \in L^\infty(\Omega)$.

As in the previous proof, we have $\nabla_i u = v_i$ for $i = 1, \dots, n$. So $u \in W^{1,\infty}(\Omega)$. $\square$

Remark VI.2.2. $C^{0,\alpha}$ does not imply weakly differentiable.

Here, we have the Cantor/Devil's staircase (Example II.1.5) is $C^{0,\alpha}$ for all $\alpha \in (0, \log_3 2)$ but has no weak derivative. $\lrcorner$

Theorem VI.2.3. Sobolev $\implies$ Differentiable Almost Everywhere.

If $\Omega \subset \mathbb{R}^n$ is open, $p \in (n, \infty]$ and $u \in W_{\text{loc}}^{1,p}(\Omega)$, then u is differentiable (classically) almost everywhere in Ω with gradient equal to its weak derivative.

Proof. By adjusting an estimate in the proof of Morrey, for $p \in (n, \infty)$ if $v \in W^{1,p}(B_{2r}(x))$, then we have:

$$|v(y) - v(x)| \leq Cr^{1-\frac{n}{p}} \left(\int_{B_{2r}(x)} |Dv(z)|^p dz \right)^{\frac{1}{p}}.$$

By the **Lebesgue Differentiation Theorem**, if we have:

$$v(y) = u(y) - u(x) - Du(x) \cdot (y - x)$$

and then we can apply the previous lemma for $r = |y - x|$. For Lebesgue points $x \in \Omega$, we have:

$$\begin{aligned} |u(y) - u(x) - Du(x)(y - x)| &\leq Cr^{1-\frac{n}{p}} \left(\int_{B_{2r}(x)} |Du(x) - Du(y)|^p \right)^{\frac{1}{p}} \\ &= \tilde{C}r \left(\int_{B_{2r}(x)} |Du(x) - Du(y)|^p \right)^{\frac{1}{p}} \rightarrow 0 \end{aligned}$$

as $y \rightarrow x$ and so u is differentiable at $x = Du(x)$.

Hence, this proves it if $p \in (n, \infty)$ But as $W_{\text{loc}}^{1,\infty}(\Omega) \subset W_{\text{loc}}^{1,p}(\Omega)$, which we have the result for $p = \infty$ also. $\square$

Remark VI.2.4. Rademaeher's Theorem.

As $W^{1,\infty} = C^{0,1}$, this implies that Lipschitz functions are differentiable almost everywhere. $\lrcorner$

Part 3

Second Order Linear PDEs

VII Ellipticity and Existence

VII.1 Preliminaries and Weak Solutions

Definition VII.1.1. Second Order Divergence Form Operator.

If $\Omega \subset \mathbb{R}^n$ is open, we define a **second order divergence form** (partial differential) operator L to be:

$$Lu = - \sum_{i,j=1}^n D_j(a^{ij}D_i u) + \sum_{i=1}^n b^i D_i u + cu,$$

where $u, a^{ij}, b^i, c : \Omega \rightarrow \mathbb{R}$ are the coefficients of L . ┘

Conceptually, a^{ij} is the diffusion term, b^i is the transport term, and c is the constant term.

Remark VII.1.2. Einstein Summation Convention.

We often use **Einstein summation convention** to sum over repeated indices (i, j) , for example:

$$Lu = -D_j(a^{ij}D_i u) + b^i D_i u + cu.$$

In particular, the divergence form refers to the term $D_j(a^{ij}D_i u)$ and if we had $a^{ij}D_j D_i u$ would be called **non-divergence form**. These coincide if a^{ij} is differentiable. ┘

Example VII.1.3. Poisson Equation.

If we set $a^{ij} = \delta_{ij} = \begin{cases} 1, & \text{when } i = j, \\ 0, & \text{when } i \neq j, \end{cases}$ and $b^i = c = 0$, then we have:

$$L = -\Delta = -\operatorname{div}(\nabla u).$$

Hence, we have retrieved the problem from [Example I.3.9](#). ◇

To make sense of the weak solutions to $\begin{cases} Lu = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$, we need a space for f to lie in. This will be $H^{-1} = (H_0^1)^*$.

Definition VII.1.4. H^{-1} .

If $\Omega \subset \mathbb{R}^n$ is open, we set:

$$H^{-1}(\Omega) = (H_0^1(\Omega))^*,$$

and we write $\langle \cdot, \cdot \rangle$ to denote the pairing between H^{-1} and H_0^1 . We define the H^{-1} norm by setting, for

$f \in H^{-1}(\Omega)$ as:

$$\|f\|_{H^{-1}(\Omega)} = \sup \left\{ \langle f, u \rangle \mid u \in H_0^1(\Omega), \|u\|_{H_0^1(\Omega)} \leq 1 \right\}.$$

Remark VII.1.5. If $f \in L^2(\Omega)$, we then have $f \in H^{-1}(\Omega)$ by pairing, and $\langle f, u \rangle = \int_{\Omega} f \cdot u$.

Also, if $f_0, f_1, \dots, f_n \in L^2(\Omega)$, we can define $f \in H^{-1}(\Omega)$ by setting:

$$\langle g, u \rangle = \int_{\Omega} \left(f_0 \cdot u - \sum_{i=1}^n f_i \cdot \nabla_i u \right) \quad \text{for } u \in H_0^1(\Omega).$$

Theorem VII.1.6. Characterization of H^{-1} .

If $\Omega \subset \mathbb{R}^n$ is open, and $f \in H^{-1}(\Omega)$, then there exists $f_0, \dots, f_n \in L^2(\Omega)$ such that:

$$\langle f, u \rangle = \int_{\Omega} \left(f_0 u - \sum_{i=1}^n f_i D_i u \right) \quad (6)$$

for $u \in H_0^1(\Omega)$ and:

$$\|f\|_{H^{-1}(\Omega)} = \inf_{f_0, \dots, f_n} \left\{ \left(\int_{\Omega} \sum_{i=0}^n |f_i|^2 \right)^{\frac{1}{2}} \mid (6) \text{ holds} \right\}$$

Proof. Consider $H_0^1(\Omega)$ with $(u, v) = \int_{\Omega} (u \cdot v + Du \cdot Dv)$ and the Riesz representation, there is some $v \in H_0^1(\Omega)$ such that $\langle f, u \rangle = (u, v)$ for each $v \in H_0^1(\Omega)$. Then (6) holds by setting $f_0 = u$, $f_i = D_i u$ for each $i = 1, 2, \dots, n$.

Now, if $g_0, \dots, g_n \in L^2(\Omega)$ with:

$$\langle f, u \rangle = \int_{\Omega} \left(g_0 u + \sum_{i=1}^n g_i D_i u \right) \quad \text{and } \|u\|_{H_0^1(\Omega)} \leq 1.$$

Then, by the Cauchy-Schwartz (or Hölder inequality), we have:

$$|\langle f, u \rangle| \leq \left(\int_{\Omega} \sum_{i=0}^n |g_i|^2 \right)^{\frac{1}{2}}.$$

Then we plug in $g_i = f_i - D_i v$ with $g_0 = f_0 = v$ for $i = 1, \dots, n$, for $\frac{v}{\|v\|_{H_0^1(\Omega)}}$ (in which the norm is then 1), we get:

$$\left(\int_{\Omega} \sum_{i=0}^n |f_i|^2 \right)^{\frac{1}{2}} \leq \left(\int_{\Omega} \sum_{i=0}^n |g_i|^2 \right)^{\frac{1}{2}},$$

with equality if and only if $f_i = g_i$ for $i = 0, \dots, n$.

Hence, by taking infimum over all such g_i for which (6) holds, which gives the result. $\square$

Remark VII.1.7. We note that $H_0^1(\Omega) \hookrightarrow L^2(\Omega) \hookrightarrow H^{-1}(\Omega)$.

Definition VII.1.8. Weak Solution.

If $\Omega \subset \mathbb{R}^n$ is open and L is as above with coefficients in $L^\infty(\Omega)$, then the associated bilinear form $B : H^1(\Omega) \times H^1(\Omega) \rightarrow \mathbb{R}$ is:

$$B(u, v) = \int_{\Omega} (a^{ij} D_i u D_j v + (b^i D_i u) v + cuv)$$

for $u, v \in H^1(\Omega)$. If $f \in H^{-1}(\Omega)$, then we say that $u \in H^1(\Omega)$ is a weak solution to $Lu = f$ on Ω if:

$$B(u, v) = \langle f, v \rangle \quad \text{for all } v \in H_0^1(\Omega).$$

J

Remark VII.1.9.

- If $u \in H_0^1(\Omega) \subset H^1(\Omega)$ is a weak solution to $Lu = f$ on Ω , we also say that it is a weak solution to:

$$\begin{cases} Lu = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

Note that we do not need $\partial\Omega$ to be C^1 here nor Ω to be bounded.

- U being a weak solution does not imply that Lu actually makes sense, unless a second weak derivative of u exists (and a^{ij} has one).

However, if $a^{ij} \in W^{1,\infty}(\Omega)$ and $u \in H^2(\Omega) \cap H^1(\Omega)$, then $a^{ij} D_i u \in H^1(\Omega)$ so $Lu \in L^2(\Omega)$.

Hence, by integration by parts, $B(u, v) = (Lu, v)_{L^2(\Omega)} = \int_{\Omega} Lu \cdot v$ for $v \in H_0^1(\Omega)$. Hence, u is a weak solution to $Lu = f$ on Ω for $f \in L^2(\Omega)$ if and only if $Lu = f$ almost everywhere on Ω by the Fundamental lemma of Calculus of variation.

J

Example VII.1.10. If $L = \Delta$, $A^{ij} = \delta_{ij}$, $b^i = c = 0$, we have:

$$B(u, v) = \int_{\Omega} \delta_{ij} D_i u D_j v = \int_{\Omega} Du \cdot Dv = (u, v)_{H_0^1(\Omega)}.$$

On a bounded domain is equivalently H^1 inner product by the Poincaré inequalities.

Hence, $B(u, v) = \langle f, v \rangle$ for $f \in L^2(\Omega) \subset H^{-1}(\Omega)$ which is equivalently the blueprint for Poisson equation (Example I.3.9) for $\int_{\Omega} Du Dv = \int_{\Omega} f v$ for all $v \in C_c^\infty(\Omega)$. $\diamond$

Remark VII.1.11. In general, we are going to denote that:

$$\begin{cases} Lu = -D_j(a^{ij} D_i u) + b^i D_i u + cu, \\ B(u, v) = \int_{\Omega} (a^{ij} D_i u D_j v + b^i (D_i u) v + cuv) \end{cases}$$

for $u, a^{ij}, b^i, c : \Omega \rightarrow \mathbb{R}$.

J

VII.2 Ellipticity

Physically, **ellipticity** captures “diffusion” of the space in *every* direction.

Definition VII.2.1. We say that L is (strictly uniformly) elliptic if there exists some $\lambda > 0$ such that for almost every $x \in \Omega$ we have:

$$\sum_{i,j=1}^n a^{i,j}(x) \xi_i \xi_j \geq \lambda |\xi|^2$$

for all $\xi \in \mathbb{R}^n$. ┘

Remark VII.2.2. Hereafter, we will just say elliptic, but the literatures for this is not consistent:

- Strictly means that $\lambda > 0$ and weak ellipticity means that $\lambda \geq 0$.
- Uniformly means that $\lambda > 0$ does not depend on $x \in \Omega$. ┘

Elliptic refers to the fact that the matrix $A = (a^{i,j})_{i,j}$, is positive definite for almost every $x \in \Omega$. Hence, if A is symmetric, then the equation:

$$\sum_{i,j=1}^n a^{i,j}(x) \xi_i \xi_j = 1$$

has a solution set given by an ellipse (if $a^{i,j} = a^{j,i}$).

Remark VII.2.3. If A is symmetric, ellipticity is equivalently that all eigenvalues of A are real and greater than or equal to λ . ┘

From now on, we will always assume that A is symmetric, which is **not** a restrictive constraint, as we can interpret by writing:

$$a^{i,j} = a_{\text{sym}}^{i,j} + a_{\text{skew}}^{i,j}.$$

Then, by integration by parts, we see that:

$$\int_{\Omega} a_{\text{skew}}^{i,j} D_i u D_j v = 0 \quad \text{for } u, v \in C_c^2(\Omega),$$

i.e., we have $a_{\text{skew}}^{i,j} = -a_{\text{skew}}^{j,i}$. Hence, the *skew-symmetric* part of A do not contribute to the bilinear form B . In particular, it will not affect the existence and regularity results.

Example VII.2.4. We will demonstrate a few cases of the equations:

- If we have $L = -\Delta$, $a^{i,j} = \delta_{i,j}$, we have:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \sum_{i,j=1}^n a^{i,j} \xi_i \xi_j = \sum_{i=1}^n \xi_i^2 = |\xi|^2.$$

Hence, $-\Delta$ is elliptic with $\lambda = 1$.

- However, if $L = \Delta$, $a^{ij} = \delta_{ij}$, and so $\sum_{i,j=1}^n a^{ij} \xi_i \xi_j = |\xi|^2$, which is not elliptic.
- If $L = -\frac{\partial^2}{\partial x^2}$ on $\mathbb{R}^2$, we have $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, which is not elliptic, and any function of y solves $Lu = 0$. $\diamond$

Theorem VII.2.5. Boundedness and Coercivity of B .

If L has bounded coefficients ($a^{ij}, b^i, c \in L^\infty(\Omega)$) on $\Omega \subset \mathbb{R}^n$ being open, and B is its bilinear form, then:

- (i) B is bounded on $H_0^1(\Omega)$, i.e., there is $\alpha > 0$ such that for $u, v \in H_0^1(\Omega)$, we have:

$$|B(u, v)| \leq \alpha \|u\|_{H_0^1(\Omega)} \|v\|_{H_0^1(\Omega)}.$$

- (ii) If L is elliptic, then for some $\beta > 0$ and $\gamma \geq 0$, we have for all $u \in H_0^1(\Omega)$ that:

$$\beta \|u\|_{H_0^1(\Omega)}^2 \leq B(u, u) + \gamma \|u\|_{L^2(\Omega)}^2,$$

which is called the **Garding inequality**.

Moreover, if Ω is bounded, $b^i = 0$ for $i = 1, \dots, n$ and $c \geq 0$, then B is coercive, i.e., we can take $\gamma = 0$ in Garding, so that for $u \in H_0^1(\Omega)$ we have:

$$\beta \|u\|_{H_0^1(\Omega)}^2 \leq B(u, u).$$

Proof. (i) We can write:

$$B(u, v) = \int_{\Omega} (a^{ij} D_i u D_j v + b^i (D_i u) v + c u v),$$

and by Hölder, we have:

$$\begin{aligned} B(u, v) &\leq \sum_{i,j=1}^n \|a^{ij}\|_{L^\infty(\Omega)} \|Du\|_{L^2(\Omega)} \|Dv\|_{L^2(\Omega)} + \sum_{i=1}^n \|b^i\|_{L^\infty(\Omega)} \|Du\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \\ &\quad + \|c\|_{L^\infty(\Omega)} \|u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \\ &\leq \alpha \|u\|_{H_0^1(\Omega)} \|v\|_{H_0^1(\Omega)}, \end{aligned}$$

since we have $\|u\|_{H_0^1}^2 (|Du|^2 + u^2)$.

- (ii) Let $\lambda > 0$ be the ellipticity constant for L , then:

$$\begin{aligned} \lambda \int_{\Omega} |Du|^2 &\leq \int_{\Omega} a^{ij}(x) D_i u D_j u, \quad \text{where } \xi = Du \\ &= B(u, u) - \int_{\Omega} b^i (D_i u) u - \int_{\Omega} c u^2 \\ &\leq B(u, v) + \sum_{i=1}^n \|b^i\|_{L^\infty(\Omega)} \int_{\Omega} |du| \cdot |u| + \|c\|_{L^\infty(\Omega)} \|u\|_{L^2(\Omega)}^2. \end{aligned}$$

Since $ab \leq \frac{a^2+b^2}{2}$ (QM-AM-GM-HM inequalities), let $\epsilon > 0$ and replace:

$$a \leftarrow \sqrt{\epsilon} a \quad b \leftarrow \frac{b}{\sqrt{\epsilon}} \quad \text{so that } ab \leq \epsilon \frac{a^2}{2} + \frac{1}{\epsilon} \frac{b^2}{2},$$

which is often called the **Peter-Paul inequality**.

Hence, by applying the Peter-Paul, we have that:

$$\int_{\Omega} |Du| \cdot |u| \leq \frac{\epsilon}{2} \int_{\Omega} |Du|^2 + \frac{1}{2\epsilon} \int_{\Omega} |u|^2.$$

Then, if we set $\epsilon \leq \frac{\lambda}{\sum_{i=1}^n \|b^i\|_{L^\infty(\Omega)}} \neq 0$, then we have:

$$\sum_{i=1}^n \|b^i\|_{L^\infty(\Omega)} \int_{\Omega} |Du| \cdot |u| \leq \frac{\lambda}{2} \int_{\Omega} |Du|^2 + \tilde{\gamma} \int_{\Omega} |u|^2.$$

Hence, we can get to:

$$\lambda \int_{\Omega} |Du|^2 \leq B(u, u) + \frac{\lambda}{2} \int_{\Omega} |Du|^2 + \tilde{\gamma} \|u\|_{L^2(\Omega)}^2 + \|c\|_{L^\infty(\Omega)} \|u\|_{L^2(\Omega)}^2.$$

Now, we have:

$$\frac{\lambda}{2} \|u\|_{H_0^1(\Omega)}^2 = \frac{\lambda}{2} \left(\int_{\Omega} |Du|^2 + \|u\|^2 \right) \leq B(u, u) + \gamma \|u\|_{L^2(\Omega)}^2,$$

where $\gamma = \tilde{\gamma} + \|c\|_{L^\infty(\Omega)} + \frac{\lambda}{2}$, so we can set $\beta = \frac{\lambda}{2}$.

If Ω is bounded, $b^i \equiv 0$, and $c \geq 0$, then the inequality becomes:

$$\lambda \int_{\Omega} |Du|^2 \leq B(u, u) - \int_{\Omega} cu^2 \leq B(u, u).$$

On bounded Ω , we have the Poincaré inequality implying that $\|u\|_{H_0^1(\Omega)} \sim \|Du\|_{L^2(\Omega)}$, so by above, we have:

$$\beta \|u\|_{H_0^1(\Omega)}^2 \leq B(u, u) \quad \text{for } \beta \text{ depending on } \lambda, c_p, \text{ and } \Omega. \quad \square$$

VII.3 The Existence Theorems

Recall the “Riesz representation theorem” to represent a linear functional by inner product with a vector, we are concerning the bilinear, and wish to represent it as the inner product with a linear functional.

Theorem VII.3.1. Lax-Milgram Theorem.

If $(\mathcal{H}, (\cdot, \cdot))$ is a Hilbert space and $B : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{R}$ is a bounded coercive (i.e., $B(u, u) \geq \beta \|u\|_{\mathcal{H}}^2$) bilinear form, and $f \in \mathcal{H}^*$, then there is a unique $u \in \mathcal{H}$ such that:

$$B(u, v) = \langle f, v \rangle \quad \text{for all } v \in \mathcal{H}.$$

Remark VII.3.2. We do not assume symmetry of B in the Lax-Milgram Theorem. ┘

If B , associated to L , was bounded, symmetric, and coercive, then B would be an inner product equivalent to the usual H^1 inner product. So $(H_0^1(\Omega), B)$ would be a Hilbert space, to which we could apply Riesz representation to find weak solution.

Proof. Note that if $v \in \mathcal{H}$, then the map $B(u, \cdot) : \mathcal{H} \rightarrow \mathbb{R}$ in which $v \mapsto B(u, v) \in \mathbb{R}$ is bounded, since B is bounded, i.e., there is $\alpha > 0$ such that:

$$|B(u, v)| \leq \alpha \|u\|_{\mathcal{H}} \cdot \|v\|_{\mathcal{H}}.$$

Hence, we have $B(v, \cdot) \in \mathcal{H}^*$.

By the Riesz representation, we get that for each $u \in \mathcal{H}$, some $A(u) \in \mathcal{H}$ such that:

$$B(u, v) = (A(u), v)_{\mathcal{H}} \quad \text{for all } v \in \mathcal{H}.$$

Consider $A : \mathcal{H} \rightarrow \mathcal{H}$ which maps $u \mapsto A(u)$, we now show that:

(i) **A is linear and bounded:** If $a, b \in \mathbb{R}$, $u, \tilde{u} \in \mathcal{H}$, for $v \in \mathcal{H}$, we have:

$$\begin{aligned} (A(Au + b\tilde{u}), v) &= B(Au + b\tilde{u}, v) = aB(u, v) + bB(\tilde{u}, v) \\ &= a(A(u), v) + b(A(\tilde{u}), v) = (aA(u) + bA(\tilde{u}), v), \end{aligned}$$

so we have $(A(Au + b\tilde{u}) - (aA(u) + bA(\tilde{u})), v) = 0$ for all $v \in \mathcal{H}$, hence, we have:

$$A(Au + b\tilde{u}) = aA(u) + bA(\tilde{u}),$$

and so A is linear.

Now, if $u \in \mathcal{H}$, we have:

$$\|A(u)\|_{\mathcal{H}}^2 = (A(u), A(u)) = B(u, A(u)) \leq \alpha \|u\|_{\mathcal{H}} \cdot \|A(u)\|_{\mathcal{H}}.$$

Note that if $A(u) = 0$, we have $\|A(u)\|_{\mathcal{H}} = 0$ so it is bounded. If not, divide by $\|A(u)\|_{\mathcal{H}}$, so we have $\|A(u)\|_{\mathcal{H}} \leq \alpha \|u\|_{\mathcal{H}}$, so A is bounded.

(ii) **A is bijective:** By coercivity of B , for $u \in \mathcal{H}$, we have:

$$\beta \|u\|_{\mathcal{H}}^2 \leq B(u, u) = (A(u), u) \leq \|A(u)\|_{\mathcal{H}} \cdot \|u\|_{\mathcal{H}},$$

hence we have $\beta \|u\|_{\mathcal{H}} \leq \|Au\|_{\mathcal{H}}$, so A is injective as if $u \neq v$, then $\|u - v\|_{\mathcal{H}} > 0$, and then we have $\|A(u) - A(v)\| > 0$. Hence, $A(u) \neq A(v)$.

Moreover, we see that A has closed range, since if $A(u_n) \rightarrow v$ in $\mathcal{H}$, since:

$$\beta \|u_n - u_m\|_{\mathcal{H}} \leq \|A(u_n) - A(u_m)\|_{\mathcal{H}},$$

which implies that $\{u_n\}$ is Cauchy, hence is convergent in $\mathcal{H}$, i.e., $u_n \rightarrow u$ in $\mathcal{H}$, but since A is bounded (i.e., continuous) so $A(u_n) \rightarrow A(u)$, so $v = A(u)$.

If A was not surjective, we could find some element $w \in (A(\mathcal{H}))^{\perp} \setminus \{0\}$. This follows since $A(\mathcal{H})$ is closed so $\mathcal{H} = A(\mathcal{H}) \oplus (A(\mathcal{H}))^{\perp}$.

Again, by coercivity:

$$0 < \beta \|w\|_{\mathcal{H}} \leq B(w, w) = (A(w), w) = 0,$$

and $0 < 0$, which is a contradiction, implying that A is surjective.

(iii) **Define $u \longleftrightarrow f \in \mathcal{H}^*$:** Again, by the Riesz representation theorem, for $f \in \mathcal{H}^*$, there is some $w \in \mathcal{H}$

such that:

$$\langle f, v \rangle = (w, v) \quad \text{for all } v \in \mathcal{H}.$$

Since $A : \mathcal{H} \rightarrow \mathcal{H}$ is a bijection, there is a unique $v \in \mathcal{H}$ such that $A(v) = w$. Then, by the construction of A :

$$B(u, v) = (A(u), v) = (w, v) = \langle f, v \rangle.$$

□

Theorem VII.3.3. Existence and Uniqueness, I.

If $\Omega \subset \mathbb{R}^n$ is open, L is elliptic with coefficients in $L^\infty(\Omega)$, then there is some $\gamma \geq 0$ such that if $\mu \geq \gamma$ and $f \in H^{-1}(\Omega)$, there exists a unique *weak solution* $u \in H_0^1(\Omega)$ to the PDE:

$$\begin{cases} Lu + \mu u = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

Proof. Let $\gamma \geq 0$ be from the **Garding's inequality** from [Theorem VII.2.5](#), i.e., $\beta \|u\|_{H_0^1(\Omega)}^2 \leq B(u, u) + \gamma \|u\|_{L^2(\Omega)}^2$.

For $\mu \geq \gamma$, we define for $u, v \in H^1(\Omega)$ that:

$$(u, v) = B(u, v) + \mu(v, u)_{L^2(\Omega)},$$

which is the associated operator to $L_u v = Lu + \mu v$.

Hence, we have:

$$\beta \|u\|_{H_0^1(\Omega)}^2 \leq B_\mu(u, u) \quad \text{for all } u \in H_0^1(\Omega),$$

which implies that B_μ is coercive.

Hence B_μ is bounded as B is (since the coefficients are bounded).

Hence, by Lax-Milgram ([Theorem VII.3.1](#)), for each $H^{-1}(\Omega) = (H_0^1(\Omega))^*$, there is a unique $u \in H_0^1(\Omega)$ such that:

$$B_\mu(u, v) = \langle f, v \rangle \quad \text{for all } v \in H_0^1(\Omega),$$

which is equivalently that u is the unique weak solution to:

$$\begin{cases} Lu + \mu u = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

□

Example VII.3.4. Poisson Equation.

Again, consider $L = -\Delta$ on $B \subset \mathbb{R}^n$, by the Poincaré inequality, we have:

$$\|u\|_{H_0^1(\Omega)} \leq B(u, u) = \int_B |Du|^2,$$

where $\gamma = 0$. So there exists unique weak solution to the differential equation:

$$\begin{cases} -\Delta u + \mu u = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

for all $f \in H^{-1}(\Omega)$ and $\mu \geq 0$. $\diamond$

Remark VII.3.5. Additionally, if $Lu = D_j(a^{ij}D_i u) + cu$, $c \geq 0$, $a^{ij}c \in L^\infty(\Omega)$, we again get unique weak solutions to $(b^i = 0)$:

$$\begin{cases} Lu + \mu v = f, & \text{on } \Omega, \\ v = 0, & \text{on } \partial\Omega. \end{cases} \quad \text{for all } f \in H^{-1}(\Omega), \mu \geq 0.$$

In particular for

$$\begin{cases} Lu = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

with $\gamma = 0$ in Garding inequality. $\lrcorner$

However, existence and uniqueness is not always guaranteed. We can, in fact, find counter examples with purely ODEs:

Example VII.3.6. Let $\Omega = (0, \pi) \subset \mathbb{R}$ and the differential equation as:

$$\begin{cases} -u'' - u = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$

where we have $f \in H^{-1}(\Omega)$.

We then have $Lu = -u'' - u$, so $a^{1,1} = 1$, $b^1 = 0$, and $x = -1$ are all in $\mathcal{L}^\infty(\Omega)$ (and it is even smooth).

But if we set $f = 0$, both $u = 0$ and $u = \sin(x)$ solves the differential equation, so there is no unique solution.

Moreover, if $f(x) = \sin(x) \in \mathcal{L}^2(\Omega) \subset H^{-1}(\Omega)$, and $u \in H_0^1(\Omega)$ is a weak solution, then we have:

$$B(u, f) = \langle f, f \rangle = \int_0^\pi f^2 = \int_0^\pi \sin^2(x) \neq 0,$$

since we have that:

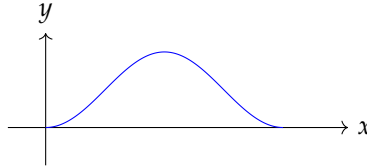


Figure VII.1. Graph of $\sin^2(x)$ on $[0, \pi]$.

On the other hand, we have:

$$B(u, f) = \int_0^\pi -u' f' + \int_0^\pi u f = \int_0^\pi u f'' + u f = \int_0^\pi u (-\sin(x) + \sin(x)) = 0.$$

Hence, this is a contradiction, so there is no weak solutions to the differential equation if $f = \sin(x)$. $\diamond$

More generally, if $f \in H^{-1}(\Omega)$ and u is the weak solution, we have:

$$u + A \sin(x) \quad \text{also being a weak solution,}$$

i.e., we have $\sin(x) \in \ker L$ if and only if $L(\sin(x)) = 0$.

VII.4 Fredholm Theory

We saw that for some $\mu > 0$ sufficiently large, we have that the operator $L_\mu : H_0^1(\Omega) \rightarrow H^{-1}(\Omega)$, defined by:

$$L_\mu u = Lu + \mu u$$

was a bounded linear bijection, which sends $u \in H_0^1(\Omega)$ to $B_\mu(u, \cdot) \in H^{-1}(\Omega)$.

Moreover, $L_\mu u^{-1} : H^{-1}(\Omega) \rightarrow H_0^1(\Omega)$ is also bounded as:

$$\beta \|u\|_{H_0^1(\Omega)}^2 \leq B_\mu(u, u) = B_\mu(u, \cdot)(u) \leq \|B_\mu(u, \cdot)\|_{H^{-1}(\Omega)} \|u\|_{H_0^1(\Omega)}.$$

Therefore, we have $\beta \|u\|_{H_0^1(\Omega)} \leq \|B_\mu(u, \cdot)\|_{H^{-1}(\Omega)}$ noting that $L_\mu(u) = B_\mu(u, \cdot)$, this is equivalently:

$$\|L_\mu^{-1}(B_\mu(u, \cdot))\|_{H_0^1(\Omega)} \leq \frac{1}{\beta} \|B_\mu(u, \cdot)\|_{H^{-1}(\Omega)},$$

we have that L_μ^{-1} being bounded. Furthermore, by Rellich-Kondrachov, we have $H_0^1(\Omega) \xhookrightarrow{c} L^2(\Omega)$ and $L^2(\Omega) \hookrightarrow H^{-1}(\Omega)$, so:

$$L_\mu^{-1} : L^2(\Omega) \hookrightarrow H^{-1}(\Omega) \xrightarrow{L_\mu^{-1}} H_0^1(\Omega) \xhookrightarrow{c} L^2(\Omega),$$

and hence, we have $L_\mu^{-1} : L^2(\Omega) \rightarrow L^2(\Omega)$ is bounded and compact. As a sanity check, we have:

$$L_\mu^{-1}(f) = u \iff u \text{ weakly solves } L_\mu(u) = f.$$

If we want general existence for weak solutions to the PDE:

$$\begin{cases} Lu = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$

for $f \in L^2(\Omega)$. Consider it equivalently as $B(u, v) = \int_\Omega f v$ for all $v \in H_0^1(\Omega)$, which is the same as $B(u, v) + \mu \int_\Omega uv = \int_\Omega f v + \mu \int_\Omega uv$ for some $\mu \geq \gamma$ from Garding inequality.

Equivalently, we have $B_\mu(u, v) = \int_\Omega (f + \mu u)u$, so we have $B_\mu(u, \cdot) = (f + \mu u, \cdot)_{L^2(\Omega)}$ in $H^{-1}(\Omega)$. Therefore, we have:

$$L_\mu u = f + \mu u \text{ in the above,} \quad \text{so } u = L_\mu^{-1}(f + \mu u).$$

This is then equivalent to say that $(\text{Id} - \mu L_\mu^{-1})u = L_\mu^{-1}(f)$.

Then, we have weakly solving the differential equation is the same as solving $(\text{Id} - \mu L_\mu^{-1})u = L_\mu^{-1}(f)$.

Abstractly, at the level of linear algebra, we want to solve equations of the form:

$$(\text{Id} - K)u = v \quad \text{for } K : \mathcal{H} \rightarrow \mathcal{H}$$

as a compact, bounded, and linear map on a Hilbert space.

Definition VII.4.1. Adjoint Operator.

If $(\mathcal{H}, (\cdot, \cdot))$ is a real Hilbert space, $A : \mathcal{H} \rightarrow \mathcal{H}$ is bounded and linear, then it's **adjoint** $A^* : \mathcal{H} \rightarrow \mathcal{H}$ is defined by setting:

$$(Au, v) = (u, A^*v) \quad \text{for all } u, v \in \mathcal{H}.$$

We say that A is symmetric if $A = A^*$, i.e., $(Au, v) = (u, Av)$ for all $u, v \in \mathcal{H}$. ┘

Remark VII.4.2.

- The adjoint exists by the Riesz representation theorem, while also being bounded and linear, which $(A^*)^* = A$.
- If the dimension of $\mathcal{H}$ is finite, then A^* has matrix corresponding to the transpose of A , and $A^\top = (a^{ij})_{i,j}$ if $A = (a^{ij})_{i,j}$.
- If $K : \mathcal{H} \rightarrow \mathcal{H}$ is bounded, linear, and compact, then so is $K^* : \mathcal{H} \rightarrow \mathcal{H}$ (since for bounded sequence $u_n \rightharpoonup u$, hence $K(u_n) \rightarrow K(u)$, and so $K^*(u_n) \rightarrow K^*(u)$ by definition). ┘

In finite dimension, all linear maps are bounded and compact, and from linear algebra, if $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is linear, we have $L(u) = v$ for $u, v \in \mathbb{R}^n$ is solvable or $\ker(L^\top) \neq \{0\}$ (but not both).

This does not hold in infinite dimensions though.

Example VII.4.3. Consider $K : L^2((0,1)) \rightarrow L^2((0,1))$ sending $f \in L^2((0,1))$ to:

$$(Kf)(x) = \int_0^x f(t)dt \in L^2((0,1))$$

being compact (by Arzelá-Ascoli).

But, consider $\ker(K^*) = \{0\}$ and $g \in \text{im } K$ implies that g is continuous, but not every $L^2((0,1))$ function is continuous, so K is not invertible.

Hence, the analogy does not hold if the dimension is infinity. ◇

Theorem VII.4.4. Fredholm Alternative.

If $K : \mathcal{H} \rightarrow \mathcal{H}$ is linear, bounded, and compact on a real Hilbert space $\mathcal{H}$, then precisely one of the following two possibilities hold:

- (i) For each $h \in \mathcal{H}$, the equation:

$$(\text{Id} - K)u = h$$

has a unique solution.

(ii) The equation:

$$(\text{Id} - K)u = 0$$

has a nonzero solution.

Moreover, the space of such solutions is finite dimensional, equal to the dimension of the solutions to:

$$(\text{Id} - K^*)u = 0.$$

Finally, $(\text{Id} - K)u = h$ is solvable if and only if $h \in (\ker(\text{Id} - K^*))^\top$.

Then, we consider the adjoint of L .

Definition VII.4.5. If $\Omega \subset \mathbb{R}^n$ is open and bounded, L is elliptic, with coefficients in $L^\infty(\Omega)$ with $b^i \in W^{1,\infty}(\Omega)$ for $i = 1, \dots, n$, then the **formal adjoint**, L^* , of L , is defined as:

$$L^*v = -D_i(a^{ij}D_jv) - b^iD_iv + (c - D_ib^i)v.$$

Remark VII.4.6. One can note the following properties of L^* :

- L^* is elliptic, since a^{ij} are the same.
- $b^i \in W^{1,\infty}(\Omega)$ for $i = 1, \dots, n$, we have that for $u, v \in H_0^1(\Omega)$, then:

$$B(u, v) = \int_{\Omega} (a^{ij}D_iuD_jv + (b^iD_iu)v + cuv) = \int_{\Omega} (a^{ij}D_iuD_jv + D_i(b^iv)u + (c - D_ib^i)uv) = B^*(v, u).$$

Hence, B^* is bilinear form associated to L^* and $B(u, v) = B^*(v, u)$.

- This is a *formal adjoint* is that while it does not map a Hilbert space to itself, it has the “form” of an adjoint since if $a^{ij}, b^i \in W^{1,\infty}$ for $i, j = 1, \dots, n$, and $u, v \in H^2(\Omega) \cap H_0^1(\Omega)$, then:

$$\langle Lu, v \rangle_{L^2(\Omega)} = B(u, v) = B^*(v, u) = \langle u, L^*v \rangle_{L^2(\Omega)}.$$

Hence, L^* looks like an adjoint with these assumptions.

Recall that $L_\mu^{-1} : L^2(\Omega) \rightarrow L^2(\Omega)$ was bounded, compact, linear for $\mu \geq 0$ large enough (from Garding inequality), But Garding then holds for L^* , so:

$$L_\mu^* = (L + \mu)^*(u) = L^*(u) + \mu(u).$$

Noting here that $\mu^* = \mu$ since $(\mu, v)_{L^2(\Omega)} = (\mu, \mu v)_{L^2(\Omega)}$. Note that since $H_\mu(u, v) = B_\mu^*(v, u)$, by applying our first existence result for L^* , we obtain unique weak solution to

$$\begin{cases} L^*u + \mu u = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega \end{cases}$$

for this choice of $\mu \geq 0$ (from Garding inequality).

Arguing as before, consider $(L_\mu^*)^{-1}: L^2(\Omega) \rightarrow L^2(\Omega)$ is bounded, compact, linear map. Hence, if $f \in L^2(\Omega)$ and $v \in H_0^1(\Omega)$, then:

$$\begin{cases} B_\mu(L_\mu^{-1}(f), v) = (f, v)_{L^2(\Omega)}, \\ B_\mu^*((L_\mu^{-1}(f), v) = (f, v)_{L^2(\Omega)}, \end{cases}$$

hence, for all $f, g \in L^2(\Omega)$, that:

$$(g, (L_\mu^*)^{-1}(f))_{L^2(\Omega)} = B_\mu(L_\mu^{-1}(g), (L_\mu^*)^{-1}(f)) = B_\mu^*((L_\mu^*)^{-1}(g), L_\mu^*(f)).$$

Theorem VII.4.7. Existence for Elliptic PDEs II.

If $\Omega \subset \mathbb{R}^n$ is open and bounded, and L is elliptic with coefficient in $a^{i,j}, c \in L^\infty(\Omega)$ and $b^i \in W^{1,\infty}(\Omega)$ for $i, j = 1, \dots, n$, then precisely one of the following holds:

(i) For each $f \in L^2(\Omega)$, there is a unique weak solution $v \in H_0^1(\Omega)$ to the PDE:

$$\begin{cases} Lu = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

(ii) There is a weak solution $u \in H_0^1(\Omega) \setminus \{0\}$ to the homogeneous PDE:

$$\begin{cases} Lu = 0, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

Moreover, the space of weak solutions to the homogeneous PDE is finite dimensional, and equal to the dimension for the space of weak solutions, $v \in H_0^1(\Omega)$, to:

$$\begin{cases} L^*v = 0, & \text{on } \Omega, \\ v = 0, & \text{on } \partial\Omega. \end{cases}$$

Finally:

$$\begin{cases} Lu = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

has a weak solution for $f \in L^2(\Omega)$ if and only if $(f, v)_{L^2(\Omega)} = 0$ for all $v \in H_0^1(\Omega)$ is weak solution to:

$$\begin{cases} Lu = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

Proof. For large enough $\mu \geq 0$, we have a unique weak solution $w \in H_0^1(\Omega)$ to:

$$\begin{cases} L^*w = f, & \text{on } \Omega, \\ w = 0, & \text{in } \partial\Omega, \end{cases}$$

for each $f \in L^2(\Omega)$.

Now, weak solutions to the first condition is equivalently $(\text{Id} - \mu L_\mu^*)u = L_\mu^{-1}(f)$, for $L_\mu^{-1} : L^2(\Omega) \rightarrow L^2(\Omega)$ being bounded, compact linear. Setting:

$$K = \mu L_\mu^{-1} \quad \text{and} \quad h = L_\mu^{-1}(f),$$

we have by Fredholm alternative that (i) holds or (ii) holds.

Moreover, we note that $K^* = (\mu L_\mu^{-1})^* = \mu (L_\mu^*)^{-1}$.

For the final assert, the first equation being weakly solvable for:

$$f \in L^2(\Omega) \iff (\text{id} - K)u = h \text{ is solvable} \iff h \in (\ker(\text{Id} - K^*))^\perp,$$

and thus, for all $v \in H_0^1(\Omega)$ weakly solving the last equation, we have $(\text{Id} - K^*)v = 0 \iff u^*(v) = v$, so:

$$\begin{aligned} 0 &= (h, v)_{L^2(\Omega)} = (L_\mu^{-1}(f), v) = \frac{1}{\mu} (K(f), v)_{L^2(\Omega)} \\ &= \frac{1}{\mu} (f, K^*(v))_{L^2(\Omega)} = \frac{1}{\mu} (f, v)_{L^2(\Omega)}. \end{aligned}$$

This is equivalently that $(f, v)_{L^2(\Omega)} = 0$ for all such $v \in H_0^1(\Omega)$ that weakly solves $\begin{cases} Lu = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$ $\square$

VII.5 Eigenvalues of Elliptic Operators

Recall that in finite dimensions, if $\dim V < \infty$, and $A : V \rightarrow V$ being a linear map. Then $\lambda \in \mathbb{R}$ is an eigenvalue of A if $Av = \lambda v$ for some $v \in V \setminus \{0\}$, which is equivalently that $(A - \lambda \text{Id})v = 0$ for $v \in V \setminus \{0\}$, so we equivalent have $\ker(A - \lambda \text{Id}) \neq \{0\}$.

In finite dimensions, injectivity is equivalent to surjectivity by **rank-nullity** theorem.

Example VII.5.1. Volterra Operator.

Note the Volterra operator for $f \in L^2((0, 1))$ as:

$$(Kf)(x) = \int_0^x f(t)dt \in L^2((0, 1)),$$

and we saw that $K : L^2((0, 1)) \rightarrow L^2((0, 1))$ was compact. But consider $Kf = \lambda f$ for $\lambda \in \mathbb{R}$, so we know $\lambda f'(x) = f(x)$ for $f \in C^1((0, 1))$, which implies that $f(x) = Ae^{\frac{x}{\lambda}}$ for $A \in \mathbb{R}$.

Since $(Kf)(0) = 0$, which implies that $\lambda f(0) = \lambda A$, and in either cases, $f \equiv 0$, and so no eigenfunctions can exist for K (so it has no eigenvalues).

Hence, $\ker(K - \lambda \text{Id}) = \{0\}$ but K is not surjective, as $(Kf) \in C^0((0, 1))$ but not every $L^2((0, 1))$. $\diamond$

Hence, we can see that **rank-nullity** no longer works for infinite dimensional space, so we need some new definition.

Definition VII.5.2. Spectrum.

If $A : \mathcal{H} \rightarrow \mathcal{H}$ is a bounded linear map on a real Hilbert space $\mathcal{H}$, then the (real) **spectrum**, $\sigma(A)$ of A , is:

$$\sigma(A) := \{\lambda \in \mathbb{R} : A - \lambda \text{Id is not invertible}\}.$$

The (real) **point spectrum** (or eigenvalues) $\sigma_p(A) \subset \sigma(A)$ is the set:

$$\sigma_p(A) := \{\lambda \in \mathbb{R} : \ker(A - \lambda \text{Id}) \neq \{0\}\}.$$

┘

With this new definition, we can have the following.

Theorem VII.5.3. Spectrum of Compact Operators.

If $K : \mathcal{H} \rightarrow \mathcal{H}$ is compact, bounded, linear map on an infinite dimensional real Hilbert space $\mathcal{H}$, then:

- (i) $0 \in \sigma(K)$, i.e., K is not invertible.
- (ii) $\sigma(K) \setminus \{0\} = \sigma_p(K) \setminus \{0\}$.
- (iii) Either $\sigma(K)$ is a finite set or $\sigma(K) \setminus \{0\} = \sigma_p(K) \setminus \{0\}$ is countable and converges to 0.

This proof appears on the appendix of the textbook. We will use this theorem to our existence for elliptic PDEs.

Theorem VII.5.4. Existence for Elliptic PDEs, III.

If $\Omega \subset \mathbb{R}^n$ is open and bounded, L is elliptic with coefficients in $L^\infty(\Omega)$, then:

- (i) There is an at most countable set, $\Sigma \subset \mathbb{R}$ such that for each $f \in L^2(\Omega)$, there is a unique solution $u \in H_0^1(\Omega)$ to the PDE:

$$\begin{cases} Lu = \lambda u + f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$

if and only if $\lambda \notin \Sigma$.

- (ii) If Σ is infinite, then $\Sigma = \{\lambda_k\}$, $\lambda_k \rightarrow \infty$, $\lambda_k \leq \lambda_{k-1}$ for $k \geq 1$.

Remark VII.5.5.

- Σ is the spectrum of L .
- From (i), there exists unique weak solutions to:

$$\begin{cases} Lu = \lambda u, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$

if and only if $\lambda \notin \Sigma$, we call λ an eigenvalue of L and u as its eigenfunction.

┘

Proof of Theorem VII.5.4. Let γ be the constant from the Garding inequality for L , and without loss of generality, let's assume $\gamma > 0$.

If $\lambda > -\gamma$, by the first existence theorem (Theorem VII.3.3), for elliptic PDE, we have a unique weak solution to
$$\begin{cases} Lu = \lambda u + f, & \text{on } \Omega, \\ v = 0, & \text{on } \partial\Omega, \end{cases}$$
 if and only if the only weak solution to
$$\begin{cases} Lu = \lambda u, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$
 is zero.

Now by adding γu , we have $u = 0$ being the only solution to:

$$\begin{cases} L_\gamma(u) = Lu + \gamma u = (\lambda + \gamma)u, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$

which is equivalently $u = (\lambda + \gamma)L_\gamma^{-1}(u)$.

By setting $K = \gamma L_\gamma^{-1}$, we have that:

$$K : L^2(\Omega) \rightarrow L^2(\Omega)$$

is bounded and compact, so now we can replace the above equivalence to $u = \frac{\lambda + \gamma}{\gamma} \cdot K(u)$ only has solution $u = 0$.

Thus, this implies that $\frac{\gamma}{\lambda + \gamma}$ is not an eigenvalue of K .

Hence, in summary:

$$\begin{cases} Lu = \lambda u + f, & \text{on } \Omega, \\ v = 0, & \text{on } \partial\Omega, \end{cases} \text{ had unique weak solution} \iff \frac{\gamma}{\lambda + \gamma} \text{ not being an eigenvalue of } K.$$

By spectral theorem for compact operators (iii), we see that eigenvalues of K are at most countable, so we have proven (i) by setting:

$$\Sigma = \left\{ \lambda \in \mathbb{R} : \frac{\gamma}{\lambda + \gamma} \text{ is an eigenvalue of } K \right\}.$$

For (ii), we have $\sigma_p(K) \setminus \{0\} = \frac{\gamma}{\lambda_k + \gamma} \rightarrow 0$ as $k \rightarrow \infty$, which implies that $\lambda_k \nearrow \infty$ as $k \rightarrow \infty$. $\square$

Theorem VII.5.6. Boundedness of L^{-1} .

With the same assumption from the last theorem, if $\lambda \notin \Sigma$, there is a constant $c > 0$ depending on λ , Ω , and L , such that $\|u\|_{L^2(\Omega)} \leq C\|f\|_{L^2(\Omega)}$ if f is in $L^2(\Omega)$ and $u \in H_0^1(\Omega)$ is the unique weak solution to:

$$\begin{cases} Lu = \lambda u + f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

Moreover, $C \rightarrow \infty$ as λ converges to eigenvalues of L .

Proof. If no such $C > 0$ exists for a given $\lambda \notin \Sigma$, we can find $\{f_k\} \subset L^2(\Omega)$ and $\{u_k\} \subset H_0^1(\Omega)$ with:

$$\begin{cases} Lu_k = \lambda u_k + f_k, & \text{on } \Omega, \\ u_k = 0, & \text{on } \partial\Omega \end{cases}$$

being weakly solved, but $\|u_k\|_{L^2(\Omega)} > K\|f_k\|_{L^2(\Omega)}$. We may assume, since L is linear, that:

$$\|u_k\|_{L^2(\Omega)} = 1 \implies f_k \rightarrow 0 \text{ in } L^2(\Omega).$$

As for $\beta > 0$ and $\gamma \geq 0$, we have:

$$\begin{aligned} \beta\|u_k\|_{H_0^1(\Omega)}^2 &\leq B_{-\lambda}(u_k, u_k) + \gamma\|u_k\|_{L^2(\Omega)}^2 \\ &= (f_k, u_k)_{L^2(\Omega)} + \gamma \leq \|f_k\|_{L^2(\Omega)} + \gamma, \end{aligned}$$

since by Cauchy-Schwartz, $|(f_k, u_k)_{L^2(\Omega)}| \leq \|f_k\|_{L^2(\Omega)} \cdot \|u_k\|_{L^2(\Omega)}$, in which $\|u_k\|_{L^2(\Omega)} = 1$, hence:

$$\|u\|_{H_0^1(\Omega)} \leq \frac{\|f_k\|_{L^2(\Omega)} + \gamma}{\beta} < A, \quad \text{where } A > 0.$$

Hence, we have $\{u_k\}$ being bounded in $H + 0^1(\Omega)$, so by weak compactness, we have $u_k \rightharpoonup u$ in $H_0^1(\Omega)$, so $u_k \rightarrow u$ in $L^2(\Omega)$, which implies that $B_{-\lambda}(u, u) = 0$ if and only if u weakly solves:

$$\begin{cases} Lu = \lambda u, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

However, since $\lambda \notin \Sigma$, so $u = 0$, so $\|u\|_{L^2(\Omega)} = 0$, but we had $\|u_k\|_{L^2(\Omega)} = 1$, so $\|u\|_{L^2(\Omega)} = 1$, which is a contradiction. $\square$

VIII Regularity Theory

VIII.1 Intuitions on Regularity

For many PDEs, we want to show that if $Lu = f$ weakly, then u has *better* regularity than f .

We will see that $f \in L^2(\Omega)$ implies that $u \in H^2(\Omega)$, so that $Lu = f$ makes sense almost everywhere (this is not just weakly).

More generally, $f \in H^k(\Omega)$ implies that $u \in H^{k+2}(\Omega)$.

This says that, if L is elliptic, controlling *some* of u 's second derivatives, in fact, allows us to control *all* of them. (This is exactly an aspect of ellipticity: Control of $Lu \rightsquigarrow$ control of D^2u if L is elliptic).

Remark VIII.1.1. The *gain* of two derivatives applies to some but not all function spaces:

- **Schaudes theory:** $f \in C^{0,\alpha}$ implies that $u \in C^{2,\alpha}$.
- **Calderon-Zygmund theory:** $f \in W^{k,p}$ implies that $u \in W^{k+2,p}$ for $p \in (1, \infty)$. ┘

Example VIII.1.2. $f \in C^0$ does not imply $u \in C^2$, since we have:

$$u(x, y) = xy \log \left(-\log \left(\frac{x^2 + y^2}{e} \right) \right)$$

is a weak solution to $-\Delta u = f$ for some $f \in C^0$, but u is not C^2 near $(0, 0)$. ◇

VIII.2 Interior Regularity

We first need to work “away” from the boundary and show that if $Lu = f$ weakly for $f \in L^2(\Omega)$, implying that $u \in H_{\text{loc}}^2(\Omega)$.

As motivation, if $u \in C_c^\infty(B)$ for $B = B_1(0) \subset \mathbb{R}^n$, then:

$$\|\Delta u\|_{L^2(\Omega)}^2 = \int_B |\Delta u|^2 = \int_B (D_i D_i u)(D_j D_j u) = \int_B (D_i D_j u)(D_i D_j u) = \int_B |D^2 u|^2 = \|D^2 u\|_{L^2(B)}^2.$$

Hence, control of Δu in L^2 implies the control of $D^2 u$ in L^2 .

Ellipticity guarantees Lu in L^2 controls $D^2 u$ in L^2 for general L and Ω .

Theorem VIII.2.1. Interior Estimate for Compact Support.

If $\Omega \subset \mathbb{R}^n$ is open and $Lu = -D_j(a^{ij}D_i u)$ is elliptic with $A^{i,j} \in W^{1,\infty}(\Omega)$ for $i, j = 1, \dots, n$, then whenever $u \in H^1(\Omega)$ is a weak solution to $Lu = f$ for some $f \in L^2(\Omega)$ with $\{u \neq 0\} \Subset \Omega$, we have

$u \in H^2(\Omega)$ and:

$$\|D^2u\|_{L^2(\Omega)} \leq C(\|f\|_{L^2(\Omega)} + \|Du\|_{L^2(\Omega)}),$$

for a constant $C > 0$ depending on n and a^{ij} .

Proof. We assume without loss of generality that $\Omega = \mathbb{R}^n$. Now, $u \in H^1(\Omega)$ weakly solves $Lu = f$, if:

$$\int a^{ij} D_i u D_j v = \int f v \quad \text{for all } v \in H_0^1.$$

If $v \in H_0^1$ is compactly supported, so is $D_u^{-h}v$ for $k = 1, \dots, n$ and $h \in \mathbb{R}$.

By the product rule and integration by parts for difference quotients:

$$\begin{aligned} \int f D_k^{-h} v &= \int a^{ij} D_i u D_j (D_k^{-h} v) = \int a^{ij} D_i u D_k^{-h} (D_j v) \\ &= - \int D_k^h (a^{ij} D_i u) D_j v = - \int D_k^h (a^{ij}) D_i u D_j v - \int a^{ij} (x + h e_k) (D_i u) D_j v. \end{aligned}$$

Recall that:

- $\|D_k^{-h} v\|_{L^2} \leq \|Dv\|_{L^2}$.
- As $a^{ij} \in W^{1,\infty}(\Omega)$, $\|D_k^h (a^{ij})\|_{L^\infty} \leq \|D a^{ij}\|_{L^\infty} \leq C$.

Therefore, from above, we have:

$$\begin{aligned} \left| \int a^{ij} (x + h e_k) D_k^h (D_i u) D_j v \right| &\leq \left| \int f D_k^{-h} v \right| + \left| \int D_k^h (a^{ij}) D_i u D_j v \right| \\ &\leq C \|f\|_{L^2} \|Dv\|_{L^2} + \|Du\|_{L^2} \|Dv\|_{L^2}. \end{aligned}$$

As L is elliptic, there is $\lambda > 0$ such that:

$$\lambda \|Du\|_{L^2} = \lambda \int |Dv|^2 \leq \int a^{ij} (x + h e_k) D_i v D_j v \quad \text{for all } v \in H_0^1 \text{ (or } C_c^\infty).$$

We choose $v = D_k^h u$ later on.

Hence, we have:

$$\lambda \|Dv\|_{L^2}^2 \leq \int a^{ij} (x + h e_k) D_i v D_j v \leq C(\|f\|_{L^2} + \|Du\|_{L^2}) \|Dv\|_{L^2}.$$

Hence, by dividing $\|Dv\|_{L^2}$ (if $\neq 0$, we are done), we have:

$$\|D_k^h (Du)\|_{L^2} \leq C(\|f\|_{L^2} + \|Du\|_{L^2}),$$

where C changes line by line only increasing.

Result for difference quotients implies that:

$$\|D^2u\|_{L^2} \leq C(\|f\|_{L^2} + \|Du\|_{L^2}).$$

□

Theorem VIII.2.2. Interior Estimate for Elliptic PDE.

If $\Omega \subset \mathbb{R}^n$ is open, L is elliptic with $a^{ij} \in W^{1,\infty}(\Omega)$, $b^i, c \in L^\infty(\Omega)$ for $i, j = 1, \dots, n$, then if $u \in H^1(\Omega)$ weakly solves $Lu = f$ for $f \in L^2(\Omega)$, we have $u \in H_{\text{loc}}^2(\Omega)$ and for each $\tilde{\Omega} \Subset \Omega$, we have:

$$\|u\|_{H^2(\tilde{\Omega})} \leq C(\|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)})$$

for a constant $C > 0$ depending on Ω , $\tilde{\Omega}$, and L .

Remark VIII.2.3. C may degenerate (i.e., blow up) as $\tilde{\Omega}$ gets larger, so we will not be able to say that $u \in H^2(\Omega)$ in general, if $\partial\Omega$ is regular enough, we can, however. $\square$

Proof of Theorem VIII.2.2. For $\tilde{\Omega} \Subset \Omega$, let $\varphi \in C_c^\infty(\Omega)$ be such that $\varphi \equiv 1$ on $\tilde{\Omega}$ and $0 \leq \varphi \leq 1$. We will see what PDE the product $\varphi \cdot u \in H_0^1(\Omega)$ solves weakly.

Since $Lu = f$ weakly, if $v \in H_0^1(\Omega)$, then $\varphi v \in H_0^1(\Omega)$ as well. Then we have:

$$\int_{\Omega} f(\varphi v) = \int_{\Omega} a^{ij} D_i u D_j (\varphi v) + b^i D_i u (\varphi v) + C u (\varphi v),$$

which is equivalently as:

$$\int_{\Omega} (f - b^i D_i u - cu) \varphi v - (a^{ij} D_i u D_j \varphi) v = \int_{\Omega} (a^{ij} D_i u D_j v) \varphi.$$

Note that $\varphi \cdot D_i u = D_i(\varphi u) - u D_i \varphi$, and we equivalently have:

$$\int_{\Omega} (f - b^i D_i u - cu) \varphi v - (a^{ij} D_i u D_j \varphi) v - D_j (a^{ij} u D_i \varphi) v = \int_{\Omega} a^{ij} D_i (\varphi u) D_j v.$$

Hence, this implies that φu weakly solves the PDE, with:

$$-D_j (a^{ij} D_i (\varphi u)) = \hat{f} = \varphi(f - b^i D_i u - cu) - a^{ij} D_i u D_j \varphi - D_j (a^{ij} u D_i \varphi) \in L^2(\Omega),$$

and φu has compact support in Ω . Hence, by the previous theorem (Theorem VIII.2.1), $\varphi u \in H^2(\Omega)$, and:

$$\|D^2(\varphi u)\|_{L^2(\Omega)} \leq C(\|\hat{f}\|_{L^2(\Omega)} + \|D(\varphi u)\|_{L^2(\Omega)}) \leq C(\|f\|_{L^2(\Omega)} + \|u\|_{H^1(\Omega)}).$$

As $\|D(\varphi u)\|_{L^2(\Omega)} \leq C\|u\|_{H^1(\Omega)}$. Since $\varphi u = u$ on $\tilde{\Omega}$, this says that:

$$\|u\|_{H^2(\tilde{\Omega})} \leq \|D^2(\varphi u)\|_{L^2(\Omega)} + \|u\|_{H^1(\Omega)} \leq C(\|f\|_{L^2(\Omega)} + \|u\|_{H^1(\Omega)}).$$

Finally, if we can prove the following:

Lemma VIII.2.4. For $\tilde{\Omega} \Subset \Omega$:

$$\|u\|_{H^1(\tilde{\Omega})} \leq C(\|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)}).$$

Now, we are just left to show that we can squeeze out a little bit more, as we can replace Ω with $\tilde{\Omega}$ so $\tilde{\Omega} \Subset \tilde{\Omega} \Subset \Omega$ for the above inequality. $\square$

Proof of Lemma VIII.2.4. Let $\varphi \in C_c^\infty(\Omega)$ with $\varphi \equiv 1$ on $\tilde{\Omega}$, $0 \leq \varphi \leq 1$ and consider $\varphi^2 \cdot u \in H_0^1(\Omega)$ in the weak formulation of $Lu = f$ to obtain:

$$\int_{\Omega} f(\varphi^2 u) = \int_{\Omega} a^{ij} D_i u D_j(\varphi^2 u) + (b^i D_i u)(\varphi^2 u) + c u^2 \varphi^2.$$

Note that $D_j(\varphi^2 u) = \varphi^2 D_j u + 2u \varphi D_j \varphi$. Rearranging and Cauchy-Schwartz gives us that:

$$\int_{\Omega} a^{ij} a^{ij} \varphi^2 D_i u D_j u \leq C(\|f\|_{L^2(\Omega)} \|u\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)} \|Du\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)}).$$

Now, ellipticity of L gives $\lambda > 0$ with:

$$\lambda \|Du\|_{L^2(\tilde{\Omega})}^2 \leq \int_{\tilde{\Omega}} a^{ij} D_i u D_j v \leq \int_{\Omega} a^{ij} \varphi^2 D_i u D_j v.$$

Hence, we have that:

$$\lambda \|Du\|_{L^2(\tilde{\Omega})}^2 \leq C(\|f\|_{L^2(\Omega)} \|u\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)} \|Du\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)}).$$

Here, we choose $\epsilon \cdot C < \frac{\lambda}{2}$ say, we have that:

$$\|Du\|_{L^2(\tilde{\Omega})}^2 \leq C(\|f\|_{L^2(\Omega)} + \|v\|_{L^2(\Omega)}).$$

□

VIII.3 Higher Interior Regularity

The idea is that if $f \in H^1(\Omega)$ and $u \in H_{\text{loc}}^2(\Omega)$ is a weak solution to $Lu = f$ (for appropriate coefficients), then we can differentiate the PDE.

Example VIII.3.1. If $f \in H^1(\Omega)$ and $u \in H_{\text{loc}}^2(\Omega)$ weakly solves $-\Delta u = f$, set $v = D_i \varphi$ for $\varphi \in C_c^\infty(\Omega)$ to get:

$$-\int_{\Omega} (D_i f) \varphi = \int_{\Omega} f (D_i \varphi) = \int_{\Omega} f v = \int_{\Omega} Du \cdot Dv = \int_{\Omega} Du (D_i \nabla \varphi) = - \int_{\Omega} \nabla_i (Du) \cdot D\varphi.$$

We can write the right hand side as $\nabla(D_i u) D\varphi$, so we have $D_i u \in H_{\text{loc}}^1(\Omega)$ weakly solves:

$$-\Delta(D_i u) = D_i f \in L^2(\Omega).$$

Then, the interior regularity implies that $D_i u \in H_{\text{loc}}^2(\Omega)$, which implies that $u \in H_{\text{loc}}^3(\Omega)$. Iterating this, we get that $f \in H^k(\Omega)$ implies that $u \in H_{\text{loc}}^{k+2}(\Omega)$.

Thus, for k being large, we have the Sobolev embedding for $u \in C_{\text{loc}}^2(\Omega)$ implying that it is the classical solution to $-\Delta u = f$ on $\tilde{\Omega} \Subset \Omega$. ◇

Theorem VIII.3.2. Higher Interior Regularity.

If $\Omega \subset \mathbb{R}^n$ is open, L is elliptic with $a^{ij} \in W^{k+1,\infty}(\Omega)$, $b^i, c \in W^{k,\infty}(\Omega)$ for $i, j = 1, \dots, n$, then if

$u \in H^k(\Omega)$ is a weak solution to $Lu = f$ for some $f \in H^k(\Omega)$, we have $u \in H_{\text{loc}}^{k+2}(\Omega)$ and for $\tilde{\Omega} \Subset \Omega$:

$$\|u\|_{H^{k+2}(\tilde{\Omega})} \leq C(\|f\|_{H^k(\Omega)} + \|u\|_{L^2(\Omega)})$$

for $C > 0$ dependent on $K, \Omega, \tilde{\Omega}$, and L .

Proof. We induct on $K \geq 0$, with the base case $k = 0$ precisely being the previous theorem (Theorem VIII.2.2). Now, we assume that:

$$\|u\|_{H^{k+2}(\tilde{\Omega})} \leq C(\|f\|_{H^k(\Omega)} + \|u\|_{L^2(\Omega)})$$

holds for some $C > 0$ dependent on $K, \Omega, \tilde{\Omega}$, and L on $k \geq 0$, and we assume that $a^{i,j} \in W^{k+2,\infty}(\Omega)$, $b^i, c \in W^{k+1,\infty}(\Omega)$, $f \in H^{k+1}(\Omega)$ and $u \in H^1(\Omega)$ weakly solves $Lu = f$. The inductive assumption tells us $u \in H_{\text{loc}}^{k+2}(\Omega)$. Fix $\tilde{\Omega} \Subset \hat{\Omega} \Subset \Omega$, with multi-indices $\alpha \in \mathbb{N}^n$ that $|\alpha| = k+1$, and $\varphi \in C_c^\infty(\Omega)$. Let $v = (-1)^{|\alpha|} D^\alpha \varphi$ in the weak formulation, so we have:

$$B(u, v) = (f, v)_{L^2(\Omega)}.$$

By **integration by parts**, we obtain that:

$$B(D^\alpha u, \varphi) = (\hat{f}, \varphi)_{L^2(\Omega)},$$

where $\hat{f} \in L^2(\Omega)$ is given by:

$$\hat{f} = D^\alpha f - \sum_{\substack{\beta \leq \alpha \\ \beta \neq \alpha}} \binom{|\alpha|}{|\beta|} [-D_j (D^{\alpha-\beta}(a^{i,j}) D^\beta(D_i u) + D^{\alpha-\beta}(b^i) D^\beta(D_i u))],$$

where $\beta \leq \alpha$ implies that $\beta_i \leq \alpha_i$ for all i . Hence, we obtain that $D^\alpha u$ is a weak solution to $L(D^\alpha u) = \hat{f} \in L^2(\tilde{\Omega})$ on $\tilde{\Omega}$.

Hence, by interior estimate, we have $D^\alpha u \in H_{\text{loc}}^2(\hat{\Omega})$ and:

$$\|D^\alpha u\|_{H^2(\hat{\Omega})} \leq C(\|\hat{f}\|_{L^2(\hat{\Omega})} + \|D^\alpha u\|_{L^2(\hat{\Omega})}).$$

Now, we have:

$$\|\hat{f}\|_{L^2(\hat{\Omega})} \leq C(\|f\|_{H^{k+1}(\Omega)} + \|u\|_{H^{k+1}(\Omega)}).$$

Now, we have:

$$\begin{aligned} \|\hat{f}\|_{L^2(\hat{\Omega})} &\leq C(\|f\|_{H^{k+1}(\Omega)} + \|u\|_{H^{k+2}(\hat{\Omega})}) \leq C(\|f\|_{H^{k-1}(\Omega)} + \|u\|_{L^2(\Omega)}), \\ \|D^\alpha u\|_{L^2(\hat{\Omega})} &\leq \|u\|_{H^{k+1}(\hat{\Omega})}. \end{aligned}$$

Now, combining all three, we see that as α is arbitrary:

$$\|u\|_{H^{k+2}(\tilde{\Omega})} \leq C(\|f\|_{H^{k+1}(\Omega)} + \|u\|_{L^2(\Omega)}),$$

which implies that $u \in H_{\text{loc}}^{k+3}(\Omega)$ (as $\tilde{\Omega}$ was arbitrary). □

Theorem VIII.3.3. Smoothness on Interior.

If $\Omega \subset \mathbb{R}^n$ is open, L is elliptic with $a^{ij}, b^i, c \in C^\infty(\Omega)$ for $i, j = 1, \dots, b$, then if $u \in H^1(\Omega)$ is a weak solution to $Lu = f$ for $f \in C^\infty(\Omega)$, we have $u \in C^\infty(\Omega)$.

Proof. By Higher Interior Regularity (Theorem VIII.3.2), we have $u \in H_{\text{loc}}^k(\Omega)$ for every $k \geq 0$, hence by Sobolev embedding, we have $u \in C^\ell(\Omega)$ for each $\ell \geq 0$ ($k - \lfloor \frac{n}{2} \rfloor > \ell$) and thus $u \in C^\infty(\Omega) = \bigcap_{k \geq 0} C^k(\Omega)$. $\square$

Remark VIII.3.4.

- $u \in H_{\text{loc}}^2(\Omega)$ implies that $Lu = f$ makes sense almost everywhere, but now we have $u \in C^2(\Omega)$ if k is large enough ($k - \lfloor \frac{n}{2} \rfloor > \ell$), so $Lu = f$ makes sense everywhere, i.e., u is a classical solution to the PDE.
- The technique of “improving” regularity from the PDE itself is called **bootstrapping**.
- We have not said anything about the regularity of u near $\partial\Omega$.

┘

VIII.4 Boundary Regularity

We aim to extend the interior estimates for weak solutions *up to* the boundary, assuming the boundary is sufficiently regular and the weak solution satisfies some boundary conditions, e.g. $u \in H_0^1$.

The question now becomes: Why do we need assumptions on $\partial\Omega$ to get regularity?

Example VIII.4.1. Consider for $\beta \in (0, 2\pi)$, we have:

$$\Omega = \{(r, \theta) : 0 < r < 1, 0 < \theta < \beta\},$$

Here, we can visualize some sectors:

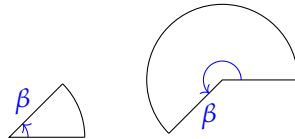


Figure VIII.1. Some sectors with angle β .

then we have $\Omega \subset \mathbb{R}^2$, and $\partial\Omega$ is Lipschitz but not C^1 .

In polar coordinates, we let:

$$u(r, \theta) = r^{\pi/\beta} \sin\left(\frac{\pi\theta}{\beta}\right) \in C^\infty(\Omega).$$

Then, if $\theta = 0$, we have $u = 0$, and writing:

$$\Delta u = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0.$$

Hence, we have $\Delta u = 0$ strongly in Ω and one can check that $u \in H^1(\Omega)$ with:

$$\int_{\Omega} |Du|^2 = \frac{\pi}{2}.$$

Now, if $\beta > \pi$, we have $u \notin H^2(\Omega)$ since:

$$\int_{\Omega} |D^2 u|^2 \geq \int_0^{\beta} \int_0^1 \left| \frac{\partial^2 u}{\partial r^2} \right|^2 r dr d\theta \geq C \int_0^1 r^{2\pi/\beta-3} dr.$$

As $\frac{2\pi}{\beta} - 3 < -1$ if and only if $\beta > \pi$, which implies that:

$$\int_0^1 r^{2\pi/\beta-3} dr = +\infty,$$

hence $u \notin H^2(\Omega)$.

If we let $r > 0$ (not < 1), we have $u \in H_0^1(\Omega)$. ◇

Now, we have an analogue of the interior estimate.

Theorem VIII.4.2. Boundary Estimate.

If $\Omega \subset \mathbb{R}^n$ is open, $\partial\Omega$ is C^2 , L is elliptic with $a^{ij} \in W^{1,\infty}(\Omega)$, $b^i, c \in L^\infty(\Omega)$ for $i, j = 1, \dots, n$, then if $u \in H_0^1(\Omega)$ is a weak solution to:

$$\begin{cases} Lu = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$

for $f \in L^2(\Omega)$, we have that $u \in H^2(\Omega) \cap H_0^1(\Omega)$ and:

$$\|u\|_{H^2(\Omega)} \leq C(\|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)})$$

for a constant $C > 0$ depending on Ω , n , and L .

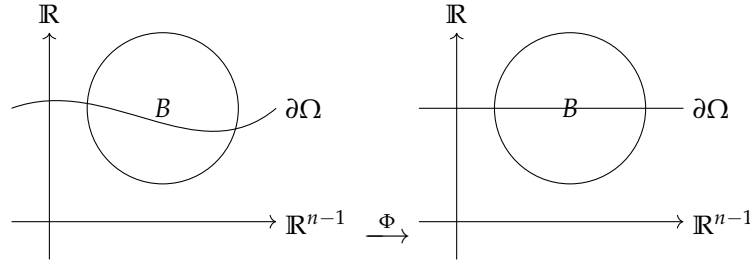
Remark VIII.4.3. If such a weak solution to the differential equation is unique, then one can show that:

$$\|u\|_{H^2(\Omega)} \leq C\|f\|_{L^2(\Omega)},$$

(since if $Lu, L\tilde{u} = f$, then we have $L(u - \tilde{u}) = 0$ and so $u = \tilde{u}$ almost everywhere.)

- We **require** $\partial\Omega$ is C^2 and $u \in H_0^1(\Omega)$. ┘

Proof of sketch. Consider that we have $\partial\Omega$ to be C^2 , we consider the straightening:

Figure VIII.2. Straighting $\partial\Omega$ and use interior ideas/estimates. □**Theorem VIII.4.4. Higher Boundary Regularity.**

If $\Omega \subset \mathbb{R}^n$ is open, $\partial\Omega$ is C^{k+2} , L is elliptic with $a^{ij} \in W^{k+1,\infty}(\Omega)$, $b^i, c \in W^{k,\infty}(\Omega)$ for $i, j = 1, \dots, n$, $k \in \mathbb{N}$, then if $u \in H_0^1(\Omega)$ is a weak solution to:

$$\begin{cases} Lu = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$

for $f \in H^k(\Omega)$, we have $u \in H^{k+2}(\Omega) \cap H_0^1(\Omega)$ and $\|u\|_{H^{k+2}(\Omega)} \leq C(\|f\|_{H^k(\Omega)} + \|u\|_{L^2(\Omega)})$ for a constant $c > 0$ depending on k, Ω, L , and n .

By repeatedly applying higher boundary regularity, we have the following.

Theorem VIII.4.5. Smoothness up to the Boundary.

If $\Omega \subset \mathbb{R}^n$ is open, $\partial\Omega$ is C^∞ , L is elliptic with $a^{ij}, b^i, c \in C^\infty(\overline{\Omega})$ for $i, j = 1, \dots, n$, then if $u \in H_0^1(\Omega)$ is a weak solution to:

$$\begin{cases} Lu = f, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$

with $f \in C^\infty(\overline{\Omega})$, we have $u \in C^\infty(\overline{\Omega})$.

Remark VIII.4.6. Now, we can entirely solve the Poisson equation on smooth domains in the classical sense. (I.e., the elliptic PDEs can be connected to the Laplacian operator.) ┘

The proofs can be found from Evans' section 6.3.2.

IX Extending from Second Order Linear PDEs to Nonlinear Cases

IX.1 Existence for a Nonlinear PDE

The aim is to think of $-\Delta u = u^p$ for $p \in [1, 2^*]$, which is a *quasi-linear* PDE, and we want to think about the existence of weak solutions using **direct method** and use regularity to show that weak solutions lead to strong solutions.

Consider $\Omega \subset \mathbb{R}^n$ being open and bounded, with $\partial\Omega$ being C^3 and for $n \geq 3$, we seek a non-negative function $u \in C^{2,\alpha}(\Omega) \cap C^0(\overline{\Omega})$ with $u > 0$ in Ω , and solving:

$$\begin{cases} -\Delta u = u^p, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases} \quad (7)$$

for $p \in (1, \frac{n+2}{n-2})$.

Remark IX.1.1. The above example is semi-linear as it is linear in its highest (2nd) order derivatives. The method we will use to find weak solutions applies more generally to PDEs of the form:

$$\begin{cases} Lu = f(u), & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

Provided that L is elliptic, and f has controlled growth.

- $(\frac{n+2}{n-2})$ has an upper bound for p will be used since:

$$p + 1 = \frac{n+2}{n-2} + 1 = \frac{2n}{2n-2} = 2^*,$$

i.e., $p + 1$ is critical exponent for the Sobolev embedding to be compact.

- To solve the equation for $p + 1 = 2^*$, one needs more refined methods, and this value of p is related to the **Yamabe problem**. ┘

Definition IX.1.2. Yamabe Problem.

Suppose (M, g) is a compact Riemannian manifold with dimension $n \geq 3$, is there a metric $\tilde{g} = e^{2u}g$, such that $\tilde{g}$ has constant scalar curvature? ┘

Existence: Let's drop the assumption $\partial\Omega$ is C^3 for now. We will now find $u \geq 0$ and $u \not\equiv 0$ over Ω . Hence, the weak solution must solve that:

$$\int_{\Omega} Du \cdot D\varphi = \int_{\Omega} u^p \varphi \quad \text{for all } \varphi \in C_c^\infty(\Omega).$$

This makes sense since $u^p \in L^1(\Omega)$ as:

$$p < \frac{n+2}{n-2} < \frac{2n}{n-2} = 2^* \implies u \in L^p(\Omega),$$

since $u \in H_0^1(\Omega)$ and the Poincaré inequality.

Let's try to use the **direct method** where we saw critical points of:

$$E(u) = \int_{\Omega} |Du|^2 \text{ gives weak solutions to } \Delta u = 0 \text{ on } \Omega.$$

We modify this and for $q \in (2, \frac{2n}{n-2})$, let:

$$E_q(u) = \frac{E(u)}{\|u\|_{L^q(\Omega)}} \text{ for } u \in H_0^1(\Omega) \setminus \{0\},$$

again, by using that $H_0^1 \hookrightarrow L^q$.

Note that:

$$E_q(u) = \frac{E(u)}{\|u\|_{L^q(\Omega)}} = \frac{\|Du\|_{L^2(\Omega)}^2}{\|u\|_{L^q(\Omega)}} \geq \frac{1}{c_p} > 0,$$

and by the Poincaré inequality (as $q < 2^*$ and Ω is bounded), we set:

$$\mathcal{E}_q = \inf \left\{ E_q(u) \mid u \in H_0^1(\Omega) \setminus \{0\} \right\} > 0.$$

We let $(u_i) \subset H_0^1(\Omega) \setminus \{0\}$ be such that $E_q(u_i) \rightarrow \mathcal{E}_q$.

As $E_q(\lambda u) = E_q(u)$ for $\lambda > 0$, $u \in H_0^1(\Omega) \setminus \{0\}$, so we may suppose (without loss of generality) that $\|u_i\|_{L^q(\Omega)} = 1$ for each $i \geq 1$, hence we have $E_q(u_i) = E(u_i)$.

So (u_i) is bounded in $H_0^1(\Omega)$ since for large $i \geq 1$, we have:

$$\|u_i\|_{H_0^1(\Omega)}^2 \leq C \|Du\|_{L^2(\Omega)}^2 = CE(u_i) = CE_q(u_i) < C(\mathcal{E}_q + 1).$$

Hence, by weak compactness in $H_0^1(\Omega)$, there is a weakly compact subsequence $u_{i(i)} \rightharpoonup u \in H_0^1(\Omega)$, and thus by Rellich-Kandrachov, this implies that $H_0^1(\Omega) \xhookrightarrow{c} L^q(\Omega)$, so $u_{i(i)} \rightarrow u \in L^q(\Omega)$, and in particular (since $\|u_{i(i)}\|_{L^q(\Omega)} = 1$), we have $\|u\|_{L^q(\Omega)} = 1$.

We also have $E(u) \leq \liminf_{i \rightarrow \infty} E(u_{i(i)})$. Since the norms are lower-semi-continuous under weak convergence, we have:

$$E_q(u) = E(u) \leq \liminf_{i \rightarrow \infty} E(u_{i(i)}) = \mathcal{E}_q,$$

and since $u \in H_0^1(\Omega) \setminus \{0\}$, we have $E_q(u) = \mathcal{E}_q$.

To ensure that $u \geq 0$ almost everywhere, note that $|u| \in H_0^1(\Omega) \setminus \{0\}$ with $|Du| = |D|u||$, so $E_q(|u|) = E_q(u) = \mathcal{E}_q$, so by replacing that $u = |u|$, we ensure that $u \geq 0$ almost everywhere.

Let's see what PDE u solves, if $\varphi \in C_c^\infty(\Omega)$, and $u + t\varphi \in H_0^1(\Omega) \setminus \{0\}$ (for small t) so that:

$$E_q(u + t\varphi) \geq E_q(u),$$

and hence $\frac{d}{dt}E_q(u + t\varphi)|_{t=0} = 0$, which is:

$$\begin{aligned}\frac{d}{dt}E_q(u + t\varphi) &= \frac{d}{dt} \left[E(u + t\varphi) \|u + t\varphi\|_{L^q(\Omega)}^{-2} \right] \\ &= 2 \int_{\Omega} Du \cdot D\varphi + E(u) \left(-2 \|u\|_{L^q(\Omega)}^{-3} \frac{d}{dt} \|u + t\varphi\|_{L^q(\Omega)} \right) \\ &= 2 \int_{\Omega} Du \cdot D\varphi - 2\mathcal{E}_q \frac{d}{dt} \left(\int_{\Omega} |u + t\varphi|^q \right)^{\frac{1}{q}} \\ &= 2 \int_{\Omega} Du \cdot D\varphi - 2\mathcal{E}_q \cdot \frac{1}{q} \left(\int_{\Omega} |u|^q \right)^{\frac{1}{q}-1} \left(\int_{\Omega} q|u|^{q-1}\varphi \right).\end{aligned}$$

Hence, we now have:

$$\int_{\Omega} Du \cdot D\varphi = \frac{\mathcal{E}_q}{q} \int_{\Omega} q|u|^{q-1}\varphi$$

for all $\varphi \in C_c^\infty(\Omega)$. Hence, we have shown that u weakly solves:

$$\begin{cases} -\Delta u = \mathcal{E}_q u^{q-1}, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

Given $p \in (1, \frac{n+2}{n-2})$, we set $q = p + 1 < 2^*$ so that u above weakly solves:

$$\begin{cases} -\Delta u = \mathcal{E}_q u^p, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega. \end{cases}$$

Consider $\mathcal{E}_q^{\frac{1}{p-1}}$, then it weakly satisfies that:

$$-\Delta(\mathcal{E}_q^{\frac{1}{p-1}} u) = \mathcal{E}_q^{\frac{1}{p-1}} (-\Delta u) = \mathcal{E}_q^{\frac{1}{p-1}} \mathcal{E}_q u^p = \mathcal{E}_q^{\frac{p}{p-1}} u^p = (\mathcal{E}_q^{\frac{1}{p-1}} u)^p,$$

so $u = \mathcal{E}_q^{\frac{1}{p-1}} u$ is ≥ 0 almost everywhere, and weakly solves:

$$\begin{cases} -\Delta u = u^p, & \text{on } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$

with $u \in H_0^1(\Omega) \setminus \{0\}$ and ≥ 0 almost everywhere.

Now, if $u^p \in L^2(\Omega)$, we have $u \in H_{\text{loc}}^2(\Omega)$ (by our interior regularity, “freezing” $f = v^p$).

Also, if $v^p \in H^k(\Omega)$ so we have $v \in H_{\text{loc}}^{k+2}(\Omega)$.

IX.2 Regularity for a Nonlinear PDE

Under the assumption that $\partial\Omega$ being C^3 , we want to then show that:

$$u \in C^\infty(\Omega) \cap C^{2,\alpha}(\overline{\Omega}).$$

Recall we saw that if $-\Delta u = f \in L^2(\Omega)$ is solved weakly by $v \in H_0^1(\Omega)$, then in fact we have $v \in H_0^2(\Omega)$.

This principle holds if $f \in L^q(\Omega)$ for $q \in (1, \infty)$ by Calderon-Zygmund estimates which then ensure that $w \in W_0^{2,q}(\Omega)$.

In our situation that $f = u^p$, we now use these estimates to argue that $u \in C^{0,\alpha}(\overline{\Omega})$.

Suppose we had $u \in W_0^{1,s}(\Omega)$ for some $s \geq 2$ (we already have $s = 2$). If $s > n$, we can apply Morrey inequality that $u \in C^{0,\alpha}(\overline{\Omega})$. If $s \in [2, n]$, however, we consider the following cases:

- First, if $s \in [2, n)$, the GNS inequality gives $u \in L^{\frac{ns}{n-s}}(\Omega)$ so we have $u^p \in L^{\frac{ns}{p(n-s)}}(\Omega)$.
As $s \geq 2$, $p \in (1, \frac{n+2}{n-2})$ and $n > 2$, so we have:

$$\frac{ns}{p(n-s)} \geq \frac{2n}{p(n-2)} > \frac{2n}{n+2} > 1.$$

So by Calderon-Zygmund, we have $u \in W_0^{2, \frac{ns}{p(n-s)}}(\Omega)$.

- Second, if $s \in [2, \frac{np}{1+p})$, we have:

$$\frac{ns}{p(n-s)} < \frac{n \left(\frac{np}{1+p} \right)}{p \left(n - \frac{np}{1+p} \right)} = \frac{n^2}{n} = n.$$

Now, by applying GNS inequality to $D_i u$ for $i = 1, \dots, n$, we have:

$$D_i u \in W^{1, \frac{ns}{p(n-s)}}(\Omega)$$

with:

$$\sigma = \frac{n \left(\frac{ns}{p(n-s)} \right)}{n - \left(\frac{np}{1+p} \right)} = \frac{ns}{(n-s)p - s} = \underbrace{\left(\frac{ns}{np - s(p+n)} \geq \frac{n}{np - 2(p+1)} \right)}_{\gamma} s = \gamma s.$$

Now, since $p \in (1, \frac{n+2}{n-2})$, so we have:

$$\gamma = \frac{n}{np - 2(p+1)} = \frac{n}{p(n-2) - 2} > \frac{n}{n+2-2} = 1.$$

Thus, we have:

$$u \in W_0^{1,s}(\Omega) \quad \text{for } s \in \left[2, \frac{np}{1+p} \right),$$

so we have $u \in W_0^{1,\gamma s}(\Omega)$ for some $\gamma > 1$.

- Starting at $s = 2$, we iteratively apply the previous step, we get $u \in W_0^{1,\gamma^k s}(\Omega)$ for all $k \geq 1$ such that $\gamma^k s < \frac{np}{1+p}$, so one application give $u \in W_0^{1,t}(\Omega)$ for some $t > \frac{np}{1+p}$, so we have $u \in W_0^{1, \frac{np}{1+p}}(\Omega)$.

As $\frac{np}{1+p} < n$, by applying GNS inequality again, we have $u \in L^{\frac{n \left(\frac{np}{1+p} \right)}{n - \left(\frac{np}{1+p} \right)}}(\Omega) = L^{np}(\Omega)$, so we have $u^p \in L^n(\Omega)$.

One further application of the Calderon-Zygmund gives $u \in W_0^{2,n}(\Omega)$, and by Sobolev embedding, as $2 > \frac{n}{p}$, we have that $u \in C^{0,\alpha}(\overline{\Omega})$ for any $\alpha \in (0, 1)$.

We now use Schauder estimates which says that $-\Delta v = f \in C^{0,\alpha}(\bar{\Omega})$ implies $v \in C^{2,\alpha}(\bar{\Omega})$.

As we have $-\Delta u = u^p \in C^{0,\alpha}(\bar{\Omega})$ as $x \mapsto x^p$ locally Lipschitz as $p > 1$ and so Schauder estimates give $u \in C^{2,\alpha}(\bar{\Omega})$. Hence, u solves (7).

Iteratively applying the Schauder estimates to $\tilde{\Omega} \Subset \Omega$ with $\partial\tilde{\Omega}$ being smooth (differentiating the PDE: $-\Delta u = u^p$ into $-\Delta D_i^k u = D_i^k(u^p) \in C^{0,\alpha}(\bar{\Omega})$, so $u \in C^{k+2,\alpha}(\bar{\Omega})$) to get $u \in C^\infty(\bar{\tilde{\Omega}})$, i.e., $u \in C^\infty(\Omega)$. Thus by Calderon-Zygmund and Schauder, we have $u \in C^{2,\alpha}(\bar{\Omega}) \cap C^\infty(\Omega)$.

Part 4

Second Order Nonlinear PDEs

X Hilbert's XIX-th Problem

X.1 Minimal Surface Equation

Given $\Omega \subset \mathbb{R}^n$ being open and bounded, with some $g : \partial\Omega \rightarrow \mathbb{R}$, consider the area of the graph of some $u : \Omega \rightarrow \mathbb{R}$ such that $u|_{\partial\Omega} = g$, then:

$$\text{Area}(\text{graph}(u)) = \int_{\Omega} \sqrt{1 + |\nabla u|^2} dx,$$

where $\text{graph}(u) = \{(x, u(x)) : x \in \Omega\} \subset \mathbb{R}^{n+1}$.

Suppose we try to minimize the area of these graphs over all such functions; one may wonder whether a minimizing function exists, and, if so, what regularity does it possess.

By similar reasoning to the **direct method** for the Dirichlet energy:

$$\int_{\Omega} |\nabla u|^2,$$

where the minimizer solves the Laplace equation, *i.e.*, $\Delta u = 0$, minimizers of the area formally solve the minimal surface equation:

$$\text{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) = \left(\delta_{i,j} - \frac{D_i u D_j u}{\sqrt{1 + |\nabla u|^2}} \right) D_{i,j} u = 0. \quad (\text{MSE})$$

Geometrically, this corresponds to the graph of u having zero *mean curvature* and provides a solution to the graphical **Plateau problem**.

Remark X.1.1.

- Existence for the Laplace's equation followed, for general $\Omega \subset \mathbb{R}^n$, by a “straightforward” direct method approach. However, for the minimal surface equation, the existence of a minimizer, in the class of smooth (or $C^{2,\alpha}$), actually fails, unless we assume Ω has geometric constraints (we need *mean convexity*).
- On small scales, where $\nabla u \approx \text{constant} = c$, the minimal surface equation looks like:

$$\text{div} \left(\frac{\nabla u}{\sqrt{1 + c^2}} \right) = 0 \iff \tilde{c} \Delta u = 0.$$

Hence, minimal surface equation is similar to the Laplacian in the locally sense. They are very different globally. ┘

Recall that, Liouville's theorem states that if $\Delta u = 0$ and u is bounded (or within polynomial growth), then u is a constant. But for the mean surface equation, this only holds if $n \leq 7$ (Bernstein problem).

Many natural processes involve minimization and can be modeled by an appropriate energy.

For such energy, one may wonder whether all minimizers are smooth under some appropriate assumptions. To this end, Hilbert posed, in the 1900, the following:

Hilbert's XIX-th Problem.

Given $\Omega \subset \mathbb{R}^n$ being open, $L : \mathbb{R}^n \rightarrow \mathbb{R}$ smooth and uniformly convex, are local minimizers, u , smooth for the energy $\mathcal{E}(u) = \int_{\Omega} L(\nabla u)$?

Remark X.1.2. The original statement was phrased for L being analytic and of the form $L : \Omega \times \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}$, asking whether local minimizers are analytic, i.e., $\mathcal{E}(u) = \int_{\Omega} L(x, u, \nabla u)$. $\lrcorner$

X.2 Existence and Uniqueness of Local Minimizer

The Hilbert's XIX Problem could be some observations with the dimensions.

Remark X.2.1.

- If $n = 1$, all minimizers are linear, hence analytic.
- If $n \geq 2$, there are smooth $L : \mathbb{R}^n \rightarrow \mathbb{R}$ with smooth but not analytic local minimizers. (This can be done via Legendre transformation). $\lrcorner$

Example X.2.2. Consider the following examples:

- $L(\nabla u) = |\nabla u|^2$ is the Dirichlet energy and $\nabla u = 0$.
- $L(\nabla u) = \sqrt{1 + |\nabla u|^2}$ corresponds to area and the minimal surface energy.

Here are also non-example:

- If $n = 1$ and $L : \mathbb{R} \rightarrow \mathbb{R}$ is smooth and has two distinct minima, $x, y \in \mathbb{R}$. Then if L is not convex, we can build/construct Lipschitz but not smooth minimizers of L .

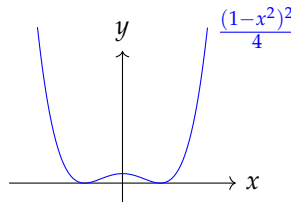


Figure X.1. Graph of $\frac{(1-x^2)^2}{4}$ being nonconvex, minimizer at ± 1 .

In this non-example, $u(x) = |x|$ is a minimizer of $\mathcal{E}(u) = \int_{\Omega} L(\nabla u) = 0$. $\diamond$

To solve the XIX-th problem, we first will establish existence and uniqueness of global minimizers of the energy and see what PDE does minimizers solve. In fact, these minimizers solve (weakly) to the divergence form second order PDE:

$$\operatorname{div}(\nabla L(\nabla u)) = 0 \quad \text{in } \Omega,$$

if $u \in C^2(\Omega)$, then it is equivalent to:

$$(D_{i,j}L)(\nabla u)D_{i,j}u = 0 \quad \text{in } \Omega.$$

Hence, the uniform convexity of L actually guarantees ellipticity as the coefficients $D_{i,j}L(\nabla u)$ will be positive definite. However, the PDE is not linear so cannot use the Sobolev theory (or divergence form).

To proceed, we use the following two steps instead:

- Show that the local minimizers of Energy/weak solutions to the divergence form are actually $C^{1,\alpha}$ regular (since with $u \in H^1$ and divergence form implies $u \in C^{1,\alpha}$), which is the **De Giorgi-Nash** form.
- Show that $C^{1,\alpha}$ minimizers of energy/weak solution to the equivalent form are actually small (together with $u \in C^{1,\alpha}$ implies $u \in C^\infty$), which is **Schauder** estimates.

Definition X.2.3. If $\Omega \subset \mathbb{R}^n$ open, $L : \mathbb{R}^n \rightarrow \mathbb{R}$ is smooth, then:

- L is **uniformly convex** if there are $0 < \lambda \leq \Lambda$ such that:

$$0 < \lambda \operatorname{Id} \leq D^2L(x) \leq \Lambda \operatorname{Id} \quad \text{for all } x \in \mathbb{R}^n,$$

where $\operatorname{Id} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the identity on $\mathbb{R}^n$.

Specifically, we consider the eigenfunctions as $\lambda|\xi|^2 \leq \xi^\top D^2L(x)\xi \leq \Lambda|\xi|^2$

- Without loss of generality, $L \geq 0$ (shift by a constant).
- We will assume $n \geq 3$ (works for $n = 1, 2$ with minor changes).
- $u \in H^1(\Omega) = W^{1,2}(\Omega)$ is a local minimizer of $\mathcal{E}(u) = \int_{\Omega} L(\nabla u)$ if $\mathcal{E}(u) \leq \mathcal{E}(u + \varphi)$ for all $\varphi \in C_c^\infty(\Omega)$. $\lrcorner$

Theorem X.2.4. If $\Omega \subset \mathbb{R}^n$ is open, bounded, $\partial\Omega$ is Lipschitz and for $g \in L^2(\partial\Omega)$, the set:

$$\mathcal{A} = \{u \in H^1(\Omega) \mid u|_{\partial\Omega} = g\} \neq \emptyset,$$

then for $L : \mathbb{R}^n \rightarrow \mathbb{R}$ smooth and uniformly convex, there exists a unique minimizer $u \in H^1(\Omega)$, with $u|_{\partial\Omega} = g$ of $\mathcal{E}(u) = \int_{\Omega} L(\nabla u)$ which weakly solves $\operatorname{div}(\nabla L(\nabla u)) = 0$ on Ω .

Remark X.2.5. $u|_{\partial\Omega} = g$ means that the trace $T(u)$, in the Sobolev sense which is well defined as $\partial\Omega$ is Lipschitz. $\square$

Proof. Step 1: (Lower semi-continuity). If $\{u_k\}, u \in W^{1,1}(\Omega)$ and we assume that $u_k \rightarrow u$ in $L^1_{\text{loc}}(\Omega)$, this holds, for example, up to a subsequence if $u_k \rightharpoonup u$ in $H^1(\Omega)$ by Rellich-Kondrachov (Theorem IV.2.10) since Ω is bounded.

For $\epsilon > 0$, we mollify with $\eta \in C_c^\infty(B_1(0))$ with $\eta \geq 0$, $\int_{B_1(0)} \eta = 1$, to set:

$$\eta_\epsilon(x) = \epsilon^{-n} \eta\left(\frac{x}{\epsilon}\right),$$

so that we have $\int_{B_\epsilon(0)} \eta_\epsilon = 1$.

We then get:

$$(u_k)_\epsilon(x) = (u_k * \eta_\epsilon)(x) = \int_{B_\epsilon(x)} u_k(x-y) \eta_\epsilon(y) dy,$$

which is similar for u_ϵ . Choosing $\tilde{\Omega} \Subset \Omega$ so that for small $\epsilon > 0$, $B_\epsilon(x) \subset \Omega$ for each $x \in \tilde{\Omega}$. Since $u_k \rightarrow u$ in $L^1_{\text{loc}}(\Omega)$ and L is smooth, we get:

$$\begin{cases} \nabla(u_k)_\epsilon(x) \rightarrow \nabla u_\epsilon(x), \\ L(\nabla(u_k)_\epsilon(x)) \rightarrow L(\nabla u_\epsilon(x)), \end{cases} \quad \text{for each } x \in \tilde{\Omega}.$$

As $L \geq 0$, Fatou's lemma gives that:

$$\int_{\tilde{\Omega}} L(\nabla u_\epsilon) \leq \liminf_{k \rightarrow \infty} \int_{\tilde{\Omega}} L(\nabla(u_k)_\epsilon).$$

Noting that $\nabla(u_k)_\epsilon = (\nabla u_k)_\epsilon$ and using the Jensen's inequality, as L is convex and $\int_{B_1(0)} \eta = 1$, we get for each $x \in \tilde{\Omega}$ that:

$$L(\nabla(u_k)_\epsilon(x)) = L\left(\int_{B_\epsilon(x)} \nabla u_k(y) \eta_\epsilon(x-y) dy\right) \leq \int_{B_\epsilon(x)} L(\nabla u_k(y)) \eta_\epsilon(x-y) dy,$$

where we consider the (unit) measure $\eta_\epsilon(x-y) dy$. Integrating over $\tilde{\Omega}$, we have:

$$\begin{aligned} \int_{\tilde{\Omega}} L(\nabla(u_k)_\epsilon) dx &\leq \int_{\tilde{\Omega}} \left(\int_{B_\epsilon(0)} L(\nabla u_k(y)) \eta_\epsilon(x-y) dy \right) dx \\ &\leq \int_{\tilde{\Omega}_\epsilon} L(\nabla u_k(y)) \left(\int_{B_\epsilon(0)} \eta_\epsilon(x-y) dx \right) dy \leq \int_{\tilde{\Omega}} L(\nabla u_k). \end{aligned}$$

By combining the above two arguments, we have:

$$\int_{\tilde{\Omega}} L(\nabla u_\epsilon) \leq \liminf_{k \rightarrow \infty} \int_{\tilde{\Omega}} L(\nabla(u_k)_\epsilon) \leq \liminf_{k \rightarrow \infty} \int_{\tilde{\Omega}} L(\nabla u_k).$$

Since $u \in W^{1,1}(\Omega)$, we get $\nabla u_\epsilon \rightarrow \nabla u$ a.e. on $\tilde{\Omega}$ so sending $\epsilon \rightarrow 0$ gives:

$$\int_{\tilde{\Omega}} L(\nabla u) \leq \liminf_{k \rightarrow \infty} \int_{\tilde{\Omega}} L(\nabla u_k),$$

and by compact exhaustion of Ω that $\mathcal{E}(u) \leq \liminf_{k \rightarrow \infty} \mathcal{E}(u_k)$.

Step 2: (Existence of a minimizer in $\mathcal{A}$). As L is uniformly convex, it has a unique minimum, $p_L \in \mathbb{R}^n$, so that $L(p) > L(p_L)$ for all $p \in \mathbb{R}^n \setminus \{p_L\}$, and $DL(p_L) = 0$. We Taylor expand L at p_L to see that for each $p \in \mathbb{R}^n$ and some $\xi \in \mathbb{R}^n$ (on the line from p_L to p) that:

$$L(p) = L(p_L) + DL(p_L)(p - p_L) + \frac{1}{2}(p - p_L)^\top D^2L(\xi)(p - p_L),$$

in which we have by the uniform convex property through:

$$\lambda|p - p_L|^2 \leq \frac{1}{2}(p - p_L)^\top D^2L(\xi)(p - p_L) \leq \Lambda|p - p_L|^2.$$

Hence, by setting $p \leftarrow p_L - p$, we get that:

$$0 < \lambda|p|^2 \leq L(p_L - p) - L(p_L) \leq \Lambda|p|^2.$$

By translation, $p_L = 0$ (or consider $\tilde{L}(p) = L(p_L - p) - L(p_L)$ since Ω is bounded, so $\int_\Omega L(p_L)$ is a constant), and so that $L(0) = 0$, $DL(0) = 0$. Thus, we have:

$$0 < \lambda|p|^2 \leq L(p) \leq \Lambda|p|^2 \quad \text{for each } p \in \mathbb{R}^n.$$

Then, we set:

$$\mathcal{E}_0 = \inf \left\{ \mathcal{E}(u) = \int_\Omega L(\nabla u) : u \in H^1(\Omega), u|_{\partial\Omega} = g \right\} \geq 0,$$

which is finite since we have $\mathcal{E}_0 \leq \Lambda \int_\Omega |\nabla u|^2 < \infty$ for some $u \in \mathcal{A} \neq \emptyset$.

Consider $\{u_k\} \subset \mathcal{A}$ with $\mathcal{E}(u_k) \rightarrow \mathcal{E}_0$, then it again gives that:

$$\frac{\lambda}{c_p} \|u_k\|_{L^2(\Omega)}^2 \leq \lambda \|\nabla u_k\|_{L^2(\Omega)}^2 \leq \int_\Omega L(\nabla u_k) = \mathcal{E}(u_k) < \infty.$$

Therefore, by the (average-free) Poincaré inequality ([Theorem III.4.6](#)), we have $\{u_k\}$ being bounded in $H^1(\Omega)$, so that by Rellich-Kondrachov ([Theorem IV.2.10](#)) again, we have $u_k \rightharpoonup u \in H^1(\Omega)$ up to a subsequence not relabeled and $u_k \rightarrow u$ in $L^2(\Omega)$. Therefore, by linear semi-continuity, we have:

$$\mathcal{E}(u) \leq \liminf_{k \rightarrow \infty} \mathcal{E}(u_k) = \mathcal{E}_0.$$

Moreover, we have $u \in \mathcal{A}$, i.e., $u|_{\partial\Omega} = g$ since the trace is continuous and bounded. Therefore, u minimizes $\mathcal{E}$ in $\mathcal{A}$.

Step 3: (Uniqueness of the minimizer). Suppose we have $u \neq v$ in $\mathcal{A}$ both minimizing $\mathcal{E}$, i.e., $\mathcal{E}(u) = \mathcal{E}(v) = \mathcal{E}_0$. As $u|_{\partial\Omega} = v|_{\partial\Omega} = g$, so we must have $\nabla u \neq \nabla v$ in $L^2(\Omega)$, so $E := \{\nabla u \neq \nabla v\}$ has strictly positive measure.

As L is uniformly convex:

$$L\left(\frac{\nabla u + \nabla v}{2}\right) \leq \frac{L(\nabla u) + L(\nabla v)}{2}.$$

So, as $|E| > 0$, we have:

$$\int_E L\left(\frac{\nabla u + \nabla v}{2}\right) \leq \int_E \frac{L(\nabla u) + L(\nabla v)}{2},$$

so that we have:

$$\mathcal{E}_0 \leq \int_\Omega L\left(\frac{\nabla u + \nabla v}{2}\right) < \frac{1}{2} \int_\Omega L(\nabla u) + \frac{1}{2} \int_\Omega L(\nabla v) = \frac{1}{2} \mathcal{E}(u) + \frac{1}{2} \mathcal{E}(v) = \mathcal{E}_0,$$

which is a contradiction, so $u = v$ almost everywhere is the unique minimizer.

Step 4: (Weak solution). We will show that local minimizers weakly solve the PDE $\operatorname{div}(DL(\nabla u)) = 0$. If $u \in H^1(\Omega)$ is a local minimizer of $\mathcal{E}$ that:

$$\mathcal{E}(u) \leq \mathcal{E}(u + t\varphi) \quad \text{for } t \in \mathbb{R}, \varphi \in C_c^\infty(\Omega),$$

and so we have $\int_\Omega L(u) \leq \int_\Omega L(\nabla u + t\nabla\varphi)$. By writing:

$$L(\nabla u + t\nabla\varphi) \leq L(\nabla u) + tDL(\nabla u)\nabla\varphi + \frac{t^2}{2} \sup_{\xi \in \mathbb{R}^n} |D^2L(\xi)| \cdot |\nabla\varphi|^2,$$

we obtain that:

$$-\Lambda \frac{t}{2} \int_\Omega |\nabla\varphi|^2 \leq \int_\Omega DL(\nabla u)\nabla\varphi \quad \text{for } t > 0 \text{ and } \varphi \in C_c^\infty(\Omega).$$

Then, by sending $t \rightarrow 0^+$, we have $0 \leq \int_\Omega \nabla L(\nabla u)\nabla\varphi$ for $\varphi \in C_c^\infty(\Omega)$.

Similarly, if $t < 0$, we get:

$$-\Lambda \frac{t}{2} |\nabla u|^2 \geq \int_\Omega DL(\nabla u)\nabla\varphi \quad \text{for } \varphi \in C_c^\infty(\Omega),$$

so that if $t \rightarrow 0^-$, $\int_\Omega DL(\nabla u)\nabla\varphi \leq 0$ for all $\varphi \in C_c^\infty(\Omega)$. Thus, $\int_\Omega \nabla L(\nabla u)\nabla\varphi = 0$ for all $\varphi \in C_c^\infty(\Omega)$, which is equivalently that u weakly solves the PDE in Ω . $\square$

X.3 Regularity for Local Minimizer

Formally, if $e \in S^{n-1} \subset \mathbb{R}^n$ and we differentiate the above PDE as:

$$\operatorname{div}(\nabla^2 L(\nabla u)\nabla(\partial_e u)) = 0 \quad \text{in } \Omega.$$

So, if we set $v = \partial_e u$, $a^{ij}(x) = D^2L(\nabla u(x))$ so that $A(x) = (a^{ij}(x))_{i,j}$, and this becomes:

$$\operatorname{div}(A(x)\nabla v) = 0 \quad \text{in } \Omega.$$

Since L is uniformly convex, this is equivalent to A being uniformly elliptic, *i.e.*, the eigenvalues of A are controlled such that:

$$0 < \lambda \operatorname{Id} \leq A(x) \leq \Lambda \operatorname{Id}.$$

This is a linear divergence form, second order, elliptic PDE but without any regularity on the coefficients A , other than bounded and measurable.

In particular, the De Giorgi-Nash theory addresses the regularity of the above PDE.

Theorem X.3.1. De Giorgi-Nash.

If $v \in H^1(\Omega)$ weakly solves the PDE:

$$\operatorname{div}(A(x)\nabla v) = 0 \quad \text{in } \Omega$$

with $0 < \lambda \text{Id} \leq A(x) \leq \Lambda \text{Id}$ for each $x \in \Omega$, then there is $\alpha > 0$ such that $v \in C^{0,\alpha}(\tilde{\Omega})$ for each $\tilde{\Omega} \Subset \Omega$ and $\|v\|_{C^{0,\alpha}(\tilde{\Omega})} \leq C\|v\|_{L^2(\Omega)}$ for some constant $c > 0$ depending on $\tilde{\Omega}$, Ω , n , λ , and Λ , and for $\alpha > 0$ depending only on λ and Λ .

If $u \in H^2(\Omega)$, one can see that $v = \partial_e u$ does weakly solves the PDE for each $e \in \mathbb{S}^{n-1} \subset \mathbb{R}^n$ so we have $v \in C^{0,\alpha}(\tilde{\Omega})$ and so $u \in C^{1,\alpha}(\tilde{\Omega})$ for each $\tilde{\Omega} \Subset \Omega$.

Lemma X.3.2. $u \in H^1$ and De Giorgi-Nash implies $u \in C^{1,\alpha}$.

If $u \in H^1(\Omega)$ is a local minimizer of $\mathcal{E}(u) = \int_{\Omega} L(\nabla u)$ in Ω , then assuming De Giorgi-Nash holds, we have that $u \in C_{\text{loc}}^{1,\alpha}(\Omega)$ for some $\alpha > 0$.

Proof. Fix $\tilde{\Omega} \Subset \Omega$, $\varphi \in C_c^\infty(\Omega)$, and $0 < |h| < \frac{1}{2} \text{dist}(\tilde{\Omega}, \partial\Omega)$. Since u weakly solves $\text{div}(DL(\nabla u)) = 0$ in Ω , we have that:

$$\int_{\tilde{\Omega}} (DL(\nabla u(x+h)) - DL(\nabla u(x)) \nabla \varphi(x)) dx = 0.$$

By the fundamental theorem of calculus, we get:

$$DL(\nabla u(x+h)) - DL(\nabla u(x)) = \underbrace{\left[\int_0^1 D^2 L(t \nabla u(x+h) + (1-t) \nabla u(x)) dt \right]}_{\tilde{A}(x)} (\nabla u(x+h) - \nabla u(x)),$$

and so we have:

$$\int_{\tilde{\Omega}} \tilde{A}(x) \nabla \left(\underbrace{\frac{u(x+h) - u(x)}{|h|}}_{D^{|h|}u} \right) \nabla \varphi(x) = 0 \quad \text{for all } \varphi \in C_c^\infty(\Omega),$$

which is the difference quotient (Definition VI.1.1) on modules, and implies that $D^{|h|}u$ weakly solves the PDE $\text{div}(\tilde{A}(x) \nabla(D^{|h|}u)) = 0$ in Ω .

By the uniform convexity of L , $\tilde{A}$ is uniformly elliptic and since $D^{|h|}(u) \in H^1(\Omega)$, we can apply De Giorgi-Nash to see that:

$$D^{|h|}u \in C_{\text{loc}}^{0,\alpha}(\Omega) \quad \text{and} \quad \|D^{|h|}u\|_{C^{0,\alpha}(\hat{\Omega})} \leq C \|D^{|h|}u\|_{L^2(\tilde{\Omega})}$$

for each $\hat{\Omega} \Subset \tilde{\Omega} \Subset \Omega$.

We saw earlier that $\|D^{|h|}u\|_{L^2(\tilde{\Omega})} \leq C\|u\|_{L^2(\Omega)}$ for $c > 0$ independent of h .

Thus, we have uniform $C^{0,\alpha}$ bounds on the $D^{|h|}u$, so by Arzelà-Ascoli, we see that $\nabla u \in C^{0,\alpha}(\hat{\Omega})$. As $\hat{\Omega}$ and $\tilde{\Omega}$ were arbitrary, this implies that $Du \in C_{\text{loc}}^{0,\alpha}(\Omega)$, so we have $u \in C_{\text{loc}}^{1,\alpha}(\Omega)$ as desired. $\square$

De Giorgi's approach is to let:

$$\mathcal{L}v = \text{div}(A(x) \nabla v),$$

where A is uniformly elliptic with constant with constants $0 < \lambda < \Lambda$. Since we can cover compact sets with finitely many balls of a given radius, it suffices to prove the result in the case $\Omega = B_1(0) \subset \mathbb{R}^n$ and $\tilde{\Omega} = B_{\frac{1}{2}}(0) \subset \mathbb{R}^n$, this is fine.

Since by rescaling and translating, if v weakly solves $\mathcal{L}v = 0$, then if $a \in \mathbb{R}^n$, $C, r > 0$, $Cv(a+r) = \tilde{v}$ weakly solves $\tilde{\mathcal{L}}\tilde{v} = 0$ for an operator of the form:

$$\tilde{\mathcal{L}}v = \operatorname{div} (A(a+r)\nabla \tilde{v})$$

with the same λ, Λ . In summary, we can prove the case $\Omega = B_1(0)$ and $\tilde{\Omega} = B_{\frac{1}{2}}(0)$.

We proceed in two, largely independent steps:

$$(i) \ v \in H^1 \implies v \in \mathcal{L}^\infty.$$

$$(ii) \ v \in \mathcal{L}^\infty \implies v \in C^{0,\alpha}.$$

Definition X.3.3. Weak Subsolution and Supersolution.

We say that $v \in H^1(\Omega)$ is a weak subsolution of $\mathcal{L}v = 0$ in Ω if $\mathcal{L}v \geq 0$ weakly in Ω , i.e., $\int_{\Omega} a^{ij} D_i v D_j \varphi \leq 0$ for all $\varphi \in C_c^\infty(\Omega)$ with $\varphi \geq 0$.

Similarly, weak supersolution is if $\mathcal{L}v \leq 0$ in Ω . ┘

The motivation is that $\Delta u \geq 0$ when $n = 1$ if $u'' \geq 0$ so u lies below solutions, as shown:



Figure X.2. Subsolution beneath linear solution (shown in blue).

Lemma X.3.4. If $v \in H^1(\Omega)$ is a weak subsolution, i.e., $\mathcal{L}v \geq 0$, in Ω , then $\mathcal{L}v_+ \geq 0$ weakly in Ω , where $v_+ = \max\{v, 0\} \geq 0$, i.e., the positive part of v is also a weak subsolution.

Proof. We approximate $\max\{x, 0\}$ by smooth, increasing, convex function F_ϵ , with uniformly bounded derivatives so that in particular $\|F_\epsilon(v)\|_{H^1(\Omega)} \leq C$ and $F_\epsilon \rightarrow \max\{x, 0\}$ in C_{loc}^0 .

Note that $v \in H^1(\Omega)$ implies that $F_\epsilon(v) \in H^1(\Omega)$, so we can compute that:

$$\int_{\Omega} a^{ij} \nabla_i (F(v)) \nabla_j \varphi = \int_{\Omega} a^{ij} F'(v) \nabla_i v \nabla_j \varphi = \int_{\Omega} a^{ij} \nabla_j \underbrace{(F'(v)\varphi)}_{\substack{\in H_0^1(\Omega) \\ \text{and } \geq 0}} \nabla_i v - \int_{\Omega} \underbrace{a^{ij} F''(v)}_{\geq 0} \varphi \nabla_i v \nabla_j v$$

for each $\varphi \in C_c^\infty(\Omega)$ and $\varphi \geq 0$.

Since $\mathcal{L}v \geq 0$, we have the right hand sign being ≤ 0 , so $F_\epsilon(v)$ is a weak subsolution.

Thus, sending $\epsilon \rightarrow 0$, we see that:

$$\int_{\Omega} a^{ij} \nabla_i (v_+) \nabla_j \varphi \leq 0 \quad \text{for all } \varphi \in C_c^\infty(\Omega) \text{ and } \varphi \geq 0,$$

i.e., v_+ is a weak subsolution. □

Thus, we now derive a “reverse” Poincaré inequality, or the Energy/Caccioppoli inequality.

Lemma X.3.5. Caccioppoli Inequality.

If $v \in H^1(\Omega)$ is such that $v \geq 0$ and $\mathcal{L}v \geq 0$ weakly in Ω , then for each $\varphi \in C_c^\infty(\Omega)$, we have:

$$\int_{\Omega} |\nabla(\varphi v)|^2 \leq C \|\nabla \varphi\|_{L^\infty(\Omega)}^2 \int_{\Omega \cap \text{supp}(\varphi)} v^2,$$

where $c > 0$ depends only on n , λ , and Λ .

Proof. Let $\varphi^2 v \in H_0^1(\Omega)$ which is such that $\varphi^2 v \geq 0$ and use the weak form of $\mathcal{L}v \geq 0$ to get:

$$\int_{\Omega} a^{ij} \nabla_i v \nabla_j (\varphi^2 v) \leq 0.$$

Note that both $\nabla(\varphi^2 v) = \varphi \nabla(\varphi v) + (\varphi v) \nabla \varphi$ and $\nabla(\varphi v) = \nabla \varphi v + \varphi \nabla v$ so from above we get that:

$$\begin{aligned} 0 &\geq \int_{\Omega} a^{ij} \varphi \nabla_i v \nabla_j (\varphi v) + \int_{\Omega} a^{ij} (\varphi v) \nabla_i v \nabla_j \varphi \\ &= \int_{\Omega} a^{ij} \nabla_i (\varphi v) \nabla_j (\varphi v) - \int_{\Omega} a^{ij} v \nabla_i \varphi \nabla_j (\varphi v) + \int_{\Omega} a^{ij} (\varphi v) \nabla_i v \nabla_j \varphi. \end{aligned}$$

The third term gives:

$$\int_{\Omega} a^{ij} (\varphi v) \nabla_i v \nabla_j \varphi = \int_{\Omega} a^{ij} v \nabla_j (\varphi v) - \int_{\Omega} a^{ij} v^2 D_j \varphi D_i \varphi.$$

By combining the previous two terms, we have:

$$0 \geq \int_{\Omega} a^{ij} \nabla_i (\varphi v) \nabla_j (\varphi v) - \int_{\Omega} (a^{ij} - a^{ji}) v D_j (\varphi v) \nabla_i \varphi - \int_{\Omega} a^{ij} v^2 \nabla_i \varphi \nabla_j \varphi.$$

For the second term above, we use the Hölder inequality and $(A - A^\top)^2 \leq 2\Lambda^2 \text{Id}$ to obtain that:

$$\begin{aligned} \int_{\Omega} (a^{ij} - a^{ji}) v D_j (\varphi v) \nabla_i \varphi &\leq \left(\beta^2 \right)^{\frac{1}{2}} \left(\int_{\Omega} |\nabla(\varphi v)|^2 \right)^{\frac{1}{2}} \leq 2\Lambda \left(\int_{\Omega} |v \nabla \varphi|^2 \right)^{\frac{1}{2}} \left(\int_{\Omega} |\nabla(\varphi v)|^2 \right)^{\frac{1}{2}} \\ &\leq \frac{2\Lambda}{\lambda^{\frac{1}{2}}} \left(\int_{\Omega} |v \Delta \varphi|^2 \right)^{\frac{1}{2}} \left(\int_{\Omega} \nabla_i (\varphi v) a^{ij} \nabla_j (\varphi v) \right)^{\frac{1}{2}} \\ &\leq \frac{(2\Lambda)^2}{\lambda} \int_{\Omega} |v \nabla \varphi|^2 + \frac{1}{2} \int_{\Omega} a^{ij} \nabla_i (\varphi v) \nabla_j (\varphi v). \end{aligned}$$

Therefore, we all together have that:

$$\frac{2\Lambda^2}{\lambda} \int_{\Omega} |v \nabla \varphi|^2 \geq \frac{1}{2} \int_{\Omega} a^{ij} \nabla_i (\varphi v) D_j (\varphi v) - \int_{\Omega} a^{ij} v^2 \nabla_i \varphi \nabla_j \varphi,$$

so we obtain that:

$$\lambda \int_{\Omega} |\nabla(\varphi v)|^2 \leq \int_{\Omega} a^{ij} \nabla_i (\varphi v) \nabla_j (\varphi v) \leq \left(2\Lambda + \frac{4\Lambda^2}{\lambda} \right) \|\nabla \varphi\|_{L^\infty(\Omega)}^2 \int_{\Omega \cap \text{supp}(\varphi)} v^2.$$

□

Proposition X.3.6. L^2 to L^∞ .

If $v \in H^1(B_1(0))$ is a weak subsolution to $\mathcal{L}v = 0$ (i.e., $\mathcal{L}v \geq 0$ weakly) in $B_1(0)$, then:

$$\|v_+\|_{L^\infty(B_{1/2}(0))} \leq C \|v_+\|_{L^2(B_1(0))}$$

for $C > 0$, dependent on n , λ , and Λ .

Proof. We actually will show that if $\mathcal{L}v \geq 0$ in $B_1(0)$ weakly, then there is $\delta > 0$, depending on n , λ , and Λ such that:

$$\int_{B_1(0)} v_+^2 < \delta \quad \implies \quad v \leq 1 \quad \text{on } B_{1/2}(0).$$

This implies the proposition by taking $\tilde{v} = \sqrt{\delta} \frac{v}{\|v_+\|_{L^2(B_1(0))}}$, where $\mathcal{L}\tilde{v} \geq 0$ weakly also, we we can take $c = \frac{1}{\sqrt{\delta}}$ in the statement.

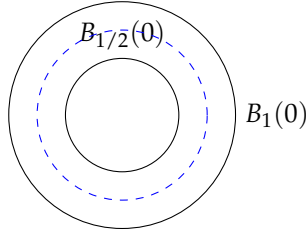


Figure X.3. Top-view of the shrinking balls.

Here, we consider the shrinking balls as:

$$\tilde{B}_k = \left\{ |x| \leq \frac{1}{2} + 2^{-k-1} \right\},$$

where we have $\tilde{B}_0 = B_1(0)$ and $\tilde{B}_k \rightarrow B_{1/2}(0)$ as $k \rightarrow \infty$.

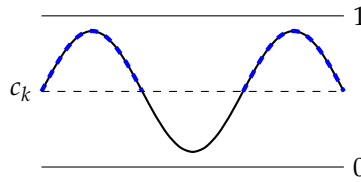


Figure X.4. Cutoff line for the side view.

Meanwhile, we let $c_k = 1 - 2^{-k}$ as the cutoff, so $c_0 = 0$ and $c_k \rightarrow 1$, and we set:

$$v_k = \max\{v - c_k, 0\} \geq 0,$$

so that v_k detects regions where $v \geq c_k$.

Then, we define the cutoff function $\varphi_k \in C_c^\infty(B_1(0))$ with values in $[0, 1]$ with:

$$\varphi_k = \begin{cases} 1, & \text{in } \tilde{B}_k, \\ 0, & \text{in } \tilde{B}_{k-1}^c, \end{cases}$$

where $\nabla \varphi_k$ is bounded. We then consider:

$$V_k = \int_{B_1(0)} \varphi_k^2 v_k^2 \left(\approx \int_{\tilde{B}_k} v_k^2 \xrightarrow{k \rightarrow \infty} \int_{B_{1/2}(0)} (v-1)_+^2 \right).$$

Hence, showing that $v_k \rightarrow 0$ implies that $\int_{B_{1/2}(0)} (v-1)_+^2 = 0$, which implies that $v \leq 1$ in $B_{1/2}(0)$. Recall Sobolev inequality for $p = \frac{2n}{n-2} > 1$ (recall that $n \geq 3$, we use $p = 4$ if $n = 1$ or 2) says that:

$$\|u\|_{L^p(\mathbb{R}^n)} \leq C \|\nabla u\|_{L^2(\mathbb{R}^n)}.$$

Then, we compute that:

$$\left(\int_{B_1(0)} |\varphi_{k+1} v_{k+1}|^p \right)^{\frac{2}{p}} \leq C \left(\int_{B_1(0)} |\nabla(\varphi_{k+1} v_{k+1})|^2 \right) \leq C 2^{2k} \int_{\tilde{B}_k} v_{k+1}^2.$$

Noting that $v_{k+1} \leq v_k$ as $c_{k+1} \geq c_k$, so that:

$$\left(\int_{B_1(0)} |\varphi_{k+1} v_{k+1}|^p \right)^{\frac{2}{p}} \leq C 2^{2k} \int_{\tilde{B}_k} v_{k+1}^2 \leq C 2^{2k} \int_{B_1(0)} \varphi_k^2 v_k^2 = C \cdot 2^{2k} \cdot V_k.$$

On the other hand, using Hölder inequality, we get:

$$V_{k+1} = \int_{B_1(0)} \varphi_{k+1}^2 v_{k+1}^2 \leq \left(\int_{B_1(0)} |\varphi_{k+1} v_{k+1}|^p \right)^{\frac{2}{p}} |\{\varphi_{k+1} v_{k+1} > 0\}|^{\frac{2}{n}}.$$

Therefore, as $v_{k+1} \leq v_k - 2^{-k-1}$, we get:

$$|\{\varphi_{k+1} v_{k+1} > 0\}| \leq |\{\varphi_k v_k > 2^{-k-1}\}| = |\{\varphi_k^2 v_k^2 > 2^{-2k-2}\}| \leq 2^{2k+1} \int_{B_1(0)} \varphi_k^2 v_k^2.$$

Note that the last inequality is by Markov's inequality that $a|\{f > a\}| \leq \int_{\{f > a\}} f$. Combining the previous inequalities, we have:

$$V_{k+1} \leq C \cdot 2^{2k} V_k (2^{2k+2} V_k)^{\frac{2}{n}} = C^{k+1} V_k^{1+\frac{2}{n}},$$

where $c > 0$ may increase line by line but depends only on n , λ , and Λ .

To conclude, if $\delta > 0$ is "small enough", then we claim that:

$$\begin{cases} 0 \leq v_0 \leq \delta, \\ 0 \leq V_{k+1} \leq c^{k+1} V^{1+\frac{1}{n}} \end{cases} \implies V_k \rightarrow 0 \text{ as } k \rightarrow \infty.$$

To prove the claim, we go by induction and show that:

$$V_k^{\frac{2}{n}} \leq \frac{C^{-k-1}}{(2c)^{n/2}},$$

which $\rightarrow 0$ as $k \rightarrow \infty$ since we can claim without loss of generality that $c > 1$.

Consider that $V_0 \leq C^{-\frac{n}{2}-\frac{n^2}{4}} = \delta$:

- Base case: $V_0^{\frac{2}{n}} \leq \frac{C^{-1}}{(2c)^{\frac{n}{2}}}$, which is equivalently:

$$V_0 \leq \frac{C^{-\frac{n}{2}}}{2^{\frac{n^2}{4}} C^{\frac{n^2}{4}}} = 2^{-\frac{n^2}{4}} C^{-\frac{n}{2} - \frac{n^2}{4}} < \delta.$$

- Induction step: Assuming it for some $k \geq 0$ that:

$$\begin{aligned} V_{k+1}^{\frac{2}{n}} &\leq \left(C^{k+1} V_k^{1+\frac{2}{n}} \right)^{\frac{2}{n}} = C^{\frac{2k+2}{n}} V_k^{\frac{2}{n} + \frac{4}{n^2}} = C^{\frac{2k_2}{n}} (V_k^{\frac{2}{n}})^{1+\frac{2}{n}} \\ &\leq C^{\frac{2k+2}{n}} \left(\frac{c^{-k-1}}{(2c)^{\frac{n}{2}}} \right)^{\frac{2}{n}} \left(\frac{c^{-k-1}}{(2c)^{\frac{n}{2}}} \right)^{\frac{2}{n}} = \frac{c^{-k-2}}{2(2c)^{\frac{n}{2}}} < \frac{c^{-(k+1)-1}}{(2c)^{\frac{n}{2}}}. \end{aligned}$$

Thus, we have $V_k \rightarrow 0$ as $k \rightarrow \infty$.

Therefore, we have:

$$V_k = \int_{B_1(0)} \varphi_k^2 v_k^2 \rightarrow \int_{B_{\frac{1}{2}}(0)} (v-1)_+^2 = 0,$$

which implies that $v \leq 1$ in $B_{\frac{1}{2}}(0)$. □

With this proposition, writing $v = v_+ + v_-$ for $v_- = \min\{v, 0\} = -\max\{-v, 0\}$, then:

$$\|v\|_{L^2(B_1(0))} = \|v_+\|_{L^2(B_1(0))} + \|v_-\|_{L^2(B_1(0))},$$

and hence as $\mathcal{L}v = 0$ weakly, so we have $\mathcal{L}v_+ \geq 0$ and $\mathcal{L}v_- \leq 0$, and this gives that:

$$\|v\|_{L^\infty(B_{\frac{1}{2}}(0))} \leq C\|v\|_{L^2(B_1(0))}.$$

Remark X.3.7. Moser Iteration.

We take $\varphi^2 v$ to get Caccioppoli but instead Moser consider $\varphi^2 v^\beta$ for $\beta > 1$ to get:

$$\int_{B_1(0)} |\nabla(v^{\frac{\beta+1}{2}} \varphi)|^2 \leq C \int_{B_1(0)} v^{\beta+1} |\nabla \varphi|^2 \quad \text{for all } \varphi \in C_c^\infty(B_1(0)).$$

Using this in place of Caccioppoli with Sobolev inequalities and shrinking balls, we get:

$$\|v\|_{L^{2\gamma^k}(B_{r_k}(0))} \leq C\|v\|_{L^2(B_1(0))} \quad \text{for } \begin{cases} \gamma > 1, \\ r_k \rightarrow \frac{1}{2}^+, \end{cases}$$

sending $k \rightarrow \infty$ so we have $\|v\|_{L^\infty(B_{1/2}(0))} \leq C\|v\|_{L^2(B_1(0))}$. ┘

Hence, we are done with the first step of De Giorgi-Nash, we will next seek a L^∞ to $C^{0,\alpha}$ estimate.

Definition X.3.8. Oscillation.

If $u : \Omega \rightarrow \mathbb{R}$, the oscillation of u in Ω is:

$$\text{osc}_\Omega(u) = \sup_\Omega(u) - \inf_\Omega(u) \in \mathbb{R} \cup \{\pm\infty\}.$$

┘

We can characterize the oscillation decay here:

Lemma X.3.9. Oscillation Decay $\implies$ Hölder.

If $u : B_2(0) \rightarrow \mathbb{R}$ is bounded and there is $\theta \in (0, 1)$ such that:

$$\operatorname{osc}_{B_{r/2}(x)} \leq (1 - \theta) \operatorname{osc}_{B_r(x)},$$

for all $r > 0, x \in B_2(0)$ with $B_r(x) \subset B_2(0)$, then:

$$\|u\|_{C^{0,\alpha}(B_{1/2}(0))} \leq C \|u\|_{L^\infty(B_2(0))},$$

where $\alpha, C > 0$ are to be decided and dependent on θ and $\operatorname{osc}(u)$.

Proof. Let $x \neq y \in B_{1/2}(0)$ and let $r = |x - y| > 0$, so that for some $k \geq 0$, we have:

$$r_k = 2^{-k} \geq r \geq 2^{-k-1} = r_{k+1}.$$

Then $x, y \in B_{r_k}(z) \subset B_2(0)$ for some $z \in B_{1/2}(0)$ and so we have:

$$|u(x) - u(y)| \leq \operatorname{osc}_{B_{r_k}(z)}(u) \leq (1 - \theta)^{k-1} \operatorname{osc}_{B_{1/2}(z)}(u) \leq (1 - \theta)^{k-1} \operatorname{osc}_{B_1(0)}(u) \leq (1 - \theta)^k \operatorname{osc}_{B_2(0)}(u).$$

Now, set $\alpha = -\log_2(10\theta) > 0$ so that:

$$(1 - \theta)^k = 2^{-\alpha k} = (2^{-k})^\alpha = (2^{-k-1} \cdot 2)^\alpha \leq 2^{\alpha} \cdot r^\alpha = 2^\alpha |x - y|^\alpha.$$

Therefore, the earlier equation yields that:

$$|u(x) - u(y)| \leq \underbrace{2^\alpha \operatorname{osc}_{B_2(0)}(u)}_C |x - y|^\alpha,$$

and hence $u \in C^{0,\alpha}(B_{1/2}(0))$, and:

$$\|u\|_{C^{0,\alpha}(B_{1/2}(0))} \leq C \|u\|_{L^\infty(B_2(0))}$$

as $\operatorname{osc}_{B_2(0)} \leq 2 \|u\|_{L^\infty(B_2(0))}$. □

Proposition X.3.10. L^∞ to $C^{0,\alpha}$.

If $v \in H^1(B_2(0))$ is a weak solution to $\mathcal{L}v = \operatorname{div}(A \nabla v) = 0$ in $B_2(0)$ for $0 \leq \lambda \operatorname{Id} \leq A \leq \Lambda \operatorname{Id}$, then:

$$\operatorname{osc}_{B_{1/2}(0)}(u) \leq (1 - \theta) \operatorname{osc}_{B_2(0)}(u)$$

for $\theta \in (0, 1)$ depending on n, λ, Λ .

Remark X.3.11. Previous lemma, rescaling/translating, and covering will give step 1 of Hilbert's XIX problem. $\square$

Lemma X.3.12. De Giorgi Isoperimetric Inequality.

Let $w \in H^1(B_1(0))$ be such that:

$$\int_{B_1(0)} |\nabla w|^2 \leq C_0 \quad \text{for some } C_0 > 0.$$

Let:

$$A = \{w \leq 0\} \cap B_1(0), \quad D = \left\{w \geq \frac{1}{2}\right\} \cap B_1(0), \quad \text{and} \quad E = \left\{0 < w < \frac{1}{2}\right\} \cap B_1(0).$$

Then, we have:

$$C_0 |E| \geq c |A|^2 |D|^2 \quad \text{for } c > 0 \text{ depending on } n.$$

Proof. We define $\bar{w} \in H^1(B_1(0))$ by setting:

$$\bar{w} = \begin{cases} w, & \text{in } E, \\ \frac{1}{2}, & \text{in } D, \\ 0, & \text{in } A. \end{cases}$$

This can be portrayed as follows:

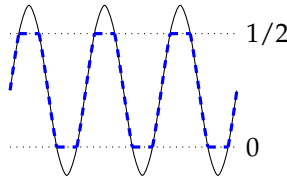


Figure X.5. Truncation of the function with A , D , and E .

Then, $\nabla \bar{w} = 0$ in $B_1(0) \setminus E$ and $\int_{B_1(0)} |\nabla \bar{w}|^2 \leq \int_{B_1(0)} |\nabla w|^2 \leq C_0$.

We compute that:

$$\begin{aligned} |A| \cdot |D| &= |A| \int_D 2|\bar{w}(x)| dx = \int_A \int_D 2|\bar{w}(x)| dx dy = \int_A \int_D 2|\bar{w}(x) - \bar{w}(y)| dx dy \\ &\leq 2 \int_{B_1(0)} \int_{B_1(0)} \left[\left| \bar{w}(x) - \frac{1}{|B_1(0)|} \int_{B_1(0)} \bar{w}(z) dz \right| + \left| \frac{1}{|B_1(0)|} \int_{B_1(0)} \bar{w}(z) dz - \bar{w}(y) \right| \right] dx dy \\ &= 4|B_1(0)| \int_{B_1(0)} \left| \bar{w}(x) - \frac{1}{|B_1(0)|} \int_{B_1(0)} \bar{w}(z) dz \right| dx \leq C \int_E |\nabla \bar{w}(x)| dx. \end{aligned}$$

Thus, by the Hölder inequality, we have:

$$|A| |D| \leq C \left(\int_{B_1(0)} |\nabla \bar{w}|^2 \right)^{1/2} |E|^{1/2} \leq C C_0^{1/2} |E|^{1/2},$$

and so by seauaring both side, we have that:

$$|A|^2|D|^2 \leq CC_0|E|.$$

□

Lemma X.3.13. Quantitative Jumping.

If $v \in H^1(B_1(0))$ is such that $v \leq 1$ in $B_2(0)$ and $\mathcal{L}v \geq 0$ weakly in $B_2(0)$, then if:

$$|\{v \leq 0\} \cap B_1(0)| \geq \mu > 0,$$

we have that for some $\gamma > 0$, depending on n, λ, Λ , and μ :

$$\sup_{B_{1/2}(0)} (v) \leq 1 - \gamma.$$

Proof. We apply the previous lemma to a sequence of functions that raise the highest value from $\frac{1}{2}$ to 1 (c.f. L^2 to L^∞ , [Proposition X.3.6](#)). Let us set, for $k \geq 1$ that:

$$w_k = 2^k [v - (1 - 2^{-k})]_+ = [2^k v - (2^k - 1)]_+ \in [0, 1],$$

so that $v \leq 1$ implies that $w_k \leq 1$ and $\mathcal{L}v \geq 0$, which implies that $\mathcal{L}w_k \geq 0$ also. We apply Caccioppoli, with a cutoff being 1 in $B_1(0)$ and 0 on $B_2^c(0)$ to get:

$$\int_{B_1(0)} |\nabla w_k|^2 \leq C \int_{B_2(0)} w_k^2.$$

As $v \leq 0$ implies that $w_k = 0$, if we set $A_k = \{w_k \leq 0\} \cap B_1(0)$, then:

$$|A_k| \geq |\{v \leq 0\} \cap B_1(0)| \geq \mu > 0.$$

Set $D_k = \{w_k \geq \frac{1}{2}\} \cap B_1(0)$ and $E_k = \{0 < w_k < \frac{1}{2}\} \cap B_1(0)$.

Since $w_{k+1} > 0$ implies that $w_k \geq \frac{1}{2}$ since $v - (1 - \frac{1}{2} \cdot 2^{-k}) > 0$, so we have:

$$v - (1 - 2^{-k}) > \frac{1}{2} \cdot 2^{-k},$$

so we have:

$$|\{w_{k+1} > 0\} \cap B_1(0)| \leq |D_k|.$$

So, if:

$$\int_{B_1(0)} w_{k+1}^2 \geq \delta^2 \quad \text{for some fixed } \delta > 0,$$

as $w_{k+1} \leq 1$, we have:

$$\delta^2 < \int_{B_1(0)} w_{k+1}^2 \leq |\{w_{k+1} > 0\} \cap B_1(0)| \leq |D_k|.$$

Hence, by applying the De Giorgi's isoperimetric inequality ([Lemma X.3.12](#)), we get:

$$C_0|E_k| \geq c|A_k^2|D_k|^2 \geq c\mu^2\delta^2 > 0,$$

and hence we have $|E_k| > \beta > 0$ where β is independent of k .

As the E_k are mutually disjoint, that is $w_{k+1} > 0$ implies $w_k \geq \frac{1}{2}$, and $|B_1(0)| < \infty$, we must find $k_0 \geq 1$ depending on n, μ, δ (on β) such that:

$$\int_{B_1(0)} w_{k_0}^2 < \delta^2.$$

Hence, our L^2 to L^∞ estimate gives that:

$$\|(w_{k_0})_+\|_{L^\infty(B_{1/2}(0))} \leq c\delta,$$

so if $c\delta \leq \frac{1}{2}$, we get $w_{k_0} \leq \frac{1}{2}$ in $B_{1/2}(0)$, but:

$$2^{k_0} [v - (1 - 2^{-k_0})]_+ \leq \frac{1}{2},$$

and so $v \leq 1 - 2^{-k_0-1} = 1 - \gamma$. Thus, we see that $v \leq 1 - \gamma$ in $B_{1/2}(0)$ for γ depending on n, λ, Λ and μ . $\square$

Then, we return to the proof of L^∞ to $C^{0,\alpha}$ inequality.

Proof of Proposition X.3.10. Recall that we assume $v \in H^1(B_2(0))$ weakly solves $\mathcal{L}v = 0$ and want $\theta \in (0, 1)$, depending on n, λ, Λ such that:

$$\text{osc}_{B_{1/2}(v)} \leq (1 - \theta) \text{osc}_{B_2(0)}(v),$$

so we have $v \in C^{0,\alpha}(B_{1/2}(0))$.

Here, we define that:

$$w(x) = \frac{2}{\text{osc}_{B_{1/2}(0)}(v)} \left[v(x) - \frac{\sup_{B_2(0)}(v) + \inf_{B_2(0)}(v)}{2} \right],$$

by doing so, we translate the function to midpoint 0 and upper/lower bounds being $\pm \frac{\text{osc}(v)}{2}$, and by rescaling, it is between ± 1 .

Therefore, we have $\text{osc}_{B_2(0)}(w) = 2$ and $\mathcal{L}w = 0$ weakly, and consider $w \in [-1, 1]$. Without loss of generality, let's suppose that:

$$|\{w \leq 0\} \cap B_1(0)| \geq \frac{|B_1(0)|}{2} = \mu > 0,$$

if not consider $-w$. Quantitative jumping lemma (Lemma X.3.13) gives $\gamma > 0$ depending on n, λ, Λ such that:

$$-1 \leq w \leq 1 - \gamma < 1 \quad \text{on } B_{\frac{1}{2}}(0),$$

and thus we have $\text{osc}_{B_{1/2}(0)} \leq 2 - \gamma$. Then $\text{osc}_{B_{1/2}(0)}(v) = \frac{\text{osc}_{B_{1/2}(0)}(w)}{2} \cdot \text{osc}_{B_{1/2}(0)}(w)$, so we have:

$$\text{osc}_{B_{1/2}(v)} \leq \left(1 - \frac{\gamma}{2}\right) \text{osc}_{B_2(0)}(v),$$

and so, we set $\theta = \frac{\gamma}{2}$ to get the oscillation decay. $\square$

Eventually, we will pick up the clues to get to the proof to De Giorgi-Nash Theorem.

Proof of Theorem X.3.1. If $v \in H^1(\Omega)$ is a weak solution to $\mathcal{L}v = 0$ in Ω , and $\tilde{\Omega} \Subset \Omega$ by rescaling/translation, we get a cover of $\tilde{\Omega}$ by balls of radius 2. So if $\tilde{\Omega} \Subset \hat{\Omega} \Subset \Omega$, we get a constant $C > 0$ depending on $\Omega, \hat{\Omega}, n, \lambda$, and Λ with $\|v\|_{L^\infty(\hat{\Omega})} \leq C\|v\|_{L^2(\Omega)}$ and then another constant $\tilde{C} > 0$ depending on $\tilde{\Omega}, \Omega, n, \lambda$, and Λ with $\|v\|_{C^{0,\alpha}(\tilde{\Omega})} \leq \tilde{C}\|v\|_{L^\infty(\hat{\Omega})}$. Hence, up to increasing $c \leftarrow C\tilde{C}$, we have $\|v\|_{C^{0,\alpha}(\tilde{\Omega})} \leq c\|v\|_{L^2(\Omega)}$, i.e., $v \in C_{\text{loc}}^{0,\alpha}(\Omega)$. $\square$

X.4 Further Results on Regularity

Historically, $C^{1,\alpha}$ to C^∞ was shown first so De Giorgi-Nash marker the chronological conclusion of Hilbert's XIX problem.

However, so far, we just have $C^{1,\alpha}$. By similar techniques one can prove Harnack's inequality for operators $\mathcal{L}$, and we will see this for Harmonic functions.

Remark X.4.1. We saw that uniform convexity of L was necessary to get $C^{1,\alpha}$ regularity. If we allow 0 or ∞ eigenvalues for $D^2L = A$ (which arise in physical problems), then minimizers are Lipschitz, but only C^1 if $n = 2$ when the set of points where eigenvalues are 0 or ∞ is finite (De Silva-Savin). If $n \geq 4$, this fails (Mooney). $\lrcorner$

Open Problem.

The problem is still open if $n = 3$ and in $n = 2$ without the finiteness assumption.

The next step ($C^{1,\alpha}$ to C^∞) will require a diversion to non-divergence form linear elliptic PDEs (e.g., $a^{ij}D_i u D_j u$), we cannot use Sobolev theory.

Here, we consider the minimizers to weakly solve (assuming $v \in C^2$), which is to relate divergence form to non-divergence form, i.e.:

$$\int \varphi \operatorname{div}(A \nabla v) = 0 \rightsquigarrow \tilde{A} D_i v D_j v = f(\nabla v).$$

XI Non-Divergence Forms

XI.1 Harmonic Functions

As usual, we have $\Delta u = \sum_{i=1}^n \frac{\partial^2 u}{\partial x_i^2} = \operatorname{div}(\nabla u)$ in $\mathbb{R}^n$.

Definition XI.1.1. Harmonic Functions.

A function $u \in C^2(\Omega)$ for $\Omega \subset \mathbb{R}^n$ open and connected (often called domain) is:

- **harmonic** if $\nabla u = 0$ in Ω .
- **subharmonic** if $\nabla u \geq 0$ in Ω .
- **superharmonic** if $\nabla u \leq 0$ in Ω .

┘

A property of harmonic function is that it equals to their averages.

Lemma XI.1.2. Mean Value Property.

If $u \in C^2(\Omega)$ is subharmonic on domain Ω , then if $\overline{B_r(x)} \subset \Omega$, we have:

$$u(x) \leq \oint_{B_r(x)} u = \frac{1}{\omega_n r^n} \int_{B_r(x)} u \quad \text{and} \quad u(x) \leq \frac{1}{n\omega_n} r^{n-1} \int_{\partial B_r(x)} u,$$

where ω_n is the measure of $B_1(0) \subset \mathbb{R}^n$.

Proof. As $\Delta u \geq 0$ in Ω , so by the divergence theorem

$$\begin{aligned} 0 &\leq \int_{B_\rho(x)} \Delta u = \int_{B_\rho(x)} \operatorname{div}(\nabla u) = \int_{\partial B_\rho(x)} \nabla u(y) \left(\frac{y-x}{\rho} \right) dy \\ &= \rho^{n-1} \int_{\mathbb{S}^{n-1}} \nabla u(\rho w + x) w dw = \rho^{n+1} \int_{\mathbb{S}^{n-1}} \frac{\partial}{\partial \rho} (u(\rho w + x)) dw, \end{aligned}$$

and this holds for all $0 \leq \rho \leq r$ as $\overline{B_r(x)} \subset \Omega$.

Thus:

$$0 \leq \int_{\mathbb{S}^{n-1}} \frac{\partial}{\partial \rho} (u(\rho w + x)) dw = \frac{\partial}{\partial \rho} \left(\int_{\mathbb{S}^{n-1}} u(\rho w + x) dw \right),$$

and hence $\int_{\mathbb{S}^{n-1}} u(\rho w + x) dw$ is increasing in ρ . Hence:

$$\frac{1}{r^{n-1}} \int_{\partial B_r(x)} u(y) dy = \int_{\mathbb{S}^{n-1}} u(rw + x) dw \geq \int_{\mathbb{S}^{n-1}} u(\rho w + x) dw \geq \int_{\mathbb{S}^{n-1}} u(x) dw = u(x) \cdot n \cdot \omega_n.$$

Thus, dividing by $n \cdot \omega_n$, we have:

$$u(x) \leq \frac{1}{n\omega_n r^{n-1}} \int_{\partial B_r(x)} u(y) dy,$$

which gives the second inequality. Hence, by integrating:

$$\int_{B_r(x)} u(y) dy = \int_0^r \int_{\partial B_\rho(x)} u(y) dy d\rho \geq \int_0^r (n\omega_n \rho^{n-1} u(x)) d\rho = u(x) \cdot \omega_n \cdot r^n,$$

and by dividing $\omega_n r^n$, we have:

$$u(x) \leq \frac{1}{\omega_n r^n} \int_{B_r(0)} u,$$

as desired. $\square$

Corollary XI.1.3. Mean Value Property applies for:

- superharmonic functions by making all inequalities into $\geq$, and
- harmonic functions by making all inequalities into equality.

Remark XI.1.4. In fact, if $u \in C^0(\Omega)$ and $u(x) = \frac{1}{\omega_n r^n} \int_{B_r(x)} u$ for all $\overline{B_r(x)} \subset \Omega$, then u is harmonic and smooth. (Reverse of the mean value property, can be shown using mollification). $\lrcorner$

Lemma XI.1.5. Strong Maximum Principle.

If $u \in C^2(\Omega)$ is subharmonic in Ω , and $u(y) = \sup_{\Omega}(u)$ for some $y \in \Omega$, then u is constant on Ω .

Proof. Let $\Sigma = \{x \in \Omega \mid u(x) = M = \sup_{\Omega}(u)\} \ni y (\neq \emptyset)$, we show that Σ is clopen, then closed conclusion follows by the continuity of u .

If $x \in \Sigma$, the mean value property gives that:

$$M = u(x) \leq \frac{1}{\omega_n r^n} \int_{B_r(x)} u,$$

but $u \leq M = \sup_{\Omega}(u)$, so we have:

$$\frac{1}{\omega_n r^n} \int_{B_r(x)} (M - u) \leq 0.$$

Therefore, since $M - u \geq 0$ in Ω , this implies that $u = M$ almost everywhere in $B_r(x)$ by continuity, so $u = M$ in $B_r(x)$.

Therefore, we have $B_r(x) \subset \Sigma$ so Σ is open, hence it is clopen and nonempty, so $\Sigma = \Omega$, and therefore, $u = \sup_{\Omega}(u) = M$ on Ω . $\square$

Corollary XI.1.6. Similarly, if u is superharmonic and $u(y) = \inf_{\Omega}(u)$ for some $y \in \Omega$, then u is constant on Ω .

If u is harmonic and $u(y) = \inf_{\Omega}(u)$ or $u(y) = \sup_{\Omega}(u)$ for some $y \in \Omega$, then u is constant on Ω .

We can consider also a weaker version of the theorem.

Lemma XI.1.7. Weak Maximum Principle.

If $\Omega \subset \mathbb{R}^n$ is a bounded domain and $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ is subharmonic in Ω , then u is maximized on $\partial\Omega$, i.e., $\sup_{\Omega}(u) = \sup_{\partial\Omega}(u)$.

Similarly, superharmonic functions are minimized on $\partial\Omega$, and harmonic functions obtain extrema on $\partial\Omega$.

Proof. As Ω is bounded, $u \in C^0(\overline{\Omega})$ both $\sup_{\Omega}(u)$ and $\inf_{\Omega}(u)$ are attained on $\overline{\Omega}$, hence the strong maximum principle gives that if u is not constant, both must be attained on $\partial\Omega$ in the respective cases. $\square$

Theorem XI.1.8. Uniqueness for Poisson with Dirichlet Boundary Conditions.

If $\Omega \subset \mathbb{R}^n$ is a bounded domain and $u, \tilde{u} \in C^2(\Omega) \cap C^0(\overline{\Omega})$ are such that $\Delta u = \Delta \tilde{u}$ in Ω , and $u = \tilde{u}$ on $\partial\Omega$, then $u = \tilde{u}$ in $\overline{\Omega}$.

Proof. As Δ is linear, $\Delta(u - \tilde{u}) = 0$ in Ω , and $u - \tilde{u} = 0$ on $\partial\Omega$, hence the weak maximum principle applied to $(u - \tilde{u})$ gives that the supremum/infimum of $(u - \tilde{u})$ occurs on $\partial\Omega$, but $u - \tilde{u} = 0$ on $\partial\Omega$, so $u - \tilde{u} = 0$ in $\overline{\Omega}$ which equivalently indicates that $u = \tilde{u}$ on $\overline{\Omega}$. $\square$

Then, we will relate this with the De Giorgi cases.

Lemma XI.1.9. Harnack Inequality for Harmonic Functions.

If $u \in C^2(\Omega)$ is harmonic, nonnegative, $u \geq 0$ on a domain Ω , then if $\tilde{\Omega} \Subset \Omega$, we have:

$$\sup_{\tilde{\Omega}}(u) \leq C \inf_{\tilde{\Omega}}(u)$$

for $C > 0$ depending on $n, \tilde{\Omega}$, and Ω .

Proof. If $y \in \Omega$ and some $r > 0$ such that $\overline{B_{4r}(y)} \subset \Omega$, then if $x_1, x_2 \in B_r(y)$, we can apply the mean value property to get that:

$$\begin{cases} u(x_1) = \frac{1}{\omega_n r^n} \int_{B_r(x_1)} u \leq \frac{1}{\omega_n r^n} \int_{B_{2r}(y)} u, \\ u(x_2) = \frac{1}{\omega_n (3r)^n} \int_{B_{3r}(x_2)} u \geq \frac{1}{\omega_n (3r)^n} \int_{B_{2r}(y)} u, \end{cases}$$

so we have $u(x_1) \leq 3^n u(x_2)$, and so we have local Harnack as:

$$\sup_{B_r(y)} u \leq 3^n \inf_{B_r(y)} (u).$$

Now, for $\tilde{\Omega} \Subset \Omega$, let $4r < \text{dist}(\tilde{\Omega}, \partial\Omega)$ and then cover $\overline{\tilde{\Omega}}$ by balls of radius r , where compactness of $\overline{\tilde{\Omega}}$ gives that there is a $N \geq 1$ and $y_1, \dots, y_N$ such that $\overline{\tilde{\Omega}} \subset \bigcup_{i=1}^N B_r(y_i)$. If $\tilde{\Omega}$ is connected and $x_1, x_2 \in \tilde{\Omega}$, then we can find a continuous path γ from x_1 to x_2 that goes through each ball at most once (e.g. piecewise linear path). Applying local Harnack along this path repeatedly, and we get:

$$u(x_1) \leq \underbrace{3^n \cdots 3^n}_{\text{at most } N} u(x_2) \leq 3^{n \cdot N} u(x_2).$$

$\square$

XI.2 Inequalities about Smooth Harmonic Functions

Lemma XI.2.1. Derivative Estimate for Harmonic Functions.

Suppose $u \in C^\infty(\Omega)$ is harmonic in Ω , $\Delta u = 0$ in Ω , then if $B_r(y) \subset \Omega$, we have:

$$|Du(y)| \leq \frac{C}{r} \sup_{\partial B_r(y)} |u|,$$

where $C > 0$ depends on n . More generally, if $\alpha \in \mathbb{N}^n$ and $\tilde{\Omega} \Subset \Omega$, then:

$$\sup_{\tilde{\Omega}} |D^\alpha u| \leq C \sup_{\Omega} |u|.$$

Proof. Differentiating $\Delta u = 0$ implies that $\Delta(D_i u) = 0$, so $D_i u$ is harmonic in Ω for all $i = 1, \dots, n$. Thus, by the mean value property, we get:

$$\begin{aligned} D_i u(y) &= \frac{1}{\omega_n r^n} \int_{B_r(y)} D_i u(x) dx = \frac{1}{\omega_n r^n} \int_{B_r(y)} \operatorname{div}(0, \dots, 0, \underbrace{u}_{i\text{-th}}, 0, \dots, 0) \\ &= \frac{1}{\omega_n r^n} \int_{\partial B_r(y)} u(x) v_i(x) dx \quad \text{where } v_i(x) = \frac{x_i - y_i}{r}, \end{aligned}$$

as the i -th component of outward unit normal to $\partial B_r(y)$.

As $|v_i(x)| \leq |v(x)| = 1$, we get:

$$\begin{aligned} |D_i u(y)| &\leq \frac{1}{\omega_n r^n} \int_{\partial B_r(y)} |u(x)| \cdot |v_i(x)| dx \\ &\leq \sup_{\partial B_r(y)} |u| \cdot \frac{n \omega_n r^{n-1}}{\omega_n r^n} = \frac{n}{r} \sup_{\partial B_r(y)} |u|, \end{aligned}$$

and hence we have:

$$|Du(y)| \leq \frac{C}{r} \sup_{\partial B_r(y)} |u|$$

for $C > 0$ depending on n .

For any $\alpha \in \mathbb{N}^n$ and $\tilde{\Omega} \Subset \Omega$, if $y \in \tilde{\Omega}$, then $B_r(y) \subset \Omega$ for all $r \in (0, \operatorname{dist}(\tilde{\Omega}, \partial\Omega))$. Applying above estimate for a covering of $\tilde{\Omega}$ of balls of radius:

$$r = \frac{\operatorname{dist}(\tilde{\Omega}, \partial\Omega)}{2} > 0,$$

which by compactness has a finite subcover, we get that:

$$\sup_{\tilde{\Omega}} |D^\alpha u| \leq C \sup_{\Omega} |u| \quad \text{where } C \approx \frac{c(n)}{r^{|\alpha|}}.$$

□

Theorem XI.2.2. Liouville.

If $u \in C^\infty(\mathbb{R}^n)$ is harmonic, which has *sub-linear growth*, i.e., $|u(x)| \leq C(1 + |x|^\gamma)$ for $\gamma \in (0, 1)$, $C > 0$ for all $x \in \mathbb{R}^n$, then u is constant.

Proof. Fix $y \in \mathbb{R}^n$, for each $r > 0$, we have $B_r(y) \subset \mathbb{R}^n$ so that the estimate gives that:

$$|\nabla u(y)| \leq \frac{C}{r} \sup_{\partial B_r(y)} |u| \leq \frac{\tilde{C}}{r} (1 + |y| + r)^\gamma,$$

and when we send $r \rightarrow \infty$, we must have $|Du(y)| = 0$, so u is constant as y is arbitrary. $\square$

Lemma XI.2.3. Weyl.

If $u \in L^1_{\text{loc}}(\Omega)$ is weakly harmonic, i.e., $\int_{\Omega} u \cdot \Delta \varphi = 0$ for all $\varphi \in C_c^\infty(\Omega)$, then $u \in C^\infty(\Omega)$ and is harmonic in Ω , i.e., $\Delta u = 0$ in Ω .

Proof. Since $u \in L^1_{\text{loc}}(\Omega)$, it's mollifications $u_\epsilon = \eta_\epsilon * u$ are smooth and $u_\epsilon \rightarrow u$ in $L^1_{\text{loc}}(\Omega)$. Moreover, $\Delta_\epsilon = 0$ in the set:

$$\Omega_\epsilon = \{x \in \Omega : \text{dist}(x, \partial\Omega) > \epsilon\},$$

which is nonempty for small $\epsilon > 0$, since:

$$\Delta u_\epsilon(x) = \int_{\Omega} u(y) \Delta_y \eta_\epsilon(x - y) dy = 0.$$

Then, by applying the gradient estimates to u_ϵ , we guarantee that if $\tilde{\Omega} \Subset \Omega_\delta \cap B_{1/\delta}(0)$, then there is $C > 0$, independent of ϵ , dependent on $n, \tilde{\Omega}, \Omega_\delta$, and α such that for all small $\epsilon > 0$, we have:

$$\sup_{\tilde{\Omega}} |D^\alpha u_\epsilon| \leq C \sup_{\Omega_\delta} |u_\epsilon| \leq \frac{C}{\omega_n \delta^n} \int_{\Omega_\delta \cap B_{1/\delta}(0)} |u_\epsilon| \leq (\tilde{C} + 1) \int_{\Omega_\delta \cap B_{1/\delta}(0)} |u|.$$

We thus have uniform bounds on $\sup_{\tilde{\Omega}} |D^\alpha u|$ independent of ϵ . Therefore, by using Arzela-Ascoli and diagonal argument, we obtain that $u_{\epsilon_i} \rightarrow \tilde{u}$ in $C^k(\tilde{\Omega})$ for all $k \geq 1$. But then $\Delta \tilde{u} = \lim_{i \rightarrow \infty} (\Delta u_{\epsilon_i}) = 0$ so $\tilde{u}$ is harmonic in Ω and $\tilde{u} \in C^\infty(\Omega)$ (that is $\tilde{\Omega} \Subset \Omega$ arbitrarily). However, $u_\epsilon \rightarrow u$ in L^1_{loc} ensures almost everywhere convergence up to a subsequence, so that $\tilde{u} = u$ almost everywhere in Ω , so $u \in C^\infty(\Omega)$ and harmonic in Ω . $\square$

Lemma XI.2.4. Existence and Uniqueness for Dirichlet Problem with Continuous Data.

If $\Omega \subset \mathbb{R}^n$ is a bounded domain with $\partial\Omega$ regular enough to solve the Dirichlet problem with smooth boundary data, then if $\varphi \in C^0(\partial\Omega)$, there exists a unique $u \in C^\infty(\Omega) \cap C^0(\overline{\Omega})$ solving the PDE:

$$\begin{cases} \Delta u = 0, & \text{in } \Omega, \\ u = \varphi, & \text{on } \partial\Omega. \end{cases}$$

Proof. Recall weak maximum principle (Lemma XI.1.7), uniqueness is guaranteed if we have solvability. We approximate φ by smooth functions $\{\varphi_i\} \subset C^\infty(\mathbb{R}^n)$, so that $\varphi_i \rightarrow \varphi$ uniformly on $\partial\Omega$ (since $\partial\Omega$ is compact). By the assumption on $\partial\Omega$, there exists a unique $\{u_i\} \subset C^0(\overline{\Omega})$ with $\Delta u_i = 0$ in Ω , and $u_i = \varphi_i$ on $\partial\Omega$. In particular, $\Delta(u_i - u_j) = 0$ in Ω for $i, j \geq 1$ and $(u_i - u_j) = (\varphi_i - \varphi_j)$ on $\partial\Omega$.

Thus, the weak maximum principle gives that:

$$\sup_{\Omega} |u_i - u_j| \leq \sup_{\partial\Omega} |u_i - u_j| = \sup_{\partial\Omega} |\varphi_i - \varphi_j| \rightarrow 0$$

as $\varphi_i \rightarrow \varphi$ uniformly, so $\{u_i\}$ is Cauchy in $C^0(\overline{\Omega})$, which is Banach, so $u_i \rightarrow u$ in $C^0(\overline{\Omega})$, thus $u = \varphi \in \partial\Omega$. By the gradient estimates, we also ensure that $u_i \rightarrow u$ in $C^k(\tilde{\Omega})$ for each $\tilde{\Omega} \Subset \Omega$, hence $\Delta u = 0$ in Ω and $u \in C^\infty(\Omega)$. Thus, $u \in C^\infty(\Omega) \cap C^0(\overline{\Omega})$, solving the given PDE problem. $\square$

XI.3 The Maximum Principles

Here, we consider $\Omega \subset \mathbb{R}^n$ being open (and often connected) and the non-divergence form operators as:

$$Lu = a^{ij}D_{ij}u + b^iD_iu + cu,$$

where $u \in C^2(\Omega)$ is a function that is classically differentiable, and $a^{ij}, b^i, c : \Omega \rightarrow \mathbb{R}$. We look primarily at the Dirichlet problem:

$$\begin{cases} Lu = f, & \text{in } \Omega, \\ u = \varphi, & \text{on } \partial\Omega. \end{cases} \quad (\text{D.P.})$$

In general, we will assume $a^{ij} = a^{ji}$ for $i, j = 1, \dots, n$.

Definition XI.3.1. Ellipticity in Non-Divergence Form.

We say L in the form $Lu = a^{ij}D_{ij}u + b^iD_iu + cu$ is:

- elliptic in Ω if $A = (a^{ij})_{i,j}$ is positive definite for each $x \in \Omega$, i.e., $0 < \lambda(x)|\xi|^2 \leq a^{ij}\xi_i\xi_j \leq \Lambda(x)|\xi|^2$ for all $\xi \in \mathbb{R}^n \setminus \{0\}$,
- strictly elliptic in Ω if it is elliptic and $-\infty < \lambda_0 < \lambda(x)$ for all $x \in \Omega$, and
- uniform ellipticity in Ω if it is elliptic and $\sup_{\Omega} \left(\frac{\Lambda}{\lambda}\right) < \infty$. $\lrcorner$

Remark XI.3.2. Uniform ellipticity does not imply strict ellipticity in general. If L is strictly elliptic, uniform ellipticity is equivalently $\sup_{\Omega}(\Lambda) < \infty$. $\lrcorner$

Example XI.3.3. For the Minimal Surface Equation, we have:

$$\operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) = 0 \quad \Longleftrightarrow \quad \sum_{i,j=1}^n \underbrace{\left(\delta^{ij} - \frac{D_i u D_j u}{1 + |\nabla u|^2} \right)}_{a^{ij}} D_{ij} u = 0.$$

Then, we have $a^{ij} = \delta^{ij} - \frac{D_i u D_j u}{1 + |\nabla u|^2} > 0$ if and only if $1 + |\nabla u|^2 \geq |D_i u|^2$, which implies that the MSE is elliptic.

Without a gradient bound, i.e., if $|\nabla u| \rightarrow \infty$, we must have $a^{ij} \rightarrow 0$ so it is not strictly/uniformly elliptic unless $|Du|$ is bounded. $\diamond$

Definition XI.3.4. If $f \in C^0(\overline{\Omega})$, we say that $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ is a:

- subsolution if $Lu = f$ if $Lu \geq f$ in Ω , and
- supersolution if $Lu = f$ if $Lu \leq f$ in Ω .

Theorem XI.3.5. Strict Maximum Principle.

If $\Omega \subset \mathbb{R}^n$ is open, L is elliptic, $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ is such that $Lu > 0$ in Ω (i.e., a strict subsolution to $Lu = 0$), then:

- (i) if $c = 0$ on Ω , u cannot attain an interior maximum in Ω ,
- (ii) if $c \leq 0$ on Ω , u cannot attain an interior nonnegative interior maximum in Ω , and
- (iii) if c is free on Ω , then u cannot attain an interior zero maximum in Ω .

Proof. For the sake of contradiction, assume $y \in \Omega$ is a maximum of u , thus $Du(y)$ is negative semi-definite with eigenvalues $\eta_i \leq 0$. As $D^2u(y)$ is symmetric, we can diagonalize it by an orthogonal matrix P (that satisfies that $P^\top = P^{-1}$) with $PDP^\top = D^2u(y)$ with $D_i = (\eta_i)_{i=1, \dots, n}$ as diagonal.

Let $A = (a^{ij})_{i,j}$, $\tilde{A} = P^\top A P$ so that for $\xi \in \mathbb{R}^n$, $v \in P\xi$, then:

$$\tilde{a}^{ij}\xi_i\xi_j = \text{trace}(\underbrace{\xi^\top P^\top}_{v^\top} A \underbrace{P\xi}_v) = \text{trace}(v^\top A v) = a^{ij}v_i v_j > 0.$$

As $\text{trace}(B) = \text{trace}(B^\top B P^\top)$, we have:

$$a^{ij}D_{i,j}u(y) = \text{trace}(AD^2u(y)) = \text{trace}(AP^\top DP) = \text{trace}(P^\top APD) = \text{trace}(\tilde{A}D) = \tilde{a}^{ij}\xi_i \leq 0.$$

Now, we have:

$$Lu(y) = a^{ij}D_{i,j}u(y) + b^i D_i u(y) + cu(y),$$

and so, we have for:

- (i) $c = 0$ implies that $Lu \leq 0$,
- (ii) $c \leq 0$ implies that $Lu(y) \leq 0$ if it is nonnegative maximum at y , and
- (iii) c is free implies that $Lu(y) \leq 0$ if $u(y) = 0$.

□

Remark XI.3.6. It is possible that u does not attain its supremum on $\overline{\Omega}$, the strict maximum principle only says that if it is attained, it must be constrained.

Theorem XI.3.7. (General) Weak Maximum Principle.

If $\Omega \subset \mathbb{R}^n$ is open and bounded, L is elliptic with $\sup_{\Omega,i} \left(\frac{|b^i|}{\lambda} \right) < \infty$, and $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ is a subsolution to $Lu = 0$, i.e., $Lu \geq 0$, then:

- (i) If $c = 0$ in Ω , then $\sup_{\Omega}(u) = \sup_{\partial\Omega}(u)$.

(ii) If $c \leq 0$ on Ω , then $\sup_{\Omega}(u) \leq \sup_{\partial\Omega}(u_+)$ where $u_+ = \max\{u, 0\}$.

Proof. For (i), we perturb u to a strict subsolution. As Ω is bounded, we find $d > 0$ so that, up to translation:

$$\Omega \subset \{(x_1, \dots, x_n) \in \mathbb{R}^n : 0 < x_1 < d\}$$

as the “slab” region.

Then, we choose $\alpha > \sup_{\Omega, i} \left(\frac{|b^i|}{\lambda} \right) = \beta \geq 0$ and set $v(x) = e^{\alpha x_1} > 1$ to compute:

$$Lv = \alpha^2 a^{1,1} e^{\alpha x_1} + \alpha b^1 e^{\alpha x_1} \geq \alpha(\alpha\lambda - \lambda\beta) e^{\alpha x_1} > 0,$$

by using $a^{1,1} > \lambda$ and $|b^1| < \lambda\beta$.

The, if $\alpha > 0$, $L(u + \epsilon v) = Lu + \epsilon Lv > 0$ so we have $u + \epsilon v$ is a strict subsolution.

So, strict maximum principle implies that $\sup_{\overline{\Omega}}(u + \epsilon v)$ is attained by $z \in \partial\Omega$. If $y \in \Omega$ is such that $u(y) > \sup_{\partial\Omega}(u)$ implies that $u(y) > u(z)$ but then we have:

$$u(y) + \epsilon < u(y) + \epsilon v(y) < u(z) + \epsilon v(z) \leq u(z) + \epsilon e^{\alpha d},$$

and so sending $\epsilon \rightarrow 0$ gives that $u(y) < u(z)$ which is a contradiction, and so $\sup_{\Omega}(u) = \sup_{\partial\Omega}(u)$ if $c = 0$ on Ω .

For part (ii), we will reduce the case to (i). Let $\Omega_+ = \{x \in \Omega : u(x) > 0\}$. Then, Ω_+ is open (as u is continuous) and bounded (as Ω is). If $\Omega_+ = \emptyset$, then $\sup_{\partial\Omega} = 0$ so $\sup_{\Omega}(u) \leq 0$ automatically, so we would be done.

Now, assume $\Omega_+ \neq \emptyset$ and consider $L_0 = L - c$, then $c_0 = 0$ and we can apply case (i) on Ω_+ as $L_0 u = Lu - cu \geq 0$ (as $c \leq 0$, $u > 0$). Then, from part (i), we have $\sup_{\Omega_+}(u) = \sup_{\partial\Omega_+}(u)$.

Now:

$$\partial\Omega_+ \subset \{x \in \Omega : u(x) = 0\} \cup \{x \in \partial\Omega : u(x) \geq 0\},$$

and so of $u \equiv 0$ on $\partial\Omega_+$ which implies that $\sup_{\Omega_+}(u) = 0$, so $\Omega_+ = \emptyset$, but we assume $\Omega_+ \neq \emptyset$, so $\sup_{\partial\Omega_+}(u)$ is attained on the set:

$$S = \{x \in \partial\Omega : u(x) \geq 0\}.$$

We then compute that:

$$\sup_{\Omega}(u) = \sup_{\Omega_+}(u) = \sup_{\partial\Omega_+}(u) \leq \sup_S(u) = \sup_{\partial\Omega}(u_+).$$

□

Remark XI.3.8.

- $c \leq 0$ is necessary as if $n = 1$, $\Omega = [0, \frac{2\pi}{k}]$, and $L = \frac{\partial^2}{\partial x^2} + k^2$ for $k > 0$, then $u(x) = \sin(kx)$ solves $Lu = 0$ but has an interior maximum.
- If $\Omega_+ = \emptyset$ in (ii), we gain no information, i.e., if $n = 1$, $\Omega = [-1, 1]$, $L = \frac{\partial^2}{\partial x^2} - 1$, for $\delta > 2\alpha > 0$, we have $u(x) = -\alpha x^2 - \delta$ is such that $Lu = u'' - u > 0$, but $u < 0$ so $\Omega_+ = \emptyset$, $\sup_{\Omega}(u) = -\delta$, hence we have $\sup_{\partial\Omega}(u) < \sup_{\Omega}(u)$. ┘

Corollary XI.3.9. Uniqueness for Dirichlet Problem.

If $\Omega \subset \mathbb{R}^n$ is open and bounded, L is elliptic with $\sup_{\Omega,i} \left(\frac{|b^i|}{\lambda} \right) < \infty$, $c \leq 0$, $\varphi \in C^0(\partial\Omega)$ and $f : \Omega \rightarrow \mathbb{R}$, then there is at most one solution $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ to the Dirichlet problem:

$$\begin{cases} Lu = f, & \text{in } \Omega, \\ u = \varphi, & \text{on } \partial\Omega. \end{cases}$$

Proof. If $u, \tilde{u} \in C^2(\Omega) \cap C^0(\overline{\Omega})$ solve the Dirichlet problem, then by linearity, we have:

$$L(u - \tilde{u}) = Lu - L\tilde{u} = f - f = 0$$

in Ω , so by the second case of the (general) weak maximum principle, we have:

$$\sup_{\Omega} (u - \tilde{u}) \leq \sup_{\partial\Omega} ((u - \tilde{u})_+) = 0,$$

thus we have $u \leq \tilde{u}$ on Ω , but by swapping the above argument, we have $\tilde{u} \leq u$, hence $u = \tilde{u}$ on Ω . $\square$

Remark XI.3.10. If $c > 0$, we cannot guarantee uniqueness. For example, if $\Omega = [0, \frac{2\pi}{k}]$ with $t \in \mathbb{R}$, then $u_t(x) = t \sin(kx) \in C^2(\Omega) \cap C^0(\overline{\Omega})$ solve $Lu_t = \frac{\partial^2 u}{\partial x^2} + k^2 u = 0$ with $y_t = 0$ on $\partial\Omega$. For $n \geq 2$, one can consider Δ eigenvalues. $\lrcorner$

Corollary XI.3.11. Comparison Principle.

If $\Omega \subset \mathbb{R}^n$ is open and bounded, L is elliptic with $\sup_{\Omega,i} \left(\frac{|b^i|}{\lambda} \right) < \infty$ and $c \leq 0$, then if $u, v \in C^2(\Omega) \cap C^0(\overline{\Omega})$ are such that $u \leq v$ on $\partial\Omega$ and $Lu \geq Lv$ in Ω , then we have that $u \leq v$ on Ω .

Proof. Consider $L(u - v) = Lu - Lv \geq 0$ in Ω and note that $u - v \leq 0$ on Ω , so the (general) weak maximum principle for case (ii) gives:

$$\sup_{\Omega} (u - v) \leq \sup_{\partial\Omega} ((u - v)_+) = 0,$$

so we have $u \leq v$ on Ω . $\square$

Remark XI.3.12. This retroactively motivates (or verifies) the terminology for subsolutions and supersolutions. $\lrcorner$

Corollary XI.3.13. Weak Maximum Principle for Supersolutions and Solutions.

If $\Omega \subset \mathbb{R}^n$ is open and bounded, L is elliptic with $\sup_{\Omega,i} \left(\frac{|b^i|}{\lambda} \right) < \infty$, and $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$, then if $Lu \leq 0$, we have that:

- (i) If $c = 0$ on Ω , then $\inf_{\Omega}(u) = \inf_{\partial\Omega}(u)$.

(ii) If $c \leq 0$ on Ω , then $\inf_{\Omega}(u) \geq \inf_{\partial\Omega}(-u_-)$, where $u_i = -\min\{u, 0\}$.

Moreover, if u is a solution, i.e., $Lu = 0$, then if $c \leq 0$, in fact:

$$\sup_{\Omega} |u| = \sup_{\partial\Omega} |u|.$$

Proof. Apply weak maximum principle to $\tilde{u} = -u$ to get the two cases as $\sup_{\Omega}(u) = -\inf_{\Omega}(-u)$, by which they apply for both subsolution and supersolutions. $\square$

XI.4 Hopf Boundary Point

Definition XI.4.1. Interior-Sphere Condition.

We say that an open set $\Omega \subset \mathbb{R}^n$ satisfies the interior-sphere condition at $y \in \partial\Omega$ if there is ball $B \subset \Omega$ with $y \in \partial B$. Similarly, we can define exterior sphere condition at $y \in \partial\Omega$. $\lrcorner$

Remark XI.4.2.

- If $\partial\Omega$ is C^2 , then both interior and exterior sphere hold at all points of $\partial\Omega$.
- $\partial\Omega$ is $C^{1,1}$ if and only if both interior and exterior sphere hold at all points of $\partial\Omega$. $\lrcorner$

Lemma XI.4.3. Hopf Boundary Point Lemma.

If $\Omega \subset \mathbb{R}^n$ is open, L is uniformly elliptic with $\sup_{\Omega,i} \left(\frac{|b^i| + |c|}{\lambda} \right) < \infty$ and $y \in \partial\Omega$ with:

$$u \in C^2(\Omega) \cap C^0(\partial\Omega \setminus \{y\}) \text{ and } u(y) > u(x) \text{ for all } x \in \Omega,$$

then if Ω satisfies the interior sphere condition at $y \in \partial\Omega$, then in all cases:

- (i) $c = 0$ in Ω ,
- (ii) $c \leq 0$ in Ω and $u(y) \geq 0$, or
- (iii) c is free in Ω and $u(y) = 0$.

We have $\frac{\partial u}{\partial \nu}(y) = \nabla u(y) \cdot \nu(y) > 0$, if it exists, where ν is the outward pointing unit vector normal to $\partial\Omega$ at y .

Proof. For case (i) and (ii) we perturb u as in the proof of the weak maximum principle. Since we have interior sphere condition at $y \in \partial\Omega$, we can find $z \in \Omega$, $R > 0$ such that $B_R(z) \subset \Omega$ and $y \in \partial B_R(z)$, we then consider:

$$v(x) = e^{-\alpha|x-z|^2} - e^{-\alpha R^2}$$

defined on $\mathcal{A} = B_R(z) \setminus B_{R/2}(z)$.

We compute that:

$$\begin{cases} D_i v(x) = -2\alpha(x_i - z_i)e^{-\alpha|x-z|^2}, \\ D_{i,j} v(x) = -2\alpha\delta^{i,j}e^{-\alpha|x-z|^2} + 4\alpha^2(x_i - z_i)(x_j - z_j)e^{-\alpha|x-z|^2}. \end{cases}$$

Hence, we have:

$$\begin{aligned} Lv(x) &= e^{-\alpha|x-z|^2} \left[-2\alpha(a^{i,i} + b^i(x_i - z_i)) + 4\alpha^2 a^{i,j}(x_i - z_i)(x_j - z_j) \right] + c(e^{-\alpha|x-z|^2} - e^{-\alpha R^2}) \\ &\geq e^{-\alpha|x-z|^2} \left[-2\alpha\Lambda(x) - 2\alpha|b^i|R + \alpha^2\lambda(x)R^2 - |c| \right] \\ &\leq \lambda(x)e^{-\alpha|x-z|^2} \underbrace{\left[-2\alpha \sup_{\Omega} \left(\frac{\Lambda}{\lambda} \right) - 2\alpha \sup_{\Omega,i} \left(\frac{|b^i|}{\lambda} \right) R + \alpha^2 R^2 - \sup_{\Omega} \left(\frac{|c|}{\lambda} \right) \right]}_{\text{quadratic in } \alpha \text{ with } \alpha^2 \text{ coefficient} > 0}, \end{aligned}$$

for $\alpha \gg 0$, we ensure that $Lv(x) > 0$ for each $x \in \mathcal{A}$.

We consider $u(y) - \epsilon v$ for $\epsilon > 0$ and since $c \leq 0$ and $u(y) \geq 0$, we have:

$$L(u(y) - \epsilon v) = cu(y) - \epsilon Lv \leq 0 \leq Lu \quad \text{on } \mathcal{A}.$$

As $u(y) > u(x)$ for all $x \in \Omega$, we have $u \leq u(y) - \epsilon v$ in $\partial\mathcal{A} = \partial B_R(z) \cap \partial B_{R/2}(z)$ whenever $\epsilon > 0$ is small enough.

So, the comparison principle applied to give $u \leq u(y) - \epsilon v$ on all of $\mathcal{A}$.

Computing that:

$$\begin{aligned} \frac{\partial u}{\partial v}(y) &= \frac{\partial(u - u(y))}{\partial v}(y) = \lim_{t \nearrow 0^-} \frac{u(y + tv) - u(y)}{t} \\ &\geq -\epsilon \lim_{t \rightarrow 0^-} \frac{v(y + \epsilon v - v(y))}{t} = -\epsilon \frac{dv}{dv}(y) \\ &= -\epsilon \nabla v(y) \cdot v = 2\epsilon\alpha(y - z)e^{-\alpha R^2} \cdot \frac{y - z}{R} = 2\epsilon\alpha R e^{-\alpha R^2} > 0. \end{aligned}$$

Hence, the proof for case (i) and (ii) are completed.

For case (iii), consider $\tilde{L} = L - C_+$ so that $\tilde{L}u = Lu - c_+u \geq 0$ in Ω as $u(x) < u(y) = 0$ in Ω . By applying (ii), we then get that $\frac{\partial u}{\partial v}(y) > 0$ as long as it exists. $\square$

Remark XI.4.4. Even if $\frac{\partial u}{\partial v}(y)$ does not exist, we have $\liminf_{x \rightarrow y} \frac{u(x) - u(y)}{|x - y|} > 0$ where the angle between $(y - x)$ and v is less than $\frac{\pi}{2} - \delta$ for some $\delta > 0$. This hints at more relaxed assumptions on $\partial\Omega$, but Δ on $\mathbb{R}^2 \cap \{x_1, x_2 > 0\}$ with $u(x_1, x_2) = -x_1x_2$ has $u < 0$, $u(0) = 0$, $\Delta u = 0$ but $\lim_{x \rightarrow 0} \frac{|u|}{|x|} = 0$. $\lrcorner$

Theorem XI.4.5. (General) Strong Maximum Principle.

If $\Omega \subset \mathbb{R}^n$ is open and connected, L is uniformly elliptic with $\sup_{\Omega,i} \left(\frac{|b^i| + |c|}{\lambda} \right) < \infty$, and $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ with $Lu \geq 0$ in Ω , then:

- (i) if $c = 0$ and u attains an interior maximum, then u is constant,
- (ii) if $c \leq 0$ and u attains a nonnegative interior maximum, then u is constant, and

(iii) if c is free and u has a zero interior maximum, then u is constant.

Proof. We set:

$$\Sigma = \{x \in \Omega \mid \min u(x) = \max_{\Omega} u\} \neq \emptyset,$$

and by the continuity of the function u , Σ is closed. We will show that Σ is also open so that $\Sigma = \Omega$ as Ω is connected.

For the sake of contradiction, we assume that $\Sigma \neq \Omega$ and choose $z \in \Omega \setminus \Sigma$, we may suppose without loss of generality that:

$$\text{dist}(z, \partial\Omega) > \text{dist}(z, \partial\Sigma).$$

To see this, we let $\tilde{z} \in \partial\Sigma \cap \Omega$ and set $\tilde{r} > 0$ such that $B_{\tilde{r}} \subset \Omega$. Then, by the triangle inequality, each $z \in B_{\tilde{r}/2}(\tilde{z}) \setminus \Sigma$ satisfies the claim.

Thus, we set $R = \sup\{r > 0 \mid B_r(z) \subset \Omega \setminus \Sigma\} > 0$, so that for some point $y \in \partial B_R(z) \cap \Sigma$. Then, y is a maximum of u so that $\nabla u(y) = 0$.

By construction $\partial(\Omega \setminus \Sigma)$ satisfies the interior sphere condition at the point y . By the Hopf lemma in each case, we have $\frac{\partial u}{\partial \nu}(y) \cdot \nu > 0$, but this contradicts the fact that $Du(y) = 0$. Hence, $\Sigma = \Omega$ and thus u is constant on all of Ω . $\square$

Corollary XI.4.6. Avoidance Principle.

With the same assumptions as the strong maximum principle, if $u, v \in C^2(\Omega) \cap C^0(\overline{\Omega})$ with $Lu \geq Lv$ and $u \leq v$ in Ω , then either $u = v$ or $u < v$ on Ω .

Proof. As $L(u - v) \geq 0$ and $(u - v) \leq 0$ in Ω . If $u(x) = v(x)$ anywhere in Ω , the strong maximum principle implies that $u = v$ on Ω . $\square$

Corollary XI.4.7. With the same assumptions as the strong maximum principle and $\partial\Omega$ is C^2 , then if $y \in \partial\Omega$, $u, v \in C^2(\Omega) \cap C^1(\Omega \cup \{y\})$ with $Lu \geq Lv$ and $u \leq v$ in Ω , $u(y) = v(y)$ and $Du(y) = Dv(y)$, then $u = v$ in Ω .

Proof. If $\partial\Omega$ is C^2 , then the interior sphere holds at $y \in \partial\Omega$. Then, the Hopf boundary point lemma case (iii) can be applied to $(u - v)$ at y , hence $D(u - v)(y) \neq 0$ if $u < v$ in Ω . Thus, Hopf fails so there exists $x \in \Omega$ with $u(x) = v(x)$, but then avoidance implies that $u = v$ in Ω . $\square$

Corollary XI.4.8. Uniqueness for the Neumann Problem.

If $\Omega \subset \mathbb{R}^n$ is open, connected, and bounded, with the interior sphere condition satisfied at all points of the boundary $\partial\Omega$, L is uniformly elliptic with $\sup_{\Omega, i} \left(\frac{|b^i| + |c|}{\lambda} \right) < \infty$ and $c \leq 0$, then if $u, \tilde{u} \in C^2(\Omega) \cap$

$C^0(\overline{\Omega})$ solve the Neumann problem:

$$\begin{cases} Lu = f, & \text{in } \Omega, \\ \frac{\partial u}{\partial \nu} = \varphi, & \text{on } \partial\Omega, \end{cases} \quad (\text{N.P.})$$

for $f : \Omega \rightarrow \mathbb{R}$, $\varphi : \partial\Omega \rightarrow \mathbb{R}$, $u = \tilde{u} + M$ for some $M \in \mathbb{R}$.

Proof. Let $v = u - \tilde{u}$, it then suffices to show that only constant functions solve (N.P.) for $\varphi, f = 0$. Let $M = \sup_{\overline{\Omega}}(v) = \max_{\overline{\Omega}}(v)$ and without loss of generality suppose $M \geq 0$ (or consider $-v$). By combining the strict maximum principle and strong maximum principle, if $v \neq M$ in $\overline{\Omega}$, we must have a $y \in \partial\Omega$ such that $u(x) < u(y)$ for all $x \in \Omega$ (as $c \leq 0$). Applying the Hopf boundary point lemma to see that $\frac{\partial v}{\partial \nu} > 0$, contradicting the assumption that v solves (N.P.) with $\varphi = 0$. Hence, $v = M$ in $\overline{\Omega}$ and so $u = \tilde{u} + M$. $\square$

Remark XI.4.9. If $f, \varphi = 0$, then only constants solve the (N.P.), then $(M) = 0$ implies that $cM = 0$ in Ω . Hence, if $c \neq 0$ in Ω , thus $M = 0$ and solutions to (N.P.) are unique. $\lrcorner$

XI.5 A-Priori Estimates

Theorem XI.5.1. Maximum Principle A Priori Estimates.

If $\Omega \subset \mathbb{R}^n$ is open, bounded, and connected, L is elliptic with $\sup_{\Omega,i} \left(\frac{|b^i|}{\lambda} \right) < \infty$ and $c \leq 0$, then if $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$, $f : \Omega \rightarrow \mathbb{R}$ we have that:

- (i) If $Lu \geq f$ in Ω , then $\sup_{\Omega}(u) \leq \sup_{\partial\Omega}(u_+) + c \sup_{\Omega} \left(\frac{|f|}{\lambda} \right)$.
- (ii) If $Lu = f$ in Ω , then $\sup_{\Omega} |u| \leq \sup_{\partial\Omega} |u| + c \sup_{\Omega} \left(\frac{|f|}{\lambda} \right)$, where $c > 0$ is a constant depending on Ω and $\sup_{\Omega,i} \left(\frac{|b^i|}{\lambda} \right)$.

Precisely, if $d = \text{diam}(\Omega)$, $\beta = \sup_{\Omega,i} \left(\frac{|b^i|}{\lambda} \right)$, then $C = \exp((\beta + 1)d) - 1$.

Proof. We only prove (i) since (ii) follows by considering $-u$.

The idea is to construct a function v , so that $u \leq v$ on $\partial\Omega$, $Lu \geq Lv$ in Ω so that we can apply the comparison principle. As Ω is bounded, as in the proof of the weak maximum principle, we assume $\Omega \subset \{0 < x_1 < d\}$ for $d = \text{diam}(\Omega)$.

By considering $a + B^{ax_1}$ for α, A, B we set:

$$v(x) = \sup_{\partial\Omega}(u_+) + (e^{\alpha d} - e^{\alpha x_1}) \sup_{\Omega} \left(\frac{|f|}{\lambda} \right) \geq 0,$$

where $\alpha \geq \beta + 1 > 0$.

Note that for $y \in \partial\Omega$, we have $u(y) \leq \sup_{\partial\Omega}(u_+) \leq v(y)$. Setting $L_0 = L - c$, we compute that:

$$\begin{aligned} L_0(e^{\alpha x_1}) &= (a^{ij}D_{ij} + b^i D_i)(e^{\alpha x_1}) = e^{\alpha x_1}(\alpha^2 a'' + \alpha b') \geq \lambda e^{\alpha x_1} \left(\alpha^2 + \alpha \frac{b^1}{\lambda} \right) \\ &\geq \lambda e^{\alpha x_1}(\alpha^2 - \alpha\beta) = \lambda e^{\alpha x_1} \alpha(\alpha - \beta) \geq \lambda e^{\alpha x_1} \geq \lambda. \end{aligned}$$

Thus, we have:

$$Lv = L_0 v + cv \leq -\sup_{\Omega} \left(\frac{|f|}{\lambda} \right) L(e^{\alpha x_1}) \leq -\lambda \sup_{\Omega} \left(\frac{|f|}{\lambda} \right).$$

So, as $Lu \geq f$ in Ω , we have:

$$L(u - v) \geq f - Lv \geq f + \lambda \sup_{\Omega} \left(\frac{|f|}{\lambda} \right) = \lambda \left(\frac{f}{\lambda} + \sup_{\Omega} \left(\frac{|f|}{\lambda} \right) \right) \geq 0$$

Thus, $Lu \geq Lv$ in Ω . Hence, comparison principle gives $u \leq v$ on Ω so that:

$$\sup_{\Omega} u \leq \sup_{\Omega} v \leq \sup_{\partial\Omega}(u_+) + (e^{(\beta+1)\alpha} - 1) \sup_{\Omega} \left(\frac{|f|}{\lambda} \right).$$

□

Remark XI.5.2. In (i) one needs only to bound $|f_-|$ but we do need consider since $u'' \geq z$ if $u(t) = 1 - t^2$ in $\Omega = (-1, 1)$. ┘

XII The Schauder Estimates

XII.1 Interior Schauder Estimates

In general, for elliptic operator L , if we solve the PDE $Lu = f$, we expect u to have two more derivatives than f .

Specifically, we have seen in IX.2 for the case with $f \in H^k$ for $k \geq 1$:

- If $f \in L^p$ this kind of estimate is called a Calderon-Zygmund estimate.
- If $f \in C^{0,\alpha}$ for $\alpha \in (0, 1)$, this kind of estimate is called a Schauder estimate.

Example XII.1.1. $f \in L^\infty \not\Rightarrow u \in W^{2,\infty}$.

By considering:

$$u(x, y) = (x^2 - y^2) \log((x^2 + y^2)^{1/2})$$

on $\mathbb{R}^2$, we have:

$$\Delta u(x, y) = \frac{8(x^2 - y^2)}{x^2 + y^2} = f(x, y) \in L^\infty(\mathbb{R}^2).$$

However, if we consider that:

$$D_1^2 u(x, y) = 2 \log(x^2 + y^2) + \frac{8x^2}{x^2 + y^2} - \frac{2(x^2 - y^2)^2}{(x^2 + y^2)^2},$$

which is not L^∞ , so $u \notin W^{2,\infty}(\mathbb{R}^2)$. ◇

Example XII.1.2. $f \in C^0 \not\Rightarrow u \in C^2$.

By considering:

$$u(x, y) = (x^2 - y^2) \log(x^2 + y^2 + 1)$$

on $\mathbb{R}^2$, we have:

$$\Delta u(x, y) = \Delta u(x, y) = (x^2 - y^2) \left(\frac{8}{x^2 + y^2 + 1} + \frac{4}{(x^2 + y^2 + 1)^2} \right) \in C^0(\mathbb{R}^2).$$

However, if we consider that:

$$D_1^2 u(x, y) = 2 \log(\log(1 + x^2 + y^2)) + \text{additional terms}$$

which is not unbounded near $(0, 0)$, so $u \notin C^2(\mathbb{R}^2)$. ◇

Here, we will recall some notations from Hölder Spaces (IV.1).

Definition XII.1.3. Hölder Continuity.

If $\Omega \subset \mathbb{R}^n$ is open, $\alpha \in (0, 1]$, $u : \Omega \rightarrow \mathbb{R}$ is uniformly Hölder continuous with exponent α if:

$$[u]_{\alpha; \Omega} = \sup_{\substack{x, y \in \Omega \\ x \neq y}} \frac{|u(x) - u(y)|}{|x - y|^\alpha} < +\infty.$$

We also say that u is locally Hölder continuous with exponent α if:

$$[u]_{\alpha;K} < +\infty \quad \text{for all } K \Subset \Omega.$$

We then set:

$$\begin{cases} C^{k,\alpha}(\overline{\Omega}) = \left\{ u \in C^k(\Omega) : D^\beta u \text{ is uniformly } \alpha\text{-Hölder for all } |\beta| \leq k \right\}, \\ C^{k,\alpha}(\Omega) = \left\{ u \in C^k(\Omega) : D^\beta u \text{ is locally } \alpha\text{-Hölder for all } |\beta| \leq k \right\}. \end{cases}$$

with norms being:

$$\|u\|_{C^{k,\alpha}(\overline{\Omega})} = |u|_{l,\alpha;\Omega} = |u|_{k;\Omega} + [D^k u]_{\alpha;\Omega},$$

where $|u|_{k;\Omega} = \|u\|_{C^k(\overline{\Omega})}$ and $[D^k u]_{\alpha;\Omega} = \sup_{|\beta|=k} [D^\beta u]_{\alpha;\Omega}$. ┘

With the given norms, the above spaces are Banach.

Theorem XII.1.4. Arzelá-Ascoli (from Theorem IV.2.6).

If $\Omega \subset \mathbb{R}^n$ is open, $k \geq 1$, $\alpha \in (0, 1]$, and $\{u_i\} \subset C^{k,\alpha}(\Omega)$ is such that:

$$\sup_i |u_i|_{k,\alpha;\tilde{\Omega}} < \infty \quad \text{for all } \tilde{\Omega} \Subset \Omega,$$

then up to a subsequence (not relabeled), we have $u_i \rightarrow u \in C^{k,\alpha}(\Omega)$ in the sense of C_{loc}^k convergence.

Lemma XII.1.5. Hölder Interpolation.

If $1 \leq k \leq \ell$, $\alpha \in (0, 1]$, and $\epsilon > 0$, then there is a constant $C > 0$, depending on n, α, ϵ , and ℓ such that:

$$R^k |D^k v|_{0;B_R(x_0)} \leq C |v|_{0;B_R(x_0)} + \epsilon R^{\ell+\alpha} [D^\ell v]_{\alpha;B_R(x_0)}$$

for all $v \in C^{\ell,\alpha}(\overline{B_R(x_0)})$ for $B_R(x_0) \in \mathbb{R}^n$ where $R > 0$.

Proof. Rescale and let $\tilde{v}(x) = v(x_0 + Rx) \in C^{\ell,\alpha}(\overline{B_1(0)})$ so it suffices to prove that the lemma with $x_0 = 0$ and $R = 1$. We compute that:

$$|D^k \tilde{v}|_{0;B_1(0)} = R^k |D^k v|_{0;B_R(0)} \quad \text{and} \quad [D^\ell \tilde{v}]_{\alpha;B_1(0)} = R^{\ell+\alpha} [D^\ell v]_{\alpha;B_R(0)}.$$

Then, by we can prove this by contradiction and use Arzelá-Ascoli (Theorem XII.1.4). □

Then, we can proceed to the **interior Sahcuder estimate** for non-divergence form, starting from the unit ball.

Theorem XII.1.6. Unit Scale Interior Schauder.

If $L = a^{ij}D_{ij} + b^iD_i + C$ is strictly elliptic ($a^{ij}\xi_i\xi_j \geq \lambda|\xi|^2$ for $\lambda > 0$) with:

$$\max_{i,j} \{|a^{ij}|_{0,\alpha;B_1(0)} + |b^i|_{0,\alpha;B_1(0)} + |c|_{0,\alpha;B_1(0)}\} \leq \beta \quad \text{for } \alpha \in (0,1),$$

then if $u \in C^{2,\alpha}(B_1(0)) \cap C^0(\overline{B_1(0)})$ and $f \in C^{0,\alpha}(\overline{B_1(0)})$ satisfies $Lu = f$ in $B_1(0)$, we have:

$$|u|_{2,\alpha,B_{1/2}(0)} \leq C(|u|_{0;B_1(0)} + |f|_{0,\alpha;B_1(0)})$$

for a constant $C > 0$, depending on n, α, λ , and β .

Remark XII.1.7.

- The half ball is not special, we could consider any ball $B_\theta(0) \subset B_1(0)$, but then C depends on $\theta \in (0,1)$.

We have to take a smaller ball since $u(x,y) = \Re(z^k)$ for $z = x + iy$, then z^k is holomorphic, we have that $\Delta u = 0$ on $\mathbb{R}^2$. Note that:

$$|D^2u| \approx k(k-1)|z|^{k-2},$$

so we have $|u|_{2,\alpha;B_1(0)} \gtrsim k^2$ but $|u|_{0,\alpha;B_1(0)} = 1$.

Then consider for $\theta = 1$, we have $k^2 \lesssim C$ but this fails for any fixed $V > 0$ by considering large $k \geq 1$.

- We cannot take $\alpha = 1$ since there are C^2 but not $C^{2,1}$ functions solving $\Delta u = f \in C^0$.
- We get compactness of solutions to $Lu = f$, indeed, if $\{u_i\} \subset C^{2,\alpha}(B_1(0)) \cap C^0(\overline{B_1(0)})$ solving $Lu_i = f$ in $B_1(0)$ and if $\gamma = \sup_i |u_i|_{0;B_1(0)} < \infty$ then we have:

$$\sup_i |u_i|_{2,\alpha;B_\theta(0)} < C_\theta$$

for $C_\theta > 0$ as a constant depending on $\gamma, n, \alpha, \beta, \lambda, f$, and θ . Then Arzelá-Ascoli gives subsequential convergence in C_{loc}^2 to $u \in C^{2,\alpha}(B_1(0))$ solving $Lu = f$ also. $\square$

Let's work on scaling. For $x_0 \in \mathbb{R}^n$, $R > 0$. Suppose that we have $Lu = f$ on $B_R(x_0)$ for $u \in C^{2,\alpha}(B_R(x_0)) \cap C^0(\overline{B_R(x_0)})$.

By setting $\tilde{u}(x) = u(x_0 + Rx) \in C^\infty(B_1(0)) \cap C^0(\overline{B_1(0)})$, we have:

$$\begin{cases} D\tilde{u}(x) = RDu(x_0 + Rx), \\ D^2\tilde{u}(x) = R^2u(x_0 + Rx). \end{cases}$$

Hence:

$$[D^2\tilde{u}]_{\alpha;B_1(0)} = \sup_{\substack{x,y \in B_1(0) \\ x \neq y}} \frac{D^2\tilde{u}(x) - D^2\tilde{u}(y)}{|x - y|^\alpha} = \frac{R^2}{R^{-\alpha}} \sup_{\substack{x,y \in B_1(0) \\ x \neq y}} \frac{D^2u(x_0 + Rx) - D^2u(x_0 + Ry)}{|(x_0 + Rx) - (x_0 + Ry)|^\alpha} = R^{2+\alpha} [D^2u]_{\alpha;B_R(x_0)}.$$

We now see what PDE $\tilde{u}$ solves. As $Lu = f$ on $B_R(x_0)$ for $x \in B_1(0)$, we have:

$$a^{ij}(x_0 + Rx) \frac{D_{ij}\tilde{u}(x)}{R^2} + b^i(x_0 + Rx) \frac{D_i\tilde{u}(x)}{R} + c(x_0 + Rx)\tilde{u}(x) = f(x_0 + Rx),$$

then we have $\tilde{L}\tilde{u} = \tilde{f}$, where $\tilde{a}^{ij} = a^{ij}(x_0 + Rx)$, $\tilde{b}^i(x) = Rb^i(x_0 + Rx)$, $\tilde{c}(x) = R^2c(x_0 + Rx)$, and:

$$\tilde{f} = R^2f(x_0 + Rx).$$

Then, $\tilde{L}$ is strictly elliptic with constant $\lambda > 0$. By computing the norms, we get:

$$|\tilde{u}|_{2,\alpha;B_1(0)} = |u|_{0,B_R(x_0)} + R|Du|_{0,B_R(x_0)} + R^2|D^2u|_{0,B_R(x_0)} + R^{\alpha+2}[D^2u]_{\alpha;B_R(x_0)} = |u|'_{2,\alpha;B_R(x_0)}.$$

Similarly, $|f|'_{0,\alpha;B_R(x_0)} = |f|_{0;B_R(x_0)} + R^d[f]_{\alpha;B_R(x_0)}$.

Then, the assumption that:

$$\max_{i,j} \left\{ |\tilde{a}^{ij}|_{0,\alpha;B_1(0)} + |\tilde{b}^i|_{0,\alpha;B_1(0)} + |\tilde{c}|_{0,\alpha;B_1(0)} \right\} \leq \beta$$

becomes:

$$\max_{i,j} \left\{ |\tilde{a}^{ij}|'_{0,\alpha;B_R(x_0)} + R|\tilde{b}^i|'_{0,\alpha;B_R(x_0)} + R^2|\tilde{c}|'_{0,\alpha;B_R(x_0)} \right\} \leq \beta.$$

Then, assuming the uniform Schauder holds, we have the following.

Theorem XII.1.8. Scale Invariant Interior Schauder.

If L is strictly elliptic (for $\lambda > 0$), with:

$$\max_{i,j} \left\{ |\tilde{a}^{ij}|'_{0,\alpha;B_R(x_0)} + R|\tilde{b}^i|'_{0,\alpha;B_R(x_0)} + R^2|\tilde{c}|'_{0,\alpha;B_R(x_0)} \right\} \leq \beta,$$

holding for $\beta > 0$, $\alpha \in (0,1)$, then if $u \in C^{2,\alpha}(B_R(x_0)) \cap C^0(\overline{B_R(x_0)})$ solves $Lu = f$ for $f \in C^{0,\alpha}(\overline{B_R(x_0)})$ in $B_R(x_0)$, we have:

$$|u|'_{2,\alpha;B_{R/2}(x_0)} \leq C(|u|_{0;B_R(x_0)} + R^2|f|_{-\alpha;B_R(x_0)})$$

for a constant $C > 0$, depending on n, λ, α , and β .

We can use the scale invariant Schauder to get to the general interior Schauder estimate.

Theorem XII.1.9. General Interior Schauder.

If $\Omega \subset \mathbb{R}^n$ is open, L is strictly elliptic, with $\max_{i,j} \{ |a^{ij}|_{0,\alpha;\Omega} + |b^i|_{0,\alpha;\Omega} + |c|_{0,\alpha;\Omega} \} \leq \beta$ for $\alpha \in (0,1)$, then if $u \in C^{2,\alpha}(\Omega) \cap C^0(\overline{\Omega})$ solves $Lu = f$ for $f \in C^{0,\alpha}(\overline{\Omega})$ we have for each $\tilde{\Omega} \Subset \Omega$ that:

$$|u|_{2,\alpha;\tilde{\Omega}} \leq C(|u|_{0;\Omega} + |f|_{0,\alpha;\Omega}),$$

for a constant $C > 0$, depending on $n, \alpha, \lambda, \beta, \tilde{\Omega}$, and Ω .

Proof. Let $d := \min\{\text{dist}(\tilde{\Omega}, \partial\Omega), 1\} \leq 1$, so that for $x \in \tilde{\Omega}$, $B_d(x) \subset \Omega$. So, by assumption:

$$\max_{i,j} \{ |a^{ij}|'_{0,\alpha;\Omega} + |b^i|'_{0,\alpha;\Omega} + |c|'_{0,\alpha;\Omega} \} \leq \beta.$$

By applying the previous theorem on $B_d(x)$, we get, in particular, that:

$$|u|_{0,B_{1/2}(x)} + d|Du|_{0,B_{1/2}(x)} + d^2|D^2u|_{0,B_{1/2}(x)} \leq C(|u|_{0,B_d(x)} + d^2|f|_{0,B_d(x)} + d^{2+\alpha}[f]_{0,\alpha;B_d(x)}).$$

Thus, by applying this at each $x \in \tilde{\Omega}$ and taking a covering, we get:

$$|u|_{2,\tilde{\Omega}} \leq C(|u|_{0,\alpha} + |f|_{0,\alpha;\Omega})$$

for $C > 0$ bigger constant depending on Ω and $\tilde{\Omega}$.

Similarly, if $i, j = 1, \dots, n$ and consider $x \neq y$, $|x - y| < \frac{d}{2}$, $x, y \in \tilde{\Omega}$, we have:

$$\sup_{\substack{x \neq y, x, y \in \tilde{\Omega} \\ |x-y| < \frac{d}{2}}} \frac{|D_{i,j}u(x) - D_{i,j}u(y)|}{|x - y|^\alpha} \leq C(|u|_{0;\Omega} + |f|_{0,\alpha;\Omega}),$$

and if $|x - y| > \frac{d}{2}$, we get that:

$$\frac{|D_{i,j}u(x) - D_{i,j}u(y)|}{|x - y|^\alpha} \leq \left(\frac{d}{2}\right)^{-\alpha} [|D_{i,j}u(x) - D_{i,j}u(y)|] \leq C(|u|_{0,\alpha} + |f|_{0,\alpha;\Omega}).$$

Combining the above results, we get:

$$|u|_{2,\alpha;\tilde{\Omega}} \leq C(|u|_{0;\Omega} + |f|_{0,\alpha;\Omega}).$$

□

Lemma XII.1.10. Baby Schauder.

With the same assumptions with the Unit Scale Interior Schauder (Theorem XII.1.6) with $\tilde{u}$, $\tilde{f}$, and $\tilde{L}$ in place of u , f , L , then for each $\delta > 0$, we have:

$$[D^2\tilde{u}]_{\alpha;B_{1/2}(0)} \leq \delta[D^2\tilde{u}]_{\alpha;B_1(0)} + C(|\tilde{u}|_{2;B_1(0)} + |\tilde{f}|_{0,\alpha;B_1(0)})$$

for a constant $C > 0$ depending on δ , n , α , β , and λ .

Remark XII.1.11. For $\tilde{u}(x) = u(x + Rx)$ for u defined on $B_R(x_0)$ as in the rescale Schauder, the Baby Schauder gives us that:

$$R^{2+\alpha}[D^2u]_{\alpha;B_{R/2}(x_0)} \leq \delta R^{2+\alpha}[D^2u]_{\alpha;B_R(x)} + C(|u|'_{2;B_R(x_0)} + R^2|f|'_{0,\alpha;B_R(x_0)}).$$

┘

By the Hölder interpolation with $k = 1$ and $\ell = 2$ applied to u , we get:

$$R|Du|_{0;B_R(x_0)} \leq C|u|_{0;B_R(x_0)} + \epsilon R^{2+\alpha}[D^2u]_{\alpha;B_R(x_0)},$$

and with $u = \ell = 2$, we get that:

$$R^2|D^2u|_{0;B_R(x_0)} \leq C|u|_{0;B_R(x_0)} + \epsilon R^{2+\alpha}[D^2u]_{\alpha;B_R(x_0)}.$$

By combining these two inequalities, we have:

$$|u|'_{2;B_R(x_0)} = |u|_{0;B_R(x_0)} + R|Du|_{0;B_R(x_0)} + R^2|D^2u|_{0;B_R(x_0)} \leq (1 + 2C)|u|_{0;B_R(x_0)} + 2\epsilon R^{2+\alpha}[D^2u]_{\alpha;B_R(x_0)}.$$

Combining this and the combination of the rescale and baby Schauder, we get:

$$R^{2+\alpha}[D^2u]_{\alpha;B_{R/2}(x_0)} \leq (\delta + 2c\epsilon)R^{2+\alpha}[D^2u]_{\alpha;B_R(x_0)} + C(|u|_{0;B_R(x_0)} + R^2|f|'_{0,\alpha;B_R(x_0)}).$$

If $\epsilon < \delta/2C$ and $R < 1$ with $B_R(x_0) \subset B_1(0)$:

$$R^{2+\alpha}[D^2u]_{\alpha;B_{R/2}(x_0)} \leq 2\delta R^{2+\alpha}[D^2u]_{\alpha;B_R(x_0)} + C(|u|_{0;B_1(0)} + |f|_{0,\alpha;B_1(0)}).$$

Lemma XII.1.12. Simon's Absorption.

Let $\lambda \in [0, \infty)$, $\theta \in (0, 1)$, and S as a nonnegative function defined on sub-balls of $B_r(y)\mathbb{R}^n$, which is sub-additive, i.e.:

$$B_\rho(x) \subset \bigcup_{i=1}^N B_{\rho_i}(x_i) \subset B_r(y) \implies S(B_\rho(x)) \leq \sum_{i=1}^N S(B_{\rho_i}(x_i)),$$

then there is some $\delta_{\text{SAL}} > 0$ depending on n, λ , and θ such that if:

$$B_\rho(x) \subset B_r(y) \implies \rho^\lambda S(B_{\theta\rho}(x)) \leq \delta_{\text{SAL}} \rho^\lambda S(B_\rho(x)) + \delta,$$

we have that in fact:

$$B_\rho(y) \subset B_r(y) \implies \rho^\lambda S(B_{\theta\rho}(x)) \leq C\gamma,$$

for a constant $C > 0$ depending on n, λ , and θ .

If we set $B_r(y) = B_1(0)$, $\lambda = 2 + \alpha$, $\theta = \frac{1}{2}$, and γ as the second term in the right hand side, $2\delta < \delta_{\text{SAL}}$ and $S(B_s(x)) = [D^2u]_{\alpha;B_\rho(x)}$ (which is positive functions on sub-balls of $B_1(0)$ and sub-additive by triangle inequality).

Since $B_1(0) \subset B_1(0)$, we precisely have Simon's Absorption giving:

$$[D^2u]_{\alpha;B_1(0)} \leq C(|u|_{0;B_1(0)} + |f|_{0,\alpha;B_1(0)}).$$

Now, consider that:

$$|u|_{2,\alpha;B_{1/2}(0)} = |u|_{0;B_{1/2}(0)} + |Du|_{0;B_1(0)} + |D^2u|_{0;B_{1/2}(0)} + [D^2u]_{\alpha;B_{1/2}(0)} \leq C(|u|_{0;B_1(0)} + |f|_{0,\alpha;B_1(0)}).$$

Proof of Lemma XII.1.12. Consider that:

$$Q := \sup_{B_\rho(x) \subset B_r(y)} \left(\rho^\lambda S(B_{\theta\rho}(x)) \right) \leq r^\lambda S(B_r(y)) < \infty.$$

Iterating the subadditivity, we see that:

$$(\theta\rho)^\alpha S(B_{\theta^2\rho}(x)) \leq \delta(\theta\rho)^\alpha S(B_{\theta\rho}(x)) + \gamma \leq \delta\theta^\alpha Q + \gamma,$$

where $B_\rho(x) \subset B_r(y)$.

We now prove a dimensional result.

Fact: For $B_\rho(x) \subset B_r(y)$, there are N points $N \leq C$ depending only on $n, \theta, \{z_1, \dots, z_n\} \subset B_{\theta\rho}(x)$ such that:

$$B_{\theta\rho}(x) \subset \bigcup_{i=1}^N B_{(1-\theta)\theta^2\rho}(z_i).$$

To see this fact, choose a maximal pairwise disjoint collection of balls $\{B_{(1-\theta)\theta^2\rho/2}(z_i)\}_{i=1}^N$ such that $z_i \in B_{\theta\rho}(x)$ for $i = 1, \dots, N$.

We claim that we can use $\{z_i\}_{i=1}^N$ as points for the covering since if:

$$z \in B_{\theta\rho}(x) \setminus \bigcup_{i=1}^N B_{(1-\theta)\theta^2\rho}(z_i),$$

we know that $|z - z_i| > (1 - \theta)\theta^2\rho$ for each $i = 1, \dots, N$. Then we would have $B_{\frac{(1-\theta)\theta^2\rho}{2}}(z) \cap B_{\frac{(1-\theta)\theta^2\rho}{2}}(z_i) = \emptyset$ for all $i = 1, \dots, N$, which contradicts the maximizing of the points $\{z_i\}_{i=1}^N$ and hence $B_{\theta\rho}(x) \subset \bigcup_{i=1}^N B_{(1-\theta)\theta^2\rho}(z_i)$.

To see that $N \leq C$ for C depending on n and θ , we note that:

$$\bigcup_{i=1}^N B_{(1-\theta)\theta^2\rho}(z_i) \subset \bigcup_{i=1}^N B_{\theta\rho + (1-\theta)\theta^2\rho}(x).$$

Hence, we have:

$$N \cdot \omega_n \left(\frac{(1-\theta)\theta^2\rho}{2} \right)^n \leq \omega_n \left(\theta\rho + \frac{(1-\theta)\theta^2\rho}{2} \right)^n,$$

so we ultimately have:

$$N \leq \left(\frac{2}{(1-\theta)\theta} + 1 \right)^n =: C.$$

Now, using the fact and the subadditivity of S , we get:

$$S(B_{\theta\rho}(x)) \leq \sum_{i=1}^N S(B_{(1-\theta)\theta^2\rho}(z_i)),$$

and we then bound that:

$$\rho^\lambda S(B_{\theta\rho}(x)) \leq \sum_{i=1}^N \rho^\lambda S(B_{(1-\theta)\theta^2\rho}(z_i)) \cdot \frac{((1-\theta)\theta)^\lambda}{((1-\theta)\theta)^\lambda}.$$

Now, by replacing $\rho \leftarrow (1-\theta)\rho$ and $z_i \in B_{\theta\rho}(x)$ so that $B_{(1-\theta)\theta\rho}(z_i) \subset B_\rho(x)$, hence:

$$\begin{aligned} \rho^\lambda S(B_{\theta\rho}(x)) &\leq ((1-\theta)\theta)^{-\lambda} \sum_{i=1}^N \theta^\lambda ((1-\theta)\rho)^\lambda S(B_{(1-\theta)\theta^2\rho}(z_i)) \\ &\leq ((1-\theta)\theta)^\lambda \sum_{i=1}^N [\theta^\lambda \delta Q + \delta] \leq \delta (1-\theta)^{-\lambda} N Q + (1-\theta)^{-\lambda} N \gamma \leq \delta \tilde{C} Q + C \gamma, \end{aligned}$$

where C and $\tilde{C}$ depend on n, λ , and θ .

By taking the supremum, we have:

$$Q \leq \delta \tilde{C}Q + C\delta,$$

so if $\delta = \frac{1}{2\tilde{C}}$, so we have $Q \leq 2C\delta$. $\square$

Then, we will prove Baby Schauder estimate ([Lemma XII.1.10](#)).

Proof of (Lemma XII.1.10). For the sake of contradiction and suppose for some $\delta > 0$ that the Baby Schauder fails. Hence, we can find, for each $k \geq 1$, operators:

$$L_k = a_k^{i,j} D_{i,j} + b_k^i D_i + C_k,$$

with:

$$\max_{i,j} \{ |a_k^{i,j}|_{0,\alpha;B_1(0)} + |b_k^i|_{0,\alpha;B_1(0)} + |c_k|_{0,\alpha;B_1(0)} \} \leq \beta,$$

and $u_l \in C^{2,\alpha}(B_1(0)) \cap C^0(\overline{B_1(0)})$ such that $L_k u_k = f_k$ but:

$$[D^2 u_k]_{\alpha;B_{1/2}(0)} > \delta [D^2 u]_{\alpha;B_1(0)} + k(|u_k|_{2;B_1(0)} + |f_k|_{0,\alpha;B_1(0)}).$$

By the pigeonhole principle, up to a subsequence (not relabeled), we may find $\ell, m \in \{1, \dots, n\}$ fixed and $\{z_k\}, \{y_k\} \subset B_{1/2}(0)$ such that:

$$\frac{|D_{\ell_m} v_k(z_k) - D_{\ell_m} u_k(y_k)|}{|z_k - y_k|^\alpha} > \frac{1}{2} [D^2 u]_{\alpha;B_{1/2}(0)}.$$

Setting $\rho_k = |z_k - y_k| \leq 0$ so that by the contradiction and above, we see that:

$$\frac{1}{2} [D^2 u_k]_{\alpha;B_{1/2}(0)} < \frac{|D_{\ell_m} u_k(z_k) - D_{\ell_m} u_k(y_k)|}{\rho_k^\alpha} \leq \frac{2|u_k|_{2;B_1(0)}}{\rho_k^\alpha} < \frac{2[D^2 u_k]_{\alpha;B_{1/2}(0)}}{k\rho_k^\alpha},$$

and hence we have $\rho_k^\alpha < \frac{4}{k}$ so we have $\rho_k \rightarrow 0$ as $k \rightarrow \infty$ (i.e., z_k and y_k gets closer).

Here, we consider the quadratic Taylor polynomial (2-jex) of u_k at z_k , i.e.:

$$q_k(x) = u_k(z_k) + \rho_k D_i u_k(z_k) + \frac{\rho_k^2}{2} x^i x^j D_{i,j} u_k(z_k),$$

where $x = (x^1, \dots, x^n)$. Then $u_k(z_k + \rho_k x) - q_k(x)$ is such that u_k , Du_k , and $D^2 u_k$ vanish at $x = 0$. We then set:

$$\tilde{v}_k(x) = \frac{u_k(z_k + \rho_k x) - q_k(x)}{\rho_k^{2+\alpha} [D^2 u_k]_{\alpha;B_1(0)}} \quad \text{on } B_{1/\rho_k} \left(-\frac{z_k}{\rho_k} \right) \supset B_{1/2\rho_k}(0).$$

Then, we have $[D^2 \tilde{v}_k]_{\alpha;B_{\rho_k/2}(0)} \leq 1$ since $D^2 q_k$ is constant so we have $[D^2 q_k]_{\alpha;B_{\rho_k/2}(0)} = 0$, and so by the fundamental theorem of calculus, we see that, for some constant $C > 0$, we have:

$$|\tilde{v}_k|_{2,\alpha;B_R(0)} \leq C \cdot R^{2+\alpha}$$

for each $R > 0$ (since $\rho_k \rightarrow 0$ as $k \rightarrow \infty$).

By applying Arzelà-Ascoli ([Theorem XII.1.4](#)), up to another subsequence (not relabeled) and a diagonal argument, we have that $\tilde{v}_k \rightarrow v$ in $C^2(B_R(0))$ for each $R > 0$ so that $[D^2 v]_{\alpha;\mathbb{R}^n} \leq 1$.

We now show that $D_{\ell_m} v \neq 0$, for a later contradiction. As:

$$\eta_k = \frac{y_k - z_k}{\rho_k}, |\eta_k| = 1 \quad \text{for } k \geq 1.$$

We compute that:

$$\begin{aligned} |D_{\ell_m} \tilde{v}_k(\eta_k)| &= \frac{\rho_k^2 |D_{\ell_m} u_k(z_k + \rho_k \eta_k) - D_{\ell_m} u_k(z_k)|}{\rho_k^{2+\alpha} [D^2 u_k]_{\alpha; B_1(0)}} = \frac{|D_{\ell_m} u_k(y_k) - D_{\ell_m} u_k(z_k)|}{|y_k - z_k|^\alpha [D^2 u_k]_{\alpha; B_1(0)}} \\ &> \frac{1}{2} \cdot \frac{[D^2 u_k]_{\alpha; B_{1/2}(0)}}{[D^2 u_k]_{\alpha; B_1(0)}} > \frac{1}{2} \delta > 0. \end{aligned}$$

Thus, up to a further subsequence (not relabeled), $\eta_k \rightarrow \eta$ with $|\eta| = 1$ and $D_{\ell_m} \tilde{v}_k(\eta_k) \rightarrow D_{\ell_m}(\eta)$ and we so have $|D_{\ell_m}| > \frac{\delta}{2}$ so we have $D_{\ell_m} v \neq 0$.

What PDE does v solve on $\mathbb{R}^n$?

Recall $\tilde{u}_k(x) = u_k(z_k + \rho_k x)$ solved $\tilde{L}_k \tilde{v}_k = \tilde{f}_k$ for $\tilde{a}_k^{i,j}(x) = a_k^{i,j}(z_k + \rho_k x)$, $\tilde{b}_k^i(x) = \rho_k b_k^i(z_k + \rho_k x)$, $\tilde{c}_k(x) = \rho_k^2 c_k(z_k + \rho_k x)$, and $\tilde{f}_k(x) = \rho_k^2 f_k(z_k + \rho_k x)$, then $\tilde{v}_k$ solves:

$$\tilde{L}_k \tilde{v}_k(x) = \frac{\tilde{f}_k(x) - \tilde{L}_k q_k(x)}{\rho_k^{2+\alpha} [D^2 u_k]_{\alpha, B_1(0)}} = \tilde{g}_k.$$

We compute that:

$$[\tilde{a}_k^{i,j}]_{\alpha; B_{\rho_k/2}(0)} \leq \rho_k^\alpha [a_k^{i,j}]_{\alpha; B_1(0)} \leq \rho_k^\alpha \beta \rightarrow 0,$$

and $|\tilde{a}_k^{i,j}| \leq |a_k^{i,j}|_{0; B_1(0)} \leq B$. Arzelá-Ascoli gives that $\tilde{a}_k^{i,j} \rightarrow a^{i,j}$ in $C_{\text{loc}}^0(\mathbb{R}^n)$ with $[a^{i,j}]_{\alpha; \mathbb{R}^n} = 0$ which implies that $a^{i,j}$ must be constant.

Similarly, we have $|\tilde{b}_k^i|_{0; B_{\rho_k/2}(0)}, |\tilde{c}_k|_{0; B_{\rho_k/2}(0)} \leq \rho_k \beta \rightarrow 0$, we have $\tilde{b}_k^i, \tilde{c}_k \rightarrow 0$ in $C_{\text{loc}}^0(\mathbb{R}^n)$.

Thus, $\tilde{L}_k \rightarrow L = a^{i,j} D_{i,j}$ and $a^{i,j} \xi_i \xi_j \geq \lambda |\xi|^2$ for L being strictly elliptic with some $\lambda > 0$.

We then bound $\tilde{g}_k$:

$$\begin{aligned} \tilde{L}_k q_k(x) &= \rho_k^2 \tilde{a}_k^{i,j}(x) D_{i,j} u_k(z_k) + (\rho_k \tilde{b}_k^i(x) D_i u_k(z_k) + \rho^2 \tilde{b}_k^i(x) x^j D_{i,j} u_k(z_k)) \\ &\quad + (\tilde{c}_k(x) u_k(z_k) + \rho_k \tilde{c}_k(x) + \frac{\rho_k^2}{2} \tilde{c}_k(x) x^i x^j D_{i,j} u_k(z_k)) \\ &= \tilde{a}_k^{i,j}(x) D_{i,j} \tilde{u}_k(0) + (\tilde{b}_k^i(x) D_i \tilde{u}_k(0) + \tilde{b}_k^i(x) D_{i,j} \tilde{u}_k(0)) \\ &\quad + \left(\tilde{c}_k(x) \tilde{u}_k(0) + \tilde{c}_k(x) x^i D_i \tilde{u}_k(0) + \frac{1}{2} \tilde{c}_k(x) D_{i,j} \tilde{u}_k(0) \right). \end{aligned}$$

We will call the undermined terms by $\mathcal{O}_k(x)$, we then bound on $B_R(0)$ via:

$$\begin{cases} |\tilde{b}_k^i(x) x^j D_{i,j} \tilde{u}_k(0)| \leq \rho_k \beta R \rho_k^2 |u_k|_{2; B_1(0)}, \\ |\rho_k \tilde{c}_k(x) x^i D_i \tilde{u}_k(0)| \leq \rho_k \rho_k^2 \beta R \rho_k |u_k|_{2; B_1(0)}, \\ \left| x^i x^j \cdot \frac{1}{2} \tilde{c}_k(x) D_{i,j} \tilde{u}_k(0) \right| \leq \rho_k^2 \beta \rho_k^2 R^2 |u_k|_{2; B_1(0)}, \end{cases}$$

and so result in:

$$\sup_{B_R(0)} |\mathcal{O}_k| \leq \rho_k^3 \beta (2R + R^2) |u_k|_{2; B_1(0)}.$$

Thus, we consider that:

$$\frac{|\mathcal{O}_k(x)|}{\rho_k^{2+\alpha}[D^2u_k]} \leq \rho_k^{1-\alpha}\beta(2R+R)^2 \frac{|u_k|_{2;B_1(0)}}{[D^2u_k]_{\alpha;B_1(0)}} < \frac{1}{k}(\rho_k^{1-\alpha}\beta(R+R^2)).$$

Also, we have:

$$\frac{|\tilde{f}_k(x) - \tilde{f}_k(0)|}{\rho_k^{2+\alpha}[D^2u_k]_{\alpha;B_1(0)}} \leq \frac{[f_k]_{\alpha;B_1(0)}R^\alpha}{[D^2u_k]_{\alpha;B_1(0)}} \leq \frac{R^\alpha}{k}.$$

Moreover, we consider:

$$\begin{aligned} & \frac{1}{\rho_k^{2+\alpha}[D^2u_k]_{\alpha;B_1(0)}} ([\tilde{a}_k^{i,j}(x) - \tilde{a}_k^{i,j}(0)]D_{i,j}\tilde{u}_k(0) + [\tilde{b}_k^i(x) - \tilde{b}_k^i(0)]D_i\tilde{u}_k(0) + [\tilde{c}_k(x) - \tilde{c}_k(0)]\tilde{u}_k(0)) \\ & \leq \frac{3\beta R^\alpha |u_k|_{2;B_1(0)}}{[D^2u_k]_{\alpha;B_1(0)}} < \frac{3\beta R^\alpha}{k} \end{aligned}$$

Then, we have $\tilde{g}_k \rightarrow 0$ in $C_{\text{loc}}^0(\mathbb{R}^n)$. Thus sending $k \rightarrow \infty$, we get $Lv = 0$ for $L = a^{i,j}D_{i,j}$ with $A = (a_{i,j})$ as a constant matrix, where $[D^2v]_{\alpha;\mathbb{R}^n} \leq 1$, $D_{\ell_m}v \neq 0$, so we have v, Dv, D^2v at $x = 0$.

Let $A = (a^{i,j})_{i,j}$, since $a^{i,j}$ are strictly elliptic, we can diagonalize A by an orthogonal matrix, P , so that:

$$PAP^\top = Q = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix} \quad \text{where } \lambda_n \geq \lambda_{n-1} \geq \cdots \geq \lambda_1 > 0.$$

By setting $\omega(x) = v(P^\top x)$, we use chain rule to see that:

$$D_i(\omega(Bx)) = D_\ell \omega(Bx) \frac{\partial (Bx)_\ell}{\partial x^i} = D_\ell \omega(Bx) B_{\ell_i},$$

so we have $D_{i,j}(\omega(Bx)) = B_{m_j} D_{m_\ell}(\omega(Bx)) B_{\ell_i}$, so if $B = P^\top$, we get that:

$$D^2v(x) = P^\top D^2\omega(Px)P.$$

Then, consider:

$$0 = a^{i,j}D_{i,j}v = \text{trace}(AD^2v) = \text{trace}(\underbrace{P^\top Q P P^\top}_{A} D^2\omega P) = \text{trace}(AD^2\omega),$$

which is equivalently that:

$$\sum_{i=1}^n \lambda_i D_{i,i} \omega(x) = 0 \quad \text{for all } x \in \mathbb{R}^n.$$

Then, by setting $\tilde{\omega}(x) = \omega(\sqrt{\lambda_1}x_1, \dots, \sqrt{\lambda_n}x_n)$, this implies that $\Delta \tilde{\omega}(x) = 0$ on $\mathbb{R}^n$, and $\tilde{\omega} \in C^2(\mathbb{R}^n)$

Recall that if $u \in C^2(\mathbb{R}^n)$ is harmonic, i.e., $\Delta u = 0$ in $\mathbb{R}^n$, then:

- By **Weyl**, $u \in C^\infty(\mathbb{R}^n)$, and
- by **Liouville**: if also u has sublinear growth, i.e., $|u(x)| \leq C(1 + |x|^\alpha)$ for $\alpha \in (0, 1)$, then u is a constant.

Therefore, by Weyl, $\tilde{\omega} \in C^\infty(\mathbb{R}^n)$, and hence $v \in C^\infty(\mathbb{R}^n)$.

Differentiating $\Delta\tilde{\omega} = 0$ gives that $D_{\ell m}(\Delta\tilde{\omega}) - \Delta(D_{\ell m}\tilde{\omega}) = 0$ on $\mathbb{R}^n$, hence $D_{\ell m}\tilde{\omega}$ is harmonic and since $\tilde{\omega}$ was defined by linear transformations, $D^2\tilde{\omega}(0) = 0$, since $D^2v(0) = 0$, so $D_{\ell m}\tilde{\omega}(0) = 0$.

Then, for some constant $c > 0$, depending on P, Q , we have:

$$|D_{\ell m}\tilde{\omega}(x)| \leq \underbrace{|D_{\ell m}\tilde{\omega}(0)|}_{=0} + |x|^\alpha [D^2\omega]_{\alpha; \mathbb{R}^n} \leq C|x|^\alpha \underbrace{[D^2v]_{\alpha; \mathbb{R}^n}}_{\leq 1} \leq C|x|^\alpha,$$

so we have $D_{\ell m}\tilde{\omega}$ has sub-linear growth, so $D_{\ell m}\tilde{\omega} = \text{constant} = D_{\ell m}\tilde{\omega}(0) = 0$.

Linearity of the transformation defining $\tilde{\omega}$ implies that $D_{\ell m}v = 0$ in $\mathbb{R}^n$, but we saw that $D_{\ell m}v \neq 0$ on $\mathbb{R}^n$.

Thus, we have a contradiction if Baby Schauder fails. $\square$

Theorem XII.1.13. Higher Interior Schauder.

If $\Omega \subset \mathbb{R}^n$ is open, L is strictly elliptic in Ω with $a^{ij}, b^i, c \in C^{k,\alpha}(\Omega)$ for $k \geq 0$ and $\alpha \in (0, 1)$, then if $u \in C^{2,\alpha}(\Omega)$ and $f \in C^{k,\alpha}(\Omega)$ are such that $Lu = f$ in Ω , it holds then $u \in C^{k+2,\alpha}(\Omega)$, and:

$$|u|_{k+2,\alpha;\tilde{\Omega}} \leq C(|u|_{0,\hat{\Omega}} + |f|_{k,\alpha,\hat{\Omega}}) \quad \text{for all } \tilde{\Omega} \Subset \hat{\Omega} \Subset \Omega,$$

where $C > 0$ is a constant depending on $n, k, \alpha, L, \tilde{\Omega}$, and $\hat{\Omega}$.

Proof. $k = 0$ is precisely general interior Schauder.

For $k = 1$, it will follow by using difference quotients. For $\ell = 1, \dots, n$ and $0 < |h| \text{dist}(\tilde{\Omega}, \partial\hat{\Omega})$, we compute that:

$$\begin{aligned} L(D_\ell^h u)(x) &= \frac{1}{h} [L(u(x + he_\ell)) - Lu(x)] = \frac{1}{h} [L(u(x + he_\ell)) - Lu(x + he_\ell) + f(x + he_\ell) - f(x)] \\ &= -(D_\ell^h a^{ij})(x) D_{i,j} u(x + he_\ell) - (D_\ell^h b^i)(x) D_i u(x + he_\ell) - (D_\ell^h c)(x) u(x + he_\ell) + (D_\ell^h f)(x) = g_\ell^h(x). \end{aligned}$$

Then, by the $k = 0$ case applied to $D_\ell^h u$, we get:

$$|D_\ell^h|_{z,\alpha;\tilde{\Omega}} \leq (|u|_{0,\hat{\Omega}} + |g_\ell^h|_{0,\alpha;\hat{\Omega}}) \leq C(|u|_{2,\hat{\Omega}}).$$

Then, by Arzelà-Ascoli, there is a subsequential convergence of $D_\ell^h u \xrightarrow{C^2} D_\ell u \in C^{2,\alpha}(\tilde{\Omega})$, but then $u \in C^{3,\alpha}(\tilde{\Omega})$ with the desired estimate to obtain the higher interior Schauder.

For $k \geq 2$, assume $u \in C^{l+1,\alpha}(\Omega)$, $a^{ij}, b^i, c, f \in C^{k,2}(\Omega)$ to get for $|\gamma| = k - 1$:

$$L(D^\gamma u) = D^\gamma f - \sum_{\beta < \gamma} \left(\frac{\gamma!}{\beta!(\gamma - \beta)!} D^{\gamma - \beta} a^{ij} D^\beta D_{i,j} u + D^{\gamma - \beta} b^i D^\beta D_i u + D^{\gamma - \beta} c D^\beta u \right) \in C^{1,\alpha}(\Omega). \quad \square$$

Theorem XII.1.14. Divergence Form Unit Scaling Interior Schauder.

If $u \in C^{1,\alpha}(B_1(0))$ is a weak solution to:

$$\text{div}(A \nabla u) = D_i(a^{ij} x D_j u(x)) = 0$$

in $B_1(0)$, where $a^{ij} \in C^{0,\alpha}(\overline{B_1(0)})$ for $\alpha \in (0,1)$ and $0 \leq \lambda \text{Id} \leq A \leq \Lambda \text{Id}$ (uniform elliptic), then:

$$|u|_{1,\alpha;B_{1/2}(0)} \leq C \|u\|_{L^\infty(B_1(0))}$$

for a constant $C > 0$, depending on $n, \alpha, \lambda, \Lambda$, and $|a_{0,\alpha;B_1(0)}^{ij}|$. Moreover, if $u \in C^{k+1,\alpha}(\overline{B_1(0)})$ solves the divergence form with $a^{ij} \in C^{k,\alpha}(\overline{B_1(0)})$ for $k \geq 0$, then:

$$|u|_{k+2,\alpha;B_{1/2}(0)} \leq c \|u\|_{L^\infty(B_1(0))},$$

where $c > 0$ depends on $k, n, \alpha, \lambda, \Lambda$, and $|a_{k,\alpha;B_1(0)}^{ij}|$.

Assuming this, we now complete Hilbert's XIX problem.

Theorem XII.1.15. Completion of Hilbert's XIX Problem.

If $\Omega \subset \mathbb{R}^n$ is open, $L : \mathbb{R}^n \rightarrow \mathbb{R}$ is smooth and uniformly convex, then local minimizers of:

$$\mathcal{E}(u) = \int_{\Omega} L(\nabla u)$$

are smooth in Ω .

Proof. We saw from De Giorgi-Nash that local minimizers are in $C^{1,\alpha}(\Omega)$, we did this by showing that:

$$\delta_h u = \frac{u(x+h) - u(x)}{|h|}$$

weakly solved the PDE $\text{div}(\tilde{A} \nabla(\delta_h u)) = 0$ for $\tilde{A} \in C^{0,\alpha}$ uniformly elliptic. Then, the Divergence form of unit scale Schauder implies that:

$$|\delta_h u|_{1,\alpha;\tilde{\Omega}} \leq C \|\delta_h u\|_{L^\infty(\Omega)} \leq C |u|_{1,\alpha;\hat{\Omega}} \quad \text{for } \tilde{\Omega} \Subset \hat{\Omega} \Subset \Omega.$$

By Arzelà-Ascoli, there is a subsequential convergence of $\delta_h u$ to $Du \in C^{1,\alpha}$, which implies that $u \in C^{2,\alpha}$. Iteratively, using $\delta_h u$, we get that $u \in C^{k,\alpha}$ for all $k \geq 1$, implying that $u \in C^\infty$. $\square$

XII.2 Boundary and Global Schauder Estimates

Theorem XII.2.1. Global Schauder Estimates.

If $\Omega \subset \mathbb{R}^n$ is open, bounded, connected, with $\partial\Omega$ being $C^{2,\alpha}$ (recall there is a some flattening function $\varphi : \partial\Omega \rightarrow \text{half ball}$), L is strictly ($\lambda > 0$) elliptic, with $a^{ij}, b^i, c \in C^{0,\alpha}(\overline{\Omega})$ with $\max_{i,j} \{|a^{ij}|_{0,\alpha;\Omega} + |b^i|_{0,\alpha;\Omega} + |c|_{0,\alpha;\Omega}\} \leq \beta$ for $\alpha \in (0,1)$, then if $u, \varphi \in C^{2,\alpha}(\overline{\Omega})$ and $f \in C^{0,\alpha}(\overline{\Omega})$ are such that:

$$\begin{cases} Lu = f, & \text{in } \Omega, \\ u = \varphi, & \text{in } \partial\Omega, \end{cases}$$

then it holds that:

$$|u|_{2,\alpha;\Omega} \leq C(|u|_{0;\Omega} + |f|_{0,\alpha;\Omega} + |\varphi|_{2,\alpha;\Omega})$$

for a constant $C > 0$ depending on $n, \alpha, \beta, \lambda$, and Ω .

Remark XII.2.2.

- One can prove, in the *usual* manner, high global Schauder estimates.
- If we instead only assumed $\varphi \in C^{2,\alpha}(\partial\Omega)$, then we could (with some extra work) extend φ to a function $\tilde{\varphi} \in C^{2,\alpha}(\Omega)$ with $\varphi = \tilde{\varphi}$ on $\partial\Omega$ and $|\tilde{\varphi}|_{2,\alpha;\Omega} \leq C|\varphi|_{2,\alpha;\partial\Omega}$ for a constant $C > 0$, depending on n, α , and Ω .

This implies that we can replace $|\tilde{\varphi}|_{2,\alpha;\Omega}$ with $|\varphi|_{2,\alpha;\partial\Omega}$ in the global Schauder case. $\lrcorner$

The ideal is to prove a “local/unit scale” boundary estimate and use a covering argument to get the global Schauder.

Let $B_R^+(x) = B_R(x) \cap \{x_n > 0\} \subset \mathbb{R}^n$.

Theorem XII.2.3. Boundary Unit Scale Schauder.

If L is strictly ($\lambda > 0$) elliptic, with $\max_{i,j} \{|a^{ij}|_{0,\alpha;B_1^+(0)} + |b^i|_{0,\alpha;B_1^+(0)} + |c|_{0,\alpha;B_1^+(0)}\} \leq \beta$, $\alpha \in (0, 1)$, then if $u, \varphi \in C^{2,\alpha}(\overline{B_1^+(0)})$ and $f \in C^{0,\alpha}(\overline{B_1^+(0)})$ such that:

$$\begin{cases} Lu = f, & \text{in } B_1^+(0), \\ u = \varphi, & \text{in } \partial B_1^+(0), \end{cases}$$

then $|u|_{2,\alpha;B_{1/2}^+(0)} \leq C(|u|_{0;B_1^+(0)} + |f|_{0,\alpha;B_1^+(0)} + |\varphi|_{2,\alpha;B_1^+(0)})$ for C depending on n, α, β , and λ .

Now, we will sketch the proof.

Proof of sketch. First, by the reverse triangle inequality, we may assume, without loss of generality (that is $u \leftarrow u - \varphi$), that $\varphi = 0$.

Second, we need a boundary version of the absorption lemma:

Lemma XII.2.4. Simon’s Boundary Absorption.

Let $\alpha \in [0, \infty)$, $\theta \in (0, 1)$, and S as a nonnegative function on sub-balls of $B_R^+(0) \subset \mathbb{R}^n$ which is sub-additive, then there exists $\delta_{\text{BSAL}} > 0$ depending on n, λ , and θ so that if both:

$$\begin{cases} B_\rho^+(x) \subset B_R^+(0) \implies \rho^\lambda S(B_{\theta\rho}^+(x)) \leq \delta_{\text{BSAL}} \rho^\lambda S(B_\rho^+(x)) + \gamma, \\ B_\rho(x) \subset B_R^+(0) \implies \rho^\lambda S(B_{\theta\rho}(x)) \leq \gamma \quad \text{for } \gamma \in \mathbb{R}_+, \end{cases}$$

then we have:

$$\rho^\lambda S(B_{\theta\rho}^+(x)) \leq C\gamma$$

for $C > 0$ depending on n, λ , and θ .

By rescaling and using the above lemma, we reduce to:

Lemma XII.2.5. Boundary Baby Schauder.

For $\delta > 0$ we have:

$$[D^2u]_{\alpha; B_{1/2}^+(0)} \leq \delta [D^2u]_{\alpha; B_1^+(0)} + C(|u|_{2; B_1^+(0)} + |f|_{0, \alpha; B_1^+(0)})$$

for $C > 0$ dependent on $n, \alpha, \beta, \lambda$, and δ .

We follow the same steps as before, assuming Boundary Baby Schauder fails a long a sequence of solutions, $\{u_k\} \subset C^{2, \alpha}(\overline{B_1^+(0)})$ with $c \leftarrow k (\rightarrow \infty)$, we choose $\{z_k\}, \{y_k\} \subset B_{1/2}^+(0)$ achieving at least half the term $[D^2u_k]_{0, \alpha; B_{1/2}^+(0)}$ and see $\rho_k = |z_k - y_k| \rightarrow 0$.

We have to consider two cases, that is:

$$\limsup_{k \rightarrow \infty} \frac{\text{dist}(z_k, B + 1/2^+(0) \cap \{x_n = 0\})}{\rho_k}$$

is either (i) ∞ or (ii) $= \mu < +\infty$.

(i) We proceed exactly as in the interior case, which is contradiction by Liouville.

(ii) We reflect v on the half-space to get harmonic function $\tilde{v}$ on $\mathbb{R}^n$, then reduced to interior case again.

Thus, this completes the sketch of the proof for both cases. $\square$

Theorem XII.2.6. Divergence form Schauder.

Consider the $C^{1, \alpha}$ weak solutions of $\text{div}(A \nabla u) = 0$ in $B_1(0)$ for $a^{i,j} \in C^{0, \alpha}$, $0 < \lambda \text{Id} \leq A \leq \Lambda \text{Id}$, so we have:

$$|u|_{1, \alpha; B_{1/2}(0)} \leq C|u|_{0; B_1(0)}$$

where C depends on n, λ, Λ, A , and moreover, if $u \in C^{k+1, \alpha}$, $a^{i,j} \in C^{k, \alpha}$, then:

$$|u|_{k+1, \alpha; B_{1/2}(0)} \leq C|u|_{0; B_1(0)},$$

where C depends on n, λ, Λ , and A .

Proof of sketch. We can loosely give the steps as follows:

(i) First reduce to the baby Schauder for the PDE:

$$[\nabla u]_{\alpha; B_{1/2}(0)} \leq \delta [\nabla u]_{\alpha; B_1(0)} + C|\nabla u|_{0; B_1(0)}.$$

(ii) We got by contradiction and choose $\{z_k\}, \{y_k\} \subset B_{1/2}(0)$ achieving at least $\frac{1}{2}[Du]_{\alpha; B_{1/2}(0)}$ with:

$$\rho_k = \frac{|z_k - y_k|}{2}, \quad x_k = \frac{z_k + y_k}{2}, \quad \rho_k \rightarrow 9,$$

and we instead define:

$$\tilde{v}_k(x) = \frac{u_k(x_k + \rho_k x) + u_k(x_k - \rho_k x) - 2u_k(x_k)}{\rho_k^{1+\alpha} [\nabla u_k]_{\alpha; B_1(0)}},$$

so we have:

- $\tilde{v}_k(0) = 0$ and $\nabla \tilde{v}_k(0) = 0$,
- $\nabla \tilde{v}_k \neq 0$ on $\mathbb{R}^n$, and
- $[\nabla \tilde{v}_k]_{\alpha; B_{1/2\rho_k}(0)} \leq 2$.

(iii) Use Arzelá-Ascoli, to get $v \in C^{1,\alpha}(\mathbb{R}^N)$ which, up to rescaling and translating, it satisfies that:

$$\int_{\mathbb{R}^n} \nabla \varphi \cdot \nabla v = 0 \quad \text{for all } \varphi \in C_c^\infty(\mathbb{R}^n),$$

so we have ∇v is weakly, hence by Weyl classically, harmonic.

Hence, the estimate pass to v hence Liouville gives a contradiction.

For the Divergence form Schauder, we go by induction and if $a^{i,j} \in C^{k,\alpha}$ and $u \in C^{2,\alpha}$, so we have:

$$\begin{cases} a^{i,j} D_{i,j} u = -D_i a^{i,j} D_j u, \\ Lu = f(a^{i,j}, \nabla v), \end{cases}$$

then we have it leading to the Divergence form. □

XII.3 The Continuity Method

We are now going to focus on solve the Dirichlet problem on $\Omega \subset \mathbb{R}^n$ on open, connected, bounded, for L strictly elliptic. In other words, given L, f, φ, Ω with suitable regularity, we want to find, at least, a C^2 function u solving:

$$\begin{cases} Lu = f, & \text{in } \Omega, \\ u = \varphi, & \text{on } \partial\Omega. \end{cases}$$

We reduce to solving the Dirichlet problem to the Poisson equation.

Theorem XII.3.1. The Continuity Method.

If $\Omega \subset \mathbb{R}^n$ is open, connected, bounded, with $\partial\Omega$ being $C^{2,\alpha}$, L is strictly elliptic with $a^{i,j}, b^i, c \in C^{0,\alpha}(\overline{\Omega})$ and $c \leq 0$, then the Dirichlet problem for L has a unique solution $u \in C^{2,\alpha}(\overline{\Omega})$ for any $f \in C^{0,\alpha}(\overline{\Omega})$ and $\varphi \in C^{2,\alpha}(\overline{\Omega})$ if and only if the Dirichlet problem for Δ has a unique solution $\tilde{u} \in C^{2,\alpha}(\overline{\Omega})$, for any $\tilde{f} \in C^{0,\alpha}(\overline{\Omega})$ and $\varphi \in C^{2,\alpha}(\overline{\Omega})$.

Remark XII.3.2. This is not an existence result. ┘

Proof. By linearity, we can assume without loss of generality that $\varphi = 0$. We thus consider $C_0^{2,\alpha}(\overline{\Omega}) = \{u \in C^{2,\alpha}(\overline{\Omega}) : u = 0 \text{ on } \partial\Omega\}$.

This is a Banach space with $C^{2,\alpha}$ norm (complete metric space).

For $t \in [0, 1]$, we defined $L_t : C_0^{2,\alpha}(\overline{\Omega}) \rightarrow C^{0,\alpha}(\Omega)$ by setting:

$$L_t u = t \cdot Lu + (1-t)\Delta,$$

where L_t is non-divergence form and is linear, so that $a_t^{ij} = ta^{ij} + (1-t)\delta^{ij}$, $b_t^i = tb^i$, and $c_t = tc \leq 0$. If $\lambda > 0$ is the ellipticity constant for L , then:

$$a_t^{ij} \xi_i \xi_j = ta^{ij} \xi_i \xi_j + (1-t)\delta^{ij} \xi_i \xi_j \geq t\lambda |\xi|^2 + (1-t)|\xi|^2 = |\xi|^2 \underbrace{(t\lambda + (1-t))}_{\geq \min\{1, \lambda\}}.$$

This implies that L_t is strictly elliptic for all $t \in [0, 1]$ with constant $\min\{1, \lambda\} > 0$.

Also $L_0 = \Delta$ and $L_1 = L$.

If $\max_{i,j} \{|a^{ij}|_{0,\alpha;\Omega} + |b^i|_{0,\alpha;\Omega} + |c|_{0,\alpha;\Omega}\} \leq \beta$, then by their definition, we have:

$$\max_{i,j} \{|a_t^{ij}|_{0,\alpha;\Omega} + |b_t^i|_{0,\alpha;\Omega} + |c_t|_{0,\alpha;\Omega}\} \leq \max\{1, \beta\}.$$

We thus apply the global Schauder estimates to get that if $u \in C_0^{2,\alpha}(\overline{\Omega})$, then (by $f \leftarrow L_t u$), we have:

$$|u|_{2,\alpha;\Omega} \leq C(|u|_{0;\Omega} + |L_t u|_{0,\alpha;\Omega})$$

where $c > 0$ depends on $n, \alpha, \min\{1, \alpha\}, \max\{1, \beta\}$, and Ω (but not t or u).

By the maximum principle a priori estimates (which needed $c \geq 0$), we have:

$$|u|_{0;\Omega} \leq |u|_{0;\partial\Omega} + C \left| \frac{|L_t u|}{\min\{1, \lambda\}} \right|_{0,\alpha;\Omega} = |u|_{0;\partial\Omega} + C \frac{|L_t u|_{0,\alpha;\Omega}}{\min\{1, \lambda\}},$$

where $c > 0$ has the same dependencies as above. Thus, for a bigger $c > 0$, we have:

$$|u|_{2,\alpha;\Omega} \leq C|L_t u|_{0,\alpha;\Omega}.$$

Thus, L_t is an injective map from $C_0^{2,\alpha}(\overline{\Omega}) \rightarrow C^{0,\alpha}(\Omega)$, i.e., we have uniqueness of solutions to the Dirichlet Problem if they exist.

Note that solvability of the Dirichlet problem and L_t , for $\varphi = 0$, for all $f \in C^{0,\alpha}(\overline{\Omega})$ is equivalent to surjectivity of L_t (i.e., L_t^{-1} is well defined).

We will prove that L_t being surjective for some $t \in [0, 1]$ implies L_t is surjective for all $t \in [0, 1]$. In other words, we show that:

$$\{t \in [0, 1] : L_t \text{ is surjective}\}$$

is closed and open on $[0, 1]$, so we have know it is either $\emptyset$ or $[0, 1]$ by connectedness.

Supposing that L_s is surjective for a fixed $s \in [0, 1]$, then we have:

$$|L_s^{-1}g|_{2,\alpha;\Omega} \leq C|g|_{0,\alpha;\Omega} \quad \text{for all } g \in C^{0,\alpha}(\overline{\Omega}).$$

Thus, for $t \in [0, 1]$ by linearity of L_s^{-1} that:

$$\begin{aligned} L_t u = f &\iff L_s u + (L_t - L_s)u = f \\ &\iff u = L_s^{-1}f + L_s^{-2}((L_s - L_t)u) = T(u). \end{aligned}$$

So, solving $L_t u = f$ is equivalently finding $u \in C_0^{2,\alpha}(\overline{\Omega})$ with $u = T(u)$.

Notice that for $f \in C^{0,\alpha}(\overline{\Omega})$, we have $T : C_0^{2,\alpha}(\overline{\Omega}) \rightarrow C_0^{2,\alpha}(\overline{\Omega})$.

We show existence of a fixed point for T , by the contraction mapping principle, we thus need to show, for t close to s , that T is a contraction, we compute for $u, v \in C_0^{2,\alpha}(\overline{\Omega})$ that (recall $L_t = tL + (1-t)\Delta$):

$$T(u) - T(v) = L_s^{-1}((L_s - L_t)(u - v)) = (s - t)L_s^{-1}((L - \Delta)(u - v)).$$

Then, we can apply the earlier inequality to see that for $g = (L - \Delta)(u - v)$ that:

$$|T(u) - T(v)|_{2,\alpha;\Omega} \leq c|s - t| \cdot |(L - \Delta)(u - v)|_{0,\alpha;\Omega} \leq \underbrace{\tilde{C}|s - t|}_{\text{contradiction if } < 1} \cdot |u - v|_{0,\alpha;\Omega}.$$

Thus, T is a contradiction if $\tilde{c}|s - t| < 1$ which is equivalently $|s - t| \leq 1/\tilde{c}$.

So, if this holds, contraction mapping principle, T has a unique fixed point $u \in C_0^{2,\alpha}(\overline{\Omega})$, that is equivalently that $L_t u = f$ having a solution, so equivalently L_t is surjective.

To summarize, if L_s is surjective, then so is L_t for $|s - t| < 1/\tilde{c}$. Thus, by covering $[0, 1]$ in intervals of length say $1/2\tilde{c}$, we see that, as $\tilde{c}$ is independent of S , that L_t is surjective for all $t \in [0, 1]$ if and only if it is surjective for some $L_s \in [0, 1]$.

Thus, if the Dirichlet problem is uniquely solvable for $\Delta = L_0$, it is equivalent that the Dirichlet problem is uniquely solvable for $L = L_1$. $\square$

Recall from earlier (VII.3).

Theorem XII.3.3. C^∞ Poisson on Balls.

If $B \subset \mathbb{R}^n$ is a ball, and $f, \varphi \in C^\infty(\overline{B})$, then there exists a unique solution $u \in C^\infty(\overline{B})$ to the Poisson equation:

$$\begin{cases} \Delta u = f, & \text{in } B, \\ u = \varphi, & \text{on } \partial B, \end{cases}$$

which also works for Ω with $\partial\Omega$ being smooth.

We cannot apply continuity method yet.

Proposition XII.3.4. $C^{2,\alpha}$ Poisson on Balls.

If $B \subset \mathbb{R}^n$ is a ball, $f \in C^{0,\alpha}(\overline{B})$ and $\varphi \in C^0(\partial B)$, then there is a unique solution $u \in C^{2,\alpha}(B) \cap C^0(\overline{B})$, to

the Poisson equation:

$$\begin{cases} \Delta u = f, & \text{in } B, \\ u = \varphi, & \text{on } \partial B. \end{cases}$$

Moreover, if $\varphi \in C^{2,\alpha}(\partial B)$, then the above solution $u \in C^{2,\alpha}(\bar{B})$.

Proof. We first extend f to $\tilde{f} \in C_c^{0,\alpha}(\mathbb{R}^n)$ and φ to $\tilde{\varphi} \in C_c^0(\mathbb{R}^n)$ and then mollify, for $\epsilon > 0$ small enough, to get $\tilde{f}_\epsilon = \tilde{f} * \eta_\epsilon$ and $\tilde{\varphi}_\epsilon = \tilde{\varphi} * \eta_\epsilon$ which are both in $C_c^\infty(\mathbb{R}^n)$.

By C^∞ Poisson on balls with data $\tilde{f}_\epsilon$ and $\tilde{\varphi}_\epsilon$, we get a unique solution, $u_\epsilon \in C^\infty(\bar{B})$, with:

$$\begin{cases} \Delta u_\epsilon = \tilde{f}_\epsilon, & \text{in } B, \\ u_\epsilon = \tilde{\varphi}_\epsilon, & \text{on } \partial B. \end{cases}$$

Note that:

$$|\tilde{f}_\epsilon|_{0;\mathbb{R}^n} \leq |\tilde{f}|_{0;\mathbb{R}^n} \quad \text{and} \quad |\tilde{\varphi}_\epsilon|_{0;\mathbb{R}^n} \leq |\tilde{\varphi}|_{0;\mathbb{R}^n},$$

and since we have:

$$|\tilde{f}_\epsilon(x) - \tilde{f}_\epsilon(y)| \leq \left| \int_{\mathbb{R}^n} |\tilde{f}(x-z) - \tilde{f}(y-z)| \eta_\epsilon(z) dz \right| \leq \left| \int_{\mathbb{R}^n} [\tilde{f}]_{0,\alpha;\mathbb{R}^n} |x-y|^\alpha \eta_\epsilon(z) dz \right|,$$

and by $[\tilde{f}_\epsilon]_{0,\alpha;\mathbb{R}^n} \leq [\tilde{f}]_{0,\alpha;\mathbb{R}^n}$ and similarly we get:

$$[\tilde{\varphi}]_{0,\alpha;\mathbb{R}^n} \leq [\tilde{\varphi}]_{0,\alpha;\mathbb{R}^n}.$$

For $0 < \sigma < \epsilon$ being sufficiently small, we have that as:

$$\begin{cases} \Delta(u_\epsilon - u_\sigma) = \tilde{f}_\epsilon - \tilde{f}_\sigma, & \text{in } B, \\ (u_\epsilon - u_\sigma) = \tilde{\varphi}_\epsilon - \tilde{\varphi}_\sigma, & \text{on } \partial B, \end{cases}$$

which, by the a priori estimate and maximum principle that:

$$|u_\epsilon - u_\sigma|_{0;\bar{B}} \leq |\tilde{\varphi}_\epsilon - \tilde{\varphi}_\sigma|_{0;\partial B} + C|\tilde{f}_\epsilon - \tilde{f}_\sigma|_{0;B}.$$

where $C > 0$ is independent of ϵ, σ , and the data.

Since both $\tilde{f}_\epsilon \rightarrow \tilde{f}$ and $\tilde{\varphi}_\epsilon \rightarrow \tilde{\varphi}$ uniformly on $\bar{B}$ and ∂B respectively, we see that $\{u_\epsilon\}$ (as a sequence) being Cauchy implies uniformly convergent, in $\bar{B}$, to u .

As $u_\epsilon = \tilde{\varphi}_\epsilon$ on ∂B , this implies that as $u_\epsilon \rightarrow u$ uniformly, we have that $u = \tilde{\varphi} = \varphi$ on ∂B . By the interior Schauder estimates on sub-balls, $\tilde{B} \Subset B$, we have:

$$|u_\epsilon|_{2,\alpha;\tilde{B}} \leq C(|u_\epsilon|_{0;\tilde{B}} + |\tilde{f}_\epsilon|_{0,\alpha;\tilde{B}}) \leq C(|\tilde{\varphi}_\epsilon|_{0;\partial \tilde{B}} + |\tilde{f}|_{0,\alpha;\tilde{B}}) \leq C(|\varphi|_{0,\partial B} + |f|_{0,\alpha;\tilde{B}}).$$

Thus, by Arzelá-Ascoli, up to a subsequence (not relabeled) $u_\epsilon \rightarrow u \in C_{\text{loc}}^{2,\alpha}(B)$ which by the uniqueness of limit must coincide with u , i.e., $v = u \in C^{2,\alpha}(B)$.

Hence, this implies that $u \in C^{2,\alpha}(B) \cap C^0(\bar{B})$. As $\Delta u_\epsilon = \tilde{f}_\epsilon$, the C_{loc}^2 convergence of $u_\epsilon \rightarrow u$ implies that $\Delta u = f$ in B also with $u = \varphi$ on ∂B .

Finally, in the case where $\varphi \in C^{2,\alpha}(\partial B)$, we can argue in the same way by replacing the interior Schauder estimate by the global Schauder estimation. $\square$

Hence, the continuity method implies that we can solve Dirichlet problem on balls for every strictly elliptic (non-divergence) operator with $C \leq 0$, in the class $C^{2,\alpha}$.

XII.4 Perron's Method

We now use solvability on balls to get solvability on general domains (*i.e.*, $\partial\Omega$ being less regular). We use the hypothesis:

$$\begin{cases} \Omega \subset \mathbb{R}^n \text{ is open, bounded, and connected,} \\ L \text{ is strictly elliptic with } a^{ij}, b^i, c \in C^{0,\alpha}(\overline{\Omega}) \text{ and } C \leq 0, \\ f \in C^{0,\alpha}(\overline{\Omega}), \varphi \in C^0(\partial\Omega). \end{cases}$$

Proposition XII.4.1. A function $u \in C^2(\Omega)$ is a subsolution to $Lu = f$, *i.e.*, $Lu \geq f$, if and only if, for every ball $B \Subset \Omega$ and $w \in C^2(B) \cap C^0(\overline{B})$ with $Lw = f$ in B , with $w \geq u$ on ∂B , then $w \geq u$ in B .

Proof. ($\implies$.) If $u \in C^2(\Omega)$ and $Lu \geq f$, if $B \Subset \Omega$ is a ball and $w \in C^2(\overline{B})$ solves $Lw = f$, with $w \geq u$ on ∂B , then the comparison principle, and then $w \geq u$ in B .

($\impliedby$.) If $Lu(y) < f(y)$ for $y \in \Omega$, then as $Lu \in C^0(\Omega)$ for a small ball $B = B_\delta(y) \Subset \Omega$, we have $Lu < f$ on B . The $C^{2,\alpha}$ Poisson on balls tells us that we can find a unique $w \in C^{2,\alpha}(B) \cap C^0(\overline{B})$ solving:

$$\begin{cases} Lw = f, & \text{in } B, \\ w = u, & \text{on } \partial B. \end{cases}$$

As $Lu < f = Lw$ in B and $w = u$ on ∂B , then the comparison principle gives that $w \leq u$ in B and strong maximum principle implies that $w < u$ in B (or else $w = u$ in B , and then $Lw = Lu = f$ is a contradiction). This contradicts our assumption (which would imply that $w \geq u$ in B). $\square$

Definition XII.4.2. A continuous function $u \in C^0(\Omega)$ is a continuous subsolution to $Lu = f$ in Ω , if for every ball $B \Subset \Omega$, and $w \in C^2(B) \cap C^0(\overline{B})$, for $Lw = f$ and since $w \geq u$ on ∂B , then $w \geq u$ in B . $\lrcorner$

Remark XII.4.3. One can similarly define continuous supersolution.

This just says that a continuous function is a subsolution, if it obeys the comparison principle on sub-balls of Ω , but this definition requires no derivative. $\lrcorner$

Lemma XII.4.4. Properties of Continuous Subsolutions.

- (i) If $u, v \in C^0(\overline{\Omega})$ are continuous sub and super solutions to $L(\cdot) = f$, and $u \leq v$ on $\partial\Omega$, then $u \leq v$ in Ω .

(ii) If $u \in C^0(\overline{\Omega})$ is a subsolution to $Lu = f$, then:

$$\sup_{\Omega}(u) \leq \sup_{\partial\Omega}(u_+) + C \sup_{\Omega}|f|,$$

for $c > 0$ depending on n, L , and Ω .

(iii) If $u, \tilde{u} \in C^0(\overline{\Omega})$ are continuous subsolution to $Lu = f$, then so is $\max\{u, \tilde{u}\} \in C^0(\overline{\Omega})$.

Proof. First, we note that (ii) follows from (i) constructing the identical supersolution to $Lu = f$ as in the proof of the maximum principle a priori estimates.

To show (i), we go by contradiction and suppose $v(x) < u(x)$ for some $x \in \Omega$. Since $u - v \leq 0$ on $\partial\Omega$, we have that $M = \sup_{\Omega}(u - v) > 0$ and attained in Ω , which we may as well suppose is attained at x . By continuity, there is a ball $B = B_{\delta}(x) \Subset \Omega$ such that:

$$u(y) - v(y) > 0 \quad \text{on } B.$$

We then solve the Dirichlet problem on B to find $\bar{u}, \bar{v} \in C^{2,\alpha}(B) \cap C^0(\overline{B})$, with:

$$\begin{cases} L\bar{u} = f, & \text{in } B, \\ \bar{u} = u, & \text{on } \partial B, \end{cases} \quad \text{and} \quad \begin{cases} L\bar{v} = f, & \text{in } B, \\ \bar{v} = v, & \text{on } \partial B. \end{cases}$$

As u, v are sub/supersolutions, we then have:

$$\begin{cases} u \leq \bar{u} \\ v \geq \bar{v} \end{cases} \quad \text{in } B,$$

and hence we have $\bar{u} - \bar{v} \geq u - v$ in B , thus $\sup_B(\bar{u} - \bar{v}) \geq M$.

As $L(\bar{u} - \bar{v}) = 0$ in B and $c \leq 9$, the weak maximum principle implies that:

$$\sup_B(\bar{u} - \bar{v}) \leq \sup_{\partial B}((\bar{u} - \bar{v})_+) = \sup_{\partial B}((u - v)_+) \leq M.$$

This implies that $\sup_B(\bar{u} - \bar{v}) = M > 0$, so $\bar{u} - \bar{v}$ attains an interior maximum in B , so $(\bar{u} - \bar{v}) = M$ in B by the strong maximum principle. However, $\bar{u} - \bar{v} \neq u - v$ at y but $(u - v)(y) < M$, which is a contradiction. For (iii), let $B \Subset \Omega$ be a ball, $w \in C^2(B) \cap C^0(\overline{B})$ such that $Lu = f$ in B with $w \geq \max\{u, \tilde{u}\}$ in ∂B , but as $\max\{u, \tilde{u}\}$, it implies that $w \geq u, \tilde{u}$ on ∂B so as $u, \tilde{u}$ are continuous subsolutions, we have $w \geq u, \tilde{u}$ in B , hence we have:

$$w \geq \max\{u, \tilde{u}\} \quad \text{in } B.$$

This implies that $\max\{u, \tilde{u}\}$ is a continuous subsolutions. □

Lemma XII.4.5. Lift of a Subsolution.

If $u \in C^0(\overline{\Omega})$ is a subsolution and $B \Subset \Omega$ is a ball, then the **lift of u in B** , defined to be:

$$\hat{u} = \begin{cases} u, & \text{in } \overline{\Omega} \setminus B, \\ \tilde{u}, & \text{in } B, \end{cases}$$

where $\tilde{u} \in C^{2,\alpha}(B) \cap C^0(\overline{B})$ is the unique solution to Dirichlet problem in B with boundary data u . Then, $\hat{u} \in C^0(\overline{\Omega})$ and $\hat{u}$ is a subsolution to $Lu = f$.

Proof. First, we have $\hat{u} \in C^0(\overline{\Omega})$ by construction as $B \Subset \Omega$ is a ball and $w \in C^2(B') \cap C^0(\overline{B'})$ is such that $Lw = f$ in B' and $w \geq \tilde{u}$ on $\partial B'$. As $\hat{u}$ is a solution on B with $\hat{u} = u$ on ∂B and u is a subsolution, then $\hat{u} \geq u$ on B and hence $\hat{u} \geq u$ on Ω (As $\hat{u} = u$ on $\Omega \setminus B$).

Thus, $w \geq \hat{u} \geq u$ on $\partial B'$ so as u is a subsolution, we have $w \geq v$ on B' ; hence $w \geq \hat{u}$ on $B' \setminus B$ (as $\hat{u} = u$ on $\Omega \setminus B$). If $B' \cap B = \emptyset$ we would be done, so we suppose not and note that we have both:

$$\begin{cases} Lw = f, & \text{in } B' \cap B \subset B' \\ w \geq \hat{u}, & \text{on } \overline{B} \cap \partial B' \subset \partial B' \end{cases} \quad \text{and} \quad \begin{cases} L\hat{u} = f, & \text{in } B' \cap B \subset B \\ w \geq \hat{u} = u, & \text{on } \partial B \cap \overline{B'} \subset \partial B. \end{cases}$$

So we have $w \geq \hat{u}$ on $\partial(B \cap B') = (\partial B \cap \overline{B'}) \cup (\overline{B} \cap \partial B')$ and $L\hat{u} = Lw = f$ on $B \cap B'$, then the comparison principle gives us that $w \geq \hat{u}$ in $B \cap B'$. $\square$

Definition XII.4.6. Subfunctions.

We say that $u \in C^0(\overline{\Omega})$ is a subfunction for Dirichlet problem:

$$\begin{cases} Lu = f, & \text{in } \Omega, \\ u = \varphi, & \text{on } \partial\Omega, \end{cases}$$

if u is a continuous subsolution and $u \leq \varphi$ on $\partial\Omega$. $\lrcorner$

We will now solve the Dirichlet Problem by taking supremums over all subfunctions. To see why this might work, if we know $\tilde{u} \in C^{2,\alpha}(\Omega) \cap C^0(\overline{\Omega})$ solved Dirichlet problem, then if u is the supremum over all subfunctions, then $\tilde{u}$ is also a subfunction so $u \geq \tilde{u}$. As $\tilde{u} = \varphi$ on $\partial\Omega$ for each subfunction, $v \in C^0(\overline{\Omega})$, we have $v \leq \varphi = \tilde{u}$ on $\partial\Omega$. By property (i) in the continuity lemma, we have thus $v \leq \tilde{u}$ in Ω . Hence, taking supremums we have $u \leq \tilde{u} \leq u$ on Ω , thus if a solution to Dirichlet problem exists, it must coincide with the supremum of all sub functions.

Theorem XII.4.7. Interior Perron.

Let $u : \Omega \rightarrow \mathbb{R}$ be defined at each $x \in \Omega$ by setting:

$$u(x) = \sup\{v(x) : v \in C^0(\overline{\Omega}) \text{ is a subfunction for Dirichlet problem}\},$$

then $u \in C^{2,\alpha}(\Omega)$ and $Lu = f$ in Ω .

Remark XII.4.8. We do not yet claim that $u \in C^0(\overline{\Omega})$ or $u = \varphi$ on $\partial\Omega$. For this, we need some assumptions

on $\partial\Omega$. ┘

Proof. We first show that the set of subfunctions for Dirichlet problem is not empty, and that u is bounded on Ω . We suppose that:

$$\Omega \subset \{0 < x_1 < d\} \text{ where}$$

$$d = \text{diam}(\Omega) < \infty,$$

and set:

$$v(x) = -\sup_{\partial\Omega} |\varphi| - (e^{\alpha d} - e^{\alpha x_1}) \sup_{\Omega} |f|,$$

with $\alpha > 0$ being large, which one can check is a subsolution to $Lu = f$ and $v \leq \varphi$ on $d\Omega$, hence v is a subfunction to the Dirichlet problem. Also, for the subsolution lemma, we have $u \leq \sup_{\partial\Omega} |\varphi| + c \sup_{\Omega} |f|$, so u is bounded, hence well defined.

Now, if $x \in \Omega$, we fix $R > 0$ so that $B_R(x) \Subset \Omega$ and a sequence $\{v_j\} \subset C^0(\overline{\Omega})$ such that $v_j(x) \rightarrow u(x)$.

For each $j \geq 1$, let $\hat{v}_j$ be the lift of v_j in $B_R(x)$, so that $\hat{v}_j$ is a subsolution to $Lu = f$ in Ω , and $\hat{v}_j \leq \varphi$ on $\partial\Omega$, i.e., $\{\hat{v}_j\} \subset C^0(\overline{\Omega})$ are subfunctions to the Dirichlet problem with $v_j \leq \hat{v}_j \leq u$ on $B_R(x)$, so we have $\hat{v}_j(x) \rightarrow u(x)$.

Now, for each $\hat{v}_j$ solves:

$$\begin{cases} L\hat{v}_j = f, & \text{in } B_R(x), \\ \hat{v}_j = v_j, & \text{on } \partial B_R(x), \end{cases}$$

so by the interior Schauder estimates, we get:

$$|\hat{v}_j|_{2,\alpha;B_{R/2}(x)} \leq C(|\hat{v}_j|_{0;B_R(x)} + |f|_{0,\alpha;B_R(x)}).$$

Moreover, we have $\sup_{\Omega}(\hat{v}_j) \leq C(\sup_{\partial\Omega} |\varphi| + \sup_{\Omega} |f|)$.

By replacing v_j with $\max\{v_j, v\}$, we ensure that $\inf_{\Omega}(\hat{v}_j) \geq \inf_{\Omega}(v) > -\infty$.

So to summarize, we can control $|\hat{v}_j|_{0;B_R(x)} \leq C$, where $C > 0$ is independent of j .

Thus, we have $|\hat{v}_j|_{2,\alpha;B_{R/2}(x)} \leq \tilde{C}$ so by Arzelà-Ascoli, given us subsequential convergence in C^2 to $v \in C^{2,\alpha}(\overline{B_{R/2}(x)})$. As $L\hat{v}_j = f$ in $B_R(x)$, this implies that $Lv = f$ in $B_{R/2}(x)$ and $v(x) = u(x)$ (since $\hat{v}_j(x) \rightarrow u(x)$).

We claim that $u = v$ in $B_{R/16}(x)$ so $u \in C^{2,\alpha}(B_{R/16}(x))$ and $Lu = f$ in $B_{R/16}(x)$. Then, we will have that $u \in C^{2,\alpha}(\Omega)$ and $Lu = f$ in Ω , since x was arbitrary.

If $u \neq v$ in $B_{R/17}(x)$, then we had $y \in B_{R/16}(x)$ with $u(y) > v(y)$, as u is supremum of all subfunctions at y , there is a subfunction w , with:

$$v(y) < w(y) < u(y).$$

Letting $w_j = \max\{w, \hat{v}_j\}$ be a sequence of subfunctions, we consider $\hat{w}_j$ the lift of w_j in $B_{R/4}(x)$.

Arguing identically as for $\hat{v}_j$ above, we get $|\hat{w}_j|_{2,\alpha;B_{R/8}(x)} \leq C$.

Thus, by the Arzelà-Ascoli again, we have a subsequential C^2 convergence to some function $\hat{w} \in C^{2,\alpha}(B_{R/2}(x))$ with $L\hat{w} = f$ in $B_{R/8}(x)$ and $\hat{w}(x) = u(x)$.

As $v(y) = w(y) \leq w_j(y) \leq \hat{w}_j(y)$, so we have $v(y) < \hat{w}(y)$.

Now, strong maximal principle (or avoidance theorem,) says that as $Lv = L\hat{w}$ in $B_{R/8}(x)$ and $v(x) = \hat{w}(x)$,

so we have $v = \hat{w}$ in $B_{R/2}(x)$ but $v(y) < \hat{w}(y)$, which is a contradiction. $\square$

Theorem XII.4.9. Global Perron.

If in addition to $\Omega \subset \mathbb{R}^n$ being open, bounded, and connected, L being strictly elliptic, $a^{ij}, b^i, c \in C^{0,\alpha}(\overline{\Omega})$ where $c \leq 0$, and $f \in C^{0,\alpha}(\overline{\Omega})$, $\varphi \in C^0(\partial\Omega)$, Ω also satisfies the exterior sphere condition at all points of $\partial\Omega$, then the interior Perron solution u , in fact, solves the Dirichlet Problem and is such that $u \in C^{2,\alpha}(\Omega) \cap C^0(\overline{\Omega})$ with $u = \varphi$ on $\partial\Omega$.

Proof. As Ω satisfies exterior sphere at all $\xi \in \partial\Omega$, we can find a ball $B_R(y) \subset \mathbb{R}^n \setminus \overline{\Omega}$ with $\partial B_R(y) \cap \partial\Omega = \{\xi\}$, we then define on $\mathbb{R}^n \setminus \{\xi\}$ for parameters $\sigma, \tau > 0$, then function:

$$w(x) = \tau \left(\frac{1}{R^\sigma} - \frac{1}{|x - y|^\sigma} \right),$$

so we have $w(\xi) = 0$ and $w > 0$ in $\overline{\Omega} \setminus \{\xi\}$.

We use w as a "barrier" to extract information about u at ξ , we compute for $i, j \in \{1, \dots, n\}$ that:

$$\begin{cases} D_i(-|x - y|^{-\sigma}) = \sigma|x - y|^{-\sigma-1} \frac{(x_i - y_i)}{|x - y|} = \sigma|x - y|^{-\sigma-2}(x_i - y_i), \\ D_j(-|x - y|^{-\sigma}) = -\sigma(\sigma + 2)|x - y|^{-\sigma-4}(x_i - y_i)(x_j - y_j) + \sigma|x - y|^{-\sigma-2}\delta_{ij}. \end{cases}$$

Using $c \leq 0$ and $w \geq 0$ on Ω , and strict ellipticity of L , we get:

$$\begin{aligned} L(R^{-\sigma} - |x - y|^{-\sigma}) &= a^{ij}D_{ij}(-|x - y|^{-\sigma}) + b^iD_i(-|x - y|^{-\sigma}) + c(R^{-\sigma} - |x - y|^{-\sigma}) \\ &\leq -\sigma(\sigma + 2)|x - y|^{-\sigma-4}a^{ij}(x_i - y_i)(x_j - y_j) + \sigma|x - y|^{-\sigma-2}a^{ij}\delta_{ij} + \sigma|x - y|^{0\sigma-2}b^i(x_i - y_i) \\ &\leq -\sigma(\sigma + 2)|x - y|^{-\sigma-2}\lambda + \sigma|x - y|^{-\sigma-2}n\Lambda + \sigma|x - y|^{-\sigma-1}\sup_i |b^i| \\ &= |x - y|^{-\sigma-2}(-\sigma(\sigma + 2)\lambda + \sigma n\Gamma + \sigma \sup_i |b^i| \cdot |x - y|). \end{aligned}$$

We now show that for appropriate $\sigma, \tau > 0$, we have $Lw \leq -1$ on $\overline{\Omega}$.

For each $x \in \overline{\Omega}$, by the triangle inequality, we have:

$$|x - y| \leq |x - \xi| + |\xi - y| \leq \text{diam}(\Omega) + R.$$

Using this we see that for $A > 0$ being large, we have:

$$Lw(x) \leq \tau|x - y|^{-\sigma-2}(A\sigma - \lambda\sigma^2) < -\tau|x - y|^{-\sigma-2} \leq -\tau(\text{diam}(\Omega) + R)^{-\sigma-2},$$

where in the above, we choose $\sigma > 0$ large to ensure $A\sigma - \lambda\sigma^2 < -1$. If we choose:

$$\tau > (\text{diam}(\Omega) + R)^{\sigma+2} > 0,$$

so we have $Lw \leq -1$ on $\overline{\Omega}$.

We now set for each $i \geq 1$:

$$w_i^\pm = \varphi(\xi) \pm \epsilon_i \pm k_i w \quad \text{on } \Omega \text{ for } \epsilon_i, k_i > 0.$$

As $w_i^\pm(\xi) = \varphi(\xi) \pm \epsilon_i$ by choosing any sequence $\epsilon_i \rightarrow 0$ we have that $w_i^\pm(\xi) \rightarrow \varphi(\xi)$ as $i \rightarrow \infty$. We then have that:

$$Lw_i^\pm = c\varphi(\xi) \pm c\epsilon_i \pm k_i Lw \quad \text{on } \Omega.$$

So, if we choose $k_i > \sup_\Omega |f - c\varphi(\xi)|$ for each $i \geq 1$, as $c \leq 0$ and $Lw \leq -1$ on $\overline{\Omega}$, we have:

$$Lw_i^+ \leq C\varphi(\xi) - k_i \leq f.$$

As $\varphi \in C^0(\partial\Omega)$, for each $\epsilon_i > 0$, we find $\delta_i > 0$ such that:

$$x \in B_{\delta_i}(\xi) \cap \partial\Omega \implies |\varphi(x) - \varphi(\xi)| \leq \epsilon_i,$$

by continuity. Hence we have $w_i^- \leq \varphi \leq w_i^+$ on $B_{\delta_i}(\xi) \cap \partial\Omega$.

As $w > 0$ on $\partial\Omega \setminus B_{\delta_i}(\xi)$, it has a strictly positive minimum on this set. Thus, up to increasing k_i , we can ensure that $w_i^- \leq \varphi \leq w_i^+$ on all of $\partial\Omega$.

Thus, $w_i^\pm$ are sub/super functions to the Dirichlet problem, respectively. As u is the supremum of all subfunctions we see that $w_i^- \leq u$ for each $i \geq 1$.

If $v \in C^0(\overline{\Omega})$ is any subfunction we have on $\partial\Omega$ that $v \leq \varphi \leq w_i^+$ for all $i \geq 1$. The first property of sub/super solution gives that $v \leq w_i^+$ on all of $\overline{\Omega}$. Hence, again, as u is the supremum of all subfunctions, we have $u \leq w_i^+$ on all of $\overline{\Omega}$. Hence we have $w_i^- \leq u \leq w_i^+$ on $\overline{\Omega}$.

As $w_i^\pm(\xi) \rightarrow \varphi(\xi)$ as $i \rightarrow \infty$, continuity of $w_i^\pm$ gives us that if $\epsilon > 0$, for large $i \geq 1$ we have $\delta > 0$, depending on i , such that both:

$$w_i^+ < \varphi(\xi) + \epsilon \quad \text{and} \quad w_i^- > \varphi(\xi) - \epsilon \quad \text{in } B_\delta(\xi) \cap \overline{\Omega}.$$

Combining with above gives that:

$$\varphi(\xi) - \epsilon < u < \varphi(\xi) + \epsilon \quad \text{in } B_\delta(\xi) \cap \overline{\Omega}.$$

Hence, u is continuous at ξ and $u(\xi) = \varphi(\xi)$ as $\xi \in \partial\Omega$ was arbitrary so we have u being continuous on $\partial\Omega$, so $u \in C^0(\overline{\Omega})$ and $u = \varphi$. $\square$

Corollary XII.4.10. If $\partial\Omega$ is $C^{2,\alpha}$ and $\varphi \in C^{2,\alpha}(\partial\Omega)$, then the perron solution, u , in fact solves the Dirichlet Problem and is such that $u \in C^{2,\alpha}(\overline{\Omega})$ with $u = \varphi$ on $\partial\Omega$.

Proof. We know that $u \in C^{2,\alpha} \cap C^0(\overline{\Omega})$. By the $C^{2,\alpha}$ regularity of $\partial\Omega$, we can find a $C^{2,\alpha}$ -diffeomorphism about each $z \in \partial\Omega$, so that for some $R > 0$ and $U \subset \mathbb{R}^n$ open, we have:

$$\psi : B_R(z) \rightarrow u, \quad \text{with } \overline{B} \subset \psi(\overline{\Omega} \cap B_R(z)) \text{ and } \psi(\Omega \cap B_R(z)) = u,$$

and we then set $T = \psi(\partial\Omega \cap B_{R/2}(z))$.

We have that, by choosing coordinates $xy = \psi(x)$, the PDE $Lu(x) = f(x)$ in Ω becomes $\tilde{L}\tilde{u}(y) = \tilde{f}(y)$ in u where $\tilde{u}(y) = \tilde{\varphi}(y)$ on T .

If we solve the Dirichlet problem:

$$\begin{cases} \tilde{L}v = \tilde{f}, & \text{in } B, \\ v = \tilde{u}, & \text{on } \partial B, \end{cases}$$

we obtain $v \in C^{2,\alpha}(B) \cap C^0(\bar{B})$ with $v \in C^{2,\alpha}(B \cup T)$ as $\tilde{u} = \tilde{\varphi} \in C^{2,\alpha}(T)$.

As $\tilde{L}(v - \tilde{u}) = 0$ in B and $v - \tilde{u} = 0$ on ∂B , since $v - \tilde{u} \in C^0(\bar{B})$ the maximum principle implies that $v = \tilde{u}$ in $\bar{B}$ so we have $\tilde{u} \in C^{2,\alpha}(B \cup T)$. After transforming back by ψ^{-1} , we get that $u \in C^{2,\alpha}(\bar{\Omega} \cap B_{R/2}(z))$. As $z \in \partial\Omega$ was arbitrary so we have $u \in C^{2,\alpha}(\bar{\Omega})$ as desired. $\square$

XII.5 Fredholm Alternative for Non-divergence Form Elliptic Operators

Recall that we had the Fredholm Alternative in Banach space in [Theorem VII.4.4](#), we will think of rephrasing this in a non-divergence form.

Theorem XII.5.1. Fredholm Alternative for Non-divergence Form Elliptic Operator.

If $\Omega \subset \mathbb{R}^n$ is open, bounded, connected, $\partial\Omega$ is $C^{2,\alpha}$, and L is a strictly elliptic, non-divergence form operator with coefficients in $C^{0,\alpha}(\bar{\Omega})$ for some $\alpha \in (0, 1)$, then precisely one of the following possibilities hold:

(i) The homogeneous Dirichlet problem:

$$\begin{cases} Lu = 0, & \text{in } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$

has a nonzero solution $u \in C^{2,\alpha}(\bar{\Omega})$.

(ii) Given any $f \in C^{0,\alpha}(\bar{\Omega})$ and $\varphi \in C^{2,\alpha}(\partial\Omega)$, the Dirichlet problem:

$$\begin{cases} Lu = f, & \text{in } \Omega, \\ u = \varphi, & \text{on } \partial\Omega, \end{cases}$$

has unique solution $u \in C^{2,\alpha}(\bar{\Omega})$.

Proof. Note that:

$$\begin{cases} Lu = f, & \text{in } \Omega, \\ u = \varphi, & \text{on } \partial\Omega, \end{cases}$$

has a unique solution if and only if:

$$Lu = f - L\varphi, \text{ in } \Omega, u = 0, \text{ on } \partial\Omega$$

has a unique solution.

Thus, without loss of generality, we reduce in the case $\varphi = 0$. Also, uniquely solving:

$$\begin{cases} Lu = f, & \text{in } \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases} \iff L : C_0^{2,\alpha}(\overline{\Omega}) \rightarrow C^{0,\alpha}(\overline{\Omega}).$$

Hence, if set $L_\sigma = L - \sigma$ for $\sigma > \sup_{\overline{\Omega}}(c)$, then we can apply the global Perron and Arzelá-Ascoli to get:

$$L_\sigma^{-1} : C^{0,\alpha}(\overline{\Omega}) \rightarrow C^{2,\alpha}(\overline{\Omega}) \xhookrightarrow{c} C^{0,\alpha}(\overline{\Omega})$$

being well defined. We use $(C^{0,\alpha}(\overline{\Omega}), |\cdot|_{0,\alpha;\Omega})$ as our Banach space. L_σ^{-1} is linear, since L is, and by global Schauder and maximum principle a priori estimates:

$$|L_\sigma^{-1}f|_{2,\alpha;\Omega} \leq C(|L_\sigma^{-1}f|_{0;\Omega} + |f|_{0,\alpha;\Omega}) \leq C(|f|_{0;\Omega} + |f|_{0,\alpha;\Omega}) \leq C|f|_{0,\alpha;\Omega},$$

for a constant $C > 0$, independent of f , thus L_σ^{-1} is bounded.

Also, L_σ^{-1} is compact by Arzelá-Ascoli. If $u \in C^{2,\alpha}(\overline{\Omega})$ and $f \in C^{0,\alpha}(\overline{\Omega})$, then:

$$Lu = f \iff L_\sigma u + \sigma u = f \iff u + \underbrace{\sigma L_\sigma^{-1} u}_K = L_\sigma^{-1} f.$$

So applying the Fredholm alternatives in Banach spaces, $K = \sigma L_\sigma^{-1}$ on $(C^{0,\alpha}(\overline{\Omega}), |\cdot|_{0,\alpha;\Omega})$, we have either:

- (i) $u + \sigma L_\sigma^{-1} u = 0$ has a nonzero solution $u \in C^{2,\alpha}(\overline{\Omega})$, or
- (ii) Each $f \in C^{0,\alpha}(\overline{\Omega})$, there is a unique solution $u \in C^{2,\alpha}(\overline{\Omega})$ to $u + \sigma L_\sigma^{-1} u = L_\sigma^{-1} f$.

By the above derivation, we have (i) and (ii) equivalent to the desired alternative. $\square$

Corollary XII.5.2. In both cases, the dimension of the subspace of solution to the Dirichlet problem is finite dimensional.

Proof. Let $H = \{u \in C^{2,\alpha}(\overline{\Omega}) : Lu = 0 \text{ in } \Omega \text{ and } u = 0 \text{ on } \partial\Omega\}$, then we have:

$$Lu = 0 \iff L_\sigma u + \sigma u = 0 \iff u = -\sigma L_\sigma^{-1} u,$$

so $|u|_{2,\alpha;\Omega} = |\sigma| \cdot |L_\sigma^{-1} u|_{2,\alpha;\Omega} \leq C|u|_{0,\alpha;\Omega}$.

Thus, by Arzelá-Ascoli, we have $(H, |\cdot|_{0,\alpha;\Omega})$ has compact unit ball if and only if H has finite dimensions. $\square$

Lemma XII.5.3. C^2 to $C^{2,\alpha}$.

If $\Omega \subset \mathbb{R}^n$ is open and L is strictly elliptic with coefficients in $C^{0,\alpha}(\Omega)$, then if $f \in C^{0,\alpha}(\Omega)$ and $u \in C^2(\Omega)$, solves $Lu = f$ in Ω , we in fact have that $u \in C^{2,\alpha}(\Omega)$.

Proof. Let $\tilde{L} = L - c$, then $\tilde{L}u = Lu - cu = f - cu \in C^{0,\alpha}(\Omega)$. Fix a ball $B \Subset \Omega$, and use Perron/ $C^{2,\alpha}$ Poisson on balls to find $v \in C^{2,\alpha}(\Omega) \cap C^0(\overline{\Omega})$ solving:

$$\begin{cases} Lu = f - cu, & \text{in } B, \\ v = u, & \text{on } \partial B. \end{cases}$$

Then, $\tilde{L}(v - u) = (f - cu) - (Lu - cu) = 0$ in B and $v - u = 0$ on ∂B , with $v - u \in C^0(\overline{B})$. Applying the weak maximum principle gives that $v = u$ in B so $u \in C^{2,\alpha}(B)$, since B was arbitrary, we have $u \in C^{2,\alpha}(\Omega)$. $\square$

Example XII.5.4. If $u \in C^2(\Omega)$ solves the minimum surface equation, i.e.:

$$\operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) = \left(\delta^{i,j} - \frac{D_i u D_j u}{1 + |\nabla u|^2} \right) D_{i,j} u = a^{i,j} D_{i,j} u = 0 \quad \text{in } \Omega,$$

then $u \in C^\infty(\Omega)$.

The reason is since $u \in C^2(\Omega)$, we have $D_i u \in C^1(\Omega)$, but then $a^{i,j} \in C^{0,\alpha}(\Omega)$, and so the lemma gives us that $u \in C^{2,\alpha}$. But then $D_i u \in C^{1,\alpha}(\Omega)$ so we have $a^{i,j} \in C^{1,\alpha}(\Omega)$ and higher interior Schauder gives use that $u \in C^{3,\alpha}(\Omega)$. Thus, inductively, $D_i u \in C^{2,\alpha}$ which implies that $a^{i,j} \in C^{2,\alpha}$, hence we have $u \in C^{4,\alpha}$, which tells us $u \in C^{k,\alpha}$ for all $k > 1$. $\diamond$

XIII Quasilinear Elliptic PDEs

XIII.1 From Elliptic to Quasilinear

We let $\Omega \subset \mathbb{R}^n$ to be open, bounded, connected, with $\partial\Omega$ being $C^{2,\alpha}$, and $a^{ij} \in C^{0,\alpha}(\overline{\Omega} \times \mathbb{R} \times \mathbb{R}^n)$ for $i, j = 1, \dots, n$ with $\alpha \in (0, 1)$ such that for each $(x, z, p) \in \overline{\Omega} \times \mathbb{R} \times \mathbb{R}^n$, the matrix $(a^{ij}(x, z, p))_{i,j}$ is positive definite (elliptic).

Definition XIII.1.1. Quasilinear Operator.

We defined the **Quasilinear operator** Q such that:

$$Qu = a^{ij}(x, u, \nabla u) D_{i,j}u + b(x, u, \nabla u) \quad \text{on } \overline{\Omega}.$$

┘

Example XIII.1.2. The minimum surface equation is quasilinear with $b \equiv 0$.

◇

For a boundary data $\varphi \in C^{2,\alpha}(\overline{\Omega})$ or $C^{2,\alpha}(\partial\Omega)$, we aim to solve the Dirichlet problem:

$$\begin{cases} Qu = 0, & \text{in } \Omega, \\ u = \varphi, & \text{on } \partial\Omega. \end{cases}$$

To solve this Dirichlet problem for Q , we use a fixed point theorem to reduce to establishing a priori estimates in an appropriate function space.

Recall that same principle as for linear non-divergence form earlier already.

Theorem XIII.1.3. Leray-Schauder Fixed Point Theorem.

If $(X, \|\cdot\|)$ is a Banach space and $T : X \rightarrow X$ is continuous and compact (not necessarily linear), then if there is a constant $M > 0$ such that whenever:

If $x \in X$ solves $x = \sigma T(x)$ for some $\sigma \in (0, 1]$, then we have $\|x\| \leq M$ (independent of σ).

We have some $\tilde{x} \in X$ with $\tilde{x} = T(\tilde{x})$, i.e., the a priori estimate guarantees a fixed point of T .

Remark XIII.1.4. $\tilde{x}$ may not be unique (unlike the contraction mapping principle).

┘

Proof of sketch (in finite dimensions). We claim, and do not prove, it suffice to assume that X is finite dimensions.

It is essentially since $T(\overline{B_M(0)})$ is compact, we may obtain a finite subcover and approximate T on this subcover and approximate T on this subcover.

Hence, suppose $X = \mathbb{R}^n$ assuming this claim and apply Brouwer's fixed point theorem (continuous self maps of closed balls have fixed points), and define (assuming without loss that $M = 1$ or Let $T_M(x) = \frac{1}{M}T(Mx)$):

$$\tilde{T}(x) = \begin{cases} T(x), & \text{if } \|T(x)\| \leq 1, \\ \frac{T(x)}{\|T(x)\|}, & \text{if } \|T(x)\| > 1. \end{cases}$$

So that $T : \overline{B_1(0)} \rightarrow \overline{B_1(0)}$ by restriction. Brouwer's theorem gives a point $x \in \overline{B_1(0)}$ with $x = \tilde{T}(x)$. If $\|T(x)\| \leq 1$, we have $\tilde{T}(x) = T(x)$, and we are done. Otherwise, $\|T(x)\| > 1$, and we let $\sigma = \|T(x)\|^{-1} \in (0, 1]$ so:

$$x = \sigma T(x),$$

and thus by the a priori estimate, we have $\|x\| < 1$, but $\|x\| = |\sigma| \cdot \|T(x)\| = 1$, which is a contradiction. $\square$

Let's apply this to the Dirichlet problem for Q and set $X = C^{1,\beta}(\overline{\Omega})$ for some $\beta \in (0, 1]$ to be chosen. We then set $T : X \rightarrow X$ to be the map sending $v \in X = C^{1,\beta}(\overline{\Omega})$ to the function $T(v) = u$ where u solves the linear Dirichlet problem:

$$\begin{cases} a^{ij}(x, v, \nabla v) D_{ij}u + b(x, v, \nabla v) = 0, & \text{in } \Omega, \\ u = \varphi, & \text{on } \partial\Omega, \end{cases}$$

i.e., use $v \in C^{1,\beta}(\overline{\Omega})$ to determine coefficients and give a strictly elliptic linear PDE of the form $a^{ij} D_{ij}u = f$ for $f = -b(x, v, \nabla v)$.

From our linear theory, we know that we can solve the above problem uniquely ($c = 0$ case) since $x \mapsto a^{ij}(x, v(x), \nabla v(x))$ and $x \mapsto -b(x, v(x), \nabla v(x))$ are in $C^{0,\alpha\beta}(\overline{\Omega})$, so we have $u \in C^{2,\alpha\beta}(\overline{\Omega}) \subset C^{1,\beta}(\overline{\Omega}) = X$, so T is well defined.

Note that fixed points of T gives solutions to the Dirichlet problem for Q .

If $v \in C^{1,\beta}(\overline{\Omega})$ is such that $v = \sigma T(v)$ for $\sigma \in (0, 1]$, then $u = T(v) = v/\sigma$ solves the above problem and so multiplying it by σ , we see that v solves that:

$$\begin{cases} a^{ij}(x, v, \nabla v) D_{ij}v + \sigma b(x, v, \nabla v) = 0, & \text{in } \Omega, \\ v = \sigma \varphi, & \text{on } \partial\Omega. \end{cases}$$

Hence, again, by applying the linear theory as above $T(v) \in C^{2,\alpha\beta}(\overline{\Omega})$, but then we have as $v = \sigma T(v)$ that $v \in C^{2,\alpha\beta}(\overline{\Omega})$ so the coefficients of the above problem are actually $C^{1,\alpha^2\beta}(\overline{\Omega}) \subset C^{0,\alpha}(\overline{\Omega})$ so we have $v \in C^{2,\alpha}(\overline{\Omega})$.

To see why T is compact, let $\{v_k\} \subset X = C^{1,\beta}(\overline{\Omega})$ be a bounded sequence so that $|v_k|_{1,\beta;\Omega} \leq C$ independently of k .

If we set $w_k = T(v_k)$, then by T 's definition, w_k solves:

$$\begin{cases} a^{ij}(x, v_k, \nabla v_k) D_{ij}w_k + b(x, v_k, \nabla v_k) = 0, & \text{in } \Omega, \\ w_k = \varphi, & \text{on } \partial\Omega. \end{cases}$$

Uniform bound on $\{v_k\}$ implies that bounds on $a_k^{ij}(x) = a^{ij}(x, v_k, \nabla v_k)$, $f_k(x) = -b(x, v_k, \nabla v_k)$ of the

form $|a_k^{ij}|_{0,\alpha\beta;\Omega}, |f_k|_{0,\alpha\beta;\Omega} \leq C$ and $a_k^{ij}(x)\xi_i\xi_j \geq \lambda|\xi|^2$ for all $x \in \overline{\Omega}$ and $\xi \in \mathbb{R}^n$.

By applying Global Schauder and the maximum principle a priori estimates, we have for all $k \geq 1$ that:

$$|w_k|_{2,\alpha\beta;\Omega} \leq C(|w_k|_{0;\Omega} + |f_k|_{0,\alpha\beta;\Omega} + |\varphi|_{2,\alpha\beta;\Omega}) \leq C.$$

Here, we apply Arzelá-Ascoli to the $\{w_k\}$ sequence, we obtain a subsequential C^2 convergence of the w_k to solve $w \in C^{2,\alpha\beta}(\overline{\Omega}) \subset C^{1,\beta}(\overline{\Omega}) = X$, i.e., T is compact.

To see why T is continuous, we let $v_k \rightarrow v$ in $C^{1,\beta}(\overline{\Omega}) = X$. By the above, $T(v_k) = w_k$ have uniform $C^{2,\alpha\beta}(\overline{\Omega})$ norm bounds and Arzelá-Ascoli implies subsequential C^2 -convergence to $w \in C^{2,\alpha\beta}(\overline{\Omega})$. Passing to the limit in $a_k^{ij}D_{ij}w_k = f_k$, agree up to subsequence, we have $w = T(v)$ by unique solvability of the system. But now, $\{T(v_k)\}$ is such that every subsequence has a subsequence that uniquely converges to $w = T(v)$, so we have:

$$T(v_k) \rightarrow T(v),$$

as well (check by contradiction), so we have T being continuous.

Now T is compact and continuous, so let Leray-Schauder tells use that the Dirichlet problem for Q is unique solvable for each $\varphi \in C^{2,\alpha}(\overline{\Omega})$ Provided we have an a priori estimate of the form:

There exists $\beta \in (0, 1]$ and $M > 0$ so that if $v \in C^{1,\beta}(\overline{\Omega})$ solves the σ -typed Dirichlet problem for $\sigma \in (0, 1]$ so we have $|v|_{1,\beta;\Omega} < M$.

Remark XIII.1.5. Such a v is actually in $C^{2,\alpha\beta}(\overline{\Omega})$ by our reasoning. ┘

XIII.2 Application to the Minimal Surface Equation

Recall that we reduced solvability of the Quasilinear Dirichlet problem, in $C^{2,\alpha}$ to establishing a priori estimates in $C^{1,\beta}$:

$$\{v \in C^{1,\beta}(\overline{\Omega}) \text{ solves Dirichlet Problem on } \sigma \text{ for } \sigma \in (0, 1] \implies |v|_{1,\beta;\Omega} < M \text{ for } \beta \in (0, 1] \text{ and } M > 0\}.$$

Remark XIII.2.1. Since Q is non-divergence form, the above regularity allows us to switch to a divergence form PDE by chain/product rule.

We saw that minimizers of energy $\mathcal{E}(u) = \int_{\Omega} L(\nabla u)$ for $L : \mathbb{R}^n \rightarrow \mathbb{R}$ smooth and uniformly convex solves the divergence form PDE $\operatorname{div}(DL(\nabla u)) = 0$ in Ω , which, if $u \in C^2$ is equivalent to $(D_{ij}L)(\nabla u)D_{ij}u = 0$ in Ω .

More generally, if one has an energy of the form:

$$\mathcal{E}(u) = \int_{\Omega} L(x, u, \nabla u) \quad \text{for } L : \overline{\Omega} \times \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R} \text{ via } (x, z, p) \mapsto L(x, z, p),$$

smooth and uniformly convex in the p variable ($D_p^2 L$ control).

Minimizers of such energies can be shown to solve the divergence form PDE:

$$\operatorname{div} (A(x, u, \nabla u)) + B(x, u, \nabla u) = 0 \quad \text{for} \quad \begin{cases} A^i(x, z, p) = \nabla_{p_i} L(x, z, p), \\ B(x, z, p) = -\nabla_z L(x, z, p). \end{cases}$$

This is a divergence form Quasilinear elliptic PDE. $\lrcorner$

Example XIII.2.2. Consider $L(x, z, p) = \sqrt{1 + |p|^2}$ as the minimum surface equation, then we have $B \equiv 0$ and $\operatorname{div} \left(\frac{\nabla u}{1 + |\nabla u|^2} \right) = 0$. $\diamond$

Remark XIII.2.3. De Giorgi-Nash theory also applies to such equations to give $C^{1,\alpha}$ regularity up to the boundary (if $\partial\Omega$ is regular enough). $\lrcorner$

We now focus on the minimum surface equation and try to establish an a priori $C^{1,\beta}$ bound. Recall:

$$\mathcal{M}(u) = \operatorname{div} \left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}} \right) = \left(\delta^{ij} - \frac{D_i u D_j u}{1 + |\nabla u|^2} \right) D_{i,j} u = 0 \text{ in } \overline{\Omega},$$

so we have $A(x, z, p) = \frac{p}{\sqrt{1 + |p|^2}}$ and $B(x, z, p) = 0$.

Given $\varphi \in C^{2,\alpha}(\overline{\Omega})$ for $\Omega \subset \mathbb{R}^n$ as open, bounded, connected set with $\partial\Omega$ being $C^{2,\alpha}$ for $\alpha \in (0, 1)$, we want to find $u \in C^{2,\alpha}(\overline{\Omega})$ solving $\begin{cases} \mathcal{M}(u) = 0, & \text{in } \Omega, \\ u = \varphi, & \text{on } \partial\Omega. \end{cases}$

Writing $\mathcal{M}(u) = \operatorname{div} (\tilde{A}(x) \nabla u) = 0$ for $\tilde{A}(x) = \frac{1}{\sqrt{1 + |\nabla u|^2}} \operatorname{Id}$. So the De Giorgi-Nash (up to the boundary) tells us we expect a priori $C^{1,\alpha}$ bounds on u if $0 < \lambda \operatorname{Id} \leq \tilde{A}(x) \leq \Lambda \operatorname{Id}$ for $x \in \overline{\Omega}$ and $\lambda, \Lambda > 0$. As $|\nabla u|^2 \geq 0$, we can take $\Lambda = 1$, but to ensure a uniform choice of $\lambda > 0$, we need upper bounds on $|\nabla u|$.

Thus an a priori $C^{1,\alpha}$ bound will follow from an a priori gradient bound on u .

It turns out that we can get an a priori gradient bound globally if and only if Ω is **mean-convex** (i.e., $\mathcal{M} \geq 0$ on $\partial\Omega$ graphs, or nonnegative mean curvature).

Lemma XIII.2.4. Maximum Principle for MSE.

If $\Omega \subset \mathbb{R}^n$ is open, bounded, connected, and $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ solves the minimum surface equation in Ω , then:

- (i) $\sup_{\Omega} |u| = \sup_{\partial\Omega} |u|$.
- (ii) $\sup_{\Omega} |\nabla u| \leq C \sup_{\partial\Omega} |\nabla u|$ for a dimensional constant $C > 0$, so we have $|u|_{1;\Omega} \leq \sup_{\partial\Omega} |u| + C \sup_{\partial\Omega} |\nabla u|$.

Proof of sketch. The first condition is just the usual weak maximum principle since $|\nabla u|$ is bounded on $\overline{\Omega}$

as $u \in C^1(\overline{\Omega})$.

The second follows by deriving an equation/PDE for $\partial_i u$ and then using a contradiction argument. $\square$

Lemma XIII.2.5. Comparison Principle for MSE.

If $\Omega \subset \mathbb{R}^n$ is open, bounded, connected, and $u, v \in C^2(\Omega) \cap C^1(\overline{\Omega})$ are such that both:

- (i) $\mathcal{M}(u) \geq \mathcal{M}(v)$ in Ω ,
- (ii) $u \leq v$ on $\partial\Omega$,

so we have $u \leq v$ on $\overline{\Omega}$.

Proof of sketch. We derive an equation/PDE for $u - v$ and use the fundamental theorem of calculus to get:

$$0 \leq D_i(a^{ij}(x)D_j(u - v)) \quad \text{for strictly elliptic } a^{ij},$$

which in non-divergence form is $L(u - v) = a^{ij}(u - v) + b^i D_i(u - v)$ so that the usual weak maximum principle gives:

$$\sup_{\Omega} |u - v| \leq \sup_{\partial\Omega} |u - v| \leq 0,$$

hence we have $u \leq v$ on $\overline{\Omega}$ as desired. $\square$

Provided we could construct barriers, $v^\pm$, which are sub/super-solutions to the MSE (i.e., $\mathcal{M}(v^+) \geq 0 \geq \mathcal{M}(v^-)$) in a neighborhood of the boundary of Ω with $v^\pm|_{\partial\Omega} = \varphi$, then if $v \in C^2(\Omega) \cap C^1(\overline{\Omega})$ solved the Dirichlet problem for the minimum surface equation with data φ , then the comparison principle allows us to control $\frac{\partial u}{\partial \nu}$. Namely, let us fix $y \in \partial\Omega$, $x \in \overline{\Omega}$ (close to $\partial\Omega$), then compute:

$$u(x) - u(y) = u(x) - \varphi(y) \leq v^+(x) - \varphi(y) = v^+(x) - v^+(y),$$

and so we have:

$$\frac{\partial u}{\partial \nu}(y) \leq \sup_{\partial\Omega} \left| \frac{\partial v^+}{\partial \nu} \right| \leq \sup_{\partial\Omega} |Dv^+|.$$

Similarly, we get a lower bound for:

$$\frac{\partial u}{\partial \nu}(y) \geq \inf_{\partial\Omega} |Dv^-| + \sup_{\partial\Omega} |Dv^-| + \sup_{\partial\Omega} |Dv^+|.$$

To summarize, constructing $v^\pm$ barriers as above yields the desired gradient/ C^1 bound on u by applying the maximum principle.